

## **Oracle® Product Hub**

Implementation Guide

Release 12.2

**Part No. E48841-10**

October 2025

Oracle Product Hub Implementation Guide, Release 12.2

Part No. E48841-10

Copyright © 1995, 2025, Oracle and/or its affiliates.

Primary Author: Laurel Dale

Contributing Author: Sangeetha Mani, Kashif Suleman

This software and related documentation are provided under a license agreement containing restrictions on use and disclosure and are protected by intellectual property laws. Except as expressly permitted in your license agreement or allowed by law, you may not use, copy, reproduce, translate, broadcast, modify, license, transmit, distribute, exhibit, perform, publish, or display any part, in any form, or by any means. Reverse engineering, disassembly, or decompilation of this software, unless required by law for interoperability, is prohibited.

The information contained herein is subject to change without notice and is not warranted to be error-free. If you find any errors, please report them to us in writing.

If this is software, software documentation, data (as defined in the Federal Acquisition Regulation), or related documentation that is delivered to the U.S. Government or anyone licensing it on behalf of the U.S. Government, then the following notice is applicable:

U.S. GOVERNMENT END USERS: Oracle programs (including any operating system, integrated software, any programs embedded, installed, or activated on delivered hardware, and modifications of such programs) and Oracle computer documentation or other Oracle data delivered to or accessed by U.S. Government end users are "commercial computer software," "commercial computer software documentation," or "limited rights data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, reproduction, duplication, release, display, disclosure, modification, preparation of derivative works, and/or adaptation of i) Oracle programs (including any operating system, integrated software, any programs embedded, installed, or activated on delivered hardware, and modifications of such programs), ii) Oracle computer documentation and/or iii) other Oracle data, is subject to the rights and limitations specified in the license contained in the applicable contract. The terms governing the U.S. Government's use of Oracle cloud services are defined by the applicable contract for such services. No other rights are granted to the U.S. Government.

This software or hardware is developed for general use in a variety of information management applications. It is not developed or intended for use in any inherently dangerous applications, including applications that may create a risk of personal injury. If you use this software or hardware in dangerous applications, then you shall be responsible to take all appropriate fail-safe, backup, redundancy, and other measures to ensure its safe use. Oracle Corporation and its affiliates disclaim any liability for any damages caused by use of this software or hardware in dangerous applications.

Oracle®, Java, MySQL, and NetSuite are registered trademarks of Oracle and/or its affiliates. Other names may be trademarks of their respective owners.

Intel and Intel Inside are trademarks or registered trademarks of Intel Corporation. All SPARC trademarks are used under license and are trademarks or registered trademarks of SPARC International, Inc. AMD, Epyc, and the AMD logo are trademarks or registered trademarks of Advanced Micro Devices. UNIX is a registered trademark of The Open Group.

This software or hardware and documentation may provide access to or information about content, products, and services from third parties. Oracle Corporation and its affiliates are not responsible for and expressly disclaim all warranties of any kind with respect to third-party content, products, and services unless otherwise set forth in an applicable agreement between you and Oracle. Oracle Corporation and its affiliates will not be responsible for any loss, costs, or damages incurred due to your access to or use of third-party content, products, or services, except as set forth in an applicable agreement between you and Oracle.

#### Documentation Accessibility

For information about Oracle's commitment to accessibility, visit the Oracle Accessibility Program website at <http://www.oracle.com/pls/topic/lookup?ctx=acc&id=docacc>.

#### Access to Oracle Support

Oracle customer access to and use of Oracle support services will be pursuant to the terms and conditions specified in their Oracle order for the applicable services.



---

# Contents

## Send Us Your Comments

## Preface

### 1 Setup Overview

|                                                |      |
|------------------------------------------------|------|
| Setting Up Oracle Applications Technology..... | 1-1  |
| Setup Prerequisites.....                       | 1-2  |
| Using Oracle iSetup.....                       | 1-2  |
| Setup Steps.....                               | 1-3  |
| Profile Options.....                           | 1-3  |
| Other Default Options.....                     | 1-15 |

### 2 Defining Responsibilities and Users

|                                          |     |
|------------------------------------------|-----|
| Defining Responsibilities and Users..... | 2-1 |
| Creating Responsibilities.....           | 2-1 |
| Creating Internal Users.....             | 2-4 |
| Setting Up Supplier Users.....           | 2-5 |
| Setting Up Customer Users.....           | 2-5 |

### 3 Setting Up the Item Catalog

|                                                               |      |
|---------------------------------------------------------------|------|
| Overview of Item Catalogs.....                                | 3-2  |
| Building the Item Catalog Text Index.....                     | 3-2  |
| Defining Item Catalog Categories.....                         | 3-2  |
| Defining Item Attachment Categories.....                      | 3-9  |
| Adding Attachment Categories to an Item Catalog Category..... | 3-10 |
| Defining Item Templates.....                                  | 3-12 |

|                                                                       |      |
|-----------------------------------------------------------------------|------|
| Managing Item Statuses.....                                           | 3-15 |
| Defining Change Policies.....                                         | 3-16 |
| Defining New Item Request Types.....                                  | 3-18 |
| Defining New Item Request Workflows.....                              | 3-21 |
| Associating New Item Requests with an Item Catalog Category.....      | 3-22 |
| Defining Item Numbers and Descriptions.....                           | 3-26 |
| Creating Pages for an Item Catalog Category.....                      | 3-28 |
| Managing Search Criteria.....                                         | 3-29 |
| Managing Display Formats.....                                         | 3-35 |
| Defining Item Catalog Import Formats.....                             | 3-40 |
| Assigning Systems to an Item Catalog Category.....                    | 3-44 |
| Publishing Item Catalog Categories.....                               | 3-45 |
| Defining Report Templates Using Business Intelligence Publisher ..... | 3-55 |
| Defining Cross Reference Types.....                                   | 3-57 |
| Importing Item Catalog Metadata.....                                  | 3-58 |

## 4 Setting Up Attributes and Functions

|                                                                 |      |
|-----------------------------------------------------------------|------|
| Defining Value Sets for User-Defined Attributes.....            | 4-1  |
| Defining Value Set Versions for Transaction Attributes.....     | 4-7  |
| Defining Table Value Sets.....                                  | 4-12 |
| Creating Independent Values.....                                | 4-14 |
| Publishing Value Sets.....                                      | 4-16 |
| Defining Item Attributes and Attribute Groups.....              | 4-21 |
| Defining Transaction Attributes.....                            | 4-33 |
| Using Seeded Attributes.....                                    | 4-36 |
| Using Additional Attributes.....                                | 4-38 |
| Creating User-Defined Functions for Item Attributes .....       | 4-38 |
| Defining Attribute Usages .....                                 | 4-44 |
| Displaying Attachments on Item Attribute Pages.....             | 4-45 |
| Associating Attribute Groups with an Item Catalog Category..... | 4-48 |
| Adding Actions to an Attribute Group.....                       | 4-52 |
| Publishing Attribute Groups.....                                | 4-54 |

## 5 Setting Up Catalogs

|                                                          |      |
|----------------------------------------------------------|------|
| Overview of Catalogs.....                                | 5-1  |
| Defining Catalog Categories.....                         | 5-4  |
| Defining Catalogs.....                                   | 5-6  |
| Assigning a Category to a Catalog.....                   | 5-9  |
| Mapping Catalog Categories.....                          | 5-10 |
| Administering a Category Hierarchy within a Catalog..... | 5-10 |

## **6 Administering Item Business Rules**

|                                                 |      |
|-------------------------------------------------|------|
| Overview of Item Business Rules.....            | 6-1  |
| Administering Rule Sets.....                    | 6-4  |
| Building Rule Expressions.....                  | 6-16 |
| Viewing Attributes Used in Rule Sets.....       | 6-37 |
| Including other Rule Sets.....                  | 6-38 |
| Viewing Rule Set Dependencies.....              | 6-39 |
| Assigning Rule Sets to Entities.....            | 6-40 |
| Using On-Demand Rule Set Execution.....         | 6-42 |
| Copying Item Rules Data to Other Instances..... | 6-43 |

## **7 Setting Up Change Categories and Types**

|                                                                     |      |
|---------------------------------------------------------------------|------|
| Overview of Change Categories and Types.....                        | 7-1  |
| Defining Change Categories.....                                     | 7-2  |
| Enabling Change Category Function Security.....                     | 7-5  |
| Creating Priority, Reason, and Classification Codes.....            | 7-13 |
| Creating Statuses.....                                              | 7-15 |
| Defining Workflow Templates.....                                    | 7-17 |
| Creating Task Templates.....                                        | 7-23 |
| Defining Header/Line Type Attributes and Attribute Groups.....      | 7-25 |
| Defining Header Types .....                                         | 7-28 |
| Defining Line Types.....                                            | 7-35 |
| Optimizing the Change Management Text Index.....                    | 7-39 |
| Defining Change Category Search Criteria and Display Formats.....   | 7-41 |
| Defining Change Category Reports.....                               | 7-42 |
| Subscribing to a Change Management Business Event.....              | 7-44 |
| Customizing Change Management Workflows.....                        | 7-47 |
| Implementing User Defined Functions for Change Type Attributes..... | 7-48 |

## **8 Setting Up Item Structures**

|                                                                  |      |
|------------------------------------------------------------------|------|
| Overview of Item Structures.....                                 | 8-1  |
| Defining Structure Types.....                                    | 8-2  |
| Associating a Structure Name with a Structure Type.....          | 8-4  |
| Defining Structure Attributes.....                               | 8-6  |
| Defining Component Attributes.....                               | 8-8  |
| Defining Display Formats and Search Criteria for Structures..... | 8-9  |
| Creating an Item Catalog Category Structure.....                 | 8-11 |

## **9 Setting Up Content Management Repositories**

|                                                                                             |     |
|---------------------------------------------------------------------------------------------|-----|
| Integrating Oracle E-Business Suite Applications with Content Management Repositories ..... | 9-1 |
| Setting Up a Content Management Repository.....                                             | 9-3 |

## **10 Setting Up E-Business Suite Attachments**

|                                               |      |
|-----------------------------------------------|------|
| Overview of E-Business Suite Attachments..... | 10-1 |
| Structuring Folders.....                      | 10-1 |
| Attachment Review and Approval.....           | 10-3 |
| Change Management for Attachments.....        | 10-5 |

## **11 Implementing Roles and Role Based Security**

|                                                         |       |
|---------------------------------------------------------|-------|
| Overview of Roles and Role Based Security.....          | 11-1  |
| Administering Roles.....                                | 11-3  |
| Administering People, Groups, and Companies.....        | 11-6  |
| Implementing Role Based Security.....                   | 11-11 |
| Implementing Change Management Role Based Security..... | 11-24 |
| Creating Custom Privileges.....                         | 11-27 |

## **12 Setting Up Inbound Product Data Synchronization and Data Quality Management**

|                                                             |      |
|-------------------------------------------------------------|------|
| Setup Overview of Inbound Product Data Synchronization..... | 12-1 |
| Creating Match Rules.....                                   | 12-2 |
| Defining Source Systems.....                                | 12-5 |

## **A Workflows for Change Management**

|                                      |     |
|--------------------------------------|-----|
| Workflows for Change Management..... | A-1 |
|--------------------------------------|-----|

## **B Item and Catalog Business Events**

|                      |     |
|----------------------|-----|
| Business Events..... | B-1 |
|----------------------|-----|

## **C Reading Data for User-Defined Attributes**

|               |     |
|---------------|-----|
| Overview..... | C-1 |
|---------------|-----|

## **D Synchronizing Item User-Defined Attributes with Item Descriptive Elements**

|               |     |
|---------------|-----|
| Overview..... | D-1 |
|---------------|-----|



## **E Change Management Business Events**

|                                                       |     |
|-------------------------------------------------------|-----|
| Description of Change Management Business Events..... | E-1 |
|-------------------------------------------------------|-----|

## **F Seeded Item Metadata Libraries**

|                                                         |      |
|---------------------------------------------------------|------|
| Sellable Product Information Library - Horizontal.....  | F-1  |
| Product Management Library - Horizontal.....            | F-23 |
| Communications Services Billing Library - Vertical..... | F-25 |
| Communications Product Details Library - Vertical.....  | F-44 |

## **G Web Services for Publishing**

|                                               |      |
|-----------------------------------------------|------|
| Oracle Product Hub Web Services Overview..... | G-1  |
| Value Set Web Service.....                    | G-2  |
| Attribute Group Web Service.....              | G-12 |
| Item Catalog Category Web Service.....        | G-18 |
| Item Web Service.....                         | G-30 |
| Controlling the Output Payload.....           | G-79 |
| Error Reporting.....                          | G-80 |

## **H Instructions for Restarting Concurrent Programs with a Manual Restart Mode**

|                                                                |     |
|----------------------------------------------------------------|-----|
| Restarting Concurrent Programs with a Manual Restart Mode..... | H-1 |
|----------------------------------------------------------------|-----|

## **Index**



---

# Send Us Your Comments

## **Oracle Product Hub Implementation Guide, Release 12.2**

### **Part No. E48841-10**

Oracle welcomes customers' comments and suggestions on the quality and usefulness of this document. Your feedback is important, and helps us to best meet your needs as a user of our products. For example:

- Are the implementation steps correct and complete?
- Did you understand the context of the procedures?
- Did you find any errors in the information?
- Does the structure of the information help you with your tasks?
- Do you need different information or graphics? If so, where, and in what format?
- Are the examples correct? Do you need more examples?

If you find any errors or have any other suggestions for improvement, then please tell us your name, the name of the company who has licensed our products, the title and part number of the documentation and the chapter, section, and page number (if available).

Note: Before sending us your comments, you might like to check that you have the latest version of the document and if any concerns are already addressed. To do this, access the new Oracle E-Business Suite Release Online Documentation CD available on My Oracle Support and [www.oracle.com](http://www.oracle.com). It contains the most current Documentation Library plus all documents revised or released recently.

Send your comments to us using the electronic mail address: [appsdoc\\_us@oracle.com](mailto:appsdoc_us@oracle.com)

Please give your name, address, electronic mail address, and telephone number (optional).

If you need assistance with Oracle software, then please contact your support representative or Oracle Support Services.

If you require training or instruction in using Oracle software, then please contact your Oracle local office and inquire about our Oracle University offerings. A list of Oracle offices is available on our Web site at [www.oracle.com](http://www.oracle.com).



---

# Preface

## Intended Audience

Welcome to Release 12.2 of the *Oracle Product Hub Implementation Guide*.

See Related Information Sources on page xiv for more Oracle E-Business Suite product information.

## Documentation Accessibility

For information about Oracle's commitment to accessibility, visit the Oracle Accessibility Program website at <http://www.oracle.com/pls/topic/lookup?ctx=acc&id=docacc>.

## Access to Oracle Support

Oracle customer access to and use of Oracle support services will be pursuant to the terms and conditions specified in their Oracle order for the applicable services.

## Structure

- 1 Setup Overview
- 2 Defining Responsibilities and Users
- 3 Setting Up the Item Catalog
- 4 Setting Up Attributes and Functions
- 5 Setting Up Catalogs
- 6 Administering Item Business Rules
- 7 Setting Up Change Categories and Types
- 8 Setting Up Item Structures
- 9 Setting Up Content Management Repositories
- 10 Setting Up E-Business Suite Attachments
- 11 Implementing Roles and Role Based Security
- 12 Setting Up Inbound Product Data Synchronization and Data Quality Management

- A Workflows for Change Management**
- B Item and Catalog Business Events**
- C Reading Data for User-Defined Attributes**
- D Synchronizing Item User-Defined Attributes with Item Descriptive Elements**
- E Change Management Business Events**
- F Seeded Item Metadata Libraries**
- G Web Services for Publishing**
- H Instructions for Restarting Concurrent Programs with a Manual Restart Mode**

## Related Information Sources

### *Oracle E-Business Suite User's Guide*

This guide explains how to navigate, enter and query data, and run concurrent requests using the user interface (UI) of Oracle E-Business Suite. It includes information on setting preferences and customizing the UI. In addition, this guide describes accessibility features and keyboard shortcuts for Oracle E-Business Suite.

### *Oracle Application Framework Personalization Guide*

This guide covers the design-time and run-time aspects of personalizing applications built with Oracle Application Framework.

### *Oracle Bills of Material User's Guide*

This guide describes how to create various bills of materials to maximize efficiency, improve quality and lower cost for the most sophisticated manufacturing environments. By detailing integrated product structures and processes, flexible product and process definition, and configuration management, this guide enables you to manage product details within and across multiple manufacturing sites.

### *Oracle Trading Community Architecture User Guide*

Oracle Trading Community Architecture (TCA) maintains information including relationships about parties, customers, organizations, and locations that belong to your commercial community in the TCA Registry. This guide enables you to use the features and user interfaces provided by TCA and by other Oracle E-Business Suite applications to view, create, and update Registry information. For example, you can import batches of party data in bulk from external source systems into the TCA Registry, merge duplicate parties, sites, and customer accounts, generate time zones for phones and locations, and run various customer reports.

### *Oracle Inventory User's Guide*

This guide describes how to define items and item information, perform receiving and inventory transactions, maintain cost control, plan items, perform cycle counting and physical inventories, and set up Oracle Inventory.

### *Oracle E-Business Suite Integrated SOA Gateway User's Guide*

This guide describes the high level service enablement process, explaining how users can browse and view the integration interface definitions and services residing in Oracle Integration Repository.

### *Oracle E-Business Suite Projects Fundamentals - Guide*

Oracle Project Fundamentals provides the common foundation shared across the Oracle Projects products (Project Costing, Project Billing, Project Resource Management, Project Planning and Control, and Project Portfolio Analysis). Use this guide to learn fundamental information about the Oracle Projects solution.

### *Oracle Workflow User's Guide*

This guide describes how users can view and respond to workflow notifications and monitor the progress of their workflow processes.

### *Oracle XML Publisher Administration and Developer's Guide*

This guide describes how to set up and administer Oracle XML Publisher as well as how to use the Application Programming Interface to build custom solutions.

## **Integration Repository**

The Oracle Integration Repository is a compilation of information about the service endpoints exposed by the Oracle E-Business Suite of applications. It provides a complete catalog of Oracle E-Business Suite's business service interfaces. The tool lets users easily discover and deploy the appropriate business service interface for integration with any system, application, or business partner.

The Oracle Integration Repository is shipped as part of the Oracle E-Business Suite. As your instance is patched, the repository is automatically updated with content appropriate for the precise revisions of interfaces in your environment.

## **Do Not Use Database Tools to Modify Oracle E-Business Suite Data**

Oracle **STRONGLY RECOMMENDS** that you never use SQL\*Plus, Oracle Data Browser, database triggers, or any other tool to modify Oracle E-Business Suite data unless otherwise instructed.

Oracle provides powerful tools you can use to create, store, change, retrieve, and maintain information in an Oracle database. But if you use Oracle tools such as SQL\*Plus to modify Oracle E-Business Suite data, you risk destroying the integrity of your data and you lose the ability to audit changes to your data.

Because Oracle E-Business Suite tables are interrelated, any change you make using an Oracle E-Business Suite form can update many tables at once. But when you modify Oracle E-Business Suite data using anything other than Oracle E-Business Suite, you may change a row in one table without making corresponding changes in related tables. If your tables get out of synchronization with each other, you risk retrieving erroneous information and you risk unpredictable results throughout Oracle E-Business Suite.

When you use Oracle E-Business Suite to modify your data, Oracle E-Business Suite automatically checks that your changes are valid. Oracle E-Business Suite also keeps track of who changes information. If you enter information into database tables using database tools, you may store invalid information. You also lose the ability to track who

has changed your information because SQL\*Plus and other database tools do not keep a record of changes.



---

## Setup Overview

This chapter covers the following topics:

- Setting Up Oracle Applications Technology
- Setup Prerequisites
- Using Oracle iSetup
- Setup Steps
- Profile Options
- Other Default Options

### Setting Up Oracle Applications Technology

This overview explains how to implement the parts of Oracle Applications. You need to complete several setup steps including:

- Perform system wide setup tasks such as configuring concurrent managers and printers
- Manage function and data security, which includes setting up responsibilities to allow access to a specific set of business data and complete a specific set of transactions, and assigning individual users to one or more of these responsibilities.

Also, if your product uses Oracle Workflow to manage the approval of change requests (for example) or to send change notifications, you need to set up Oracle Workflow.

### Related Topics

*Oracle Workflow Guide*

## Setup Prerequisites

Before setting up this application, you must complete the *required* setup steps for the following:

- In Oracle Inventory, see "Setting Up" in the *Oracle Inventory User's Guide*
- In Oracle Bills of Material, see "Setting Up" in the *Oracle Bills of Material User's Guide*
- In Oracle Engineering, see "Setting Up" in the *Oracle Engineering User's Guide*
- In Oracle Sourcing, if you intend to collaborate with suppliers complete the setup steps for defining suppliers see the *Oracle Sourcing Implementation and Administration Guide*
- In Oracle Receivables, if you intend to collaborate with customers complete the setup steps for defining customers see "Setting Up Receivables" in the *Oracle Receivables User's Guide*

## Using Oracle iSetup

If you are implementing a new instance of the Oracle e-Business Suite and have relatively standard implementation requirements, you can use Oracle iSetup to rapidly configure the following applications:

- Oracle Purchasing
- Oracle Inventory
- Oracle Bills of Material
- Oracle Order Management
- Oracle Shipping
- Oracle Costing
- Oracle Planning
- Oracle Engineering
- Oracle Work in Process
- Oracle General Ledger
- Oracle Payables

- Oracle Receivables
- Oracle Cash Management
- Oracle Assets

Oracle iSetup simplifies the setup process by providing templates and predefined setup steps that may or may not apply to your enterprise. You can use Oracle iSetup to migrate the following data:

- Item catalog category headers.
- Alternate catalog headers.
- Items and associated data.
- Roles and privileges.
- Structure names, structures, structure components, reference designators, and substitute components.

For more information and examples of using Oracle iSetup to migrate data from one instance of the Oracle E-Business Suite to another, refer to My Oracle Support ID 1504980.1.

## Setup Steps

Following is a summary of the setup steps:

- Defining Responsibilities and Users
- Defining the Item Catalog
- Defining Catalogs
- Defining Change Categories and Types
- Defining Item Structures
- Setting Up Content Management Repositories
- Defining Roles
- Implementing Role-Based Security

## Profile Options

Profile options enable or change the behavior of certain features in the application.

Following is a description of each of the system profiles used in this application, and their default values.

| Profile                                            | Description                                                                                                                                                                                            | Default Value                                                                                                                                                                          |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNA: Display mode for HTML-Based Product Workbench | Choose either a Standard or Compact layout. Both modes display the same information, but the compact layout displays more items or attribute rows per page in the HTML-based Product Workbench.        | Standard                                                                                                                                                                               |
| DNA: Hide Change Action column                     | Choose either Yes or No. If you add, delete, or change a component in a structure, an add, delete, or change icon appears next to the component, so the Change Action column information is redundant. | No                                                                                                                                                                                     |
| EGO: Attribute Group Sync WSDL URL                 | Profile for Installation Specific Attribute Group Sync WSDL URL. This is used only when publishing Attribute Groups to other Oracle Product Hub systems.                                               | Null<br><br><b>Additional Information:</b><br>Modify this profile value when publishing to other Oracle Product Hub systems as explained in Step 12 of My Oracle Support ID 1336146.1. |
| EGO: Enable exporting to Excel                     | Governs the action of exporting search results.<br><br>Changing the value to Yes enables users to export certain search results data.                                                                  | No                                                                                                                                                                                     |

| Profile                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Default Value |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| EGO: Enable GTIN Validation | <p>Governs whether GTIN validations are performed.</p> <p>Changing the value to No disables all GTIN validations.</p> <p><b>Additional Information:</b><br/>GTIN validations include:</p> <ul style="list-style-type: none"> <li>• Length is 14.</li> <li>• Contains a valid check digit.</li> <li>• The GTIN is unique and only assigned to one item.</li> <li>• Does not contain more than 6 leading zeros.</li> <li>• The third digit from the left in a GTIN must be 0, 1, 3, 6, 7, 8, or 9 when the second digit from the left is 0.</li> <li>• If a GTIN in a RCI message has six leading zeros, digits 7-9 must be between 301-968.</li> <li>• GTINs submitted in RCI messages cannot contain values 0980-0989 or 099 in the first 4 digits.</li> <li>• GTINs submitted in RCI messages cannot contain values 02, 04, 05, and 10-29 in the second and third</li> </ul> | Yes           |

| Profile                                                   | Description                                                                                                                                                            | Default Value |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                                           | digits from the left.                                                                                                                                                  |               |
| EGO: Enable GTIN Validation                               | Validates whether an item has a unique GTIN.<br><br>Changing the value to No enables users to specify the same GTIN for multiple items.                                | Yes           |
| EGO: Enable Oracle Collaborative Development              | Enables edit options for structures in HTML.<br><br>Oracle recommends leaving the value as No.                                                                         | No            |
| EGO: Enable Product Information Management Data Librarian | Enables the use of Product Hub-specific features such as Import Workbench if the value is Yes. See: Setup Overview of Inbound Product Data Synchronization, page 12-1. | No            |
| EGO: Enable Product Lifecycle Management                  | Enables the use of Product Lifecycle Management-specific features if the value is Yes.                                                                                 | No            |
| EGO GDSN Enabled                                          | Obsolete profile option.                                                                                                                                               | No            |

| Profile                                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Default Value   |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| EGO: Internal User Default Role for Items                              | <p>Governs the default role that is set up for internal users who access items. This role is assigned to internal users for all items in the organizations to which the users have access.</p> <p>Design Reviewer is seeded for Forms interoperability.</p> <p><b>Additional Information:</b><br/>Forms does not have data security, so users must be given additional privileges on specific items (either directly or via inheritance) to perform actions on itemsC.</p> <p>Users with the Design Reviewer role can view/access all item information. If you want to restrict access further, define a custom role and then reset the default role for all enterprise internal users.</p> <p>It is recommended that you always specify a role that minimally has the View Item privilege.</p> | Design Reviewer |
| EGO: Item Catalog Category required for Create Item using Spreadsheet. | <p>Select <b>Yes</b> to require the selection of an ICC when importing new items using a spreadsheet. Since attribute groups and other specifications are associated with an ICC, this ensures all item specifications are associated with the ICC upon creation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Null            |

| Profile                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Default Value                                                                                                                                                                          |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EGO: Item Catalog Category Sync WSDL URL | Profile for Installation Specific Item Catalog Category Sync WSDL URL. This is used only when publishing Item Catalog Categories to other Oracle Product Hub systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Null<br><br><b>Additional Information:</b><br>Modify this profile value when publishing to other Oracle Product Hub systems as explained in Step 12 of My Oracle Support ID 1336146.1. |
| EGO: ITEM RULE PROCESS SIZE              | <p>During item import, all item business rules are applied at one time to a certain number of items, specified in this profile option, during one iteration. If the profile option value is left blank (null), the number of items defaults to 100 internally.</p> <p>Adjust the number to achieve the best item import performance for your system configuration. This profile option applies to the following item import methods:</p> <ul style="list-style-type: none"> <li>• Importing Items Using a Spreadsheet</li> <li>• Import Workbench</li> <li>• Import Catalog Items</li> </ul> <p>See: Managing Item Imports, <i>Oracle Product Information Management User's Guide</i>.</p> | Null                                                                                                                                                                                   |



| Profile                                            | Description                                                                                                                                                                                                                                                                                                                                                                   | Default Value |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| EGO: Maximum Match Results                         | In Import Workbench while importing items from external source systems, users can run Match Rules to discover existing items in PIM Data Hub that match based on the criteria defined in the Match Rule. This option restricts the number of matches the matching process finds for every item. The matches display in the Matching Items section within the Unconfirmed tab. | 50            |
| EGO: Maximum number of attribute groups in Search. | <p>This site level profile option specifies the maximum number of attribute groups you can use as search criteria and in display formats.</p> <p><b>Caution:</b> Increasing the value could result in slow item search performance.</p>                                                                                                                                       | 5             |

| Profile                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Default Value |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| EGO: Maximum number of attribute groups in UDA extension framework cache. | <p>This site level profile option defines the maximum number of cached attribute groups. This ensures that there will be no performance issues while creating new display formats at the parent ICC level even when there are more than 100 attribute groups and valuesets involved.</p> <p>Changes made to this profile value are visible to all applications using the User Defined Attributes Framework. Entering a large value improves performance in Display Format, Criteria Template, and Search pages, but it also leads to larger memory utilization. Verify that your hardware has sufficient memory before increasing the value of this profile option.</p> | 100           |
| EGO: Maximum number of rows for sorting                                   | <p>Specifies the maximum number of rows to sort during an item search. If a search retrieves more than the maximum number of rows, the search results are displayed without sorting (a warning is displayed).</p> <p><b>Additional Information:</b><br/>Specifying a value greater than 2000 may have a negative impact on search performance.</p>                                                                                                                                                                                                                                                                                                                      | 2000          |
| EGO: Number of recent links to display                                    | Specifies the number of links to display in the Recently Visited list                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 20            |

| Profile                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Default Value |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| EGO: Restrict actions in Item form | <p>Governs the Forms actions for items.</p> <p>Set to Yes to ensure that data level role-based security is honored in Forms. When set to Yes users will be unable to access the following functionality in Forms:</p> <ul style="list-style-type: none"> <li>• Categories</li> <li>• Catalog</li> <li>• Revisions</li> <li>• Organization Assignments</li> <li>• Attachments (disables attachments)</li> </ul> <p>Keep the default value No if you wish to make all actions available in Forms.</p> | No            |
| EGO: Stats Collection Threshold    | This option sets the threshold for the number of records at which the statistics collection program will be executed on the interface table.                                                                                                                                                                                                                                                                                                                                                        | 100           |
| EGO: Thesaurus for Matching        | <p>Enables the use of a synonym search when matching incoming external items to Oracle Product Information Management Data Hub (PIMDH) items if the value is Yes. See: Creating Match Rules, page 12-2.</p> <p>A thesaurus is not included by default in the database. If you load a thesaurus into the database, then set this option to Yes.</p>                                                                                                                                                  | No            |

| Profile                        | Description                                                                                                                                                                                                                                                                                                                                   | Default Value                                                                                                                                                                          |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EGO: User Organization Context | <p>This option stores the organization selected by a user. When the user logs in to the system, the system retrieves the organization from this option and uses it for the session.</p> <p>Change the organization stored in this profile option by using the Change Organization function within the Development Manager responsibility.</p> | -1                                                                                                                                                                                     |
| EGO: Value Set Sync WSDL URL   | <p>Profile for Installation Specific Value Set Sync WSDL URL. This is used only when publishing Value Sets to other Oracle Product Hub systems.</p>                                                                                                                                                                                           | Null<br><br><b>Additional Information:</b><br>Modify this profile value when publishing to other Oracle Product Hub systems as explained in Step 12 of My Oracle Support ID 1336146.1. |
| Enable PIM for Telco Features  | <p>This site (organization) profile option enables the use of telecommunications industry features such as item catalog category/value set versions and transaction attributes if the value is set to Yes.</p>                                                                                                                                | No                                                                                                                                                                                     |

| Profile                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Default Value |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| ENG: Internal User Default Role for Changes | <p>Governs the default role that is set up for internal users who access change objects. This role is assigned to internal users for all change objects in the organizations to which the users have access.</p> <p>Reviewer is seeded for Forms interoperability.</p> <p><b>Additional Information:</b><br/>Forms does not have data security, so users must be given additional privileges on specific change objects (either directly or via inheritance) to perform actions on change objects.</p> <p>Users with the Reviewer role can view/access all information for a change object. If you want to restrict access further, define a custom role and then set reset the default role for all enterprise internal users.</p> <p>It is recommended that you always specify a role that minimally has the View Item privilege.</p> | Reviewer      |

| Profile                             | Description                                                                                                                                                                                                                                                                                                                                                                   | Default Value |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| ENG: ALLOW AUTO-SUBMIT WORKFLOW     | <p>This option enables the change order creation PL/SQL API to automatically explode the new change order workflow routings and submit the change orders for approval.</p> <p>Setting this option to No requires users to manually explode new change order workflow routings and submit the change orders for approval after using the change order creation PL/SQL API.</p> | No            |
| INV:Sync iProcurement Item Index    | <p>Setting this option to Yes causes the Item Catalog Text Index to run each time an item is created or updated.</p> <p>Setting this option to No prevents the index from automatically running. The concurrent program 'Item Catalog Text Index Build' must be manually scheduled to run.</p>                                                                                | Yes           |
| Item Simple Search Delimter         | <p>Enables the use of item delimiters when performing a simple search. For example, an item's name is Item A. The description for Item A is X\Y\Z. if you enter "\" as a delimiter, then you can enter the search criteria X\ and the search result is X\Y\Z.</p>                                                                                                             | Null          |
| PWB: Copy BOM Default Org Selection | <p>This option automatically selects the common structure for any selected organization when copying a structure in the Product Workbench.</p>                                                                                                                                                                                                                                | Select All    |

## Other Default Options

The EGO\_DEFAULT\_OPTIONS table enables you to specify default behavior during the item creation process. Your database administrator can use SQL to access this table and set the default value for each option. The following table lists all of the options available in EGO\_DEFAULT\_OPTIONS:

| Option                   | Description                                                                                                                                                                                                                                | Default Value |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| EGO_ASSIGN_PACK_SUPPLIER | Controls whether or not all lower levels of a packaging hierarchy are assigned the same suppliers as those assigned to the parent pack level. See: <i>Assigning Suppliers to Items, Oracle Product Hub User's Guide.</i>                   | Y             |
| EGO_ASSIGN_PACK_SUP_SITE | Controls whether or not all lower levels of a packaging hierarchy are assigned the same supplier sites as those assigned to the parent pack level. See: <i>Assigning Suppliers to Items, Oracle Product Hub User's Guide.</i>              | Y             |
| EGO_ASSIGN_PACK_SS_ORG   | Controls whether or not all lower levels of a packaging hierarchy are assigned the same supplier site organizations as those assigned to the parent pack level. See: <i>Assigning Suppliers to Items, Oracle Product Hub User's Guide.</i> | Y             |
| EGO_ASSIGN_PACK_CATEGORY | Controls whether or not all lower levels of a packaging hierarchy are assigned the same item catalog category as the parent pack level. See: <i>Assigning an Item to a Category, Oracle Product Hub User's Guide.</i>                      | Y             |

| Option                             | Description                                                                                                                                                                                                                             | Default Value |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| EGO_UNASSIGN_PACK_CATEGORY         | Controls whether or not you must individually assign an item catalog category to each level of a packaging hierarchy. See: <i>Assigning an Item to a Category, Oracle Product Hub User's Guide.</i>                                     | Y             |
| EGO_ASSIGN_PACK_ORG                | Controls whether or not any packs associated with an item are assigned to the same organizations as the item. See: "To assign a pack to an organization:" in <i>Enabling Organization Assignments, Oracle Product Hub User's Guide.</i> | Y             |
| EGO_DEFAULT_STYLE_SUP_AND_SUP_SITE | Controls whether or not any suppliers and supplier sites associated with a style item are assigned during item creation to the style's SKU items. See: <i>Assigning Suppliers to Items, Oracle Product Hub User's Guide.</i>            | Y             |
| EGO_DEFAULT_STYLE_ITEM_ORG         | Controls whether or not the SKU items associated with a style item are assigned to the same organizations as the style item during item creation. See: <i>Enabling Organization Assignments, Oracle Product Hub User's Guide.</i>       | Y             |
| EGO_DEFAULT_STYLE_SUP_SITE_ORG     | Controls whether or not any supplier site organizations associated with a style item are assigned during item creation to the style's SKU items. See: <i>Assigning Suppliers to Items, Oracle Product Hub User's Guide.</i>             | Y             |



| Option                                  | Description                                                                                                                                                                                                                                                                                                                                                                                | Default Value |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| EGO_DEFAULT_STYLE_ALT<br>ERNATE_CATALOG | Controls whether or not any alternate catalogs associated with a style item are assigned during item creation to the style's SKU items. See: <i>Assigning an Item to a Category, Oracle Product Hub User's Guide.</i>                                                                                                                                                                      | Y             |
| EGO_DEFAULT_STYLE_LC_<br>PHASES         | Controls whether or not a lifecycle and phases associated with a style item are assigned during item creation to the style's SKU items. See: <i>Managing Item Phases, Oracle Product Hub User's Guide.</i>                                                                                                                                                                                 | Y             |
| EGO_DEFAULT_STYLE_LC_<br>PROJS          | Controls whether or not a lifecycle tracking project associated with a style item is assigned during item creation to the style's SKU items.                                                                                                                                                                                                                                               | Y             |
| EGO_DEFAULT_STYLE_PEO<br>PLE            | Controls whether or not roles assigned to a style item are assigned during item creation to the style's SKU items. See: <i>Implementing Role Based Security, page 11-11.</i>                                                                                                                                                                                                               | Y             |
| EGO_SKU_CONCAT_VA_D<br>ELIM             | This delimiter value is used when creating function generated SKU item numbers and descriptions by concatenating the variant attribute values. For example, if the delimiter value is 3, then a shirt SKU item number with the variant attribute values of Color = Blue and Size = Medium might be shirtBluMed. See: <i>Creating Style and SKU Items, Oracle Product Hub User's Guide.</i> | (blank)       |



---

# Defining Responsibilities and Users

This chapter covers the following topics:

- Defining Responsibilities and Users
- Creating Responsibilities
- Creating Internal Users
- Setting Up Supplier Users
- Setting Up Customer Users

## Defining Responsibilities and Users

You can securely collaborate with internal users, suppliers, and customers. You can set up users and grant them responsibilities to enable specific business functions. You can further restrict access to specific items, catalogs, or change management objects with role-based security.

| Task                      | Required? |
|---------------------------|-----------|
| Creating Responsibilities |           |
| Creating Internal Users   | Yes       |
| Setting Up Supplier Users |           |
| Setting Up Customer Users |           |

## Creating Responsibilities

The top-level menu for the seeded responsibilities (Development Manager,

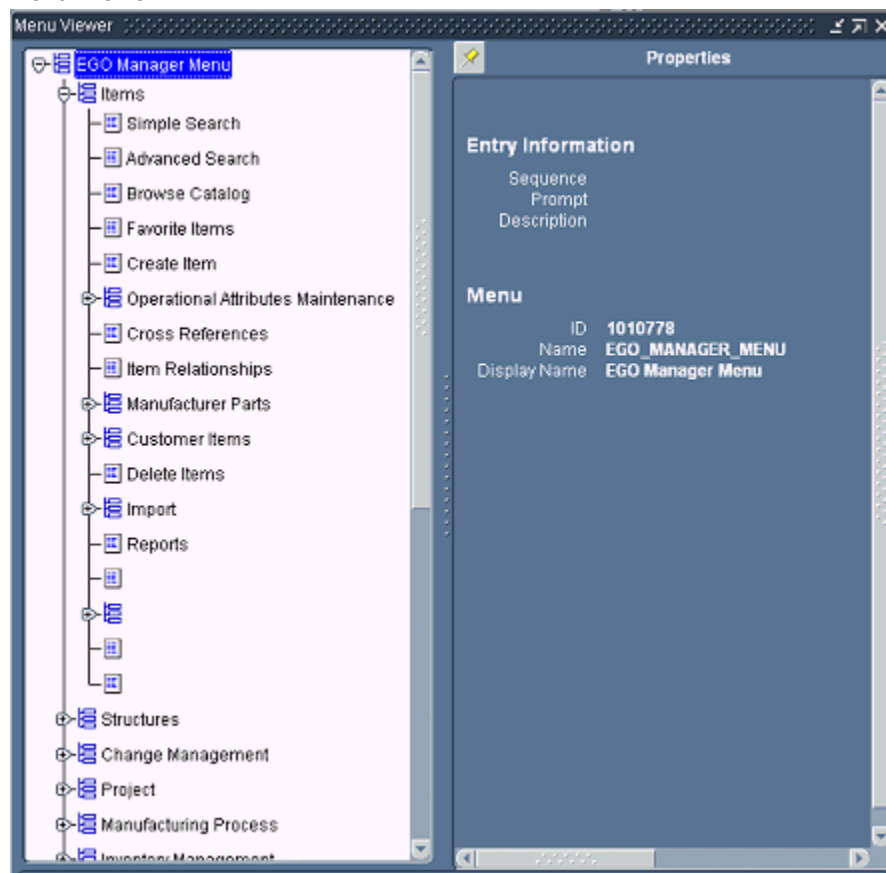
Development Engineer, and Supplier Engineer) is the EGO Manager Menu (EGO\_MANAGER\_MENU); you should view the seeded menus and functions before creating ones of your own. To view the seeded menus and functions, open the EGO Manager Menu in the Application Developer's Menu form and click the View Tree button. In the Menu Viewer you can expand each menu as you decide which menus to include in your custom responsibilities. If a node appears with no prompt under a menu, then you should include that function to ensure that all the menu items work correctly. You can use menu exclusion rules when defining your responsibilities if you need to exclude any specific menus or functions for a user.

#### ***Top level menu with seeded responsibilities***

The screenshot shows the 'Menus' form in Oracle. The 'Menu' field is set to 'EGO\_MANAGER\_MENU'. The 'User Menu Name' is 'EGO Manager Menu'. The 'Menu Type' and 'Description' fields are empty. A 'View Tree...' button is visible. Below the form fields is a table with 10 rows of seeded responsibilities.

| Seq | Prompt          | Submenu                | Function | Description | Grant                               |
|-----|-----------------|------------------------|----------|-------------|-------------------------------------|
| 1   | Items           | EGO Items              |          |             | <input checked="" type="checkbox"/> |
| 2   | Structures      | EGO Structures         |          |             | <input checked="" type="checkbox"/> |
| 3   | Change Manage   | EGO Change Managem     |          |             | <input checked="" type="checkbox"/> |
| 4   | Project         | EGO Projects           |          |             | <input checked="" type="checkbox"/> |
| 5   | Manufacturing F | EGO Routings           |          |             | <input checked="" type="checkbox"/> |
| 6   | Inventory Manag | EGO Inventory Manag    |          |             | <input checked="" type="checkbox"/> |
| 7   | Cost Manageme   | EGO Cost Management    |          |             | <input checked="" type="checkbox"/> |
| 8   | Quality Manage  | EGO Quality Manageme   |          |             | <input checked="" type="checkbox"/> |
| 9   | CADView-3D      | CADView-3D User All M  |          |             | <input checked="" type="checkbox"/> |
| 10  |                 | EGO Product Intelligen |          |             | <input checked="" type="checkbox"/> |

### Menu Viewer



### Defining responsibilities with menu exclusions

The screenshot shows the 'Responsibilities' configuration window in Oracle. The 'Supplier Engineer' responsibility is defined for the 'Oracle Engineering Online' application with the key 'EGO\_SUPPLIER\_ENGINEER'. It is available from Oracle Applications, Self Service Web Applications, and Mobile Applications. The menu is set to 'EGO Manager Menu'. The data group is 'Standard' and the request group is empty. The 'Menu Exclusions' tab is active, showing a table of excluded items.

| Type     | Name                                  | Description                               |
|----------|---------------------------------------|-------------------------------------------|
| Function | Engineering Create Change Order       |                                           |
| Function | Projects: Project Sets: Create and De | Projects: Project Sets: Create and Delete |
| Menu     | Project Manager Responsibility Main   |                                           |
| Menu     | Project Manager Role Menu             | Project Manager Role Menu                 |

## Creating Internal Users

You can define E-Business Suite users and assign them responsibilities that grant them access to specific E-Business Suite applications. After creating an employee, customer or supplier, create a user account to which you then link the employee, customer, or supplier. Each internal user should have an employee ID with an e-mail address specified so he or she can receive notifications. You can assign internal users a custom responsibility or one of the following seeded responsibilities:

- Development Engineer
- Development Manager
- Manufacturing Engineer
- Product Manager
- Program Manager
- Supplier Engineer

## Setting Up Supplier Users

You can securely collaborate with suppliers. You can set up E-Business Suite users for each of the suppliers with whom you collaborate on product and component designs.

Assuming the supplier company has already been defined in Oracle Purchasing or Oracle Sourcing, you can register a supplier user in Oracle Sourcing using the Sourcing Super User responsibility. You need to specify the supplier's e-mail address as the user ID so that the system can e-mail the supplier his or her password. After the supplier user is registered via Oracle Sourcing, you can assign him or her a responsibility such as Supplier Engineer, which specifies the functions available to that supplier user. For example, the seeded responsibility Supplier Engineer provides a restricted set of functions that enable the supplier to view specific items (secured with item roles), create issues and change requests for an item, but not create or view change orders.

For more details about registering and setting up supplier users, see the *Oracle Sourcing Implementation and Administration Guide* and *Oracle E-Business Suite Security Guide*.

## Setting Up Customer Users

You can securely collaborate with your customers. You can set up E-Business Suite users for each of the customers with whom you collaborate on product and component information.

Assuming that you have already defined the customer company in the Oracle E-Business Suite, create customer users by selecting the Trading Community Manager responsibility. In the Customers – Standard form create a new Customer Type of Person. You must specify a valid Address.

Then navigate to the Party Relations window and assign the person to the customer company as follows:

### **Relationship Type**

EMPLOYMENT

### **Relation**

Employee Of

### **Object**

Company name

### **Start Date**

Specify a start date. After assigning a person to the customer company, select the System Administrator responsibility to define the customer user. Then, in the Users form, enter the required information and specify the customer person name in the Customer field.

For more details about setting up customer users, see the *Oracle Receivables Users Guide*

and *Oracle E-Business Suite Security Guide*.

### Creating a new customer person

The screenshot shows the 'Customers - Standard' form in Oracle. The form is divided into several sections. The top section contains fields for 'Customer Type' (set to 'Person'), 'Active' (checked), 'Person Number' (26655), 'Prefix' (Ms.), 'First Name' (Jane), 'Customer Number' (3148), 'Middle Name' (empty), 'Last Name' (Williams), 'Suffix' (empty), 'Tax Registration Num' (empty), 'Alternate Name' (empty), and 'Account Name' (empty). Below this is a tabbed interface with tabs for 'Addresses', 'Classification', 'Order Management', 'Marketing', 'Communication', 'Contacts', and 'Contacts : Roles'. The 'Addresses' tab is selected, showing a list of addresses. The first address is '620 Main St., Redwood City, San Mateo, CA, 94062, United States'. There is a checkbox for 'Show Only Active Addresses'. At the bottom of the form are buttons for 'Duplicate Check', 'Party Relationships', 'New', and 'Open'.

| Field                | Value                               |
|----------------------|-------------------------------------|
| Customer Type        | Person                              |
| Active               | <input checked="" type="checkbox"/> |
| Person Number        | 26655                               |
| Prefix               | Ms.                                 |
| First Name           | Jane                                |
| Customer Number      | 3148                                |
| Middle Name          |                                     |
| Last Name            | Williams                            |
| Suffix               |                                     |
| Tax Registration Num |                                     |
| Alternate Name       |                                     |
| Account Name         |                                     |

Addresses | Classification | Order Management | Marketing | Communication | Contacts | Contacts : Roles

Address ☐ Show Only Active Addresses

| Address                                                         |
|-----------------------------------------------------------------|
| 620 Main St., Redwood City, San Mateo, CA, 94062, United States |
|                                                                 |
|                                                                 |
|                                                                 |
|                                                                 |
|                                                                 |
|                                                                 |
|                                                                 |

Duplicate Check | Party Relationships | New | Open



### Setting up party relations for a new customer person

Party Relations

Party Type: **PERSON** Party Number: **26655**

Party Name:

**Party Relations**

| Relationship Type | Relation           | Object                | Object Type   | Start Date         | End Date | Comments |
|-------------------|--------------------|-----------------------|---------------|--------------------|----------|----------|
| <b>EMPLOYMENT</b> | <b>Employee Of</b> | <b>A. C. Networks</b> | <b>ORGANI</b> | <b>03-SEP-2003</b> |          |          |
|                   |                    |                       |               |                    |          |          |
|                   |                    |                       |               |                    |          |          |
|                   |                    |                       |               |                    |          |          |
|                   |                    |                       |               |                    |          |          |
|                   |                    |                       |               |                    |          |          |
|                   |                    |                       |               |                    |          |          |
|                   |                    |                       |               |                    |          |          |
|                   |                    |                       |               |                    |          |          |

### Defining a customer user

**Users**

User Name: **JWILLIAM**

Description:

Password:

**Password Expiration**

- ☒ Days
- ☐ Accesses
- ☐ None

Person:

Customer: **Jane Williams**

Supplier:

E-Mail:

Fax:

**Effective Dates**

From: **03-SEP-2003**

To:

**Responsibilities** **Securing Attributes**

| Responsibility              | Application               | Description | Security Group  | Effective Dates    |
|-----------------------------|---------------------------|-------------|-----------------|--------------------|
|                             |                           |             |                 | From To            |
| <b>Development Engineer</b> | <b>Oracle Engineering</b> |             | <b>Standard</b> | <b>03-SEP-2003</b> |
|                             |                           |             |                 |                    |
|                             |                           |             |                 |                    |
|                             |                           |             |                 |                    |
|                             |                           |             |                 |                    |

---

## Setting Up the Item Catalog

This chapter covers the following topics:

- Overview of Item Catalogs
- Building the Item Catalog Text Index
- Defining Item Catalog Categories
- Defining Item Attachment Categories
- Adding Attachment Categories to an Item Catalog Category
- Defining Item Templates
- Managing Item Statuses
- Defining Change Policies
- Defining New Item Request Types
- Defining New Item Request Workflows
- Associating New Item Requests with an Item Catalog Category
- Defining Item Numbers and Descriptions
- Creating Pages for an Item Catalog Category
- Managing Search Criteria
- Managing Display Formats
- Defining Item Catalog Import Formats
- Assigning Systems to an Item Catalog Category
- Publishing Item Catalog Categories
- Defining Report Templates Using Business Intelligence Publisher
- Defining Cross Reference Types
- Importing Item Catalog Metadata

## Overview of Item Catalogs

Oracle enables you to define hierarchies of items using catalogs and catalog categories. The Item Catalog contains all of the items in the system; all other catalogs are subsets of the Item Catalog. For example, the Product Catalog contains all of the items in the Item Catalog that are sold. Oracle provides several predefined catalogs, such as the Product Catalog, Asset Catalog, and the Service Catalog. Customers can also define additional catalogs for browsing and reporting purposes. Each catalog includes a hierarchy of categories, with each catalog item assigned to a catalog category.

This chapter describes many tasks that enable you to use item catalog features. You only need to complete the following two tasks to use the item catalog; perform the other tasks only if you want the ability to use additional item catalog features.

- Building the Item Catalog Text Index
- Defining Item Catalog Categories

## Related Topics

Using the Item Catalog chapter, *Oracle Product Hub User's Guide*

## Building the Item Catalog Text Index

To take advantage of the Oracle Text search features available such as keyword search, stemming search, and fuzzy search— run the concurrent program Item Catalog Text Index Build. Any user with the Development Manager responsibility can submit this concurrent program from the Submit Request window. Select the concurrent request parameter Action=CREATE when submitting the request.

## Defining Item Catalog Categories

The *item catalog* contains all items defined for your company or organization. Within the item catalog, *item catalog categories* provide a way to logically classify items that have similar characteristics. For example, you can classify a computer monitor in an item catalog category named Computer Parts and Components. Item catalogs are hierarchical, and relationships between the various item catalog categories (such as an item's catalog category attributes) are inherited. You can define a rich hierarchical taxonomy of item catalog categories that enable you to set up item attribution, lifecycles, and their change policies, security, attachment categories, results formats, criteria templates, and import formats.

When defining item catalog categories, do the following:

- Create a list of all your items.

- Classify the items in unique item catalog categories and sub-categories that suit your business needs.
- Consider which category is a sub-classification within a category (for example, SRAM, DRAM could be subcategories of the item catalog category Memory).
- List the required and optional item attributes for each item catalog category.

**Additional Information:** Catalog Groups defined in Oracle Inventory appear as Item Catalog Categories.

Item catalog categories provide metadata common to all items that share the category. For example, the item catalog category "Engine" describes attributes, functions and other characteristics common to several item numbers, such as Engine M10000 and Engine M20000. Item catalog categories are hierarchically structured, with characteristics (such as attribute groups, functions, pages, lifecycles, criteria templates, and result formats) inherited throughout the hierarchy.

Following are some of the basic item catalog category terms:

- **Item Catalog Categories**

Item catalog categories inherit attributes from the parent category; so consider which attributes to associate at higher levels in the item catalog category. Inheritance helps to simplify maintenance.

- **Item Lifecycles**

Item lifecycles enable you to track and control the lifecycle phases of items. You can specify an item lifecycle for each item catalog category. Item lifecycles are inherited from the parent item catalog category.

- **Change Policy**

Item behavior is described by defining a change policy for each lifecycle phase in a given lifecycle.

- **Security**

You can control item creation, viewing and update access by assigning a role on the item catalog category to a user or group of users.

- **Attachment Categories**

You can classify a file that is attached to an item using an attachment category. Attachment categories are inherited from the parent catalog category, and those most commonly used can be defined at a higher node. You can then add more specific attachment categories for the child item catalog categories.

## Creating Versions of Item Catalog Categories

You can choose to create a different version of an item catalog category (ICC) when you change a transaction attribute or structure component in the ICC if you set the profile option "Enable PIM for Telco Features" to Yes. Enabling the use of ICC versions enables the use of transaction attributes and ICC structures. Transaction attributes differ from other types of attributes such as operational or user-defined attributes because the choices of transaction attribute values for an item can change depending on the date. For more information about transaction attributes, see *Defining Transaction Attributes*, page 4-33.

ICC structures enable you to create a default structure for use by all items created within a certain item catalog category. For more information about ICC structures, see *Creating an Item Catalog Category Structure*, page 8-11.

**Caution:** When you enable the use of ICC versions, the system automatically creates a draft version for every new ICC created. You cannot choose to create versions for some ICCs, but not others.

Once you set the profile option "Enable PIM for Telco Features" to Yes, you must create a released version for each existing ICC. Oracle provides an upgrade API that automatically performs this task. Your database administrator can invoke the upgrade API from the back end once you enable the Telco features.

**Warning:** You must turn on the "Enable PIM for Telco Features" profile option before invoking the upgrade API. Do not disable the "Enable PIM for Telco Features" profile option by changing the value from 'Y' to 'N' after it has been enabled. If it is disabled and enabled again, you must invoke the upgrade API again to maintain data integrity and consistent functionality.

### Upgrade API Sample Call

```
DECLARE
  uid NUMBER;
  rid NUMBER;
  rad NUMBER;
  sgid NUMBER;
BEGIN
  SELECT
    USER_ID, RESPONSIBILITY_ID, RESPONSIBILITY_APPLICATION_ID,
    SECURITY_GROUP_ID
  INTO
    uid, rid, rad, sgid
  FROM
    FND_USER_RESP_GROUPS
  WHERE
    USER_ID = (SELECT USER_ID FROM FND_USER WHERE USER_NAME =
    'SYSADMIN')
    AND RESPONSIBILITY_ID =
    (SELECT RESPONSIBILITY_ID FROM FND_RESPONSIBILITY_VL WHERE
    RESPONSIBILITY_KEY = 'SYSTEM_ADMINISTRATOR');
  FND_GLOBAL.apps_initialize (uid, rid, rad, sgid);
  ego_p4t_upgrade_pvt.upgrade_to_pim4telco(null);
END;
```

When creating a child ICC version, the Draft version inherits the transaction attributes and structure components defined in the latest released versions of any of its parent categories. You can then define additional attributes or structure components for the child ICC, in addition to the attributes and structure components of the parent ICC. Any new transaction attributes and structure components added to a parent ICC draft version and released in a new version appear in all child ICC draft versions unless the new parent ICC version is released with a start date in the future.

When an item is assigned to an ICC that contains transaction attributes or an ICC structure, links to the **Transaction Attributes** and **Structure** pages appear in the item's **Overview** page in the same way as links to other ICC pages containing operational and user-defined attributes appear. The **Transaction Attributes** page shows the attribute details in read-only mode, but you cannot assign an attribute value since transaction attribute values are assigned when a transaction takes place. Any items created at any given time within an ICC with versions inherits the ICC version effective during item creation.

### To create an item catalog category:

The following steps apply to creating an item catalog category with the profile option "Enable PIM for Telco Features" set to either Yes or No.

1. On the item **Search: Item Catalog Categories** page, click the Create Item Catalog Category link.
2. On the **Create Item Catalog Category** page, enter the following:

#### Catalog

Defines the category name. The number of segments (and respective labels) for item

catalog category depends on how the item catalog group flexfield is defined in Oracle Inventory setup.

**Note:** Any leading or ending white spaces in the key flexfield segments are trimmed automatically.

### Description

Optionally enter a brief description of the item catalog category.

### Parent Item Catalog Category

Specify the immediate parent of the item catalog category you are creating, thereby determining the item catalog category hierarchy.

### Default Template

Optionally specify an item template to use to populate the **Create Item** page for any item of the item catalog category you are creating. On the **Create Item** page, you can modify the template applied upon item creation or choose not to apply a template.

### Item Creation Allowed

Select to specify that items of the item catalog category can be created (if you do not select, the category only serves as a placeholder in the catalog hierarchy). If item creation is not allowed, the category typically specifies characteristics (attribute groups, etc.) that are inherited by its descendants.

### Inactivate

Optionally specify a date on which to inactivate the item catalog category. You cannot specify an inactive date that is later than the inactive date of an item catalog category's parent, nor can you specify an inactive date that has already passed. Also, all children of a parent catalog category with an inactive date should be made inactive at the same time or earlier. Making a category inactive has the following implications:

- You cannot create items in that item catalog category or any of its descendants.
- You cannot use that item catalog category or any of its descendants as the parent catalog category upon creation of an item catalog category.

### 3. Click Apply.

The system returns you to the **Basic Information** page. From here, you can perform all remaining item catalog category setup steps described in this chapter.

This is the last step when creating an item catalog category with the profile option "Enable PIM for Telco Features" set to No. If this profile option is set to Yes, continue with the following steps.



## To release an item catalog category draft version

**Important:** You can only perform the following steps if the profile option "Enable PIM for Telco Features" is set to Yes and from within the Setup Workbench (HTML interface). You cannot access item catalog category versions from the Forms interface.

Clicking Apply in the previous step creates a locked item catalog category draft version. Only the user listed in the Locked By field can release a draft version. Add all transaction attributes and associate a structure before releasing the draft version to create a numbered version. Once you release an item catalog category, you cannot update the released version. The following information associated with the item catalog category is not included in a version. If you make any changes to the following information, the changes apply to all versions.

- Basic information
- People
- Attribute groups, both operational and user-defined
- Item pages
- Lifecycles
- Attachment categories
- Search criteria
- Display formats
- Import formats
- Templates
- Report templates
- Item creation details

**Additional Information:** The system chooses which numbered version of the item catalog category to use for an item based on the versions' start and end dates. Only an item catalog category with a numbered version that has a date range which encompasses the item's effective date can be assigned to an item.

4. In the **Basic Information** page, click Release.

### To lock and unlock a draft item catalog category version

A version enabled item catalog category is automatically locked upon creation of the draft. Only the user listed in the Locked By field can update the item catalog category transaction attributes or structure or release the item catalog category to create a new version. To enable another user to update the item catalog category, unlock it. The other user can then lock it and make updates or release the item catalog category. Locking the item catalog category only locks the draft version's transaction attributes, structure, version description, and start date. You can update the item catalog category primary information, such as description and the inactivate date, regardless of the lock status. This primary information applies to all versions.

5. On the **Basic Information** page, if the draft is locked and you are the user who locked it, the Unlock button is visible. Click Unlock.

The Locked By field displays the name of the person who locked it. Only the person displayed in the Locked By field can unlock the draft.

6. If the draft is unlocked, the Locked By field is blank and the Lock button is visible. Click Lock.

### To update an item catalog category

Before updating an item catalog category draft version, you must lock it.

To update an item catalog category with no versions, refer to Updating Item Catalog Categories, *Oracle Product Hub User's Guide*.

7. To change the item catalog category primary (header) information for all item catalog category versions, in the **Basic Information** page, click the Update button.
8. To change the draft version's description or start date, in the **Basic Information** page, click the draft version's Update icon.
9. To change the draft version's transaction attributes or structure, in the **Basic Information** page, click the Draft link.

On the **Transaction Attributes** page, you can update, delete, or add transaction attributes to the item catalog category. See: Defining Transaction Attributes, page 4-33.

On the **Structure** page, you can update, add, or delete components and validate the structure. See: Creating an Item Catalog Category Structure, page 8-11.

### To revert the draft item catalog category to an earlier version

The Revert Draft to button enables you to refresh the draft version's characteristics with a released version's characteristics.

10. Select a released version that contains the characteristics that you want the draft version to contain.

11. Click Revert Draft to.

The draft version now contains the same characteristics as the released version selected.

## Related Topics

Updating Item Catalog Categories, *Oracle Product Hub User's Guide*

Defining Transaction Attributes, page 4-33

Defining Item Attributes and Attribute Groups, page 4-21

Creating an Item Catalog Category Structure, page 8-11

## Defining Item Attachment Categories

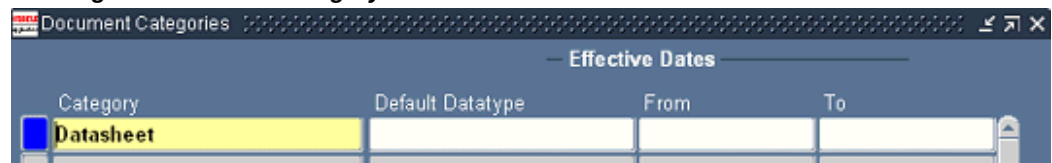
Whenever users add attachments to an item, they specify an *attachment category*. Item attachment categories provide a way to classify attachments—this classification identifies the purpose of the attachment to the business object, and can also be used as one of the search criteria within an attachment list. If users do not specify an attachment category for an item catalog category, and none are inherited from its parent catalog category, they always have the option of using the attachment category Miscellaneous.

**Additional Information:** Attachment categories inherited from a parent catalog category cannot be deleted at the child catalog category level.

Attachment categories are defined by the system administrator and are available across the various applications of the E-Business Suite. Attachment categories that you define can be made available to any E-Business Suite application. Thus, attachment categories are shared across the E-Business Suite applications and this should be kept in mind when defining any new attachment categories. Rules defining the usage of attachment categories are defined independently by each application.

You can define attachment categories in the Document Categories window. Before defining a new attachment category, verify that it does not already exist. If it already exists, then you can enable it for item catalog categories.

### Defining an attachment category



Related Topics

Adding Attachment Categories to an Item Catalog Category, page 3-10

Adding Attachment Categories to an Item Catalog Category

Once you define an attachment category (also known as a document category), you can enable it for an item by associating it with an item catalog category.

All item catalog categories inherit attachment categories (in the same way they inherit item attributes) from their parent catalog category. So, before associating attachment categories with item catalog categories, create a list of all attachment categories available for use in your item catalog. Consider which attachment categories are common among item catalog categories within each branch of the item catalog hierarchy. If an attachment category is applicable to all the child catalog categories, then the attachment category should be associated with the parent catalog category.

When associating attachment categories with item catalog categories, you can specify the effectivity of this association by providing a Start Date and an End Date. If an attachment category association has an end date, an attachment in that category cannot be used for the items belonging to the category beyond the specified date.

Associating an attachment category with an item catalog category

Attachment Categories

Item Catalog Category: Motherboards

Select Object: Delete | Update Add

Previous 1-10 Next 10

Select All | Select None

| Select Attachment Category                  | Item Catalog Category | Start Date | End Date |
|---------------------------------------------|-----------------------|------------|----------|
| <input type="checkbox"/> Document Reference | PLM High Tech         |            |          |
| <input type="checkbox"/> Specification      | Motherboards          |            |          |
| <input type="checkbox"/> Document           | Motherboards          |            |          |
| <input type="checkbox"/> Datasheet          | Motherboards          |            |          |
| <input type="checkbox"/> Design             | Motherboards          |            |          |
| <input type="checkbox"/> Collateral         | Motherboards          |            |          |
| <input type="checkbox"/> To Buyer           |                       |            |          |
| <input type="checkbox"/> To Mobile Receiver |                       |            |          |
| <input type="checkbox"/> To Mobile Putaway  |                       |            |          |
| <input type="checkbox"/> To Mobile Picker   |                       |            |          |

Select Object: Delete | Update Add

Previous 1-10 Next 10

To add attachments to an item, users need to specify an attachment category. Attachment categories sort and organize attachments that users might attach to an item or items. They provide a way to classify attachments—this classification identifies the purpose of the attachment to the business object, and can also be used as one of the search criteria within an attachment list. If users do not specify an attachment category for an item catalog category, and none are inherited from its parent catalog category,

they always have the option of using the attachment category "Miscellaneous".

When moving attachment categories, only those attachments whose categories are common between the source and target catalog categories will be retained.

By adding attachment categories, you can classify all attachments that you add to specific item catalog categories.

**Note:** Attachment categories inherited from a parent catalog category cannot be deleted at the child catalog category level. Attachment categories can be inherited from a parent catalog category *and* explicitly added to the catalog category itself. Administrators should not associate the same attachment category with an item catalog category in this manner.

The following attachment categories are available for all item catalog categories:

- Description
- Image
- Item Internal
- Miscellaneous
- To Buyer
- To Supplier
- To Receiver
- To Mobile Picker
- To Mobile Putaway
- To Mobile Receiver

**To add an attachment category to an item catalog category:**

1. On the **Search: Item Catalog Categories** page, select the item catalog category to which you wish to attach an attachment category.
2. On the **Basic Information** page of the item catalog category, click the Attachment Categories link.
3. On the **Attachment Categories** page, click Add.
4. On the **Search and Select: Attachment Categories** page, select an attachment

category from the list. You can also search within the list for a specific attachment category and then select it.

5. Click Apply.

## Related Topics

Defining Item Attachment Categories, page 3-9

## Defining Item Templates

You can specify one or more item templates during the item creation process. The system applies the templates in the order listed, so later templates overwrite attribute values applied with earlier templates. An item template is a defined set of attribute values used during item creation. When you apply a template to an item, you overlay or default in the set of attribute values to the item definition. For example, every time users in a particular organization create new items, the attributes as defined and approved by the organization appear in the appropriate fields--no user guesswork is required, and much time is saved during the creation of items with a similar form, fit and function.

You can create item templates that contain the operational attributes for an item, which are visible to either one or all organizations and either one or all item catalog categories. You can also include user-defined attributes in a template after associating the template to a specific item catalog category. When you explicitly associate a template with an item catalog category, it is made available for all create and update item actions in that item catalog category. Also, the values for the user-defined attributes that are specified in the template are inherited from parent to child within the particular item catalog category. In order to specify user-defined attributes in a template, you must associate it with an item catalog category.

### Benefits:

- Default operational attributes, user-defined attributes, or both to speed up the process of specifying item attribute values.
- Enforce consistency of item attribute values for items belonging to certain item catalog categories or item types. For example, you may have an item template for all purchased computer hard-drive assemblies.

For more details about creating item templates that use operational attributes, as well as a listing of the Oracle provided templates, see *Item Templates, Oracle Inventory User's Guide*.

### To define an item template and specify operational attributes:

1. Navigate to the Setup Workbench. In the Items tab, select the Templates subtab.

2. On the **Search: Templates** page, click Create.
3. In the Oracle Forms **Find Item Templates** window, click New.
4. In the Item Template window, provide the item template name and description, and then specify the operational attributes and the default values. When finished, save the template.

#### **To associate a template with an item catalog category:**

1. Navigate to the Setup Workbench. In the Items tab, Item Catalog Categories subtab, search for and select an item catalog category to which to associate an item template.
2. In the **Basic Information** page of the item catalog category, from the vertical tabs on the left side of the page, select the Templates tab.
3. On the **Templates** page, click Add Template.  
  
Associate item templates with any item catalog category. Item templates are inherited down through the item catalog hierarchy.
4. On the **Add Templates to Item Catalog Category** page, locate a template, select it and click Apply.

#### **To add user-defined attributes to an item template:**

1. On the **Templates** page for an item catalog category, click the template name to which you wish to add user-defined attributes.
2. On the **Update Template** page, select the user-defined attributes you wish to add to the template:
  - Select an attribute group, then click Go.
  - Select the user-defined attributes to add to the template. You can also choose to set the default attribute values.
  - Click Enable.

### Updating item template for user-defined item category attributes

[Items: Item Catalog Categories](#) > [Basic Information](#) > [Templates](#) >

#### Update Template

Item Catalog Category:

Template **Purchased Item**  
Description **Purchased Item**  
Organization  
☒ Set as Default

Attribute Group

Multi Row **No**

#### Attribute Values

| Select Attributes and ... <input type="button" value="Enable"/> <input type="button" value="Clear"/> |                                  |                                      |                       |                                     |
|------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------|-----------------------|-------------------------------------|
| <a href="#">Select All</a>   <a href="#">Select None</a>                                             |                                  |                                      |                       |                                     |
| Select Attribute                                                                                     | Value                            |                                      | Item Catalog Category | Enabled                             |
| <input type="checkbox"/> Optimal Temperature                                                         | <input type="text" value="100"/> | <input type="text" value="Celsius"/> | Motherboards          | <input checked="" type="checkbox"/> |
| <input type="checkbox"/> Humidity                                                                    | <input type="text" value="10"/>  | <input type="text" value="%"/>       | Motherboards          | <input checked="" type="checkbox"/> |
| <input type="checkbox"/> Maximum Wet Bulb Temperature                                                | <input type="text" value="20"/>  | <input type="text" value="Celsius"/> | Motherboards          | <input checked="" type="checkbox"/> |
| <input type="checkbox"/> Minimum Dew Point                                                           | <input type="text" value="65"/>  | <input type="text" value="Celsius"/> | Motherboards          | <input checked="" type="checkbox"/> |
| <input type="checkbox"/> Chassis Fan Speed                                                           | <input type="text"/>             |                                      | Motherboards          | <input type="checkbox"/>            |

You can also make this the default item template for all new items created in this item catalog category by selecting Set as Default.

Click Apply.

### How to Use Item Templates:

1. During the new item creation process you can specify one or more item templates or use the default if one was set up at the item catalog category level. See: *Creating New Items, Oracle Product Hub User's Guide*

**Caution:** If applying more than one template, specify the order to apply them. The last template's attribute values override any previously applied template attribute values.

2. The operational attribute values specified in an item template are not set for 'unapproved' items (items that require a new item request). Apply the item template manually once the item is approved.

### Related Topics

*Item Templates, Oracle Inventory User's Guide*

*Defining Item Templates, Oracle Inventory User's Guide*

*Copying Item Templates, Oracle Inventory User's Guide*



## Managing Item Statuses

An item status is assigned to a lifecycle phase; in fact, each lifecycle phase can have multiple item statuses associated with it. Create or edit item statuses before associating them with lifecycle phases.

### To create an item status:

1. On the **Search: Item Catalog Categories** page, click the Lifecycle Phases tab.
2. On the **Search: Lifecycle Phases** page, click the Item Statuses link.
3. On the **Search: Item Statuses** page, click the Create Item Status link.
4. On the **Create Item Status** page, provide the following information:

#### Status

The name of the item status.

#### Description

The description of the item status.

#### Inactive Date

The date on which the item status is no longer valid and available for use.

5. Select the Value checkboxes that correspond to the attributes that you wish to associate with the item status.
6. Click Apply.

### To edit an item status:

1. In the Applications tree menu, click the Setup Workbench link.
2. On the **Search: Item Catalog Categories** page, click the Lifecycle Phases tab.
3. On the **Search: Lifecycle Phases** page, click the Item Statuses link.
4. On the **Search: Item Statuses** page, select the item status that you wish to edit.
5. On the **Item Status Details** page, click Update.
6. On the **Edit Item Status** page, update the Description, Inactive Date, or, select an attribute you wish to apply.
7. Click Apply.

## Defining Change Policies

Administrators can define and update item change policies for item catalog categories. These item change policies determine the rules for how and when an item's attributes, attachments, and associations are changed. For example, a company manufacturing an engine that has hundreds of specifications may wish to define item change policies for the different phases of development. So when the engine is in the concept or design phase, many of the attributes are allowed to change without formal approval, and the lifecycle policy is not very restrictive. When the engine progresses to the prototype phase, the company might wish to place all attributes, attachments, and associations under stricter change control, perhaps requiring a change order for all modifications. Later, when the engine is in the production phase, the company will likely want every facet of the engine's production under tight control.

Defining item change policies enables a company to:

- Specify whether or not item changes are allowed in a particular lifecycle phase of an item lifecycle.
- Specify what types of changes (attributes, attachments, associations, structures) are allowed for an item in each lifecycle phase.
- Specify whether or not a change order is required to make particular changes to an item in a specific lifecycle phase.

**Note:** Change policies inherited from a parent catalog category cannot be edited at the child catalog category level.

If you do not define a change policy for an item, then by default changes are allowed in all lifecycle phases.

Item change policies only take effect after the item has been approved.

On the **Update Change Policy for Attributes (or Associations or Attachments)** page, select the appropriate attribute group (or association or attachment category) and the change policy you wish to apply for the particular phase. The valid change policies are:

- Allowed
- Not Allowed
- Change Order Required

## Setting lifecycle change policy

### Update Change Policy for: Attributes

Item Catalog Category: Motherboards Lifecycle: Computer Component Lifecycle

| Concept                                                                                 | Design                    | Prototype            | Pre-Production                       | Production | Retirement |
|-----------------------------------------------------------------------------------------|---------------------------|----------------------|--------------------------------------|------------|------------|
| <b>Prototype</b>                                                                        |                           |                      |                                      |            |            |
| Select an attribute and... <a href="#">Remove</a> <span>Previous 1-4 of 6 Next 2</span> |                           |                      |                                      |            |            |
| Select All   Select None                                                                |                           |                      |                                      |            |            |
| Select Attribute Group                                                                  |                           |                      |                                      |            |            |
| <input type="checkbox"/>                                                                | Assembly Clearances       | <a href="#">Info</a> | Change Policy: Allowed               |            |            |
| <input type="checkbox"/>                                                                | Benchmark Ratings         | <a href="#">Info</a> | Change Policy: Allowed               |            |            |
| <input type="checkbox"/>                                                                | Operating Conditions      | <a href="#">Info</a> | Change Policy: Change Order Required |            |            |
| <input type="checkbox"/>                                                                | Motherboard Specification | <a href="#">Info</a> | Change Policy: Change Order Required |            |            |
| <input type="checkbox"/>                                                                | Memory Configurations     | <a href="#">Info</a> | Change Policy: Change Order Required |            |            |
| <a href="#">Add Another Row</a>                                                         |                           |                      |                                      |            |            |
| Change Policy: Allowed                                                                  |                           |                      |                                      |            |            |
| Change Policy: Change Order Required                                                    |                           |                      |                                      |            |            |
| Change Policy: Not Allowed                                                              |                           |                      |                                      |            |            |

[Cancel](#) [Apply](#)

**Additional Information:** Before establishing a change policy for attributes or attachments, you must first associate the attribute group or attachment category with the item catalog category.

### To update an item change policy:

1. On the **Search: Item Catalog Categories** page, search for the item catalog category (see: Browsing Item and Alternate Catalogs , *Oracle Product Hub User's Guide*) for which you would like to define an item change policy, and click its name link.
2. On the **Basic Information** page for the item catalog category, click the Lifecycles link.
3. On the **Lifecycles** page, locate the lifecycle for which you wish to define a change policy. If no lifecycles are listed, add the lifecycle for which you wish to define item change policies.
4. From the View pulldown list (for attributes, associations, attachments, structures, or promotion/demotion), select the item change policy defined for the selected lifecycle, and then select the lifecycle phase you want to update. Click Go.
5. From the Update pulldown list, select the lifecycle phase you want to update. Click Go.
6. On the **Update Change Policy for: Attributes** (or Associations, Attachments, Structures, or Promote and Demote) page, select the appropriate attribute group (or association, attachment category, structure, or promotion, demotion, or revision) and the change policy you wish to apply to the particular phase. The valid change

policies are:

- Allowed
- Not Allowed
- Change Order Required

**Important:** Before establishing a change policy for attributes or attachments, you must first associate the attribute group or attachment category with the item catalog category.

You can switch back and forth among the various lifecycle phases via the tabs on the **Update Change Policy for: Attributes** (or Associations, Attachments, Structures, or Promote and Demote) page.

## Defining New Item Request Types

New item request is a workflow process, which enables you to route the definition and approval of a new item. When creating a new item, various people in the organization can define various aspects of an item, such as operational attributes, user-defined attributes, AML, and so on. using a workflow process. You can also combine the definition routing and approval routing in a new item request workflow.

*New Item Request* is a seeded Change Management category for which you must define a type before you can set up new item requests for an item catalog category. new item request (NIR) types are defined like any other change type. The topic Defining Header Types, page 7-28 provides additional type definition details.

Once a specific NIR type is created, the administrator can assign additional information, such as the associating NIR-related attributes. Examples of such attributes are Cost Impact and Implementation Cost. You can then organize these attribute groups in different pages, similar to Associating Attribute Groups with an Item Catalog Category, page 4-48.

When users create items in item catalog categories that require an item request, the new item request type specified for the item catalog category determines how the request is routed for distribution and approval, and identifies the approvers for the new item. For example, the new item request type determines what item request attributes and sections are available on pages; it determines the valid priorities and reasons available to users; it also determines the valid statuses for the request (for example, Open, Approval, and Closed). In essence, the new item request change type defines the process for routing the collection of item attributes and approval requests from various people who are internal (such as Engineering and Manufacturing) or external to your organization (such as Suppliers).

## Prerequisites

- ☐ Define one or more workflow templates. See:
  - Defining Workflow Templates, page 7-17
  - Defining New Item Request Workflows, page 3-21

### To define a new item request type:

1. On the Change Management **Categories** page in the Setup Workbench, select the New Item Request category, then select the Types tab in the lower half of the page. When the page refreshes click Create.

#### Create New Item Request Type

**Create New Item Request Type**

\* Indicates required field

\* Name: New item Request for Outsourced Components

Description: New item Request for Outsourced Components

\* Start Date: 01-Apr-2005 (example: 17-Mar-2005)

End Date: (example: 17-Mar-2005)

**Autonumbering**

\* Number Generation: Sequence Generated

\* Prefix: NIR\_EXT

\* Next Available Number:

**Subject Selection Criteria**

\* Subject: New Item

**Default Assigned To**

Assignee Type: Item Role at Catalog Category Level

Assignee:

Cancel Apply

2. On the **Create New Item Request Type** page, enter information in the required fields.

The Number Generation field provides multiple methods for item number request generation, but you can only select either the Sequence Generated or Function Generated method for the new item request type. When using the sequence generated item request numbering method, all item number requests automatically generate based on the sequence generation rules specified here. Specify a prefix, starting number, and increment. For the function generated method, specify a function and select whether or not to enable key attributes.

**Note:** The Number Generation field on the **Create New Item Request Type** page applies to the number generation for new item requests, not for new items. Item number generation is defined at the item catalog category level.

You must have already defined a custom function before using the function generation method. For details, see *Creating User Defined Functions*, page 4-38.

You can set up the default Assignee for all new item requests of this type in the Default Assigned To section.

3. After creating the new item request type you can continue the setup tasks via the side navigation links on the **Basic Information** page.

Click the Attribute Groups link to associate attribute groups with the new item request type.

Click the Pages link to organize the associated attribute groups on one or more **New Item Request** pages.

#### Defining User Defined Attributes for a New Item Request Type

| Attribute Groups                                         |             |                        |                                |                |
|----------------------------------------------------------|-------------|------------------------|--------------------------------|----------------|
| Change Category                                          |             | New Item Request       |                                |                |
|                                                          |             | Type                   | Standard New Item Request Type |                |
| Select Attribute Groups and...                           |             | <a href="#">Delete</a> |                                |                |
| <a href="#">Select All</a>   <a href="#">Select None</a> |             |                        |                                |                |
| Select Display Name                                      | Description | Data Level             | Pages                          | Update Actions |
| <input type="checkbox"/> Cost Impact                     | Cost Impact | Change                 |                                |                |
| <input type="checkbox"/> Implementation Costs            |             | Change                 |                                |                |

Click the Codes link to associate valid priorities and reasons for new items that are created with this request type.

Click the Configuration link to select the display sections and primary attributes that you wish to make available for this request type.

Click the Workflow link to define the statuses of the workflow. Click the Update Properties icon for a particular status to set up valid phases for promotion, demotion and the associated workflows. In the Workflows region, you can add many workflow templates, if desired. When creating a new item request using this new item request type, the user can select which workflow template to use. Select which workflow template to appear as the default during the new item request creation process.

You can then select each workflow template one at a time and specify the following:

- Auto Promote

Indicates that when the workflow completes successfully and all required

approvers have approved, the item request will automatically be promoted to the specified valid phase.

- Auto Demote

Indicates that when the workflow completes successfully and all required approvers have approved, the item request will automatically be demoted to the specified valid phase.

- Enable Digital Signature

Indicates that users who are expected to approve or reject the item request must enter their username and password to reauthenticate themselves.

## Related Topics

Defining Header Types, page 7-28

Associating New Item Requests with an Item Catalog Category, page 3-22

Defining New Item Request Workflows, page 3-21

Creating New Item Requests, *Oracle Product Hub User's Guide*

## Defining New Item Request Workflows

One of the most important features of new item requests is the ability to associate one or more workflows to each status of a new item request. The administrator can add workflow steps and specify the status of the NIR in which to execute a workflow. The administrator can add multiple workflows for each status, then the user can choose from this list of workflows which one to use for the status in a particular new item request. The administrator specifies the step number, status of the NIR process in which one of the workflows will be initiated, and finally associates the workflows by choosing them from the workflow templates.

To define a workflow template to assign to a new item request, see *Defining Workflow Templates*, page 7-17. NIR workflows are typically Definition and Approval workflows. An example of a NIR workflow is outlined in the following figure.

### NIR Workflow example

[Change Management Workflow >](#)  
**Update Workflow Template**  
• Indicates required field

Name   
Description   
Start Date   
End Date   
Type **Definition and Approval**

[Show All Details](#) | [Hide All Details](#)

| Details                | Step | Workflow Process        | Response Required | Type        | Assigned To                | Days to Respond | Update | Delete |
|------------------------|------|-------------------------|-------------------|-------------|----------------------------|-----------------|--------|--------|
| <a href="#">▶ Show</a> | 10   | Definition and Approval | One Assignee      | Item Role   | Commodity Part Manager     | 3               |        |        |
| <a href="#">▶ Show</a> | 20   | Definition and Approval | One Assignee      | Item Role   | CAD Librarian              | 3               |        |        |
| <a href="#">▶ Show</a> | 30   | Definition and Approval | One Assignee      | Item Role   | Buyer                      | 3               |        |        |
| <a href="#">▶ Show</a> | 40   | Definition and Approval | One Assignee      | Item Role   | Commodity Part Coordinator | 3               |        |        |
| <a href="#">▶ Show</a> | 50   | FYI                     | None              | Item Role   | Commodity Part Manager     |                 |        |        |
| <a href="#">▶ Show</a> | 50   | FYI                     | None              | Change Role | Requestor                  |                 |        |        |
| <a href="#">▶ Show</a> | 50   | FYI                     | None              | Group       | Vision Engineering         |                 |        |        |

Once you finish associating workflow templates with the NIR type, you need to specify the category association for the NIR.

## Related Topics

Associating New Item Requests with an Item Catalog Category, page 3-22

Defining Workflow Templates, page 7-17

## Associating New Item Requests with an Item Catalog Category

**Important:** Before associating new item requests with an item catalog category, you must set up *both* a workflow template and a new item request type. For details, see *Defining Workflow Templates*, page 7-17 and *Defining New Item Request Types*, page 3-18.

You can define a new item approval request for all items created in a particular item catalog category. Defining a new item approval request enables an enterprise to enforce standard business processes during item creation, thereby avoiding costly mishaps, such as duplicate parts. New item approval requests enable your enterprise to have different people from various lines of business help define the many item attributes required during item creation.

You can set up a new item request for an item catalog category to request approval



from several people in sequence or parallel. Each person may also be required to enter certain item attributes during the approval process.

To associate a specific new item request type with a category, you need to navigate to the item catalog category and specify the new item request type in the **New Item Request** page.

In the Workflow/Approval section, you can specify which attributes the people associated with a workflow step are required to enter. For example, when the new item request has a workflow status of Open, one person in the workflow may be responsible for some technical specifications associated with the new item being created; another might need to provide details about the new item's safety standards. In this case, associate the attribute groups related to technical specifications and safety standards so that the appropriate people can add the information necessary to define all aspects of the new item.

Since a NIR is a change, it has a status type. Only certain workflow types are allowed for a given NIR status type as tabulated below.

| NIR Status Type | Valid Workflow Type                  |
|-----------------|--------------------------------------|
| Open            | Definition                           |
| Approval/Review | Definition and Approval, or Approval |
| Others          | Generic                              |

## Prerequisites

- ☐ Defining Workflow Templates, page 7-17
- ☐ Defining New Item Request Workflows, page 3-21
- ☐ Setting Up New Item Request Types, page 3-18

### To associate new item requests with an item catalog category:

1. Navigate to the Setup Workbench Item Catalog Categories tab. Find and select the item catalog category to which to associate a new item request type.
2. Click the New Item Request link. In the **New Item Request** page, click Update.  
The **Update New Item Request** page opens.

### Associating a New Item Request with an Item Catalog Category

Items Documents Structures Change Management Security

Item Catalog Categories Cross Reference Types Lifecycle Phases At

Items: Item Catalog Categories >

**Update New Item Request**

Cancel Apply

Catalog Category **Apparel**

\* New Item Request Required Specify for Category

\* Request Type None Specify for Category Inherit from Parent

Cancel Apply

Items Documents Structures Change Management Security Functions Catalogs

About this Page Privacy Statement

3. In the New Item Request Required field, specify one of the following:
  - None - no new item request is required for items created within this item catalog category.
  - Specify for Category - if you select this option, you can select a value in the Item Request Type field.
  - Inherit from Parent - use the same new item request type as the parent item catalog category.

You can change the value of the New Item Request Required field at any time. For example, if you no longer want to inherit the NIR type from the parent ICC, you can specify a new NIR type or no longer require new item requests.

Click Apply. The **New Item Request** page appears.

4. If the item catalog category is associated with an item request type, either through inheritance or directly, then the Workflow/Approval section appears in the **New Item Request** page.

If there is more than one workflow associated with an item request type, select each workflow for each status using the pull down lists in the Workflow and Workflow Status fields. For each workflow, you can choose to associate attribute groups to each workflow step only if the ICC is associated with an item request type directly. Otherwise, the attribute group definitions are inherited from the parent ICC or

nonexistent.

To update the attribute groups for a workflow step, click the Associate Attribute Groups icon for the workflow step.

### Example of When to Associate Attribute Groups

When the new item request is in the Open workflow phase, one person in the workflow may be responsible for some technical specifications associated with the new item being created; another might need to provide details about the new item's safety standards. In this case, associate the attribute groups related to technical specifications and safety standards so that the appropriate people can add the information necessary to define all aspects of the new item.

You can set up the collection of user-defined attributes during the new item request process for any attribute group associated with the item catalog category.

**Note:** People associated with the workflow must also have sufficient privileges on the item for which you are creating a request. To ensure the appropriate people have sufficient privileges to view and update the attributes on the item for which they are responsible, you should confirm that they have the appropriate role at the organization or item catalog category level.

5. In the **Associate Attribute Groups** page, search for and select the user-defined attribute groups associated with the ICC for which you want the workflow assignees to add attribute values.

Click Apply.

### Associating Item Attribute groups with Definition Steps

**Associate Attribute Groups**

• Indicates required field

Step **20** Cancel Apply

Workflow Process **Definition and Approval**

Assignee Type **Item Rule**

Name **CAD Librarian**

| *Attribute Groups       | Delete                                                                                |
|-------------------------|---------------------------------------------------------------------------------------|
| Technology              |  |
| Layout Specifications   |  |
| Symbol                  |  |
| Footprint               |  |
| <span>Add 5 More</span> |                                                                                       |

Cancel Apply

## Related Topics

Defining New Item Request Workflows, page 3-21

## Defining Item Numbers and Descriptions

You can define new item numbers and descriptions so that they are automatically generated when users create new items. Setting up autonumber generation ensures that new items created in item catalog categories have a consistent numbering scheme. You can set up item number and description generation so that numbers/descriptions are either:

- Sequence generated

If the item request numbering method is sequence generated, then all item number requests automatically generate based on the sequence generation rules specified here. You can specify a prefix, starting number, increment, and suffix.

**Note:** Sequence generation does not apply to Item Description generation--it is for item number generation only.

- Function generated

If the item request numbering/description method is function generated, then all number/descriptions for new items in the item catalog category automatically generate based on a custom function. When specifying that an item number/description is function generated, select the function that you want to use, and then map the function parameters to the user-defined attributes that you want included in the item number/description. When creating a new item, users receive prompts to enter the user-defined attributes used in generating the item number/description.

**Note:** You must have already defined a custom function before using this item number/description generation method. For details, see *Creating User-Defined Functions*, page 4-38.

- Inherited from parent

If the item request numbering/description generation method is inherited from a parent, then the item number/descriptions for the items generate using the same method as the parent item catalog category.

**Note:** All item catalog category subcategories have, by default, a number/description scheme that is inherited from their parent.

- User entered

If the item request numbering/description method is user entered, then the user is required to manually enter a number or a description for new items.

To set up automatic change number generation using user-defined functions, follow these steps:

1. Update the header type.
2. Select **Function Generated** from the Number Creation drop-down list.
3. Associate the function.
4. Associate the function parameters to the attributes.

### To define automatic item number/description generation:

1. On the **Basic Information** page of an item catalog category, click either the Number Generation or Description Generation link. On the **Auto Generation** page for the generation method you selected, click Update.

#### Item Number Generation

##### Auto Generation

Catalog Category: **Capacitors**

##### Number Generation

Item Numbering **Sequence Generated**

Update

Prefix VI

Starting Number 60000

Increment 1

Suffix CAP

#### Setting up Item Number Generation

##### Item Number Generation

Cancel Apply

Catalog Category: **Capacitors**

Item Numbering Sequence Generated Go

Function Generated

Inherited From Parent

Prefix Sequence Generated

• Starting Number User Defined 500

• Increment 1

Suffix CAP

Cancel Apply

2. On either the **Item Description Generation** or **Item Number Generation** page, select the generation method you wish to use and fill in the required fields.

### Description Generation Sample

#### Auto Generation

Catalog Category: Capacitors

#### Description Generation

Item Description    Function Generated

Update

#### Generation Function

\* Indicates required field

Function    GenCapacitorItemDesc

#### Mappings For Function

| Function Parameter | Attribute Group           | Attribute         |
|--------------------|---------------------------|-------------------|
| Tolerance          | Capacitor Specifications  | Tolerance         |
| Capacitance Value  | Capacitor Specifications  | Capacitance Value |
| Size               | Capacitor Packaging Specs | Size              |
| Package Type       | Capacitor Packaging Specs | Package Type      |
| Case Size          | Capacitor Packaging Specs | Case Size         |
| Dielectric         | Capacitor Specifications  | Dielectric        |
| Voltage            | Capacitor Specifications  | Voltage           |

## Creating Pages for an Item Catalog Category

Item pages provide a mechanism to customize the user interface. The administrator can combine and sequence attribute groups into pages. There is no limit on the number of attribute groups associated with a page. However, attribute groups contained in a page must have the same association level, for example, they must all be at one of the following attribute group levels:

- Item
- Item Revision
- Item Supplier
- Item Supplier Site
- Item Organization
- Item Supplier Site Organization

**Note:** Pages inherited from a parent catalog category cannot be deleted at the child catalog category level. Pages can be inherited from a parent catalog category *and* explicitly added to the catalog category itself. Administrators should not associate the same pages with an item catalog category in this manner.

### To create a page for an item catalog category:

1. On the **Search: Item Catalog Categories** page, search for the item catalog category (see: Browsing Item and Alternate Catalogs , *Oracle Product Hub User's Guide*) and click its corresponding name link.
2. On the **Basic Information** page for the item catalog category, click the Item Pages link.
3. On the **Item Pages** page, click Add.
4. On the **Create Item Catalog Category Page** page, enter the following:

#### **Display Name**

Enter the name to be displayed in the user interface.

#### **Internal Name**

Enter the internal name. The internal name cannot be updated once a page has been created.

#### **Description**

Enter a description of the catalog category page.

#### **Sequence**

Enter the sequence number. For each page associated with a catalog category, a corresponding link is displayed to the user on the Item Detail page. The sequence number determines the order for the page links on the **Item Detail** page.

#### **Data Level**

Select Data Level and click Go. The data level constrains applicable attribute groups. You cannot update the data level after a page has been created.

**Note:** If you change the data level after you have already associated attribute groups with a page, those attribute groups will no longer be associated with the page.

5. Click Add Another Row if you wish to add an attribute group to the page being created.
6. Click Apply.

## Managing Search Criteria

Search criteria templates offer a convenient way to save frequently used search criteria. Search criteria can be created either by administrators or end users. Administrator-

created search criteria templates are available for all users. User-created search criteria are available only to the particular user who creates it. When a default search criteria template for a particular object is defined by a user and administrator, the user-defined search criteria template takes precedence.

You can create item search criteria and import formats for the following business entities:

- Item
- Item Organization
- Item Revision
- Item Supplier
- Item Supplier Site
- Item Supplier Site Organization

Because search criteria templates contain primary, operational, and user-defined attributes, they are always defined in the context of an item catalog category, structure type, or change category (depending on the object for which a criteria template is created).

The following figure shows the **Update Search Criteria** page, with the criteria template for the item catalog category Capacitors. Begin by listing the common attributes a typical user might search for when using the item catalog category Capacitors. Here, attributes such as Capacitor Tolerance and Capacitor Voltage, among others, are specified. After adding attributes, you can specify operators and values to further restrict the scope of your search.



## Defining Item Catalog Search Criteria

| Items                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Documents         | Structures                | Change Management   | Security    | Functions | Catalogs | Value Sets |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|---------------------|-------------|-----------|----------|------------|--------|--------|-----------------|-----------|----------|-------|--------------------------|------|--------------------------|-----------|----|--|--------------------------|------|--------------------------|---------------------|----|--|--------------------------|------|---------|--------|-------------|--|--------------------------|-------------------|------------------|------------------|----|--|--------------------------|------|--------------------------|-----------|----|---------|--------------------------|-------------------|-----------|----------------|----|--|--------------------------|------|---------------------------|----------------|----|----|
| <a href="#">Item Catalog Categories</a>   <a href="#">Cross Reference Types</a>   <a href="#">Lifecycle Phases</a>   <a href="#">Attribute Groups</a>   <a href="#">Templates</a>   <a href="#">GTIN</a>   <a href="#">Item Type</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                           |                     |             |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <a href="#">Items: Item Catalog Categories</a> > <a href="#">Basic Information</a> > <a href="#">Search Criteria</a> >                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |                           |                     |             |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <b>Update Search Criteria</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                           |                     |             |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| * Indicates required field                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |                           |                     |             |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <div> <div>Item Catalog Category</div> <div>Capacitors</div> </div> <div> <div>* Name</div> <div>SM Capacitors</div> </div> <div> <div>Description</div> <div>Capacitors with SM package type</div> </div> <div> <input type="checkbox"/> Set as Default </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |                           |                     |             |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <div> <div>Attribute Criteria</div> <div> <p> <b>TIP</b> If you duplicate an attribute and use an operator to include criteria such as 'starts with', 'is', 'ends with', 'greater than', etc., then the application applies the 'OR' operator to each duplicate attribute. If you duplicate an attribute and use an operator to exclude criteria such as 'is not', 'none of which is', etc., or you do not duplicate attributes, then the application applies the 'AND' operator to each attribute. </p> <p> <b>TIP</b> ^ indicates the attribute is indexed. </p> </div> <div> <div>Select Criteria:</div> <div> <input type="button" value="Clear"/> <input type="button" value="Delete"/> <input type="button" value="Duplicate"/> <input type="button" value="Add Criteria"/> </div> </div> <div> <div>Select All</div> <div>Select None</div> </div> <table border="1"> <thead> <tr> <th>Select</th> <th>Entity</th> <th>Attribute Group</th> <th>Attribute</th> <th>Operator</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td><input type="checkbox"/></td> <td>Item</td> <td>Capacitor Specifications</td> <td>Tolerance</td> <td>is</td> <td></td> </tr> <tr> <td><input type="checkbox"/></td> <td>Item</td> <td>Capacitor Specifications</td> <td>^ Temp. Coefficient</td> <td>is</td> <td></td> </tr> <tr> <td><input type="checkbox"/></td> <td>Item</td> <td>Primary</td> <td>^ Item</td> <td>starts with</td> <td></td> </tr> <tr> <td><input type="checkbox"/></td> <td>Item Organization</td> <td>Order Management</td> <td>Customer Ordered</td> <td>is</td> <td></td> </tr> <tr> <td><input type="checkbox"/></td> <td>Item</td> <td>Capacitor Specifications</td> <td>^ Voltage</td> <td>is</td> <td>10 Volt</td> </tr> <tr> <td><input type="checkbox"/></td> <td>Item Organization</td> <td>Inventory</td> <td>Inventory Item</td> <td>is</td> <td></td> </tr> <tr> <td><input type="checkbox"/></td> <td>Item</td> <td>Capacitor Packaging Specs</td> <td>^ Package Type</td> <td>is</td> <td>SM</td> </tr> </tbody> </table> </div> |                   |                           |                     |             |           |          |            | Select | Entity | Attribute Group | Attribute | Operator | Value | <input type="checkbox"/> | Item | Capacitor Specifications | Tolerance | is |  | <input type="checkbox"/> | Item | Capacitor Specifications | ^ Temp. Coefficient | is |  | <input type="checkbox"/> | Item | Primary | ^ Item | starts with |  | <input type="checkbox"/> | Item Organization | Order Management | Customer Ordered | is |  | <input type="checkbox"/> | Item | Capacitor Specifications | ^ Voltage | is | 10 Volt | <input type="checkbox"/> | Item Organization | Inventory | Inventory Item | is |  | <input type="checkbox"/> | Item | Capacitor Packaging Specs | ^ Package Type | is | SM |
| Select                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Entity            | Attribute Group           | Attribute           | Operator    | Value     |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Item              | Capacitor Specifications  | Tolerance           | is          |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Item              | Capacitor Specifications  | ^ Temp. Coefficient | is          |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Item              | Primary                   | ^ Item              | starts with |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Item Organization | Order Management          | Customer Ordered    | is          |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Item              | Capacitor Specifications  | ^ Voltage           | is          | 10 Volt   |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Item Organization | Inventory                 | Inventory Item      | is          |           |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Item              | Capacitor Packaging Specs | ^ Package Type      | is          | SM        |          |            |        |        |                 |           |          |       |                          |      |                          |           |    |  |                          |      |                          |                     |    |  |                          |      |         |        |             |  |                          |                   |                  |                  |    |  |                          |      |                          |           |    |         |                          |                   |           |                |    |  |                          |      |                           |                |    |    |

## To create administrator level search criteria:

Administrators can create search criteria from the respective item catalog category definition page for an item, the structure type definition page for a structure, or the change category list page for a change object. You can access these pages via the Setup Workbench link on the Applications page.

1. From the object's definition page (either the **Basic Information** page for an item catalog category or structure type or the **Categories** page for a change object), click the Search Criteria link.

**Note:** On the change management **Categories** page, click the Search Criteria icon in the row of the category for which you wish to create search criteria.

**Note:** The Packaging BOM structure type **Basic Information** page has no link to Search Criteria. You can check Display Formats and Import Formats from the Setup Workbench Structures tab.

2. On the **Search Criteria Template** page, click Create.
3. On the **Create Search Criteria** page, provide a name and description for the search criteria. You can set this search criteria as the default search criteria for the object by clicking Set as Default. Note that the name of the respective item catalog category, structure type, or change category already appears. You have the option of copying an existing item catalog category, structure type, or change category search criteria and combining the respective attributes with those you wish to add to this search criteria.
4. Click Add Criteria to add search criteria.

If you simply wish to locate a single attribute, enter its name and click Find. To search for attributes by business entity (for item catalog categories only) and attribute group, select the business entity, click Go, then select the attribute group and click Go again.

For change categories, search for attributes by change category.

For structures, search for attributes either by structure type (you do not need to specify a structure type because you are always in the context of a structure and structure type) or by catalog category. To search for attributes by structure type, select Component. To search for attributes by catalog category, select Item.

The screenshot shows the 'Add Criteria' dialog box. At the top, there are tabs for 'Items', 'Documents', 'Structures', 'Change Management', 'Security', 'Functions', 'Catalogs', and 'Value Sets'. Below these are sub-tabs for 'Item Catalog Categories', 'Cross Reference Types', 'Lifecycle Phases', 'Attribute Groups', 'Templates', 'GTIN', and 'Item Type'. The breadcrumb trail indicates the path: 'Items: Item Catalog Categories > Basic Information > Search Criteria > Create Search Criteria > Add Criteria'.

The dialog is divided into two main sections: 'Available Criteria' on the left and 'Selected Criteria' on the right. In the 'Available Criteria' section, there are input fields for 'Attribute Name', 'Business Entity' (set to 'Item Supplier'), and 'Attribute Group' (set to 'Primary'). There are 'Find', 'Go', and 'Go' buttons. Below these is a list of available criteria, including 'Item Supplier:Primary:Primary Supplier Site', 'Item Supplier:Primary:Primary Supplier Site Status', 'Item Supplier:Primary:Primary Supplier Site Organization Status', 'Item Supplier:Primary:Primary Supplier Number', 'Item Supplier:Primary:Primary Supplier', and 'Item Supplier:Primary:Primary Supplier Status'. A tip at the bottom left states: 'TIP ^ indicates the attribute is indexed.'

In the 'Selected Criteria' section, there is a list of selected criteria: '^ Item Supplier:Primary:Primary Supplier Name', '^ Item Supplier:Primary:Primary Supplier Site', 'Item Organization:Physical Attributes:Container', 'Item Organization:Purchasing:Purchasable', and '^ Item:Primary:Item Status'. Between the two lists are buttons for 'Move', 'Move All', 'Remove', and 'Remove All'. At the bottom right, there are 'Cancel' and 'Apply' buttons.

5. On the **Add Criteria** page, you can move search criteria back and forth from the Available Criteria list to the Selected Criteria list via the Move, Move All, Remove, and Remove All shuttles. When finished, click Continue.
6. Back on the **Create Search Criteria** page after adding criteria, you can specify operators and values to further refine the search criteria. For example, when searching for Capacitors, you may always want to include Capacitors with a Capacitance value that is between 5 and 30 uF (microfarads)--in this case you would

add the attribute "Capacitance Value" to your search criteria, then select the operator "between," and specify a minimum value of "5" uF and a maximum value of "30" uF.

**Note:** For multi-row attributes with a value set, the distinction between the operators "is not" and "none of which is" is as follows:

- *is not*

Returns *any* record containing at least one attribute value that is not the value specified. For example, suppose you have a multi-row attribute for "Color." If you specify an operator and value "is not" as "Red," then the search results will include records where there is at least one attribute value for "Color" that is not "Red."

- *none of which is*

Returns any record where none of the attribute values is the value specified. For example, suppose you have a multi-row attribute for "Color." If you specify an operator and the value "none of which is" as "Red," then the search results will include records where none of the attribute values for "Color" are equal to "Red."

To clear the criteria values, select the criteria and click Clear. To delete criteria attributes, select the criteria and click Delete. To copy a criteria attribute and its value, select the criteria and click Duplicate.

**Tip:** All of the operators except for "contains" are case sensitive.

7. Click Apply to save your search criteria definition.

### To create user level search criteria:

1. Navigate to the **Advanced Search** page for items, change objects, or structures.

For items, click on the Item Advanced Search link in the Applications tree menu. For change objects, click on the Issue, Idea, or Change Advanced Search link in the Applications tree menu. See *Searching for Items in a Structure, Oracle Product Hub User's Guide* for instructions on how to navigate to the **Advanced Search** page for structures.

2. On the **Advanced Search** page, select an Item Catalog Category or Change Category (whichever is appropriate for your context) and click Go.

Skip this step for structures.

3. Click Personalize in the Search Criteria section.
4. Once in the **Personalize Search Criteria** page, follow the steps for "To create administrator level search criteria:" listed above. Begin at the step where you are in the **Search Criteria Template** page.

### To update user level search criteria:

1. Navigate to the **Advanced Search** page for items, change objects, or structures.  
  
For items, click on the Item Advanced Search link in the Applications tree menu.  
For change objects, click on the Issue, Idea, or Change Advanced Search link in the Applications tree menu. See *Searching for Items in a Structure, Oracle Product Hub User's Guide* for instructions on how to navigate to the **Advanced Search** page for structures.
2. On the **Advanced Search** page, select an Item Catalog Category or Change Category (whichever is appropriate for your context) and click Go.  
  
**Note:** For structures, you do not have to specify a Structure Type because you are always in the context of a structure and structure type.
3. Click Personalize in the Search Criteria section.
4. On the **Personalize Search Criteria** page you are provided with a list of search criterion that are available for the item catalog category/structure type/change category you originally selected. You can only edit search criteria that you created. Click the update icon for the search criteria you wish to update.
5. On the **Update Search Criteria** page, you can update the name, description, set as default, and other attributes of the search criteria. You can also add/delete/copy/clear/modify criteria, corresponding search operators, and values.
6. Click Apply to save your search criteria definition.

## Related Topics

Managing Display Formats, page 3-35

Searching for Items, *Oracle Product Hub User's Guide*

Searching for Ideas, Issues, and Changes, *Oracle Product Hub User's Guide*

Defining Change Category Search Criteria and Display Formats, page 7-41

## Managing Display Formats

Display Formats enable you to predefine search display views. You can use these views to look at different sets of attributes of the items, change objects, or structure components that are returned by the search. Item display formats, by default, always include the item number, item description, item catalog category, and revision level. Change display formats always include, by default, the change number. Structure display formats always include, by default, name, description, and component revision.

You can create item display formats using primary, operational, and user-defined attributes. Select user-defined attributes from attribute groups associated with any of the following business entities:

- Item
- Item Organization
- Item Revision
- Item Supplier
- Item Supplier Site
- Item Supplier Site Organization

## Creating a Display Format

Items | Documents | Structures | Change Management | Security | Functions | Catalogs | Value Sets

Item Catalog Categories | Cross Reference Types | Lifecycle Phases | Attribute Groups | Templates | GTIN | Item Type

Items: Item Catalog Categories > Basic Information > Display Formats > Create Display Format

\* Indicates required field

[Personalize Stack Layout](#)  
[Personalize "Test"](#)  
[Personalize Table Layout: \(EgoEditResultsFormatInfoTL\)](#)  
[Personalize Table Layout: \(EgoEditResultsFormatInfoAttributesTL\)](#)

Item Catalog Category: 1 Retailers Ltd

\* Name:

Description:

Number of Rows Displayed:

☐ Set as Default

[Personalize Stack Layout](#)  
[Personalize "View Columns"](#)

**View Columns**

[Personalize Table Layout: \(EgoRenameColumnsTL\)](#)  
[Personalize Table Layout: \(EgoResultsFormatViewColumnsTL\)](#)

Available Columns

| Attribute Name | Business Entity   | Attribute Group                 |
|----------------|-------------------|---------------------------------|
|                | Item Organization | Item                            |
|                | Item Organization | Item Organization               |
|                | Item Organization | Item Revision                   |
|                | Item Organization | Item Supplier                   |
|                | Item Organization | Item Supplier Site              |
|                | Item Organization | Item Supplier Site Organization |

Find:  Go:

Selected Columns

| Attribute Name                       |
|--------------------------------------|
| Item:Primary:Item                    |
| Item:Primary:Description             |
| Item:Primary:Item Status             |
| Item:Primary:Primary Unit of Measure |

Move:   
Move All:   
Remove:   
Remove All:

Both administrators and users can create display formats. Administrator-created display formats are available to all users. User-created display formats are available only to the users who created them.

### To create an administrator level display format:

1. Administrators can create display formats from the respective item catalog category definition page for an item, the structure type definition page for a structure, or the change category list page for a change object. You can access these pages via the Setup Workbench link in the Applications tree menu.
2. From the object's definition page (either the **Basic Information** page for an item catalog category or structure type, or the **Categories** page for a change object), click the Display Formats link.

**Note:** On the change management **Categories** page, click the Display Format icon in the row of the category for which you wish to create a display format.

3. On the **Display Formats** page, click Create.

**Note:** When viewing display formats for structures, the name of this page is **Search Results Format**.

Alternatively, you can click Copy to select an existing display format and copy its columns into a new display format. You can add additional, new columns into this copied display format, too.

4. On the **Create Display Format** page, provide the name and a description for the display format. Also, select the number of rows you want to display per page.

You can set this display format as the default display format for the object by clicking Set as Default. Note that the name of the respective item catalog category, structure type, or change category already appears.

In the View Columns region, specify the columns you want to display in the search results display. If you simply wish to locate a single attribute, enter its name and click Find. To search for attributes by business entity (for item catalog categories only) and attribute group, select the business entity, click Go, then select the attribute group and click Go again.

**Additional Information:** If you want to include the attribute Default Material Status in your display format, you must also include the attributes Lot:Control and Generation. You can find these attributes within the Item Organization business entity and the Inventory attribute group. For more information about these attributes, refer to the *Oracle Inventory User's Guide*.

For structures, search for attributes either by structure type (you do not need to specify a structure type because you are always in the context of a structure and structure type) or by catalog category. To search for attributes by structure type, select Component. To search for attributes by catalog category, select Item. If you select Item, you must specify a catalog category.

You can move display attributes back and forth from the Available Columns list to the Selected Columns list via the Move, Move All, Remove, and Remove All shuttles.

Note that you can rename the columns that will appear in your display format by clicking Rename Columns. On the **Rename Columns** page, the original column names that you have already selected are listed. Enter the new column names in the New Column Name field and click Continue.

In the Sort Order section, you can select up to three columns upon which to sort the display results. You can also specify whether or not the search results in these columns be displayed in ascending or descending order. Only indexed and non-secured attributes (attributes that are not secured by a view or edit privilege) are available for sorting. Thus, you cannot sort secured attributes.

Under "Display Sections," (available only for items and change objects) you can select which sections will be displayed in your search results. Sections are displayed as a column containing an icon with a direct link to the section. For example, to include an Attachments link on your search results page, select Attachments from the Available Sections list and move it to the Selected Sections list.

5. Click Apply to save your display format definition.

**Note:** Set up a display format to include item multi-row attributes the same way you set it up to include single attributes. After the list of items that met your search criteria appears, click the Show link for a particular item. The Show region displays the multi-row attributes.

### To update an administrator level display format:

1. Administrators can update display formats from the respective item catalog category definition page, the structure type definition page, or change category list page--all of which are accessible via the Setup Workbench link in the Applications tree menu.
2. From the object's definition page (either the **Basic Information** page for an item catalog category or structure type, or the **Categories** page for a change object), click the Display Format link.

**Note:** On the change management **Categories** page, click the Display Format icon in the row of the category for which you wish to update a display format.

3. On the **Display Format** page, click the Update icon in the row of the display format you wish to update. You can only edit the display formats that are explicitly associated with the item catalog category, structure type, or change category originally selected.
4. On the **Update Display Format** page, you can update the name, description and other columns of the display format. You can also add/remove/rename columns and modify sort criteria.
5. Click Apply to save your display format definition.

### To create a user level display format:

1. Navigate to the **Advanced Search** page for items, change objects, or structures. For items, click on the Advanced Search link in the Applications tree menu. For



change objects, click on the Issue, Idea, or Change Advanced Search link in the Applications tree menu. See *Searching for Items in a Structure, Oracle Product Hub User's Guide* for instructions on how to navigate to the **Advanced Search** page for structures.

2. On the **Advanced Search** page, select an Item Catalog Category or Change Category (whichever is appropriate for your context) or, alternatively, an Alternate Catalog, and click Go.

Optionally, you can limit your search further to a certain organization, revision, and, if you selected an Alternate Catalog, alternate category.

3. Click Personalize in the Display Format section.
4. On the **Personalize Display Formats** page, click Create or click the Copy icon for one of the existing display formats.
5. Once in the **Personalize Display Formats** page, follow the steps for "To create an administrator level display format:" listed above. Begin at the step where you are in the **Create Display Format** page.
6. Click Apply to save your display format definition.

#### **To update a user level display format:**

1. Navigate to the **Advanced Search** page for items, change objects, or structures.

For items, click on the Advanced Search link in the Applications tree menu. For change objects, click on the Issue, Idea, or Change Advanced Search link in the Applications tree menu. See *Searching for Items in a Structure, Oracle Product Hub User's Guide* for instructions on how to navigate to the **Advanced Search** page for structures.

2. On the **Advanced Search** page, select an Item Catalog Category or Change Category (whichever is appropriate for your context) or, alternatively, an Alternate Catalog, and click Go.

Optionally, you can limit your search further to a certain organization, revision, and, if you selected an Alternate Catalog, alternate category.

For structures, you do not need to specify a structure type because you are always in the context of a structure and structure type.

3. Click Personalize in the Display Format section.
4. On the **Personalize Display Formats** page, click the Update icon of the display format you wish to update. You can only update your user display formats that are explicitly associated with the item catalog category, alternate catalog, structure type, or change category originally selected.

5. On the **Update Display Format** page, you can update all of the display format fields.
6. Click Apply to save your display format definition.

## Related Topics

Searching for Items, *Oracle Product Hub User's Guide*

Searching for Ideas, Issues, and Changes, *Oracle Product Hub User's Guide*

Defining Change Category Search Criteria and Display Formats, page 7-41

Searching for Items in a Structure, *Oracle Product Hub User's Guide*

Defining Display Formats and Search Criteria for Structures, page 8-4

## Defining Item Catalog Import Formats

An import format identifies the primary and user-defined attributes (but not the operational attributes) in an item catalog category that is imported into the application using a spreadsheet. You can import operational attributes; however, to do this, you must specify an item template and item status in your import format. Both item templates and item statuses can be used to set operational attributes when importing items.

### Seeded Business Entities for Importing

You can define import formats for the following business entities:

- Item
- Item Revision
- Item Supplier
- Item Supplier Site
- Item Organization
- Item Supplier Site Organization

An import format can also import values for a multi-row attribute group.

To import any of the above business entities from the system into an Excel spreadsheet, you must use an import format. An import format identifies the operational and user defined attributes in an item catalog category that will be imported into the spreadsheet. Consequently, when you import item business entities from an Excel spreadsheet, the items are all imported to a particular item catalog category defined in the import format. These imported item business entities inherit all the attribute groups

defined for the specific item catalog category. You can also copy, update and delete import formats.

If users do not select an item catalog category when importing business entities, the default System business entity import format is available for use. If an import format has not yet been defined for an item catalog category that a user selects, then the default System business entity import format is available for use. The System business entity import format does not appear if a user selects an item catalog category that already has import formats defined. You cannot modify the System business entity import formats.

Items uploaded to the system from a spreadsheet become, by default, Engineering items. If you wish to upload a non-Engineering item, then include an available primary attribute column called "Engineering Item Flag" in your import format. When uploading the spreadsheet identify such items in this column by specifying the value "Yes" or "No" in this column. If you leave the column blank, the item once again defaults to an Engineering item.

### **To create an item import format:**

1. In the Applications tree menu, click the Setup Workbench link.
2. On the **Search: Catalog Categories** page, locate the item catalog category for which you wish to create the import format, and click its name link.
3. On the **Basic Information** page for the item catalog category, click the Import Formats link.
4. On the **Import Formats** page, click Create.
5. On the **Create Import Format** page, provide the following information:

#### **Name**

Provide the name of the import format you are creating for the item catalog category.

#### **Description**

Provide a description of the import format.

#### **Set as Default**

Select the checkbox if you wish for this import format to be the default import format for the item catalog category in which it is created.

**Note:** There can be only one default import format for each item catalog category.

6. In the View Columns region of the **Create Import Format** page, find the attribute columns to appear on the import format using the following fields:

- Attribute Name

If you know the partial name of an attribute and the business entity it belongs to, you can use this field to find it.

For example, if you are looking for a status attribute assigned to the Item Supplier business entity, select Item Supplier in the Business Entity field and click Go. Enter %stat% in the Attribute Name field and click Find. The Item Supplier attributes containing "stat" in their names are listed below and are available for selection.

- Business Entity

To view all attributes assigned to a business entity, select one of the following entities and click Go.

- Item
- Item Organization
- Item Revision
- Item Supplier
- Item Supplier Site
- Item Supplier Site Organization

- Attribute Group

Select an attribute group assigned to the selected business entity, then click Go. All attributes within the attribute group are now available for selection.

7. Use the Move and Remove buttons to select and remove attributes from the import format.

You can select either single or multi-row attributes to include in the import format.

**Tip:** You can rename the columns that will appear in your import format by clicking Rename Columns. On the **Rename Columns** page, the original column names that you have already selected are listed. Enter the new column names in the New Column Name field and click Apply.

8. Once you have moved all of the attribute columns you want in the import format into the Selected Columns box, click Apply.

Your new import format now appears in the list of available import formats for the selected Item Catalog Category.

**Note:** When importing data using Excel, you can use a combination of 220 character fields, 200 numeric fields, and 50 date fields, subject to an overall 255 field limit. For example, an import format can include 220 character columns and 35 numeric columns for a total of 255 columns. If you use a display format to export and import data, the same restriction applies.

### **To copy an import format:**

1. Navigate to the **Import Format** page. See: To create an item import format, page 3-41.
2. On the **Import Formats** page, locate the import format you wish to copy and click its Copy icon in the Copy column.
3. On the **Copy Import Format** page, note that a new name is defaulted, as well as the description. You can change either, and also add or remove attributes in the Selected Columns box.

Note that you can rename the columns that will appear in your import format by clicking Rename Columns. On the Rename Columns page, the original column names that you have already selected are listed. Enter the new column names in the New Column Name field and click Continue.

If there are more columns/attributes than can be easily scrolled in the list, you can filter the list by providing the attribute group and clicking Go, or providing the attribute name and clicking Find.

**Note:** You can select either single or multi-row attributes to include in the import format.

4. Click Apply.

### **To update an import format:**

1. Navigate to the **Import Format** page. See: To create an item import format, page 3-41.
2. On the **Import Formats** page, locate the import format you wish to update and click the Update icon in the Update column.
3. On the **Edit Import Format** page, you can change the name, description, default setting and add or remove attributes/columns as needed.

Note that you can rename the columns that will appear in your import format by clicking Rename Columns. On the **Rename Columns** page, the original column

names that you have already selected are listed. Enter the new column names in the New Column Name field and click Continue.

If there are more columns/attributes than can be easily scrolled in the list, you can filter the list by providing the attribute group and clicking Go, or providing the attribute name and clicking Find.

**Note:** You can select either single or multi-row attributes to include in the import format. Also, you cannot modify the System Item import format.

4. Click Apply.

#### To delete an import format:

1. Navigate to the **Import Format** page. See: To create an item import format, page 3-41.
2. On the **Import Formats** page, locate the import format you wish to delete, select it and click Delete.

## Related Topics

Importing Items and Related Objects Using a Spreadsheet, *Oracle Product Hub User's Guide*.

## Assigning Systems to an Item Catalog Category

You can publish value sets, item catalog categories, items, and structures to other systems that use this information. The following steps describe how to define default systems for an item catalog category. Systems defined for a parent item catalog category are inherited by the entire ICC hierarchy below the parent. If you choose to define default systems for an item catalog category, these systems are automatically selected during the publishing process when publishing item catalog categories, items, and structures.

1. From the **Oracle Applications Home Page**, select an appropriate responsibility, then Setup Workbench. Select the Items tab.
2. In the **Item Catalog Categories** page, search for and select an item catalog category.
3. In the **Basic Information** page, click Systems.
4. In the **Systems** page, select an organization in which to associate systems.
5. Click Update, then Add Another Row to add systems.

6. In the Systems field, select a system from the list.

If the system you want to select is not listed, then see *Defining Source Systems*, page 12-5.

If you want to remove a system from the item catalog category and organization, select the system, then click Remove.

7. Once you have added all of the systems for the item catalog category and organization, then click Apply.

Repeat this task for as many item catalog category, organization, and system combinations as needed.

## Related Topics

Publishing Item Catalog Categories, page 3-45

## Publishing Item Catalog Categories

You can search for and select an item catalog category (ICC) and its hierarchy, then publish it to one or more systems, such as Siebel Customer Relationship Management, Oracle Billing and Revenue Management, or another Oracle Product Hub instance. You have the option to publish:

- only the selected ICC.
- the selected ICC and its parent ICC.
- the selected ICC and its children ICCs.
- the selected ICC, its parent, and its children (recommended).

You must publish the ICC with its parent unless the parent ICC has been previously published. Republish the ICC when you release a new version of the ICC. Publishing an ICC also publishes all attribute groups and transaction attributes associated with the ICC. When releasing a new version of an ICC, create a new revision for any item associated with the ICC to make the ICC changes available at the item level.

**Additional Information:** Alternatively, you can use a PL/SQL Public API to publish ICCs. See: *Using PL/SQL Public APIs to Publish Objects*, *Oracle Product Hub User's Guide*.

### Attributes Published with an Item Catalog Category

The following types of attributes are published along with an ICC:

- Seeded attributes.

- User-defined attributes.
- Transaction attributes.

The sections below provide more information about publishing each type of attribute.

### **Seeded Attributes**

You can assign seeded attribute groups (see: Using Seeded Attributes, page 4-36 and Seeded Item Metadata Libraries, *Oracle Product Hub Implementation Guide*) to an ICC. These are published with the ICC and any assigned attribute values are published with the items. Do not change the seeded attribute group and attribute metadata. Changes prevent correct synchronization with the target system.

### **User-Defined Attributes**

You can add new user-defined attribute groups and attributes to a new version of an ICC at any time. In order to publish user-defined attributes, use the following naming conventions:

- Use the format "attribute group.attribute" as the internal name for the attribute.  
For example, a new attribute SupplierName added to the attribute group "Suppliers" has the internal name Suppliers.SupplierName. This prevents duplicate attributes across the attribute groups.
- When publishing to Siebel, the internal name for all user-defined attributes must begin with "XX\_".  
For example, a new attribute SupplierName added to the attribute group "Suppliers" has the internal name XX\_Suppliers.SupplierName.

**Additional Information:** In Siebel, the user-defined attributes are not associated with a value set, but the values are stored and displayed as free-form text.

All user-defined attributes published without an internal name prefix of "XX\_" are sent to the target application connector service. The target application connector service ignores these attributes unless the target application connector service is extended to map these attributes to the corresponding attributes in the target application.

### **Transaction Attributes**

When publishing transaction attributes (see: Defining Transaction Attributes, page 4-33) along with ICCs to a Siebel system, you must associate a value set with each transaction attribute assigned to an ICC. If you select the Check Eligibility field for a transaction attribute in Oracle Product Hub, then verify that a defined eligibility procedure for the corresponding transaction attribute exists in the target Siebel system. This preserves the enrichment of the transaction attributes in Siebel.

If a parent ICC and a child ICC both have a transaction attribute with the same name, the transaction attribute from the child ICC takes precedence.



### Value Sets Published with an Item Catalog Category

You can publish value sets associated to attributes with an ICC or you can publish value sets independently. For more information about publishing value sets, refer to *Publishing Value Sets*, page 4-16.

### Structures Published with an Item Catalog Category

If a default ICC structure is defined for an ICC (see: *Creating an Item Catalog Category Structure*, page 8-11), the structure and any associated component attributes are published along with the ICC. You must publish the component items in the ICC structure before publishing the ICC.

## Prerequisites

- ☐ Before publishing any data, you must set up Oracle Data Integrator Artifacts and Web Services for Oracle Product Hub. For set up instructions, refer to My Oracle Support ID 1336146.1. For more information about the Web Services used during publishing, see: *Oracle Product Hub Web Services Overview*, page G-1.
- ☐ It is possible to create duplicate ICCs if multiple records with the same ICC name are simultaneously imported. To prevent creation of duplicate ICCs, create a custom unique index for the `MTL_ITEM_CATALOG_GROUPS_B` table. To create this index, determine which segments are enabled for the Item Catalogs flexfield (Flexfield Code MICG). If, for example, a two segment flexfield is used, with `SEGMENT8` and `SEGMENT12` enabled, then use the following command:  

```
SQL> create unique index MTL_ITEM_CATALOG_GROUPS_B_UC2 ON  
MTL_ITEM_CATALOG_GROUPS_B(SEGMENT8, SEGMENT12)
```
- ☐ Optionally, define the `oracle.apps.ego.item.publishItemCatalogCategory` event subscription if an external system requires an email notification when an ICC is published. When using Oracle Product Hub for Communications integrated with Oracle Communications Design Studio, Oracle recommends defining this event subscription before publishing an ICC to Design Studio.

### To define an event subscription:

You only need to perform this task if you want to send an email notification to a system owner when an ICC is published. In order to send an email notification to a user or a group of users, you must:

- Create a user and, optionally, a group of users, with associated e-mail addresses.
- Define a system.
- Define a subscription for the business event `oracle.apps.ego.item.publishItemCatalogCategory`.

Once an item catalog category is published to a system, an email is sent to the user or group associated with the `oracle.apps.ego.item.publishItemCatalogCategory` event

subscription.

**Tip:** If you plan to publish an ICC to two or more systems, you only need to define an event subscription for the systems that require an email notification upon ICC publication. Not every system requires notification.

#### **To create a user**

1. Refer to Maintaining People and Users, *Oracle E-Business Suite Security Guide* for instructions on how to create a user. Associate the person's e-mail address with their user account.

#### **To create a group of users**

2. Refer to Administering People, Groups, and Companies, page 11-6 for instructions on how to create a group.

#### **To define a system**

3. Refer to Defining Source Systems, page 12-5 for instructions on how to define a source system.

#### **To create an event subscription**

Refer to Defining Event Subscriptions, *Oracle Workflow Developer's Guide* for instructions on how to define an event subscription. The following steps explain how to define an event subscription specifically for oracle.apps.ego.item.publishItemCatalogCategory.

4. From the Workflow Administrator responsibility, navigate to Web Application > Administrator Workflow > Business Events.
5. Search for the event oracle.apps.ego.item.publishItemCatalogCategory.
6. For the event oracle.apps.ego.item.publishItemCatalogCategory, click the 'Subscription' icon, then click 'Create Subscription'.
7. In the **Create Event Subscriptions** page, specify the following field values:
  - System = Local System (for more information about the local system properties, refer to "Local System" in the Managing Business Events chapter, *Oracle Workflow Developer's Guide*).
  - Source Type = Local
  - Event Filter = oracle.apps.ego.item.publishItemCatalogCategory
  - Phase = Set this between 1-99. The execution of the subscription is deferred for phases larger than 100.

- Status = Enabled
  - Rule Data = Key
  - Action Type = Custom
8. Click Next.
9. In the **Create Event Subscription: Custom** page, specify the following field values:
- PL/SQL Rule Function = WF\_PUB\_ICC.SEND\_NOTIFICATION. When executing the event subscription, Oracle Workflow calls this procedure to handle the event message.
  - Enter the following Subscription Parameters:
    - PUB\_SYSTEM: Mandatory parameter. This is the internal name of the system interested in the published ICC.
    - WF\_ADMINISTRATOR: Optional parameter. This can be either a person or a group of persons. The administrator is notified of any errors during the workflow process (for example, if the receiver of the notification is invalid).
    - MAIL\_TO: Mandatory parameter. This can be either a person or a group of persons. The person listed here is the receiver of the notification about the published ICC.

**Important:** Input the values for WF\_ADMINISTRATOR and MAIL\_TO using the following format: For a single person, use the value PER:<User\_Name>. User\_Name is the user name used to log in to the application. For a group of people, use the value HZ\_GROUP:<GROUP\_NAME>. GROUP\_NAME represents the internal name of the group.

#### **To publish an item catalog category:**

1. From the **Oracle Applications Home Page**, select an appropriate responsibility, then Setup Workbench. Select the Items tab.
2. In the **Item Catalog Categories** page, search for and select an item catalog category.
3. In the **Basic Information** page, click Publish.
4. In the **ICC Publish Parameters** page, enter the following:
  - Batch Name. Enter a unique name.

- **Version.** Select one of the existing item catalog category versions to publish. When publishing to an Oracle Product Hub system, you must select a current or future effective version.
- **Publish to PIM Systems Only.** Check this box to publish to other Oracle Product Hub systems. Leave it unchecked to publish to external systems. You cannot publish to both Product Hub and external systems within the same Publish action.
- **Publish Parent ICC(s).** Select this box to publish any parent ICCs along with the selected ICC (item catalog category).
- **Publish Children ICC(s).** Select this box to publish any children ICCs along with the selected ICC.

5. Click Continue.

The **Publish** page appears. This page contains different fields depending on whether the Publish to PIM Systems Only field is checked or not.

**To publish an item catalog category to an external system (when the Publish to PIM Systems Only field is NOT selected)**

6. In the **Publish** page, enter the following:

- **Workspace Name.** Required. This field is used by Siebel for grouping multiple batches into a single group. Enter the name of the Siebel workspace project. If you are not using Siebel, enter any value.
- **Auto-Release.** Required. Select Default, Yes, or No from the list. This field refers to auto-releasing a Siebel workspace project.
  - Default - the workspace project auto-releases based on the setting of Auto Release within Siebel.
  - Yes - once the items are published to Siebel, the workspace project is automatically released.
  - No - once the items are published to Siebel, you need to manually release the workspace project.
- **Systems.** Systems previously associated with the ICC are listed, but you can add additional systems or remove any existing systems.

7. Click Save as Draft or Publish.

The **Publish Batch Search** page opens, displaying the status of the batch submitted. From here you can delete, copy, or update the batch, as well as view the batch's

history. See: Managing Publishing Batches, *Oracle Product Hub User's Guide*.

**Additional Information:** Possible statuses for the batch include:

- Batch Submitted - The batch has been submitted for processing, but the EGO Import Metadata concurrent program has not run yet.
- Draft Batch - The batch has been saved as a draft. You must update the batch, then select Publish to submit the batch.
- Failed - The ICC failed the import process for the specified target systems.
- Ignored - The system either does not intend or does not have the capability to consume the published entity.
- In Process - The Publish action is in process.
- Not Published - The ICC has previously been submitted for publishing to this system. It remains unpublished, in Draft batch status.
- Partially Succeeded - The ICC and some of the associated attribute groups and value sets have been successfully passed to the target systems. One or more associated attribute groups or value sets have failed the import process for one or more specified target systems.
- Succeeded - The ICC and all associated attribute groups and value sets have been successfully passed to the target systems..
- Warning - This is an aggregated status for a publish batch. An ICC publish batch could contain one ICC and multiple systems. Each ICC and system combination has an individual publish status.. Among all of the individual publish statuses, if one or more individual publish statuses is Ignored, Not Published, Fail, Partial Fail, or Partially Succeeded and none have the In Progress status, the aggregated status for the batch is Warning.

**To publish an item catalog category to an Oracle Product Hub system (when the Publish to PIM Systems Only field is selected)**

**Important:** Verify that the profile option "Enable PIM for Telco Features" is set to the same value in both the source and target Oracle

Product Hub systems.

8. In the **Publish** page, optionally select Auto Import.

Checking the Auto Import box automatically launches the EGO Import Metadata (EGOIMDCP) concurrent program in the target Oracle Product Hub system.

If you leave this box unchecked, you must manually run the EGO Import Metadata concurrent program in the target Oracle Product Hub system. When manually running this program, you can choose to import or not import the attribute groups and value sets associated with the ICC.

The following entities and relationships associated with the ICC are also published and imported, regardless of whether you check the Auto Import box or not:

- The selected ICC Version (if ICC Versions are in use).
- Item pages.
- Attribute groups and their attributes, including value sets associated to the attributes.
- Transaction attributes.
- Number Generation setting along with the function reference if function generated.
- Description Generation setting along with the function reference if function generated.

**Important:** Verify that the function definition referenced in the Number Generation or Description Generation settings (if function generated) in the source system already exists in the target systems. The function name, function parameters, and function parameter mappings are published, but not the function definition (for example, PL/SQL or a Java code).

- Parent ICCs (If selected in the **ICC Publish Parameters** page earlier).
- Child ICCs (If selected in the **ICC Publish Parameters** page earlier).

**Warning:** The publish and import process can create and update entities associated with an ICC (such as attribute groups, attributes, and value sets), but cannot delete entities already associated in the target system. For example:

- Publish ICC1 and its associated attribute group, AG1, to a target system. ICC1 and AG1 now exist in the target system.
- Delete attribute group AG1 from the source system.
- Associate attribute group AG2 to ICC1 in the source system.
- Publish ICC1 and its associated attribute group, AG2, to the target system.
- ICC1 in the target system now has two attribute groups, AG1 and AG2, associated with it.

To remove attribute group AG1 from the target system, you must manually delete it in the target system.

9. In the Select Systems region, click the Add Another Row icon as many times as necessary to select the target systems to which you want to publish.

Systems previously associated with the ICC are listed, but you can add additional systems or remove any existing systems.

10. Click Save as Draft or Publish.

The **Publish Batch Search** page opens, displaying the status of the batch submitted. From here you can delete, copy, or update the batch, as well as view the batch's history. See: Managing Publishing Batches, *Oracle Product Hub User's Guide*.

**Additional Information:** Possible statuses for the batch include:

- Batch Submitted - The batch has been submitted for processing, but the EGO Import Metadata concurrent program has not run yet.
- Draft Batch - The batch has been saved as a draft. You must update the batch, then select Publish to submit the batch.
- Failed - The ICC failed the import process for the specified target systems.
- Ignored - The system either does not intend or does not have the capability to consume the published entity.
- In Process - This status appears for the following reasons:
  - When Auto Import is selected - The Publish action is in

process. If there are any errors or time out issues, the system appends the status with an appropriate message indicating that either the EGO Import Metadata concurrent program did not process the updated records or the concurrent program is still running. Check the status of the EGO Import Metadata concurrent program in the target system and take appropriate action.

- When Auto Import is not selected - The Publish action is in process. The target interface tables were loaded successfully. Manually invoke the EGO Import Metadata concurrent program in the target system.
- Not Published - The ICC has previously been submitted for publishing to this system. It remains unpublished, in Draft batch status.
- Partially Succeeded - The ICC and some of the associated attribute groups, value sets, and attributes have been successfully passed to a particular target system. One or more associated attribute groups, value sets, or attributes have failed the import process for the target system.
- Succeeded - The ICC and all associated attribute groups and value sets have been successfully passed to the target systems.
- Warning - This is an aggregated status for a publish batch. An ICC publish batch could contain one ICC and multiple systems. Each ICC and system combination has an individual publish status.. Among all of the individual publish statuses, if one or more individual publish statuses is Ignored, Not Published, Fail, Partial Fail, or Partially Succeeded and none have the In Progress status, the aggregated status for the batch is Warning.

## Related Topics

Publishing Overview, *Oracle Product Hub User's Guide*

Assigning Systems to an Item Catalog Category, page 3-44

Publishing Value Sets, page 4-16

Publishing Attribute Groups, page 4-54

Publishing Items and Structures, *Oracle Product Hub User's Guide*

Using PL/SQL Public APIs to Publish Objects, *Oracle Product Hub User's Guide*



Viewing the Publishing History, *Oracle Product Hub User's Guide*

Oracle Product Hub Web Services Overview, page G-1

For more information about how product data synchronizes with participating applications using the publishing process, see Chapter 2: Oracle Product Master Data Management Integration Base Pack, *Oracle Product Master Data Management Integration 2.5 - Implementation Guide*. This guide is part of the Oracle Application Integration Architecture Release 2.5 Media Pack , which you can download from <http://edelivery.oracle.com>.

## Defining Report Templates Using Business Intelligence Publisher

Creating a template file consists of two basic steps:

1. Design your template layout - Use the formatting features of a word processing application and save the file as RTF.
2. Mark up your template layout - Insert the Business Intelligence Publisher simplified tags.

To create a template, you need to create or use any of the existing seeded data definitions and associate or assign the template file with the corresponding data definition file. For example, if you have a Change Order template file, associate it to a Change Order data definition file.

When you create a template, you need to assign it a data definition and upload the RTF. There are several seeded "Summary" templates available in the system. These cannot be updated. Initially, upload one template file for a specific language and territory combination. This file will become the Default Template File. To upload additional template files or to change the Default Template File, use the **View Template** page.

### To create a template:

1. Using the XML Publisher Administrator responsibility, navigate to the **Templates** page.
2. Select the Create Template button.
3. Provide the following information:

#### **Name**

Enter a user-friendly name for your template.

#### **Code**

Assign a template code using the product short name and a descriptive ending.

#### **Application**

While creating templates, select the correct application from the LOV. This

application should be the same as the application of the Data definition. For Change Templates, the supported application is "Engineering".

#### **Data Definition**

Select the product's data definition from the LOV. Oracle Engineering (Application Short Name = ENG) and Oracle Product Hub (Application Short Name = EGO) come seeded with many different data definitions for changes and items. For example, for an Issue Report Template, select the seeded Oracle Engineering Data Definition for Issues. This will result in the Template being available for Issue Reports. Select the input template type (RTF, PDF, or XSL-FO) from the LOV.

#### **Start Date**

Enter the date from which the template will be active.

#### **End Date**

To make the template inactive, enter an end date.

#### **Subtemplate**

If this is a subtemplate, select the check box.

**Additional Information:** A subtemplate is referenced by other templates, but cannot run on its own.

#### **File**

Use the Browse button to upload your RTF or PDF template layout file.

#### **Language**

Select the template language from the LOV. Add more language template files to your template definition from the View Template page.

#### **Territory**

Select the language territory from the LOV.

**Important:** After the template definition is created, the following fields cannot be updated: Application, Code, and Type. You can update the template from the View Template page.

In the template file, you can include simplified XML tags to refer to the Item information you want Business Intelligence Publisher to print while generating the report.

For further information, see: *Oracle Fusion Middleware Report Designer's Guide for Oracle Business Intelligence Publisher*.

### **Associating Report Templates to Item Catalog Categories:**

Associating a Report Template with an item catalog category makes these templates available for all items in the item catalog category. Report Templates are inherited throughout the item catalog category hierarchy—child categories inherit the report templates from the parent category.

#### **To associate a report template to an item catalog category**

1. On the **Search: Item Catalog Categories** page, locate the item catalog category to which you wish to associate a report template.
2. On the **Basic Information** page of the item catalog category, click the Report Templates link.
3. In the **Report Templates** page, click Add Report Templates.
4. In the Add Report Templates page, search for templates to add using the Name or Description field.  
  
Enter a search term in either or both fields. Use the percent (%) sign before and/or after a search term as needed to perform a Wildcard search.
5. Select all report templates that you want to add to the item catalog category.
6. Click Apply.

### **Related Topics**

Generating Reports, *Oracle Product Hub User's Guide*

## **Defining Cross Reference Types**

Items can have many cross-references. Examples of cross-references include customer part numbers and superseding/preceding item numbers. An administrator can define their own cross-reference types based on their business need. User-defined flex-fields and value sets can be associated with each cross-reference type. You can also activate or inactivate cross-reference types using effectivity dates.

#### **To create a cross reference type:**

1. On the **Search: Item Catalog Categories** page, click the Cross Reference Types tab.
2. In the **Search: Cross Reference Types** page, click Create.

## Defining Cross Reference Types

Search : Cross Reference Types

### Search

To find your cross-reference type, select a filter in the poplist, and enter a word in the text field, then click the "Go" button. To see a list of **all** the cross-reference types, clear all the text fields, and click the "Go" button.

\* Search by

### Results

| <input type="button" value="Create"/> |                           |               |        |                       |        |
|---------------------------------------|---------------------------|---------------|--------|-----------------------|--------|
| Cross Reference Type /                | Description               | Inactive Date | Update | Additional Attributes | Delete |
| ChargeMaster                          | Charge Description Master |               |        |                       |        |
| Customer                              | Customer part number      |               |        |                       |        |
| Substitute                            | Substitute part number    |               |        |                       |        |
| Vendor                                | Vendor part number        |               |        |                       |        |

3. On the **Create/Update Cross-Reference Type** page, provide the following information:
  - Type - the name of the cross reference type. Required.
  - Description. Optional.
  - Inactive On - the date the cross reference type becomes inactive. Optional.
4. Click Apply.

## Importing Item Catalog Metadata

Oracle provides the following methods for importing item catalog metadata:

- FNDLOAD
- iSetup
- Interfaces and Public APIs
- Import from Excel
- Webservices

Refer to My Oracle Support ID 1504980.1 for a description each method's benefits and limitations, as well as examples of how to use each method. You can find additional information about some of the above methods in the following topics:

- Using Oracle iSetup, page 1-2
- Import from Excel

- Importing Item Categories Using a Spreadsheet, *Oracle Product Hub User's Guide*
- Importing Items Using a Spreadsheet, *Oracle Product Hub User's Guide*
- Importing Item Revisions Using a Spreadsheet, *Oracle Product Hub User's Guide*
- Importing Item Organization Assignment Using a Spreadsheet, *Oracle Product Hub User's Guide*
- Importing Item People Assignment Using a Spreadsheet, *Oracle Product Hub User's Guide*
- Web Services for Publishing, page G-12

The rest of this topic provides more information about using FNDLOAD and the open interface tables used to import item catalog metadata.

## Using FNDLOAD to Import Item Catalog Metadata

The Generic Loader (FNDLOAD) is a concurrent program that can move Oracle E-Business Suite data between database and text file representations. The loader reads a configuration file to determine what data to access. It can download data from an application entity into a portable, editable text file. You can then upload the file into any other database to copy the data. Conversion between database store and file format is specified by a configuration file read by the loader. FNDLOAD downloads data from a database according to a configuration (.lct) file, and converts the data into a data file (.ldt file). The Generic Loader can then upload this data to another database using a configuration file. Use FNDLOAD to import and export item catalog categories and the following related entities:

- Attribute Groups, page 4-21
- Value Sets, page 4-1
- Attribute Group Associations, page 4-48
- Item Pages and Item Page entries, page 3-28
- Display Formats, page 3-35
- Import Formats, page 3-40
- Search Criteria, page 3-29

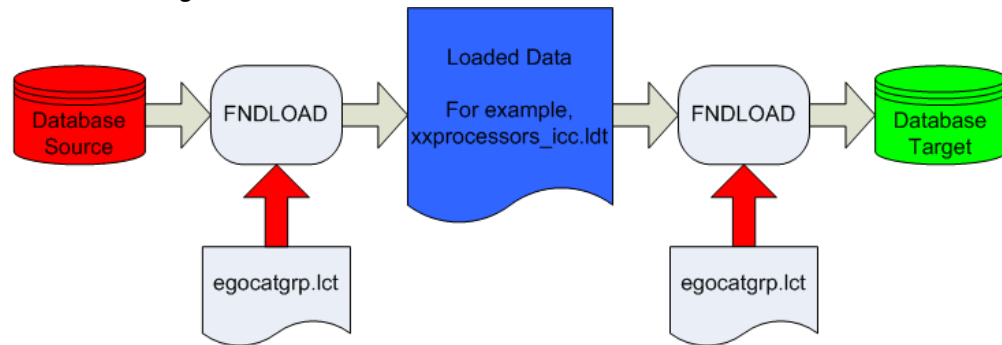
In addition, you can use FNDLOAD to import and export the following entities:

- Alternative Catalogs, page 5-4

- Roles, Privileges, and Role Mappings, page 11-3
- Structure Types, Structure Names, and Structure Name Associations, page 8-4

Refer to My Oracle Support (<https://support.oracle.com>) ID 1504980.1 for specific instructions on how to import and export each entity using FNDLOAD.

#### **Data Transfer Using FNDLOAD**



### **Using Open Interface Tables to Import Item Catalog Metadata**

Use open interface tables to import the following item catalog metadata:

- Item catalog categories (ICCs)
  - ICC definition  
See: Defining Item Catalog Categories, page 3-2
  - Attribute group association  
See: Associating Attribute Groups with an Item Catalog Category, page 4-48
  - Item pages association  
See: Creating Pages for an Item Catalog Category, page 3-28
  - Item number and description generation methods  
See: Defining Item Numbers and Descriptions, page 3-26
  - New item request association  
See: Associating New Item Requests with an Item Catalog Category, page 3-22
  - ICC versions
  - Transaction attributes

See: Defining Transaction Attributes, page 4-33

- Value sets

See: Defining Value Sets for User-Defined Attributes, page 4-1

- Non-versioned value sets and its values
- Versioned value sets and its values

See: Defining Value Set Versions for Transaction Attributes, page 4-7

- Child value sets and its values

- Attribute groups and attributes

See: Defining Item Attributes and Attribute Groups, page 4-21

- Functions

See: Creating User-Defined Functions for Item Attributes, page 4-38

- Function header
- Function parameter

Using the concurrent program EGO Import Metadata (EGOIMDCP), you can choose to validate and import the above data from the open interface tables into the production tables collectively or one at a time by selecting one of the following parameters:

- Import Value Sets
- Import Attribute Groups
- Import Item Catalog Categories
- Batch Id (Null for All).

Selecting the Batch Id parameter imports all batch data into the production tables.

- Delete Successfully Processed Records

**Tip:** When troubleshooting this import process, leave this parameter set to No.

### ***Item Catalog Metadata Open Interface Tables***

| <b>Table Name</b>            | <b>Description</b>                                                                                                                                                                                                       |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EGO_FLEX_VALUE_SET_INTF      | This table stores intermediate information about value sets. It stores information for both value set versions and non-versioned value sets, as well as the parent-child relationship for a value set.                   |
| EGO_FLEX_VALUE_INTF          | This table stores intermediate values for an associated value set. It also stores table and column information for a table type value set. It stores values corresponding to each version for a value set with versions. |
| EGO_FLEX_VALUE_TL_INTF       | This table stores translations for value set values in each of the installed languages.                                                                                                                                  |
| EGO_ATTR_GROUPS_INTERFACE    | This table stores the definitions for attribute groups.                                                                                                                                                                  |
| EGO_ATTR_GROUPS_DL_INTERFACE | This table stores the definitions for attribute group business entities (data levels).                                                                                                                                   |
| EGO_ATTR_GROUP_COLS_INTF     | This table stores the definitions for attributes.                                                                                                                                                                        |
| EGO_PAGES_INTERFACE          | This table stores the definitions for Item Catalog Category pages.                                                                                                                                                       |
| EGO_PAGE_ENTRIES_INTERFACE   | This table stores the definitions for Item Catalog Category page entries.                                                                                                                                                |
| EGO_FUNCTIONS_INTERFACE      | This table stores the function header information.                                                                                                                                                                       |
| EGO_FUNC_PARAMS_INTERFACE    | This table stores the function parameters information.                                                                                                                                                                   |
| MTL_ITEM_CAT_GRP_INTERFACE   | This table stores the ICC header record mimicking the base table MTL_ITEM_CATALOG_GROUPS_B.                                                                                                                              |



| Table Name                    | Description                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EGO_ATTR_GRP5_ASSOC_INTERFACE | This table stores the Attribute Group associations with ICCs.                                                                                                                 |
| EGO_ICC_VERS_INTERFACE        | This table stores version related information. It links to the Transaction Attributes interface table (EGO_TRANS_ATTRS_VERS_INTF) by ICC name/ID and version sequence number. |
| EGO_TRANS_ATTRS_VERS_INTF     | This table stores information about the transaction attributes for a specific ICC version.                                                                                    |
| EGO_FUNC_PARAMS_MAP_INTERFACE | This table stores function parameters mapping for number generation and description generation for ICCs.                                                                      |

**Warning:** It is possible to create duplicate ICCs if multiple records with the same ICC name are simultaneously imported. To prevent creation of duplicate ICCs, create a custom unique index for the MTL\_ITEM\_CATALOG\_GROUPS\_B table. To create this index, determine which segments are enabled for the Item Catalogs flexfield (Flexfield Code MICG). If, for example, a two segment flexfield is used, with SEGMENT8 and SEGMENT12 enabled, then use the following command:

```
SQL> create unique index MTL_ITEM_CATALOG_GROUPS_B_UC2 ON
MTL_ITEM_CATALOG_GROUPS_B (SEGMENT8, SEGMENT12)
```

**Warning:** The following columns in the open interface tables are either used by the system during record processing or are not supported at this time. Do not populate these columns with values.

| Column         | Appears in Table              |
|----------------|-------------------------------|
| ASSOCIATION_ID | EGO_ATTR_GRP5_ASSOC_INTERFACE |
| DATA_LEVEL     | EGO_ATTR_GRP5_ASSOC_INTERFACE |

| Column            | Appears in Table                                                                                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| DATA_LEVEL_ID     | EGO_ATTR_GRP_ASSOC_INTERFACE                                                                                                                 |
| END_DATE          | EGO_ICC_VERS_INTERFACE                                                                                                                       |
| VERSION_SEQ_ID    | EGO_ICC_VERS_INTERFACE                                                                                                                       |
| INVENTORY_ITEM_ID | EGO_TRANS_ATTRS_VERS_INTF                                                                                                                    |
| ORGANIZATION_ID   | EGO_TRANS_ATTRS_VERS_INTF                                                                                                                    |
| REVISION_ID       | EGO_TRANS_ATTRS_VERS_INTF                                                                                                                    |
| ASSOCIATION_ID    | EGO_TRANS_ATTRS_VERS_INTF                                                                                                                    |
|                   | <b>Caution:</b> Do not populate this column if the Transaction Type = Create. You must populate this column for all other transaction types. |
| ATTR_ID           | EGO_TRANS_ATTRS_VERS_INTF                                                                                                                    |
|                   | <b>Caution:</b> Do not populate this column if the Transaction Type = Create. You must populate this column for all other transaction types. |
| CREATED_BY        | Appears in all item catalog metadata open interface tables                                                                                   |
| CREATION_DATE     | Appears in all item catalog metadata open interface tables                                                                                   |
| LAST_UPDATE_LOGIN | Appears in all item catalog metadata open interface tables                                                                                   |

| Column                 | Appears in Table                                           |
|------------------------|------------------------------------------------------------|
| LAST_UPDATED_BY        | Appears in all item catalog metadata open interface tables |
| LAST_UPDATE_DATE       | Appears in all item catalog metadata open interface tables |
| REQUEST_ID             | Appears in all item catalog metadata open interface tables |
| PROGRAM_APPLICATION_ID | Appears in all item catalog metadata open interface tables |
| PROGRAM_ID             | Appears in all item catalog metadata open interface tables |
| PROGRAM_UPDATE_DATE    | Appears in all item catalog metadata open interface tables |
| TRANSACTION_ID         | Appears in all item catalog metadata open interface tables |
| START_DATE_ACTIVE      | MTL_ITEM_CAT_GRP_INTERFACE                                 |
| END_DATE_ACTIVE        | MTL_ITEM_CAT_GRP_INTERFACE                                 |
| SUMMARY_FLAG           | MTL_ITEM_CAT_GRP_INTERFACE                                 |
| ITEM_DESC_ACTION_ID    | MTL_ITEM_CAT_GRP_INTERFACE                                 |
| ITEM_NUM_ACTION_ID     | MTL_ITEM_CAT_GRP_INTERFACE                                 |

### To import item catalog metadata using open interface tables:

1. Load the item catalog metadata open interface tables with the data you want to import. For detailed help on loading the Open Interface tables, refer to the

Integration Repository. See: Oracle Integration Repository User's Guide.

## What's Next

Here are some suggestions if you encounter problems while importing item catalog metadata using the open interface tables:

- Consider writing errors into the MTL\_INTERFACE\_ERRORS table.
- Perform the following steps before logging a bug:
  - Set the profile option 'INV: Debug Trace' to Yes. This will write detailed debug messages into the concurrent program log file.
  - Turn ON the trace for the concurrent program 'EGO Import Metadata' (short name EGOIMDCP).
  - While launching the concurrent program, set the parameter 'Delete Successfully Processed Records' to 'No'.
- In the bug, provide the following information:
  - Concurrent Program log file
  - Raw Trace file. If it is a performance issue, then generate the tkprof file for this raw trace file and provide both the raw trace file as well as the tkprof file.
  - Set of records from the table mtl\_interface\_errors for this specific request\_id.
  - Set of records from the specific entity's interface table before and after launching the concurrent program.
  - List of invalid objects in EGO schema.
- If a running EGO Import Metadata concurrent program terminates unexpectedly, then refer to Instructions for Restarting Concurrent Programs with a Manual Restart Mode, page H-1.

---

## Setting Up Attributes and Functions

This chapter covers the following topics:

- Defining Value Sets for User-Defined Attributes
- Defining Value Set Versions for Transaction Attributes
- Defining Table Value Sets
- Creating Independent Values
- Publishing Value Sets
- Defining Item Attributes and Attribute Groups
- Defining Transaction Attributes
- Using Seeded Attributes
- Using Additional Attributes
- Creating User-Defined Functions for Item Attributes
- Defining Attribute Usages
- Displaying Attachments on Item Attribute Pages
- Associating Attribute Groups with an Item Catalog Category
- Adding Actions to an Attribute Group
- Publishing Attribute Groups

### Defining Value Sets for User-Defined Attributes

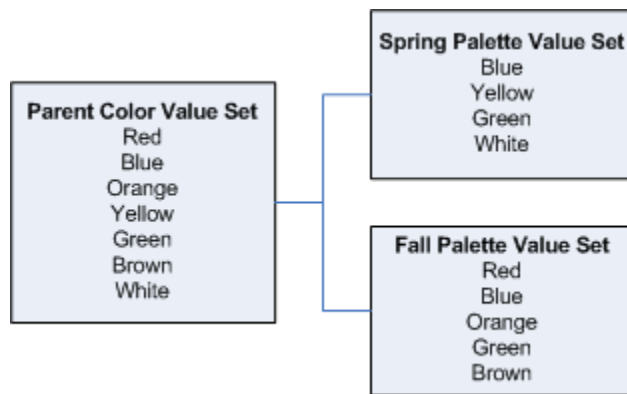
User-defined attributes capture all the detailed information (such as cost information) about an object (items, change requests or change orders). The item catalog enables you to create user-defined attributes with validation logic to support the needs of your organization. To do so, create value sets and associate the value sets with user-defined attributes. Attributes can have a static or dynamic list of valid values, or a range of values.

For each user-defined attribute, you can optionally specify a value set with data type and validation rules to be applied when the user inputs data. Once created, value sets can be reused for different attributes. It is recommended that you create your value set before defining your attribute groups.

You can also create child value sets that are subsets of a parent value set. The parent value set includes all possible values while the child value sets include some of the values from the parent value set. Use parent and child value sets for variant attributes when creating SKUs from style items (see: Overview of Style and SKU Items, *Oracle Product Hub User's Guide*).

### Parent and Child Value Set Example

A clothing retailer sells a certain style of shirt in different colors for different seasons. Define all colors in the parent value set, then define a child value set for each season.



When creating the color variant attribute, specify the parent value set for the attribute. When creating the style item, you can choose either the parent or one of the child value sets for the color variant attribute.

Prior to creating value sets for user-defined attributes establish the validation rules for each attribute as follows:

- Choose the data type for the values.
- Know the range of the values.
- Know which set of actual values will satisfy the required condition for the attribute. For static values, specify a fixed set of values. Dynamic values are retrieved from a database table or view.
- Select the manner in which the values will be displayed: list of values (LOV), pop list (dropdown) or radio group.

The following table shows the validation rules for the attribute group Benchmark Rating.

| Attribute Group  | Attributes           | Data Type | Value Range     | Values                                          | Display    | Create Value Sets |
|------------------|----------------------|-----------|-----------------|-------------------------------------------------|------------|-------------------|
| Benchmark Rating | Risk Rating          | Number    | 1 to 5          | Static: 1,2,3,4,5                               | Pop list   | Rating            |
| Benchmark Rating | Quality Rating       | Number    | 1 to 5          | Static: 1,2,3,4,5                               | Pop list   | Rating            |
| Benchmark Rating | I/O Subsystem Rating | Number    | 1 to 5          | Static: 1,2,3,4,5                               | Pop list   | Rating            |
| Benchmark Rating | Video System         | Number    | 1 to 5          | Static: 1,2,3,4,5                               | Pop list   | Rating            |
| Benchmark Rating | Overall Rating       | Number    | 1 to 5          | Static: 1,2,3,4,5                               | Pop list   | Rating            |
| Benchmark Rating | Failure Rate         | Number    | Min 0<br>Max100 | Within the range                                | Text Field | Percent Range     |
| Benchmark Rating | Supplier             | Char      | N/A             | Dynamic:<br>Queried from tables in the database | LOV        | Primary Supplier  |

If you need a Yes or No value set, you can use the seeded value set EGO\_YES\_NO. This value set automatically displays when you set the Display As field to "Checkbox" for the attribute. You can see some of the seeded value sets on the **Value Set** page; these are used to support images and attachments in the long description on the **Item Summary** pages.

**Additional Information:** You can import value sets using open interface tables and the concurrent program EGO Import Metadata (EGOIMDCP). For more information, refer to Importing Item Catalog Metadata, page 3-58

For more information about User Defined Attributes in Oracle Purchasing and Oracle Contract Lifecycle Management, see:

- Oracle Purchasing Help Index, *Oracle Purchasing User's Guide*

- Oracle CLM Purchasing Help Index, *Oracle Contract Lifecycle Management for Public Sector Purchasing User Guide*

**To create a value set:**

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Catalog Categories** page, click the Value Sets tab.
3. On the **Maintain Value Sets** page, click Create.
4. On the **Create Value Set** page, enter the following information:

**Value Set Name**

The name by which the system and users keep track of the value set. You can only enter alphanumeric (a, b, c,..., 1, 2, 3,...) and the underscore ( \_ ) characters for the value set name. You cannot use spaces or special characters. The length is limited to 15 characters.

**Important:** Once specified, you cannot edit the value set name.

**Description**

Optionally, enter a description of the value set.

**Data Type**

The data type of the value set. The data type that you select determines the values that are available in the value set. An attribute's data type must match the data type specified for that attribute's value set. The different data types are:

- Character
- Number
- Standard Date
- Standard Date Time

**Important:** You cannot edit the data type once the value set has been created.

**Warning:** You cannot create SKUs with the following variant attribute data types:

- Standard Date



- Standard Date Time
- Translatable Text

### Maximum Size

Specify the value for Maximum Size if you wish to limit the user's input in the attribute text field. For example, in some cases you may wish to limit the number of characters in the attribute Date to 10 characters, or the number of characters in Name to 50. Keep the default value of 0 if you wish to omit this particular validation.

### Validation Type

A set of values against which the values entered by users are validated. Choices are:

- None

There is no explicit set of values against which the user's input is validated.

- Independent

The explicit values against which the user's input is validated are defined here. To create explicit values, select Table and click Apply and Continue to edit information on the **Enter Validation Table Information** page. If you select this validation type, see: Creating Independent Values, page 4-14. Select the "Poplist" radio button to display valid values as a drop-down list. Select the "List of Values" radio button to display valid values as a searchable list of values.

- Translatable Independent

This validation type behaves the same as Independent, but enables the display of values in another language.

- Table

The explicit values against which the user's input is validated comes from a database table. To create explicit values, select Independent and click Add Values or click Table and then click Edit Table Information. If you select this validation type, see: Defining Table Value Sets, page 4-12.

**Additional Information:** Display value sets using the Table validation type as a list of values (LOV) or as a pop list.

### Display Type

- Poplist

**Additional Information:** Display as Radio Group can be set while creating or editing attributes. See: Defining Item Attributes and Attribute Groups, page 4-21. Oracle recommends that when you use the value set type Independent, you save it as a pop list.

- List of Values

**Important:** You can associate the Value Set to an attribute only if the above conditions are satisfied and the Security Type is 'No Security'. Value sets created in forms with a security type are not applicable for PLM/PIM and are not visible in the HTML user interface.

5. Optionally specify a Minimum Value and Maximum Value in the "Value Range" section. If you selected the Validation Type "None" and a data type of "Number," "Standard Date," or "Standard Date Time," you can enter a minimum and maximum value for your value set. If the data type is "Standard Date" or "Standard Date Time," you can substitute the \$SYSDATE\$ token for the current date. \$SYSDATE\$ is a min/max value placeholder that is replaced by the current day on the day the user enters a value for the attribute. You can also add or subtract whole numbers from \$SYSDATE\$. For example, "\$SYSDATE\$ - 4" would be four days prior to the current date.
6. Click Apply to stop after creating one value set or click Apply and Continue to create another value set.

**Important:** The system's user-defined attributes framework does not support the following value set options. These options can be defined using the flex-fields value set forms.

- Uppercase only
- Number only
- Precision

The **Value Set Details** page is displayed upon completion. You can edit certain settings after creation by clicking Update. You can find all existing value sets on the **Value Sets** page. You can search for value sets using the criteria Name, Description, Data type, or Validation type, as well as search for values within value sets. Optionally, add a description to a value.

### To create a child value set:

After creating a value set, the system returns you to the **Maintain Value Sets** page. You can create a child value set directly from this page or from the **Value Set Details** page.

You can only create a child value set when the parent value set has a validation type of Independent or Translatable Independent.

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Catalog Categories** page, click the Value Sets tab.
3. On the **Maintain Value Sets** page, click the Create Child Value Set icon for a value set.

Alternatively, select the value set name. From the **Value Set Details** page, in the Child Value Sets region, click Create.

4. In the **Create/Edit Child Value Set** page, enter the following information:

- **Child Value Set Name**

The name by which the system and users keep track of the child value set. You can only enter alphanumeric (a, b, c,..., 1, 2, 3,...) and the underscore ( \_ ) characters for the child value set name. You cannot use spaces or special characters. The length is limited to 15 characters.

**Important:** Once specified, you cannot edit the child value set name.

#### Description

Optionally, enter a description of the child value set.

#### Select Values

Select a value from the parent value set and use the arrow buttons to move (or remove) the value into the child value set.

5. Once you have selected all of the values for the child value set, click Apply.

## Related Topics

Defining Table Value Sets, page 4-12

Creating Independent Values, page 4-14

## Defining Value Set Versions for Transaction Attributes

You can represent date-dependent item characteristics using transaction attributes

applied to the item during a transaction (not during item creation, like user-defined attributes). The value sets used by the transaction attributes are also date-dependent. This date dependency is enabled by creating versions of a value set. Each released version of the value set is valid for a different time period. Only transaction attributes use value set versions; all other attributes use non-version value sets (See: Defining Value Sets for User-Defined Attributes, page 4-1). The following example shows how multiple value set versions might work in the cellular phone industry.

#### **Mobile Phone Service Plan Example**

Currently (February 1), when a customer purchases a mobile phone service plan, they can choose from the following amounts of included minutes defined in the TALKTIME value set, version 1:

- 100 minutes
- 200 minutes
- 500 minutes

There is a promotion scheduled for a limited time, through (March 1-31). During this promotion, customers can select one of the following TALKTIME, version 2, values:

- 200 minutes
- 500 minutes
- Unlimited minutes

When the promotion ends (April 1), the following options are available indefinitely by selecting one of the following TALKTIME, version 3, values:

- 100 minutes
- 200 minutes
- 500 minutes
- 1000 minutes

**Additional Information:** You can import value set versions using open interface tables and the concurrent program EGO Import Metadata (EGOIMDCP). For more information, refer to Importing Item Catalog Metadata, page 3-58

## **Prerequisites**

- ☐ Set the profile option "Enable PIM for Telco Features" to Yes.

**To create a value set draft version:**

1. From the Oracle Applications Home Page, click the Setup Workbench link.
2. On the **Item Catalog Categories** page, click the Value Sets tab.
3. On the **Maintain Value Sets** page, click Create.
4. On the **Create Value Set** page, enter the following information:

**Value Set Name**

The name by which the system and users identify the value set. You can only enter alphanumeric (a, b, c,..., 1, 2, 3,...) and the underscore ( \_ ) characters for the value set name. You cannot use spaces or special characters. The length is limited to 15 characters.

**Important:** Once you have created the value set, you cannot edit the value set name.

**Description**

Optionally, enter a description of the value set.

**Data Type**

The data type of the value set. The data type that you select determines the values that are available in the value set. An attribute's data type must match the data type specified for that attribute's value set. The different data types are:

- Character
- Number
- Standard Date
- Standard Date Time

**Important:** You cannot edit the data type once the value set has been created.

**Maximum Size**

Specify the value for Maximum Size if you wish to limit the user's input in the attribute text field. For example, in some cases you may wish to limit the number of characters in the attribute value field to 10 characters. Enter the value of 0 if you wish to omit this particular validation. The default value is 150.

**Note:** The maximum number of characters for a character type attribute is 150 characters. Translatable text fields have a limit of 4,000 characters. Number type fields have a maximum size and precision of 22.5 (22 digits to the left of the decimal and 5 digits to the right).

### Enable Versioning

Optionally, you can enable the creation of value set versions for use with transaction attributes only by selecting Enable Versioning. This box only appears when the profile option "Enable PIM for Telco Features" is set to Yes.

**Important:** If you choose not to enable versioning, then follow the instructions for creating value sets at Defining Value Sets for User-Defined Attributes, page 4-1 .

### Validation Type

Select the type of values against which the values entered by users are validated. Choices are:

- Independent  
This option requires the specification of explicit values against which the user's input is validated. If you select this validation type, see: Creating Independent Values, page 4-14. .
- Translatable Independent  
This validation type behaves the same as Independent, but enables the display of values in another language.

### Display Type

- Poplist  
Select the "Poplist" radio button to display valid values as a drop-down list.
- List of Values  
Select the "List of Values" radio button to display valid values as a searchable list of values

5. Click Apply to stop after creating one value set or click Apply and Continue to create another value set.

The **Value Set Details** page is displayed upon completion. You can edit certain settings after creation by clicking Update. You can find all existing value sets on the **Value Sets** page. You can search for value sets using the criteria Name, Description,

Data type, or Validation type, as well as search for values within value sets. Optionally, add a description to a value.

#### **To add values to the value set**

6. After creating a value set for transaction attributes, you must define the values. See: Creating Independent Values, page 4-14.

You must add values to the value set in order to release the value set for use by transaction attributes.

#### **To release a version of the value set**

Only the user listed in the Locked By field can release a version of a value set. Once you release a value set, you cannot update the released version. Only released versions can be associated to transaction attributes.

7. On the **Value Set Details** page, select the Draft version, then click Release.
8. Optionally, on the **Release Value Set Version** page, you can update the version description and enter a start date. The start date defaults to the current date if this field is left blank.
  - If you release another version with a later start date, the system assigns an end date to the previous version that is 1 second before the next version's start date. You cannot have two versions that are valid at the same time.
  - Once you release a version, the new Draft version has the same definition as the latest released version. In order to release the draft, you must change the definition.
9. Click Apply.

#### **To lock and unlock a draft value set**

A version enabled value set is automatically locked upon creation of the draft. Only the user listed in the Locked By field can update the value set details or release the value set to create a version. To enable another user to update the value set details, unlock the value set. The other user can then lock it and make updates or release the value set. Locking the value set only locks the draft version's value set details. You can update the value set primary information, such as description, regardless of the lock status. This primary information applies to all versions.

10. On the **Value Set Details** page, select the Draft version.
11. If the draft is locked, and you are the user who locked it, the Unlock button is visible. Click Unlock.

The Locked By field displays the name of the person who locked it. Only the person displayed in the Locked By field can unlock the draft.

12. If the draft is unlocked, the Locked By field is blank and the Lock button is visible. Click Lock.

### To update a value set

Before updating a value set draft version, you must lock it.

**Warning:** Only update value set versions from the Setup Workbench (HTML interface). Updating value set versions from the Forms interface could lead to data corruption.

13. To change the value set primary (header) information for all value set versions, in the **Value Set Details** page, click the Update button.
14. To change the draft version's description or start date, in the **Value Set Details** page, select the draft version, then click the draft version's Update icon.
15. To change the draft version's values, you can click the Update or Delete icon for a particular value or click Create to create a new value.

### To revert the draft value set to an earlier version

The Revert Draft to button enables you to refresh the draft version's values with a released version's values.

16. Select a released version that contains the values you want the draft version to contain.
17. Click Revert Draft to.

The draft version now contains the same values as the released version selected.

## What's Next

After defining and releasing a value set version, you can use this version to define a transaction attribute. See: *Defining Transaction Attributes*, page 4-33.

## Defining Table Value Sets

When creating a value set, you must specify a validation type. If you select the validation type "Table", then you must define the table whose values comprise the value set.

Table value sets build queries that draw valid values from a table/view. For example, you have identified the table EMP as having a value column EMP\_ID, and the ID



column EMP\_NAME. If a user searches for a person named "Smith", the query is directed to the table EMP. The search value entered is "Sm", so the actual search value used is "Sm%". The value set builds a search of the EMP table to find all employees whose names begin with "Sm". A list of values containing all employees whose last names begin with "Sm" is returned to the user. Then the user selects the correct employee, and that employee ID is the value that gets stored for the attribute.

**Note:** Identifying a table is part of the process of creating a value set. Before identifying a table, you must create a value set. See: Defining Value Sets for User-Defined Attributes, page 4-1.

### **To identify a table for a value set:**

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Catalog Categories** page, click the Value Sets tab. Find and select the value set for which you want to enter validation table information.
3. In the **Value Set Details** page, Validation Table Information region, click Update.
4. On the **Enter Validation Table Information** page, enter the following information:

#### **Application Name**

The name of the application in which the table is located.

#### **Table Name**

The name of the database table or view in the schema.

5. In the Value Column section, enter the following information:

#### **Name**

The name of the column that stores the value displayed to the user.

#### **Type**

The data type of the value column.

#### **Size**

The size of the value column.

6. In the Meaning Column section, enter the following information:

#### **Name**

The name of the column that stores the description of the value.

#### **Type**

The data type of the meaning column.

**Size**

The size of the meaning column.

7. In the ID Column section, enter the following information:

**Name**

The name of the column that stores the value internally in the table.

**Type**

The data type of the ID column.

**Size**

The size of ID column.

8. In the "Where Clause" section, enter an additional Where clause to further constrain a query. For example, go back to the example above. Say you wish to further constrict the search results by only searching for current employees. In this case, you would add an additional Where clause defining the JOB\_STATUS as ACTIVE.

You can also use bind values in Where clauses in the following ways:

- You can refer to other attributes in the same attribute group as the attribute that uses this value set by using the following syntax:

**:\$ATTRIBUTEGROUP\$.<your attribute's internal name>**

For example:

**(lookup\_type='EGO\_EF\_Industry\_TYPE' and instr(:\$ATTRIBUTEGROUP\$.Attr1 , tag) > 0 )**

**Important:** In the example above, you must include a space between the comma and the word "tag" in order for the sql string to parse correctly.

- You can refer to primary keys for the object to which the attribute group is associated by using the following syntax:

**:\$OBJECT\$.<the object's primary key column name>**

9. Click Apply.

## Creating Independent Values

When creating a value set, you must specify a validation type. If you select the validation type "Independent", then you must define the independent values that comprise the value set.

## Prerequisites

- ☐ Creating an independent value is part of the process of creating a value set. Before creating an independent value, you must create a value set. See: *Defining Value Sets for User-Defined Attributes*, page 4-1.
- ☐ When you initially install or upgrade to Release 12.2.2 or later, no users are allowed to view, insert or update any value set values. You must explicitly set up access for specific users by enabling appropriate grants and roles for those users. For more information about setting up access, refer to *Flexfield Value Set Security*, *Oracle E-Business Suite Flexfields Guide*.

### To create an independent value type:

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Catalog Categories** page, click the Value Sets tab.  
Find and select the value set for which you want to enter values.
3. In the **Value Set Details** page, Values region, click Create.
4. On the **Create Value** page, enter the following information:

#### **Value (Internal Name)**

The internal name of the value; this is a valid value in your value set. The internal name must be of the same data type as the value set data type.

#### **Translated Value**

The display name of the value in another language. This field only appears if the value set validation type is Translatable Independent.

#### **Description**

The description of the value. The description does not have to be the same data type as the value set data type.

#### **Enabled**

Specify whether or not the value is enabled (and available for use) or disabled (and not available for use).

#### **Sequence**

The order or sequence in which the independent values are displayed.

#### **Start Date**

The date on which this value becomes a valid value in the value set.

#### **End Date**

The date on which this value is no longer a valid value in the value set.

5. Click Apply or Add Another. Both buttons save the value to the database.

## Publishing Value Sets

You can search for and select one or more value sets and publish them to one or more systems, such as Siebel Customer Relationship Management, Oracle Billing and Revenue Management, or another Oracle Product Hub instance. Alternatively, you can publish value sets with their associated item catalog category (see: Publishing Item Catalog Categories, page 3-45). When you publish a value set, all associated child value sets are also published. You can only publish value sets associated with attributes:

- Associate value set versions to transaction attributes in the ICC.
- Associate value sets without versions to user-defined, seeded, or component attributes in the ICC.

Oracle Product Hub does not support value sets with a data type of Boolean or Integer, yet Siebel Customer Relationship Management uses value sets with these data types. In order to enable the process integration between Oracle Product Hub and Siebel, use the suffixes `_BOOL` and `_INT` in the Value Set Name field as appropriate when creating a value set. The process integration to Siebel parses the name to create the corresponding value sets in Siebel with data type Boolean and Integer, respectively.

**Additional Information:** Alternatively, you can use a PL/SQL Public API to publish value sets. See: Using PL/SQL Public APIs to Publish Objects, *Oracle Product Hub User's Guide*.

For more information about creating value sets, refer to:

- Defining Value Sets for User-Defined Attributes, page 4-1
- Defining Value Set Versions for Transaction Attributes, page 4-7

For more information about publishing, refer to the Publishing Overview, *Oracle Product Hub User's Guide*.

## Prerequisites

- ☐ Before publishing any data, you must set up Oracle Data Integrator Artifacts and Web Services for Oracle Product Hub. For set up instructions, refer to My Oracle Support ID 1336146.1. For more information about the Web Services used during publishing, see: Oracle Product Hub Web Services Overview, page G-1.

### To publish value sets:

1. From the **Oracle Applications Home Page**, select an appropriate responsibility, then Setup Workbench. Select the Value Sets tab.

Alternatively, navigate to the **Publish Batch Search** page (Setup > Setup Workbench (T) Publish). In the View field, select Value Sets. Click Create. See: To create a draft publication batch, *Managing Publishing Batches, Oracle Product Hub User's Guide*.

2. In the **Maintain Value Sets** page, search for one or more value sets to publish.

3. Select the value sets to publish, then click Publish.

4. In the **Valueset Publish Parameters** page, enter a batch name.

You can assign any unique name to the batch. The Batch ID also uniquely identifies the batch. The Batch ID and Created By fields are automatically filled when you click Publish.

5. If the value set has versions, select the version number to publish.

**Additional Information:** By default, the system selects the current effective version. You can publish value set versions or value sets without a version. You cannot publish a draft version.

6. Select the Publish to PIM Systems Only box to publish to other Oracle Product Hub systems. Leave it unchecked to publish to external systems.

You cannot publish to both Product Hub and external systems within the same Publish action.

7. If you want to remove a value set from the batch, select the value set, then click Remove.

8. Click Continue.

The **ValueSet Publish Systems** page appears. This page contains different fields depending on whether the Publish to PIM Systems Only field is checked or not.

### To publish value sets to an external system (when the Publish to PIM Systems Only box is NOT selected)

9. In the **ValueSet Publish Systems** page, Enter values in the following fields:

- **Workspace Name.** Required. This field is used by Siebel for grouping multiple batches into a single group. Enter the name of the Siebel workspace project. If you are not using Siebel, enter any value.

- Auto-Release. Required. Select Default, Yes, or No from the list. This field refers to auto-releasing a Siebel workspace project.
    - Default - the workspace project auto-releases based on the setting of Auto Release within Siebel.
    - Yes - once the items are published to Siebel, the workspace project is automatically released.
    - No - once the items are published to Siebel, you need to manually release the workspace project.
10. Select the systems where you want to publish the value sets by clicking Add Another Row as many times as needed.
  11. In the Value Sets: Select Systems region, select a system name for each row added. If you need to delete a system, select the system's row and click Remove.
  12. Once you are satisfied with the batch details selected, click Finish to publish the batch.  
  
If you need to review or change any batch details, click Back to return to the **Valueset Publish Parameters** page.
  13. Click Save as Draft or Publish.

The **Publish Batch Search** page opens, displaying the status of the batch submitted. From here you can delete, copy, merge, or update the batch, as well as view the batch's history or create a new batch. See: *Managing Publishing Batches, Oracle Product Hub User's Guide*.

**Additional Information:** Possible statuses for the batch include:

- Draft Batch - The batch has been saved as a draft. You must update the batch, then select Publish to submit the batch.
- Batch Submitted - The batch has been submitted for processing, but the EGO Import Metadata concurrent program has not run yet.
- In Process - This status appears for the following reasons:
  - When Auto Import is selected - The Publish action is in process. If there are any errors or time out issues, the system appends the status with an appropriate message indicating that either the EGO Import Metadata concurrent

program did not process the updated records or the concurrent program is still running. Check the status of the EGO Import Metadata concurrent program in the target system and take appropriate action.

- When Auto Import is not selected - The Publish action is in process. The target interface tables were loaded successfully. Manually invoke the EGO Import Metadata concurrent program in the target system.
- Succeeded - The value set has been successfully passed to the target systems.
- Failed - The value set failed the import process for the specified target systems.
- Partially Succeeded - The some of the value sets have been successfully passed to the target systems. One or more value sets have failed the import process for one or more specified target systems.

**To publish value sets to an Oracle Product Hub system (when the Publish to PIM Systems Only box is selected)**

**Important:** Verify that the profile option "Enable PIM for Telco Features" is set to the same value in both the source and target Oracle Product Hub systems.

**14. In the ValueSet Publish Systems page, enter the following:**

- Auto Import. Checking this box automatically launches the EGO Import Metadata (EGOIMDCP) concurrent program in the target Oracle Product Hub system. If you leave this box unchecked, you must manually run the EGO Import Metadata concurrent program.

**Important:** Before publishing a value set with a Table validation type, verify that the table referenced in the query of a table type value set in the source system already exists in the target system. The referenced table is not published. If the table does not exist in the target system, the value set import process fails.

The following entities and relationships associated with the value set are

published and imported, regardless of whether you check the Auto Import box or not:

- Value Set definition.
- Value Set values.
- Associated child value sets, if any.
- Systems. Add systems or remove added systems as desired.

**15. Click Save as Draft or Publish.**

The **Publish Batch Search** page opens, displaying the status of the batch submitted. From here you can delete, copy, merge, or update the batch, as well as view the batch's history. See: Managing Publishing Batches, *Oracle Product Hub User's Guide*.

**Additional Information:** Possible statuses for the batch include:

- Draft Batch - The batch has been saved as a draft. You must update the batch, then select Publish to submit the batch.
- Batch Submitted - The batch has been submitted for processing, but the EGO Import Metadata concurrent program has not run yet.
- In Process - This status appears for the following reasons:
  - When Auto Import is selected - The Publish action is in process. If there are any errors or time out issues, the system appends the status with an appropriate message indicating that either the EGO Import Metadata concurrent program did not process the updated records or the concurrent program is still running. Check the status of the EGO Import Metadata concurrent program in the target system and take appropriate action.
  - When Auto Import is not selected - The Publish action is in process. The target interface tables were loaded successfully. Manually invoke the EGO Import Metadata concurrent program in the target system.
- Succeeded - The value set has been successfully passed to the target systems.
- Failed - The value set failed the import process for the specified



target systems.

- Partially Succeeded - The some of the value sets have been successfully passed to the target systems. One or more value sets have failed the import process for one or more specified target systems.

## Related Topics

Publishing Overview, *Oracle Product Hub User's Guide*

Assigning Systems to an Item Catalog Category, page 3-44

Publishing Item Catalog Categories, page 3-45

Publishing Attribute Groups, page 4-54

Publishing Items and Structures, *Oracle Product Hub User's Guide*

Using PL/SQL Public APIs to Publish Objects, *Oracle Product Hub User's Guide*

Managing Publishing Batches, *Oracle Product Hub User's Guide*

Viewing the Publishing History, *Oracle Product Hub User's Guide*

Oracle Product Hub Web Services Overview, page G-1

For more information about how product data synchronizes with participating applications using the publishing process, see Chapter 2: Oracle Product Master Data Management Integration Base Pack, *Oracle Product Master Data Management Integration 2.5 - Implementation Guide*. This guide is part of the Oracle Application Integration Architecture Release 2.5 Media Pack , which you can download from <http://edelivery.oracle.com>.

## Defining Item Attributes and Attribute Groups

Every catalog item has a set of operational attributes that determines the behavior of features in other E-Business Suite applications. In addition, you can create user-defined attributes that more specifically identify items, their characteristics and specifications, and capture business process information. User-defined attributes can also possess certain validation logic (for example, value sets) and indexes.

Attributes are defined by their names and values, and are saved within attribute groups. You can associate attribute groups with the following business entities (see: *Assigning Suppliers to Items, Oracle Product Hub User's Guide*):

- item
- item revision

- item organization
- item supplier
- item supplier site
- item supplier site organization

Item business entity attribute values default from entity to entity in the following order when an attribute group is associated with multiple business entities:

- item to item supplier
- item supplier to item supplier site
- item supplier site to item supplier site organization
- item to item organization

Attribute values only default from the parent to the child level during item business entity creation. You can override the defaulted attribute value at the child level. For example, define an initial cost attribute value at the parent or style item level and it defaults to the child item business entities and/or SKU items. You can update the initial cost at the child level later.

You can set up as many attribute groups as necessary to define an item business entity, with the following limitations on the number of attributes within each attribute group:

- 40 character attributes
- 20 number attributes
- 10 date attributes
- 40 translatable attributes

Later, users can enter the values for the attributes on the item business entity pages. Configure how attribute groups are displayed on these pages to improve usability (see: *Associating Attribute Groups with an Item Catalog Category*, page 4-48).

**Note:** You can also create user-defined attribute groups and attributes for change objects (see: *Defining Header/Line Type Attributes and Attribute Groups*, page 7-25).

For each user-defined attribute, you can optionally specify a value set (see: *Defining Value Sets for User-Defined Attributes*, page 4-1) with data type and validation rules that are applied when the user inputs data. Once created, you can reuse value sets for different attributes. Create value sets before defining your attribute groups.

An attribute group can be single-row, multi-row, or variant. Multi-row attribute groups enable you to associate multiple sets of attribute values with the same object instance. For example, if your item is a book, you can create an attribute group called "chapters" with attributes "chapter number," "name" and "number of pages." Multiple rows of "chapters" can be associated with a book, while "name" and "number of pages" each require a single row. The attribute "chapter number" is identified as Part of Unique Key (see: Part of Unique Key, page 4-30 later in this section). Use variant attribute groups only with style and SKU items (see: Using Style and SKU Items, *Oracle Product Hub User's Guide*)

**Caution:** Style items can have an unlimited number of attribute groups. The restrictions on the number of variant attributes within each group are the same as for any other type of attribute. Oracle Retail only allows 4 variant attributes per style item, however. When integrating Oracle Product Hub with Oracle Retail, limit the number of variant attributes to four per style item.

Third-party systems integrators can easily generate a database view of existing attributes and attribute groups. These views are particularly useful when users wish to read the Oracle-provided data as they write code for integration with Oracle applications. To generate database views, on the **Search: Attribute Groups** page, select the attribute groups for which you wish to generate the view and click Generate Database View.

**Additional Information:** You can import attribute groups and attributes using open interface tables and the concurrent program EGO Import Metadata (EGOIMDCP). For more information, refer to Importing Item Catalog Metadata, page 3-58.

For more information about User Defined Attributes in Oracle Purchasing and Oracle Contract Lifecycle Management, see:

- Oracle Purchasing Help Index, *Oracle Purchasing User's Guide*
- Oracle CLM Purchasing Help Index, *Oracle Contract Lifecycle Management for Public Sector Purchasing User Guide*

## Prerequisites

- ☐ Create value sets. See: Defining Value Sets for User-Defined Attributes, page 4-1.
- ☐ Group related attributes within the same attribute group. The following table shows some examples of attribute groups.

| Attribute Group           | Attributes          | Data Type |
|---------------------------|---------------------|-----------|
| Capacitor Specifications  | Dielectric          | Char      |
|                           | Voltage             | Number    |
|                           | Tolerance           | Number    |
|                           | Minimum Temperature | Number    |
|                           | Maximum Temperature | Number    |
| Capacitor Packaging Specs | Package Type        | Char      |
|                           | Size                | Number    |
|                           | Case Size           | Number    |

- ☐ Create custom privileges. See: *Creating Custom Privileges*, page 11-27.
- ☐ Create an object role (item) with custom privileges to define attribute group security (if necessary). You can secure the attribute group by setting Edit/View privileges. Later on, only users with certain roles can view or edit those attributes. See: *To implement attribute group security*, page 11-19.
- ☐ For attributes with a data type of Number, set up the Unit of Measure Classes (for example, Currency) and Units of Measure (for example, US Dollars). See: *Defining Unit of Measure Classes, Oracle Inventory User's Guide* and *Defining Units of Measure, Oracle Inventory User's Guide*.  
Also, see *Display As*, page 4-29 later in this section for related information.
- ☐ Establish the validation rules for each attribute, if necessary. See: *Defining Value Sets for User Defined Attributes*, page 4-1.

#### To create and maintain attribute groups:

1. From the Home page, click the Setup Workbench link.
2. On the **Search: Item Catalog Categories** page, click the Attribute Groups subtab.
3. On the **Attribute Groups** page, click Create.
4. On the **Create Attribute Group for Item Management** page, enter the following:

**Internal Name**

Enter the internal name of the attribute group.

**Display Name**

Enter the name of the attribute group as it will be displayed in the user interface.

**Description**

Optionally, enter the description of the attribute group.

**Behavior**

Select the Behavior of the attribute group:

- Multi-Row to associate multiple sets of attribute values with the same object instance.
- Single-Row to associate one attribute value with each object instance.
- Variant to use with style and SKU items. Variant attribute groups contain attributes that differentiate SKUs within one style item. See: Using Style and SKU Items, *Oracle Product Hub User's Guide*

**Caution:** You can only associate a variant attribute group with an item business entity.

**Number of columns in the page layout**

Enter the number of attribute columns to appear when you view the attribute group from within an item's user defined attribute page. The default value is 2. This field does not appear when defining a variant attribute group.

**Number of rows in the page layout**

This field only appears for multi-row attribute groups. Enter the number of columns to appear within an item's user defined attribute page. The default value is 5.

Items Documents Structures Change Management Security Functions Catalogs Value Sets

Item Catalog Categories Cross Reference Types Lifecycle Phases **Attribute Groups** Templates GTIN Item Type Attr

Items: Attribute Groups > Attribute Group Details >

**Edit Attribute Group for Item Management**

\* Indicates required field

Internal Name **Marketing**

\* Display Name **Marketing**

Description **Marketing**

Behavior **Multi-Row**

\* Number of columns shown in the table **5**

\* Number of rows shown in the table **5**

**Business Entities**

[Select All](#) | [Select None](#)

| Select                              | Name                     | Style to SKU | View | Privilege | Raise Attribute Change Event                      |
|-------------------------------------|--------------------------|--------------|------|-----------|---------------------------------------------------|
|                                     |                          |              |      | Edit      | Pre Post                                          |
| <input type="checkbox"/>            | Item Supplier Site       | Defaulting   |      |           | <input type="checkbox"/> <input type="checkbox"/> |
| <input type="checkbox"/>            | Item Revision            |              |      |           | <input type="checkbox"/> <input type="checkbox"/> |
| <input type="checkbox"/>            | Item Supplier            | Defaulting   |      |           | <input type="checkbox"/> <input type="checkbox"/> |
| <input type="checkbox"/>            | Item Supplier Site Store | Defaulting   |      |           | <input type="checkbox"/> <input type="checkbox"/> |
| <input type="checkbox"/>            | Item Organization/Store  | Defaulting   |      |           | <input type="checkbox"/> <input type="checkbox"/> |
| <input checked="" type="checkbox"/> | Item                     | Defaulting   |      |           | <input type="checkbox"/> <input type="checkbox"/> |

[Cancel](#) [Apply](#)

- Select the business entities to which you want to apply the attribute group. For each business entity selected, specify the following:

- Style to SKU**

Select one of the following:

- Defaulting - when an attribute defaults from a style item to a SKU during SKU creation, you can change the attribute at the SKU level later without affecting the style item. Conversely, if you change a defaulted attribute at the style item level after SKU creation, the change does not automatically appear at the SKU level. You must manually make the change at the SKU level or use a mass update function to change the SKUs.
- Inheritance - when a SKU inherits an attribute from its style item, the attribute is always controlled at the style item level. You can only change an inherited item level user defined attribute at the style item level. Once the style item attribute is changed, all related SKUs reflect the new attribute value.

**Tip:** When updating an attribute group, you can change the Style to SKU field as long as the attribute group is not associated with any item catalog category. If you need to update the Style to SKU field, delete the association and then make the update.

- **View Privilege**

Select a viewing privilege. Users must have a view privilege for any object to which this attribute group is associated. Then, you can narrow the definition of viewing privileges to meet the needs of your enterprise. For example, you may create a privilege called View Item Cost and associate it with an attribute group called "Costs." Users who have view privileges for the item can view the item, but not necessarily the cost information associated with it. To view the costs associated with the item, users must have the additional View Item Cost privilege.

**Note:** The lock icon (viewed on the **Search Results** page) indicates that a user does not have the privilege required to view a particular attribute group. If you wish to create a custom privilege, see *Creating Custom Privileges*, page 11-27.

- **Edit Privilege**

Select an edit privilege. Users must have an edit privilege to edit information in this particular attribute group. Privileges are granted by roles assigned to users. If no view or edit privilege is specified for the attribute group, then users' ability to view and edit the attribute group is controlled by the view and edit privileges on the object to which the attribute group is associated. You can define editing privileges to meet the needs of your enterprise.

- **Raise Pre Attribute Change Event**

Click Raise Pre Attribute Change Event to raise a business event every time you want to test a proposed attribute change. This enables you to test a proposed attribute change against validating criteria before committing the attribute value to the database. See: *Item Business Events*, page B-1 .

- **Raise Post Attribute Change Event**

Click Raise Post Attribute Change Event to raise a business event every time an attribute is changed. Based on this event, you can choose to execute other functions or workflows to implement your company's business processes. See: *User-Defined Attributes Business Event within the Synchronizing Item User-Defined Attributes With Item Descriptive Elements Overview*, page D-1.

6. Click Apply and Add Attributes to add user-defined attributes to your attribute group. Note that clicking Apply and Add Attributes saves the attribute group and commits it to the database, even if you click Cancel while on the **Create Attribute** page. Alternatively, click Apply to only save the attribute group and stop the process.

If choosing Apply and Add Attributes, continue on to the next step.

7. On the **Create Attribute** page, provide the following information:

**Internal Name**

The internal name of the attribute. The name by which the attribute is tracked internally.

**Display Name**

The name of the attribute as it appears within the user interface.

**Sequence**

The sequence number for the attribute. The sequence determines the order in which the attribute is displayed on the page, and also determines the order in which the attribute is processed.

**Tip**

The description of the attribute; this description also appears as tip text on pages that have attributes that can be updated.

**Data Type**

Lists the available data types. The data type that you select determines the values that are available in the column and value set. An attribute's data type must match the data type specified for that attribute's value set. The list of values for a value set only displays value sets whose data type matches the data type of the attribute. Note that selecting a data type always clears the column and value set. For example, say you selected the Number data type. Then you select your column and value set--remember that your column and value set choices are determined by the data type you chose. Then you decide to change the data type from Number to Date. Notice that after you change the data type, your column and value set are cleared; you need to select new ones based on your new data type. Additionally, the data type determines the values that are available in the Display As field. You cannot change the data type once an attribute is created.

**Note:** The maximum number of characters for a character type attribute is 150 characters. Translatable text fields have a limit of 4,000 characters. Number type fields have a maximum size and precision of 22.5 (22 digits to the left of the decimal and 5 digits to the right).

**Column**

A list of values that enables you to specify the column in which the attribute is stored in the database table. The column list of values only returns columns with the data type you specified in Data Type. The list of values also indicates whether or not the column is indexed. If you want the attribute to be searchable, then select a column in the database. Ensure that the Indexed checkbox is selected before you complete the attribute definition. If no indexed columns are available in the



database, and you still want a searchable attribute, select a non-indexed column, and ensure the Indexed checkbox is selected before completing the attribute definition; then the database column is automatically indexed. If you do not want the attribute to be searchable, and the only columns available in the database are indexed, then ensure that the Indexed checkbox is not selected before completing the attribute definition.

### **Enabled**

Specify whether or not the attribute is enabled (and available for use) or disabled (and not available for use). If the attribute is enabled, specify whether or not the attribute is optional or required. If required, the user cannot save data for an object using the attribute group without entering a value for that attribute. You can disable attributes, unless the attributes are mapped to functions. However, you cannot delete attributes after an attribute group has been associated with an item catalog category.

### **Required**

Specify whether or not the user must enter an attribute value.

### **Display As**

Determines how the attribute appears within the user interface. For example, if you select Text Field for an attribute called "Cost Center", then "Cost Center" appears in the user interface as a text field. The available values for Display As are determined by the data type selected.

If you choose Checkbox, then the value set defaults to EGO\_YES\_NO.

If you select Radio Group, then you must choose an independent value type set (in other words, the value set has a discrete set of values that you have already specified). If, while updating the value set, you select the Long List of Values validation type, attributes will be displayed as a text field with a list of values, thereby ignoring your choice of Radio Group.

If you select Text Field, *and* choose a value set whose type is independent, then your display would be either a poplist or list of values (whether or not you get the poplist or list of values is determined by the way you define the value set). Also, when the data type is Number, and Display As is a Text Field, a Unit of Measure list of values is presented. You can choose either the Unit of Measure or the Value Set; they are mutually exclusive.

Hidden attributes do not show up in the user interface; this attribute is primarily populated via user-defined functions.

Selecting Dynamic URL refreshes the page with a new section for specifying the dynamic URL. Enter a URL and use any attribute Internal Name in the attribute group, enclosed between \$\$ symbols, as a token for the value of a parameter. When users click on the URL, the value for that attribute will replace the token in the URL's query string.

Selecting Static URL enables you to input a web page address.

The following table describes which display types are supported for the various data types.

***Display Types Supported by Data Types***

| <b>Display Type</b> | <b>String</b> | <b>Number</b> | <b>Date</b> | <b>Date Time</b> | <b>Translatable Text</b> |
|---------------------|---------------|---------------|-------------|------------------|--------------------------|
| Attachment          | No            | Yes           | No          | No               | No                       |
| Check Box           | Yes           | No            | No          | No               | No                       |
| Dynamic URL         | Yes           | Yes           | No          | No               | Yes                      |
| Hidden              | Yes           | Yes           | Yes         | Yes              | Yes                      |
| Radio Group         | Yes           | Yes           | Yes         | Yes              | No                       |
| Static URL          | Yes           | No            | No          | No               | Yes                      |
| Text Area           | Yes           | No            | No          | No               | Yes                      |
| Text Field          | Yes           | Yes           | Yes         | Yes              | Yes                      |

### **Indexed**

Specify whether or not you want the attribute to be indexed. If you choose to make this an indexed attribute, it will appear as an indexed attribute on the criteria template page. Only indexed attributes are available as sort criteria in result formats.

### **Part of Unique Key**

If the attribute group is multi-row, the Part of Unique Key checkbox determines whether or not the attribute is part of the key that uniquely identifies a row. A unique key is any set of attributes whose values can be used to uniquely identify a row within the attribute group. You can define the unique key on the attribute group **Detail** page. You can add/edit the unique key as long as doing so does not destroy the uniqueness (creating duplicates) of existing records.

**Note:** You can specify that each attribute is part of a unique key when creating the attribute, or--more conveniently--you can specify

all unique key attributes in the group via the attribute group **Detail** page.

### **Value Set**

Select a value set that will serve as a set of constraints for an attribute. For details about creating value sets, see *Defining Value Sets for User-Defined Attributes*, page 4-1.

**Caution:** If versions are enabled, assign a value set to an attribute from the Setup Workbench (HTML interface) rather than from the Forms interface to avoid data corruption. Forms behavior has not been enhanced to support versions.

### **Default Value**

The default value of this attribute. If you've selected a value set, the value set's constraints apply to the default value. This value defaults upon object creation.

**Caution:** In an attribute group with at least one required attribute, no default values are applied for any attributes when a required attribute does not have an assigned default value.

**Create Attribute**

\* Indicates required field

\* Internal Name Campaign

\* Display Name Marketing Campaign

\* Sequence 40

Tip

Data Type Char

\* Column C\_EXT\_ATTR4

☒ Enabled

☐ Required

Display as Text Field

☐ Indexed

**Value Set**

Value Set Name ProductFamily

**Default Value**

Default Value

Apply and Add Another

Cancel Apply

8. Click Apply and Add Another to save and create another attribute or click Apply to save and stop the process.

### To copy an attribute group and its attributes:

When you copy an attribute group, all of the attribute group's field values and attributes default to the new attribute group. You can override all defaulted values and attributes except for the attribute group Behavior field.

1. On the **Attribute Groups** page, search for the attribute group to copy.
2. In the attribute group search results, click the Copy icon for the attribute group.
3. On the **Create Attribute Group for Item Management** page, the field values from the original attribute group appear. You must enter a new internal name for the copied attribute group, but you have the option to change all field values except for the Behavior field; you cannot change the original Single-Row, Multi-Row, or Variant value in the Behavior field.
4. Click Apply.

The **Attribute Details** page appears. The attributes from the original attribute group are attached to the newly copied attribute group. From this page, you can edit or delete the existing attributes and add new attributes.

## Defining Transaction Attributes

Transaction attributes enable you to capture the attribute value for an item at the time of order or at the time of the transaction. They differ from user-defined attributes, which capture the attribute value for an item at the time of item creation. Examples of transaction attributes include:

- The cut length of a piece of vinyl extrusion in a custom window assembly.
- The number of megabytes or gigabytes of email storage on a DSL account.
- The monogram text on a shirt pocket.

## Prerequisites

- ☐ Create a draft version of an item catalog category. See: Defining Item Catalog Categories, page 3-2
- ☐ Optionally, create value sets with versions enabled. See: Defining Value Set Versions for Transaction Attributes, page 4-7. Transaction attributes cannot use non-version value sets.
- ☐ For attributes with a data type of Number, set up the Unit of Measure Classes (for example, Currency) and Units of Measure (for example, US Dollars). See: Defining Unit of Measure Classes, *Oracle Inventory User's Guide* and Defining Units of Measure, *Oracle Inventory User's Guide*.
- ☐ You must have the Edit Transaction Attribute privilege in order to make transaction attribute updates and to enter a transaction attribute value for an item. See: Implementing Role Based Security, page 11-11.

### To add a transaction attribute to an item catalog category:

1. From the Oracle Applications Home Page, click the Setup Workbench link.
2. In the Items tab, on the **Item Catalog Categories** page, search for and select the item catalog category to which you want to add a transaction attribute.

**Note:** You can only add transaction attributes to an item catalog category when the profile option "Enable PIM for Telco Features" is set to Yes.

3. In the Versions region of the **Basic Information** page for the item catalog category, click the Draft link.

**Note:** The start date of the draft ICC version defaults to the current date. To select a future start date, select the Update icon for the version before selecting the Draft link. The transaction attribute uses a value set version with a start date on or after the ICC version's start date.

4. On the **Transaction Attributes** page, click Add Attribute.
5. On the **Add Transaction Attribute** page, enter the following:

**Additional Information:** The following field values are inherited by any child item catalog categories, but can be changed at the child level.

#### **Internal Name**

Enter the internal name of the transaction attribute.

#### **Display Name**

Enter the name of the transaction attribute as it will be displayed in the user interface.

#### **Sequence**

Enter the sequence number for the attribute. The sequence determines the order in which the attribute is displayed on the page, and also determines the order in which the attribute is processed.

#### **Data Type**

Select an available data type. The data type that you select determines the values that are available in the value set. An attribute's data type must match the data type specified for that attribute's value set. The list of values for a value set only displays values whose data type matches the data type of the attribute. You cannot change the data type once an attribute is created.

**Note:** The maximum number of characters for a character type attribute is 150 characters. Translatable text fields have a limit of 4,000 characters. Number type fields have a maximum size and precision of 22.5 (22 digits to the left of the decimal and 5 digits to the right).

#### **Value Set Name**

Select a value set that will serve as a set of constraints for the attribute. For details about creating value sets, see *Defining Value Set Versions for Transaction Attributes*, page 4-7.

### **Default Value**

The default value of this attribute. If you have selected a value set, the value set's constraints apply to the default value. This value defaults upon transaction creation.

**Note:** You must enter a default value for an attribute that is required or read only.

### **Rejected Value**

This display only field is relevant when an attribute is marked as required. If the attribute is required, the value in this field is the value that the Siebel Configurator interprets as not valid. The system marks this attribute as not entered if the rejected value is entered as the attribute's value.

### **Required**

Specify whether or not the user must enter an attribute value.

### **Read Only**

Selecting Read Only prevents the user from updating the default value of the attribute.

### **Hidden**

Specify whether or not the user can see the transaction attribute when performing a transaction. This check box prevents the attribute from displaying in quote, agreement, order, or asset views.

### **Searchable**

Enables use of the attribute in parametric searches within Siebel software.

### **Check Eligibility**

Select this checkbox to require a Siebel system to check the customer's eligibility to buy this product.

6. Click Apply to stop after creating one attribute or click Add Another to create another attribute.

### **To update a transaction attribute:**

1. On the **Transaction Attributes** page, click the Update icon for a particular attribute.  
You can also click the Delete icon for an attribute on this page.
2. In the **Update Transaction Attribute** page, change field values in the Attribute Details and Attribute Data regions as necessary.

**Tip:** You can also update, but cannot delete, transaction attributes at the item and item revision level.

3. Click Apply.

The system returns you to the **Transaction Attributes** page. Select the changed attribute, then the Attribute Data tab. If you changed an inherited attribute (the Inherited field value is Yes), the Modified field value is Yes to indicate that this attribute has changed from its original inherited state. Once you change an inherited attribute, changes to the attribute in the parent item catalog category are not inherited in the child item catalog category.

## What's Next

Release the draft version of the item catalog category containing the transaction attributes. See: "To release an item catalog category draft version" in Defining Item Catalog Categories, page 3-2.

## Using Seeded Attributes

Oracle Product Hub provides many seeded attribute groups, attributes, and value sets for different industries and uses. Seeded attribute groups behave similarly to user-defined attribute groups; you can associate them to item catalog categories, modify them and their attributes, and delete them if you have no need for them. The seeded attributes are grouped into the seeded attribute groups, and some seeded attribute groups are even grouped into seeded item pages. The attribute groups with seeded item pages automatically attach to every item catalog category created unless you remove them from the seeded item page or delete the item page itself. For example, when you create an item catalog category, operational attributes (for example, order management, purchasing, and inventory attribute groups) are automatically associated at the item organization level.

You can customize seeded attributes within the seeded attribute groups. For example:

- Create a value set and associate it to the attribute as appropriate.
- Customize a seeded value set. If you only use certain values, you can customize the value set by disabling the unused values or you can add a new value.
- Specify a default value for a seeded attribute.
- Disable unused seeded attributes from within an item catalog category.
- Delete unused seeded attributes or attribute groups

For more information about specific types of seeded attributes, see the appendix of



seeded item metadata libraries, which includes the:

- Sellable Product Information Library - Horizontal, page F-1
- Communications Services Billing Library - Vertical, page F-25
- Product Management Library - Horizontal, page F-23
- Communications Product Details Library - Vertical, page F-44

## Understanding Seeded Operational Attributes

When you create a new item catalog category that does not have a parent node (meaning it is a top node), the system associates all operational attribute groups and pages with the item catalog category at the item organization level. You cannot delete the attribute associations, but you can delete the pages. When you create a new item catalog category that has a parent node, all operational attributes and pages are inherited from the parent. If a page is deleted at the parent node, then it is also deleted within all its child nodes. So if you wish to display pages in a particular item catalog category, but they have been deleted at the parent node, you must explicitly add them in the child item catalog category.

For a complete listing of the seeded operational attributes associated with new item catalog categories, see *Item Attributes Listed in Alphabetical Order*, *Oracle Inventory User Guide*.

The following operational attribute pages are configured for all new item catalog categories:

- Physical Attributes
- Sales and Order Management
- Planning
- Purchasing
- Inventory
- Manufacturing
- Service

## Managing Seeded Attribute Changes

You can use change orders to manage changes to seeded attributes, regardless of your role or privileges. This change control process ensures that the proposed attribute values have been reviewed, before they are changed.

**Note:** Before you use change orders to manage seeded attributes, you must set up change policies for the attribute group.

For more information about change orders, see: *Creating Change Orders, Oracle Product Hub User's Guide*.

## Related Topics

Managing Item Specifications, *Oracle Product Hub User's Guide*

Associating Attribute Groups with an Item Catalog Category, page 4-48

Defining Item Attributes and Attribute Groups, page 4-21

Associating Component Attribute Groups with a Structure Type, page 8-6

## Using Additional Attributes

On the **Basic Information** page for an item catalog category, a region named Additional Attributes appears if the Item Catalog Groups descriptive flexfield is enabled. The following list provides details about this descriptive flexfield:

- Application - Inventory
- Title - Item Catalog Groups
- Name - MTL\_ITEM\_CATALOG\_GROUPS
- Table Name - MTL\_ITEM\_CATALOG\_GROUPS

You can use descriptive flexfields to track additional information, important and unique to your business, that would not otherwise be captured on a page. For more information about setting up the Item Catalog Groups descriptive flexfield, refer to the *Oracle E-Business Suite Flexfields Guide*.

## Creating User-Defined Functions for Item Attributes

To define your own custom logic, you can add user-defined functions and actions to existing pages in the Item Catalog and in Change Management without having to customize the entire page. By first setting up user-defined attributes, you can then execute user-defined functions with those attributes.

Using different algorithms, you can calculate values by passing attribute values to functions. User-defined functions can be Java, URL, or PL/SQL functions. Functions use input and/or output parameters of various data types such as string, integer, or Boolean. You can also map these parameters to attributes and object primary key values. Actions are trigger points for functions displayed as buttons or links on the

page. You can determine the conditional visibility of the button and the label displayed on the button itself. You can also prompt the user based on the user's input. See: Adding Actions to an Attribute Group, page 4-52

**Additional Information:** You can import functions using open interface tables and the concurrent program EGO Import Metadata (EGOIMDCP). For more information, refer to Importing Item Catalog Metadata, page 3-58

User-defined functions can be Java, URL, or PL/SQL functions. Prior to setting up user-defined functions and actions you should:

For more information about User Defined Attributes in Oracle Purchasing and Oracle Contract Lifecycle Management, see:

- Oracle Purchasing Help Index, *Oracle Purchasing User's Guide*
- Oracle CLM Purchasing Help Index, *Oracle Contract Lifecycle Management for Public Sector Purchasing User Guide*

## Prerequisites

- ☐ Set up user defined attributes. See: Defining Item Attributes and Attribute Groups, page 4-21
- ☐ Create item pages associated with item catalog categories. See: Creating Pages for an Item Catalog Category, page 3-28
- ☐ Determine which attributes are functions of other attributes. Create user-defined functions to perform the necessary calculations (for example, summation of costs, efforts, ratings).
- ☐ Create custom privileges and roles if you need to secure access and control who can/cannot execute the function/action. See: To implement attribute group security, page 11-19
- ☐ Determine the conditions for displaying or changing the prompt of the button/link that executes the function. For example, the button may not be displayed until certain required attributes have values entered. The name of the button may change depending on the values of certain attributes. See: Adding Actions to an Attribute Group, page 4-52

### To create a user-defined function:

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Item Catalog Categories** page, click the Functions tab.

3. On the **Search and Select: Functions** page, click Create Function.
4. On the **Create Function** page, enter the following information:

**Internal Name**

The internal name of the function.

**Display Name**

The name of the function as it appears in the user interface.

**Description**

The description of the function.

**Function Type**

Whenever you define a function, there must be an underlying implementation of that function. The type specifies the manner of implementation to which the function maps. The supported function types are:

*Java*

Specifies that the implementation is via a Java method. When you select Java, the page refreshes and you must specify the following:

- Class

The class in which the method resides.

- Method

The method that implements your custom logic.

**Note:** Place the Java class files in any directory and append this directory to the Apache servlet classpath.

*PL/SQL*

Specifies that the implementation is via a PL/SQL stored procedure. When you select PL/SQL, the page refreshes and you must specify the following:

- Package

The package in which the procedure resides.

- Procedure

The procedure that implements your custom logic.

**Note:** Run the PL/SQL package in your custom schema and then create a synonym for this package in the APPS schema.

### *URL*

Specifies that the implementation is a simple URL link. When you select URL, the page refreshes and you must specify the following:

- URL

Specify either absolute or relative URLs. For absolute URLs, begin the URL with the protocol (in most cases, the protocol will be http://).

5. Click Apply.

**Note:** You can only delete a function when it is no longer associated with an action.

### **To create parameter mappings for a function:**

After creating a user-defined function, specify the parameters to pass when that function is called.

1. On the **Search and Select: Functions** page, click the name of the function you just created.
2. On the **Function Details** page, click Add.
3. On the **Create Function Parameter** page, enter the following information:

#### **Internal Name**

The internal name of the parameter.

#### **Display Name**

The name of the parameter as it appears in the user interface.

#### **Sequence**

The order in which this parameter appears relative to other parameters associated with this function. Sequence must be unique among all parameters associated with this function.

Specify the order sequence of the function parameters; this is the order in which the parameters are passed to a function or procedure. For example, you need to calculate time duration in days by using the attribute group "Duration in days" where Duration = End Date – Start Date.

| Sequence | <b>Attribute Group:</b><br><i>Duration in Days</i> | <b>Attribute Group:</b><br><i>Duration in Days</i> | Mapping<br>Attribute &<br>Parameters | <b>Java Function:</b><br><i>Duration</i> | <b>Java Function:</b><br><i>Duration</i> | <b>Java Function:</b><br><i>Duration</i> |
|----------|----------------------------------------------------|----------------------------------------------------|--------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| Sequence | Attribute Name                                     | Data Type                                          | -----<br>@-----                      | Parameter Name                           | Data Type                                | Parameter Type                           |
| 1        | Start Date                                         | Standard Date                                      | -----<br>@-----                      | Date 1                                   | Date                                     | Input                                    |
| 2        | End Date                                           | Standard Date                                      | -----<br>@-----                      | Date 2                                   | Date                                     | Input                                    |
| 3        | Duration                                           | Number                                             | -----<br>@-----                      | Result                                   | Integer                                  | Return Value                             |

## Data Type

Lists the available data types. The values available are dependent on the type of function for which you are defining parameters.

## Parameter Type

Select the parameter type for each parameter based on whether the corresponding attribute is providing an input parameter to the function or expecting a return value from the function (for example, input, output, input/output).

The parameter options for a parameter depend on the function type and parameter data type you have already selected. For example, if the Function Type is URL, then the Data Type is constrained to String, and the Parameter type is constrained to Input.

The valid parameters for Java functions are:

- Boolean
- Standard Date
- Error Array
- Float
- Integer
- Long

- Double
- Transaction
- String
- Standard Date Time

The valid parameters for PL/SQL functions are:

- Date
- Error Array
- Number
- Varchar

The valid parameter for URL function is:

- String

The system supports Java function parameters to be used as input, output, input/output or return. For output or input/output parameter types, you have to pass back the changed value as the same object. But you cannot change the values for immutable data types. Therefore, wrapper classes are created for these data types.

Boolean - oracle.apps.ego.common.EgoBoolean

Double - oracle.apps.ego.common.EgoDouble

Float - oracle.apps.ego.common.EgoFloat

Integer - oracle.apps.ego.common.EgoInteger

Long - oracle.apps.ego.common.EgoLong

Date - oracle.apps.ego.common.EgoDate

Timestamp - oracle.apps.ego.common.EgoTimestamp

All these wrapper classes have `getValue()` and `setValue()` that return/take the basic parameters respectively. For example, `EgoInteger - setValue(Integer i) Integer getValue()`

Other Java parameters supported are:

String - java.lang.StringBuffer

Transaction - oracle.jbo.Transaction

**Additional Information:** ErrorArray data type is a java.util.Vector to

which you can add translated Error messages. This is an output parameter and the errors are displayed on the rendering page.

While writing Java Custom functions, use the above data types. When a user enters a decimal value for an attribute and this attribute is mapped to EgoInteger or EgoLong, the value is truncated before it passes to the function. For EgoBoolean, the value "Y" is considered TRUE. All other values are considered FALSE.

4. Click Apply.

## Related Topics

Adding Actions to an Attribute Group, page 4-52

Creating Pages for an Item Catalog Category, page 3-28

Defining Item Attributes and Attribute Groups, page 4-21

## Defining Attribute Usages

You can export user-defined attributes to other applications for their use. For example, export user-defined attributes to Oracle Configurator for use as item properties (See: Item Types and Imported BOM Properties within Imported BOM Models, *Oracle Configurator Developer User's Guide*). Oracle Configurator uses item properties to create generic rules for configured items, such as this rule:

### Rule

The voltage of the adapter must be equal to the voltage supplied in the country of installation.

In the rule above, an Item Property of Voltage could be created either:

- directly, in Oracle Configurator Developer.
- in Oracle Inventory, as a Descriptive Element, and imported into Configurator Developer as an Item Property (see: Defining Descriptive Elements, *Oracle Inventory User's Guide*).
- in Oracle Product Hub, as a user-defined attribute, and exported to Configurator Developer, where it becomes an Item Property.

### To export user-defined attributes to Oracle Configurator Developer:

1. From the Setup Workbench **Item Catalog Categories** page, Items tab, click the Attribute Usages sub-tab.



The screenshot shows the Oracle Item Catalog Administration interface. At the top, there's a navigation bar with 'ORACLE' and 'Item Catalog Administration'. Below it, a series of tabs: 'Items', 'Documents', 'Structures', 'Change Management', 'Security', 'Functions', 'Catalogs', and 'Value Sets'. Under 'Value Sets', there are sub-tabs: 'Item Catalog Categories', 'Cross Reference Types', 'Lifecycle Phases', 'Attributes', 'Templates', 'GTIN', 'Item Type', and 'Attribute'. The 'Attribute' sub-tab is selected, leading to the 'Attribute Usage' page. This page has a 'Hide Filters' checkbox and a search form with 'Application' and 'Attribute Group' fields, a 'Go' button, and an 'Update' button. Below the form is a table with columns 'Attribute Group' and 'Attribute'. The table currently displays 'No results found.'

2. In the Attribute Group field, search for the attribute groups that you want to export to Oracle Configurator Developer.
3. Click Go to list all of the attribute groups matching your search.
4. Select the attributes that you want to export.
5. Click Update to export the attributes.

## Related Topics

Defining Item Attributes and Attribute Groups, page 4-21

## Displaying Attachments on Item Attribute Pages

You can display items or change attachments as attributes on an item attribute page. Select a specific file from the existing item/change attachment list to display the file as a link within an attribute group on an item or change object page. You can then set up a Display Format that includes the attachment attribute as a column. Consequently, the link to the attachment is displayed in the search results.

### To display an attachment on an item attribute page:

1. Create a value set that queries the values for the object primary key and retrieves the information for the object attachment data. See: Defining Table Value Sets, page 4-12

When creating a value set to display item attachments, select the Table validation type and the Number data type (as shown in the following table).

|                                                           |                                                           |                                                           |                                                           |
|-----------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| <b>Attribute Group:</b><br><i>Computer Specifications</i> | <b>Attribute Group:</b><br><i>Computer Specifications</i> | <b>Attribute Group:</b><br><i>Computer Specifications</i> | <b>Attribute Group:</b><br><i>Computer Specifications</i> |
| <b>Attribute:</b> <i>Release Notes</i>                    | <b>Attribute:</b> <i>Release Notes</i>                    | <b>Value Set:</b> <i>Item Attachment</i>                  | <b>Value Set:</b> <i>Item Attachment</i>                  |
| <b>Data Type</b>                                          | <b>Display as</b>                                         | <b>Data Type</b>                                          | <b>Validation Type</b>                                    |
| Number                                                    | Attachment                                                | Number                                                    | Table                                                     |

### Create a value set for an attribute to display an item attachment

Value Sets: Value Sets >

Value Set Details

Value Set Name: **Item\_Attachments**  
Description: **GT3\_Item Attachments**  
Data Type: **Number**  
Maximum Size: **127**  
Validation Type: **Table**  
List Type: **List of Values**

**Validation Table Information**

Table Application:  
Table Name: **FND\_ATTACHED\_DOCUMENTS A, FND\_DOCUMENTS\_TL B, FND\_DOCUMENTS C**

**Value Column**

Name: **B.TITLE**  
Type: **Char**  
Size: **50**

**ID Column**

Name: **A.ATTACHED\_DOCUMENT\_ID**  
Type: **Number**  
Size: **40**

**Meaning Column**

Name:  
Type:  
Size:

**Where Clause**

**B.LANGUAGE = userenv('LANG') AND (A.ENTITY\_NAME = 'MTL\_ITEM\_REVISIONS' or A.ENTITY\_NAME = 'MTL\_SYSTEM\_ITEMS') AND A.DOCUMENT\_ID = B.DOCUMENT\_ID AND C.DOCUMENT\_ID = A.DOCUMENT\_ID AND A.PK1\_VALUE = :\$OBJECT\$.ORGANIZATION\_ID AND A.PK2\_VALUE = :\$OBJECT\$.INVENTORY\_ITEM\_ID**

**Important:** In the above screenshot, the value set table refers to Value Column = FND\_DOCUMENTS\_TL.TITLE. Alternatively, the value set table could refer to Value Column = FND\_DOCUMENTS\_TL.DESCRPTION. Remember, though, that the file description is not a mandatory field when adding attachments. If an attachment has no description, then no name appears for the attachment in the value set if Value Column =

## FND\_DOCUMENTS\_TL.DESCRPTION.

2. Create an attribute group with an attribute where Display As is set to Attachment. See: Defining Item Attributes and Attribute Groups, page 4-21

To see Attachments in the Display As dropdown, use the Number data type for your attribute.

3. Associate the value set to the attribute.

### Associate value set to the attribute

#### Attribute Group Details

[Up](#)

Internal Name **Computer\_Specs**

Display Name **Computer Specifications**

Description

Behavior **Single-Row**

Number of columns in the page layout

Number of columns shown in the table

Number of rows shown in the table

| Business Entities |              |           |      |                              |      |  |
|-------------------|--------------|-----------|------|------------------------------|------|--|
| Business Entities |              |           |      |                              |      |  |
| Name ▲            | Style to SKU | Privilege |      | Raise Attribute Change Event |      |  |
|                   |              | View      | Edit | Pre                          | Post |  |
| Item              | Defaulting   |           |      | No                           | No   |  |

#### Attributes

[Add Att](#)

Select Object: [Delete](#)

[Previous](#)
 1-10 of 12

[Select All](#) | [Select None](#)

| Select                   | Sequence | Internal Name | Display Name                  | Data Type | Display As | Value Set Name                   | Enabled | Required | Indexed | Column   |
|--------------------------|----------|---------------|-------------------------------|-----------|------------|----------------------------------|---------|----------|---------|----------|
| <input type="checkbox"/> | 5        | Release_Notes | <a href="#">Release Notes</a> | Number    | Attachment | <a href="#">Item Attachments</a> | Yes     | No       | No      | N_EXT_AT |
| <input type="checkbox"/> | 10       | CPU_Type      | <a href="#">CPU Type</a>      | Char      | Text Field |                                  | Yes     | No       | No      | C_EXT_AT |
| <input type="checkbox"/> | 20       | CPU_Speed     | <a href="#">CPU Speed</a>     | Char      | Text Field |                                  | Yes     | No       | No      | C_EXT_AT |
| <input type="checkbox"/> | 30       | Memnrv        | <a href="#">Memnrv</a>        | Char      | Text Field |                                  | Yes     | No       | No      | C_EXT_AT |

4. Associate the attribute group to an item catalog category. See: Associating Attribute Groups with an Item Catalog Category, page 4-48
5. Create an item page to display the attribute group. See: Creating Pages for an Item Catalog Category, page 3-28
6. Add attachments to the item attachment list. See: Adding Attachments, *Oracle Product Hub User's Guide*
7. Select an attachment for display on the item attribute page. For more information about updating attributes, see: Managing Item Specifications, *Oracle Product Hub User's Guide*

### Select attachments to be displayed on attribute page

Item: CN82441 Revision: ZX9000  
Organization: Vision Operations (V1) Recent Home Logout

Item Lifecycle Change Management Configuration  
Specifications Revisions Attachments Organizations People Associations Syndication

[Update Technical Specifications](#)

---

**Computer Specifications**

|                  |                                      |                       |                 |
|------------------|--------------------------------------|-----------------------|-----------------|
| Release Notes    | ProductSpecificationSheet.pdf        | CPU Type              | AMD AthlonXP    |
| CPU Speed        | 900MHz                               | Memory                | 128M            |
| Storage          | 100 G                                | Media Drives          | 16x/48x DVD-R   |
| Graphics Monitor | 15" LCD Flat Panel Display (15" view | Audio Speakers        | Integrated Ster |
| Power            | Worldwide Voltage                    | Communication Devices | Integrated 10/1 |
| Keyboard & Mouse | Multifunction Keyboard and Logitec   | RAM_Memory            |                 |

---

**Operating Conditions**

|                              |     |      |         |                   |  |
|------------------------------|-----|------|---------|-------------------|--|
| Optimal Temperature          | 78  | Unit | Celsius | Humidity          |  |
| Maximum Wet Bulb Temperature | 110 | Unit | Celsius | Minimum Dew Point |  |

### Item attribute page displays link to one of the attached files

Item: CN82441 Revision: ZX9000  
Organization: Vision Operations (V1) Recent

Item Lifecycle Change Management Configuration  
Specifications Revisions Attachments Organizations People Associations Syndication

Overview Search Item   [Advanced Search](#)

Classification and Categories

**Technical Specifications**

Expansion Slots  
Sales and Order Management  
Planning  
Connectors  
Purchasing  
Operating Systems  
Benchmark Ratings  
RoHS / WEEE  
Market Research  
Manufacturing  
Service  
Marketing  
Inventory/WMS  
Descriptive Attributes

---

**Computer Specifications**

|                  |                                                             |
|------------------|-------------------------------------------------------------|
| Release Notes    | ProductSpecificationSheet.pdf                               |
| CPU Speed        | 900MHz                                                      |
| Storage          | 100 G                                                       |
| Graphics Monitor | 15" LCD Flat Panel Display (15" viewable)                   |
| Power            | Worldwide Voltage                                           |
| Keyboard & Mouse | Multifunction Keyboard and Logitech USB Optical Wheel Mouse |

---

**Operating Conditions**

|                              |     |         |
|------------------------------|-----|---------|
| Optimal Temperature          | 78  | Celsius |
| Maximum Wet Bulb Temperature | 110 | Celsius |

## Associating Attribute Groups with an Item Catalog Category

Attributes are details that further define the item and specify the item's behavior. For example, Physical item attributes identify the size, shape, and color of an item.

Attribute groups represent the logical grouping of similar item attributes. They collect characteristics (attributes) that can be used to record specifications or represent the properties of an item belonging to an item catalog category. Attribute groups can be associated with an item catalog category at the item or item revision levels, thereby determining whether attribute values change over revisions.

**Note:** Attribute groups inherited from a parent catalog category cannot be deleted at the child catalog category level. Attribute groups can be inherited from a parent catalog category *and* explicitly added to the catalog category itself. Administrators should not associate the same attribute group with an item catalog category in this manner.

**Additional Information:** Common attribute group values will be retained even after moving to the target item catalog category.

To display user-defined item attributes, associate the corresponding attribute groups with an item catalog category and specify the item page upon which the attribute group should appear.

**Note:** You can re-use the same attribute group throughout the item catalog for different item catalog categories. Children categories inherit attribute groups from the parent category. You cannot edit inherited data.

Consider which item catalog categories in the item catalog hierarchy need to display the attribute groups.

Associate an attribute group to a parent category only if you need to view those attributes in all items within that hierarchy. For example, the Benchmark Rating attribute group is associated with the Computer System item catalog category. The Computer System category is the parent category for Desktop and Laptop. All items belonging to the parent category Computer System, as well as to its children Desktop and Laptop, will have Benchmark Rating attributes displayed on the items pages.

### **To associate an attribute group with an item catalog category:**

1. From the Home page, click the Setup Workbench link.
2. On the **Search: Item Catalog Categories** page, search for the item catalog category (see: *Browsing Item and Alternate Catalogs*, *Oracle Product Hub User's Guide*) and click its corresponding name link.
3. From the **Basic Information** page, click the Attribute Groups link.
4. On the **Attribute Groups** page, click Add Attribute Groups.
5. In the **Add Attribute Groups to Catalog Categories** page, search for and then select an attribute to add. Click Apply.

The system automatically adds the attribute group to all business entities eligible for that attribute group. Possible eligible business entities include:

- Item
- Item Organization/Store
- Item Revision
- Item Supplier
- Item Supplier Site
- Item Supplier Site Store

**Additional Information:** Item level attributes are set only in the Master Organization and copied to all child organizations in the organization hierarchy. Users cannot edit item level attributes in child organizations.

Item organization/store level attributes are organization specific. They default to the child organization during organization assignment. You can override the defaulting values at the child level.

Item revision level attributes are revision and organization-specific. For example, the Benchmark Rating attribute group is an item level attribute group and Desktop Specifications is associated at the item revision level. Item VI1004 is assigned to the master organization Vision Operations and to Seattle Manufacturing. Benchmark Rating attributes remain the same across the Vision Operations hierarchy while Desktop Specifications attributes vary from organization to organization and across different revisions in each organization.

**Note:** When you search for attribute groups, the system only returns those attributes eligible for the particular business entity selected in the View Attribute Groups For field. Define which attribute groups are eligible for which business entities when defining or editing an attribute group. See: Defining Item Attributes and Attribute Groups, page 4-21

#### **To delete an attribute group from an item catalog category:**

You can delete an attribute group from an item catalog category as long as the attribute group is not associated with item pages, display formats, or import formats. Once you remove any of these associations, you can delete the attribute group from the item

catalog category. Deleting the attribute group deletes it from all business entities within the item catalog category.

1. View the attributes for any business entity, select the attribute group to delete, then click Delete.

### To define the item pages for attribute groups:

You can specify one or more attribute groups on a page or create a separate page for each attribute group. Item pages are also inherited from the parent item catalog categories.

1. On the **Basic Information** page of the item catalog category, click the Item Pages link.

The **Item Pages** page lists all of the item pages created for the item catalog category by business entity.

#### Item Pages

Items | Documents | Structures | Change Management | Security | Functions | Catalogs | Value Sets

Item Catalog Categories | Cross Reference Types | Lifecycle Phases | Attribute Groups | Templates | GTIN | Item Type | Att

Items: Item Catalog Categories > Basic Information >

**Item Pages**

Item Catalog Category: Desktop

View Attribute Groups For: Item

**Search**

Search By: Display Name

Create

Select Item Pages: Delete

Select All | Select None

| Select                   | Sequence | Display Name             | Internal Name            | Description       | Classification   |
|--------------------------|----------|--------------------------|--------------------------|-------------------|------------------|
| <input type="checkbox"/> | 10       | Technical Specifications | Technical_Specifications |                   | Computer Systems |
| <input type="checkbox"/> | 20       | Expansion Slots          | Expansion_Slots          | Expansion Slots   | Computer Systems |
| <input type="checkbox"/> | 30       | Connectors               | Connectors               |                   | Computer Systems |
| <input type="checkbox"/> | 40       | Operating Systems        | Operating_Systems        | Operating Systems | Computer Systems |
| <input type="checkbox"/> | 50       | Benchmark Ratings        | Benchmark_Ratings        |                   | Computer Systems |
| <input type="checkbox"/> | 50       | RoHS / WEEE              | RoHS_WEEE                | RoHS / WEEE       | PLM High Tech    |

2. In the View Attribute Groups For field, select the business entity for which you want to create an item page.
3. Click Create Page to add a page to the item catalog category and business entity selected.

**Note:** Navigate to the item **Overview** page to view the associated

attribute group item pages.

#### Item pages for attributes on the item Overview page

Item: CN82441 Revision: ZX9000  
Organization: Vision Operations (V1)

Recent Home Logout Preferences Help

Item Lifecycle Change Management Configuration Transactions

Specifications Revisions Attachments Organizations People Associations Syndication

Overview Search Item Go Advanced Search Shortcuts Browse Catalog

Overview Actions Update

Item Catalog Category [PLM High Tech](#) > [Computer Systems](#) > [Desktop](#)

Description Sentinel System

Primary Attributes

|                            |                                 |                           |               |
|----------------------------|---------------------------------|---------------------------|---------------|
| Lifecycle                  | Computer Systems Lifecycle      | Lifecycle Phase           | Concept       |
| Item Status                | Active                          | User Item Type            | ATO model     |
| Approval Status            | Approved                        | Engineering Item          | No            |
| Primary Unit Of Measure    | Each                            | Secondary Unit of Measure | Primary       |
| Tracking                   | Primary                         | Pricing                   | 0             |
| Defaulting                 |                                 | Positive Deviation Factor | 0             |
| Conversions                | Both standard and item specific | Negative Deviation Factor |               |
| Style                      |                                 | Style Item                |               |
| Trade Item Unit Descriptor |                                 | Creation Date             | 19-Sep-2003 2 |
| Created By                 | Jonathan Smith                  |                           |               |

Actions Update

**Important:** This note applies only to Oracle Product Hub users upgrading from a release prior to Release 12.1.1.

Prior to Release 12.1.1, users were unable to define item pages by business entity, so all item and item organization attribute groups appeared on item pages together. Any item organization attribute groups in existence prior to Release 12.1.1 must be added to item pages for the item organization business entity.

## Adding Actions to an Attribute Group

You can associate user-defined actions with an attribute group in the context of an item catalog category. A user-defined action describes a function that, at run-time, can be invoked by the user. Attributes of the attribute group can be mapped to function parameters, so that their values are dynamically passed during function invocation.



**To add an action to an attribute group associated with an item catalog category:**

1. From the **Home** page, select the Setup Workbench link.
2. On the **Search: Item Catalog Categories** page, locate the appropriate item catalog category and click its name link.
3. On the **Basic Information** page, click the Attribute Groups link.
4. On the **Attribute Groups** page, click the Update Actions icon.
5. On the **View Actions** page, click Create Actions.
6. On the **Create Action** page, provide the following information:

**Sequence**

If there are multiple actions associated with an attribute group, the sequence number determines the order in which they are displayed to the user.

**Action Name**

The name of the action.

**Description**

The description of the action.

**Function**

Associates an existing function with the action. The function has a name, type, and specified parameters.

**Security Privilege**

The privilege required by a user in order for that user to execute this action.

**Execution Method**

The manner in which the action is executed.

**Display Style**

Defines the user interface element that executes the action.

**Prompt Application**

The application of the prompt for the action if the prompt is defined in the database.

**Prompt Message Name**

The message name of the prompt if the prompt is defined in the database. Otherwise, the actual prompt.

**Dynamic Prompt Function**

Selects an existing function whose return value will define the prompt of the action. Must have parameter of type Return, with data type String.

#### **Dynamic Visibility Function**

Selects an existing function whose value will determine whether or not the action is rendered.

7. Click Apply.

#### **To create parameter mappings for an action:**

After creating an action, specify the values to pass to its associated functions when that action is executed. You can have up to three associated functions: the main function, the dynamic prompt function and the dynamic visibility function.

1. On the **View Actions** page, click the name of the action you just created.
2. On the **Action Details** page, within the appropriate function heading, click Edit.
3. On the **Create Mappings** page, for each function parameter select the following:

##### **Mapping Group Type**

The type of attribute that will be mapped to this parameter.

##### **Mapped Attribute**

The actual attribute that is mapped. Only attributes with the data type that matches the parameter data type are available.

4. Click Apply.

## **Related Topics**

Creating User-Defined Functions for Item Attributes, page 4-38

## **Publishing Attribute Groups**

You can search for and select one or more attribute groups and publish them (and their associated attributes and value sets) to one or more other Oracle Product Hub systems. You cannot publish attribute groups to external systems. For more information about publishing in general, refer to *Publishing Overview*, *Oracle Product Hub User's Guide*.

**Additional Information:** Alternatively, you can use a PL/SQL Public API to publish attribute groups. See: *Using PL/SQL Public APIs to Publish Objects*, *Oracle Product Hub User's Guide*.

## Prerequisites

- ☐ Before publishing any data, you must set up Oracle Data Integrator Artifacts and Web Services for Oracle Product Hub. For set up instructions, refer to My Oracle Support ID 1336146.1. For more information about the Web Services used during publishing, see: Oracle Product Hub Web Services Overview, page G-1.

### To publish attribute groups:

**Important:** Verify that the profile option "Enable PIM for Telco Features" is set to the same value in both the source and target Oracle Product Hub systems.

1. From the **Oracle Applications Home Page**, select an appropriate responsibility, then Setup Workbench. On the **Search: Item Catalog Categories** page, click the Attribute Groups subtab.

Alternatively, navigate to the **Publish Batch Search** page (Setup > Setup Workbench (T) Publish). In the View field, select Attribute Groups. Click Create. See: To create a draft publication batch, Managing Publishing Batches, *Oracle Product Hub User's Guide*.

2. In the **Attribute Groups** page, search for one or more attribute groups to publish.
3. Select the attribute groups to publish, then click Publish.
4. In the **Attribute Group Publish Parameters** page, enter a batch name.

You can assign any unique name to the batch. The Batch ID also uniquely identifies the batch. The Batch ID and Created By fields are automatically filled when you click Publish. The Publish to PIM Systems Only box is automatically checked. You can only publish attribute groups to another Oracle Product Hub system.

If you want to remove an attribute group from the batch, select the attribute group, then click Remove.

5. Click Continue.
6. In the **Attribute Group Publish Systems** page, enter the following:
  - Auto Import. Checking this box automatically launches the EGO Import Metadata (EGOIMDCP) concurrent program in the target Oracle Product Hub system.

If you leave this box unchecked, you must manually run the EGO Import Metadata concurrent program in the target Oracle Product Hub system. When manually running this program, you can choose to import or not import the

value sets associated with the attribute groups.

The following entities and relationships associated with the attribute group are published and imported by the EGO Import Metadata concurrent program:

- Attribute group definition.
- Associated Business entity relationships.
- Attributes of the attribute group.
- Value sets associated with the attributes.

**Warning:** The publish and import process can create and update an attribute group entity relationship (such as attribute group business entities Item, Item Organization, Item Supplier, and so on), attributes, and value sets, but cannot delete already existing entity relationships, attributes, and value sets in the target system.

- Systems. Add systems or remove added systems as needed.

7. Click Save as Draft or Publish.

The **Publish Batch Search** page opens, displaying the status of the batch submitted. From here you can delete, copy, merge, or update the batch, as well as view the batch's history and create a batch. See: *Managing Publishing Batches, Oracle Product Hub User's Guide*.

**Additional Information:** Possible statuses for the batch include:

- Draft Batch - The batch has been saved as a draft. You must update the batch, then select Publish to submit the batch.
- Batch Submitted - The batch has been submitted for processing, but the EGO Import Metadata concurrent program has not run yet.
- In Process - This status appears for the following reasons:
  - When Auto Import is selected - The Publish action is in process. If there are any errors or time out issues, the system appends the status with an appropriate message indicating that either the EGO Import Metadata concurrent program did not process the updated records or the concurrent program is still running. Check the status of the

EGO Import Metadata concurrent program in the target system and take appropriate action.

- When Auto Import is not selected - The Publish action is in process. The target interface tables were loaded successfully. Manually invoke the EGO Import Metadata concurrent program in the target system.
- Succeeded - The attribute group has been successfully passed to the target systems.
- Failed - The attribute group failed the import process for the specified target systems.
- Partially Succeeded - Some attribute groups and some associated value sets have been successfully passed to the target systems. One or more attribute groups and associated value sets have failed the import process for one or more specified target systems.

## Related Topics

Publishing Overview, *Oracle Product Hub User's Guide*

Assigning Systems to an Item Catalog Category, page 3-44

Publishing Item Catalog Categories, page 3-45

Publishing Value Sets, page 4-16

Publishing Items and Structures, *Oracle Product Hub User's Guide*

Using PL/SQL Public APIs to Publish Objects, *Oracle Product Hub User's Guide*

Managing Publishing Batches, *Oracle Product Hub User's Guide*

Viewing the Publishing History, *Oracle Product Hub User's Guide*

Oracle Product Hub Web Services Overview, page G-1



---

## Setting Up Catalogs

This chapter covers the following topics:

- Overview of Catalogs
- Defining Catalog Categories
- Defining Catalogs
- Assigning a Category to a Catalog
- Mapping Catalog Categories
- Administering a Category Hierarchy within a Catalog

### Overview of Catalogs

A *catalog* is a collection of items classified within a hierarchical taxonomy. A catalog consists of a hierarchy of catalog categories, each containing items assigned to the catalog category. Catalogs provide a convenient way to organize and present your items under various taxonomies. For example, you can set up a Product Catalog of sales items that the Sales and Marketing organizations, as well as your customers, can browse to find products. You can set up a Purchasing Catalog, which contains all the purchasable items that the Engineering, Manufacturing and Procurement organizations can browse to locate parts. You can also set up a Service Catalog that contains all serviceable items and service products for your field repair and customer service departments. The following figure shows the catalog hierarchies for a Product catalog, Purchasing catalog, and Service catalog. You can define any number of catalogs and assign an item to more than one catalog. Depending on your setup, you can also assign an item to more than one catalog category within a catalog.

**Additional Information:** The *category sets* defined in the E-Business Suite appear as *catalogs*. The categories within each of these category sets appear as catalog categories in the respective catalog.

### ***Example of Catalog Hierarchies***



| ▼ Product Catalog               | ▼ Purchasing Catalog           | ▼ Service Catalog               |
|---------------------------------|--------------------------------|---------------------------------|
| ▼ Computer Systems              | ▼ Electromechanical Components | ▼ Computer Parts and Components |
| Desktop Products                | Audio Devices                  | Hard Drives                     |
| Laptop Products                 | Circuit Breakers               | Monitors                        |
| ▼ Computer Parts and Components | Fans                           | Holders                         |
| Motherboards                    | Industrial Controls & Sensors  | Mounts                          |
| PCMCIA                          | Knobs & Dials                  | Memory                          |
| Keyboards                       | ▶ Power                        | Motherboards                    |
| Mouse                           | Relays & I/O Modules           | PCMCIA                          |
| Memory                          | Switches                       | Keyboards                       |
| Hard Drives                     | ▼ Electronic Components        | Mouse                           |
| Monitors                        | ▼ Discretes                    | ▶ Electromechanical Components  |
| Holders                         | Diodes                         | ▶ Mechanical Components         |
| Mounts                          | Gates                          | ▼ Networking Components         |
|                                 | Rectifiers                     | Ethernet Controllers            |
|                                 | Transistors                    | Ethernet Cards                  |
|                                 | ▼ IC's                         | ▼ Consulting Services           |
|                                 | Memory                         | Enterprise Solutions            |
|                                 | Analog IC's                    | Small Business Solutions        |
|                                 | Digital IC's                   | Helpdesk Outsourcing            |
|                                 | Microcontrollers               | ▼ Support                       |
|                                 | Microprocessors                | ▼ Business Support              |
|                                 | ▶ Optoelectronics              | Gold Support                    |
|                                 | ▼ Passives                     | Silver Support                  |
|                                 | Capacitors                     | Bronze Support                  |
|                                 | Circuit Protection             | ▼ Consumer Support              |
|                                 | Encoders                       | On Call Support                 |
|                                 | Frequency Controllers          | Warranty                        |
|                                 | Inductors                      | Extended Warranty               |
|                                 | RC Networks                    |                                 |
|                                 | Resistors                      |                                 |
|                                 | Signal Transformers            |                                 |
|                                 | ▶ Interconnect components      |                                 |
|                                 | ▼ Mechanical Components        |                                 |
|                                 | Enclosures                     |                                 |
|                                 | Fasteners                      |                                 |
|                                 | Heat Sinks                     |                                 |
|                                 | ▼ Networking Components        |                                 |
|                                 | Ethernet Controllers           |                                 |
|                                 | Ethernet Cards                 |                                 |
|                                 | ▶ Test, Tools & Supplies       |                                 |

The following tasks described in this chapter are required in order to use a catalog. All other tasks in this chapter are optional.

- Defining Catalog Categories.
- Defining Catalogs.

## Related Topics

Defining Catalog Categories, page 5-4

Defining Catalogs, page 5-6

Assigning a Category to a Catalog, page 5-9

Mapping Catalog Categories, page 5-10

Administering a Category Hierarchy within a Catalog, page 5-10

## Defining Catalog Categories

A category is a logical classification of items that have similar characteristics. Before creating a catalog, you must have first created the appropriate category.

Use a *catalog category* to classify items within a catalog. You can define unlimited catalog categories and reuse catalog categories in more than one catalog. Before creating catalog categories, consider all the catalogs you need to define throughout your enterprise (for example, Product catalog, Purchasing catalog, Service catalog). Then create a list of all the catalog categories and sub-categories you will need to correctly classify each item within your catalogs. For example, you may have a catalog category for Hard Drives in each of your catalogs since you sell hard drives to customers, purchase hard drives from your supplier, and service hard drives. You should use the same flex structure for catalog categories so you can reuse them in different catalogs.

**Additional Information:** You can automatically trigger actions based on creating, updating, or deleting a category. See: Item Business Events, page B-1 and Managing Business Events, *Oracle Workflow Developer's Guide*.

### To create a category:

1. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
2. On the **Search: Catalogs** page, click the Categories tab.
3. On the **Search: Categories** page, click the "Create Category" link.
4. On the **Create Category** page, select a Flex Structure and click Continue.

You must enter a flexfield structure. You can define multiple segment structures for the item categories flexfield structure. Each segment in the structure can have its own display name and its own prompt.

If you choose a multi-segment flexfield structure you can assign a specific meaning to each segment. For example, if you want to group items according to product line

and product information, then you can use the first segment to represent product line and the second segment to represent the product.

Note that the categories that you assign to a catalog must have the same flexfield structure as the catalog itself. This is true even if you choose not to validate the category list.

The "Category" section of the **Create Category** page is refreshed and populated with fields defined by the flexfield structure you have chosen. Provide the information in the remaining fields of the "Category" section.

5. In the "Category Details" section, provide the following information:

**Description**

Enter a meaningful description of the category.

**Inactive on**

Select an inactive date for this category. As of the inactive date, you can no longer assign this category:

- As the default category of a new catalog.
- As a valid category of a catalog.
- To an item.
- As a valid category of an item catalog category.

You cannot set the inactive date to a date in the past. However, you can change the inactive date to one that is in the future. You cannot assign an inactive date to a category that is the default for a mandatory catalog set.

**Iprocurement**

The default is disabled (unchecked); this field cannot be modified unless the category is the default category for the purchasing functional area.

**Viewable By Supplier**

The default is enabled (checked); this field cannot be modified unless it is the default category for the purchasing functional area.

6. Click Apply

## Related Topics

Overview of Catalogs, page 5-1

Defining Catalogs, page 5-6

Assigning a Category to a Catalog, page 5-9

## Defining Catalogs

A *catalog* is a hierarchy of catalog categories used to classify items defined in your item catalog. Once you have defined all of your catalog categories you can create each of your catalogs. When you construct your catalog category hierarchy keep in mind that only the leaf nodes of the catalog can contain items.

Before creating a catalog, it is a good idea to first understand its purpose and function. It may be easiest if you first understand what makes up an item catalog. The item catalog basically contains all items defined for your company or organization. Item catalogs are hierarchical, and relationships between the various item catalog categories (such as an item's catalog category attributes) are inherited. Catalogs are also hierarchical; however, relationships between the categories are not inherited. Instead, relationships between the categories are explicitly defined by the administrator. A catalog contains some user-defined subset of items that exist in the item catalog. The following table highlights some of the key differences between an item catalog and a catalog.

**Important:** *Category sets* (in Oracle Inventory) are available for use in the system as *catalogs*. You can add people to catalogs and enable them to browse. You can also create a hierarchy to reflect the category set taxonomy (which appears in the system as a flat list using categories and subcategories).

### *Comparing the Item Catalog to a Catalog used for classification purposes only*

| Area of Comparison | Item Catalog                                                              | Catalog                                                                                     |
|--------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Purpose            | Set up all item-level attributes, lifecycles, attachment categories, etc. | Create taxonomies related to lines of business (such as Assets, Purchasing, Service, etc.). |
| Contains           | All items.                                                                | Logical subsets of items that are typically based on lines of business.                     |
| Security           | Control inherited roles for items.                                        | Control access to catalog (basically, who can view or edit the catalog).                    |

| Area of Comparison | Item Catalog                       | Catalog |
|--------------------|------------------------------------|---------|
| Lifecycle          | Set up valid lifecycles for items. | N/A     |

Before creating a catalog, you should have already created the necessary categories. Note that a category is a logical classification of items that have similar characteristics, while a catalog is a distinct grouping scheme and consists of categories.

### To create a catalog:

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
3. On the **Search: Catalogs** page, click the "Create Catalog" link.
4. On the **Create Catalog** page, provide the following information:

#### Name

Enter the name of the catalog.

#### Description

Enter a meaningful description of the catalog. A description is required so that users can browse the catalog by either the name or description.

#### Flex Structure

Select a flexfield structure. Flexfield structures contain multiple segment structures for the product catalog flexfield; each segment structure has its own display prompts and fields.

When you install the system, Oracle provides two flexfield structures by default: Item Categories and PO Item Category. You can define your own segment structures that have their own display prompts and fields.

Note that the categories that you assign to a catalog must have the same flexfield structure as the catalog itself. This is true even if you choose not to validate the category list.

#### Controlled At

Select one of the two available control levels.

- *Master Level*

Items assigned to this catalog have the same catalog category value in all organizations in which the item is assigned.

- *Org Level*

**Important:** Once you assign the control level to the Organization Item level, you cannot change it back to Item level.

### **Default Category**

Select a default category. This is the default category used when you assign an item to a catalog. You can override the default category and replace it with a more relevant category for each item.

### **Allow Multiple Item Category Assignments**

Selecting this checkbox indicates that an item can be assigned to multiple categories within this catalog. If you do not select this checkbox, an item can be assigned to exactly one category in the catalog.

**Note:** If you select this checkbox, you can assign multiple items to multiple categories within this catalog, but if multiple assignments exist, then you cannot deselect this checkbox until you delete the assignments and are left with only one.

### **Enforce List of Valid Categories**

Select this checkbox if you wish to assign items only to those categories defined as valid categories for this catalog. When selected, the default category is automatically added to the list of valid categories for this catalog. If you do not select this checkbox, then you can assign an item to any defined category that uses the same flexfield structure as this catalog.

You can specify flex structures for both category sets and categories. You can also associate different categories with a category set; if you do this, they must share the same flex structure. However, this does not make a valid category set. The category set and category must also be associated. Once the catalog and category set share the same flex structure and are associated, you have a valid category set.

### **Enable Hierarchy for Categories**

Selecting this checkbox enables you to define explicit hierarchies of categories within a catalog. Once made, you cannot revise this selection.

### **Business Events for Catalogs**

Enable the appropriate business event to raise notifications for changes to the catalog information:

- Raise Item Catalog Assignment Change Event
- Raise Catalog Category Change Event

- Raise Alternate Catalog Hierarchy Change Event

See:

- Item and Catalog Business Events, page B-1
- Managing Business Events, *Oracle Workflow Developer's Guide*

5. Click Apply.

## Related Topics

Overview of Catalogs, page 5-1

Defining Catalog Categories, page 5-4

Assigning a Category to a Catalog, page 5-9

Mapping Catalog Categories, page 5-10

Administering a Category Hierarchy within a Catalog, page 5-10

## Assigning a Category to a Catalog

You can assign different categories to a catalog; however, the only categories that you can assign to a catalog are categories with the same flexfield structure as the catalog. When you assign a category to a catalog, you can also define a hierarchical category structure, page 5-10.

**Additional Information:** You can automatically trigger actions based on assigning a category to a catalog. See: Item Business Events, page B-1 and Managing Business Events, *Oracle Workflow Developer's Guide*.

### To assign a category to a catalog:

1. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
2. On the **Search: Catalogs** page, select the catalog to which you wish to assign a category.
3. On the **Basic Information** page for the catalog, click the "Categories" link.
4. On the **Categories** page, click Add Category.
5. On the **Search: Categories** page, locate the category you wish to assign to the catalog. Note that you search for the category by Name, Description, Flex Structure or Inactive On date. Select the category and click Apply.

## Related Topics

Overview of Catalogs, page 5-1

Defining Catalog Categories, page 5-4

Defining Catalogs, page 5-6

Mapping Catalog Categories, page 5-10

Administering a Category Hierarchy within a Catalog, page 5-10

## Mapping Catalog Categories

Users can use Catalog Category mapping to map categories of different category sets to the reporting category set.

**To view the Catalog Category Mapping page:**

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
3. On the **Search: Catalogs** page, click the "Catalog Category mapping" link. The **Catalog Category Mapping** page is displayed.

## Related Topics

Overview of Catalogs, page 5-1

Defining Catalog Categories, page 5-4

Defining Catalogs, page 5-6

Assigning a Category to a Catalog, page 5-9

Administering a Category Hierarchy within a Catalog, page 5-10

## Administering a Category Hierarchy within a Catalog

You can define a hierarchy of categories within the context of a catalog, thereby making it easier to browse all the categories within a particular catalog. Creating category hierarchies within a catalog eases use because you are logically grouping categories even further within the context of a catalog. You can define multiple hierarchies within a catalog.

When you define a hierarchy, you add categories to a catalog. You can choose to add a category as a top level category, or as a child of an existing category. Note that the categories appearing in the category list all have the same flexfield structure as the catalog.



You can update a category hierarchy within a catalog by changing parent categories in the list of categories for that catalog.

**Note:** You cannot assign the parent category of a valid category to a child (also known as a descendant) because it would create loops in the hierarchy.

You can delete categories from a catalog as long as you have not already assigned items to it. In other words, if you have already assigned items to a category within a catalog, then you cannot delete the category.

**Additional Information:** You can automatically trigger actions based on deleting a category to a catalog. See: Item Business Events, page B-1 and Managing Business Events, *Oracle Workflow Developer's Guide*.

### **To create a category hierarchy within a catalog:**

1. In the Applications tree menu, click the "Setup Workbench" link.
2. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
3. On the **Search: Catalogs** page, select the catalog in which you wish to create a category hierarchy.
4. On the **Basic Information** page for the catalog, click the "Categories" link.
5. On the **Categories** page, you can add a parent category by clicking Add Category, or you can add sub-categories by selecting an existing category and then selecting Add Sub-category from the Actions pulldown.
6. On the **Search: Categories** page, locate the category you wish to assign as a sub-category. Select the category and click Apply.

### **To update a category hierarchy within a catalog:**

1. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
2. On the **Search: Catalogs** page, select the catalog in which you wish to update the category hierarchy.
3. On the **Basic Information** page for the catalog, click the "Categories" link.
4. On the **Categories** page, select the name link of the category whose parent category you wish to update.
5. On the **Category Details** page, click Update.

6. On the **Edit Catalog Category** page, select a new Parent Category, and then click Apply.

### **To delete a category from a catalog:**

You can only delete a category from a catalog other than the Item Catalog using this procedure. You cannot delete an item catalog category from the Item Catalog; you can only inactivate an item catalog category.

1. Navigate to the **Search: Catalogs** page using the following navigation path from the Development Manager responsibility:  
Setup > Setup Workbench (Tab) Catalogs
2. On the **Search: Catalogs** page, select the catalog in which you wish to delete a category.  
The catalog's **Basic Information** page opens.
3. On the **Basic Information** page for the catalog, click the "Categories" link.  
A hierarchical list of categories in the catalog displays on the **Categories** page.
4. On the **Categories** page, select the category that you wish to delete, and then select Delete from the **Select Category:** field's list of actions.
5. Click Go.

## **Related Topics**

Overview of Catalogs, page 5-1

Defining Catalog Categories, page 5-4

Defining Catalogs, page 5-6

Assigning a Category to a Catalog, page 5-9

Mapping Catalog Categories, page 5-10

---

# Administering Item Business Rules

This chapter covers the following topics:

- Overview of Item Business Rules
- Administering Rule Sets
- Building Rule Expressions
- Viewing Attributes Used in Rule Sets
- Including other Rule Sets
- Viewing Rule Set Dependencies
- Assigning Rule Sets to Entities
- Using On-Demand Rule Set Execution
- Copying Item Rules Data to Other Instances

## Overview of Item Business Rules

Oracle Product Hub enables you to define and manage items and bills of material (BOMs) in multiple organizations and organization hierarchies. Item business rules define integrity constraints on the attributes of those items. You can define integrity constraints on primary, operational, and user-defined attributes (UDAs). A typical constraint, known as a validation expression, might check that a certain attribute is less than another. For example:

```
PhysicalAttributes.UNIT_WEIGHT <= Logistics.Shipping_Weight
```

In other words, the net weight of an item must be less than or equal to the shipping weight.

You can also define assignment expressions. For example, use the following expression for the "Daily Waste Percent", where the daily waste percent equals the total waste percent divided by the shelf life in days.

```
Inventory.Total_Waste_Percent / Inventory.Shelf_Life
```

In order to use item business rules, you must define:

1. Rules and rule sets. See: Administering Rule Sets, page 6-4
2. Validation and assignment expressions. See: Building Rule Expressions, page 6-16

To activate a rule set, you must assign it to a business entity. See: Assigning Rule Sets to Entities, page 6-40

## Rules and Rule Sets

Rules encapsulate a single integrity constraint or assignment expression, for example:

- Constraint:  
`PhysicalAttributes.UNIT_WEIGHT <= Logistics.Shipping_Weight`
- Assignment:  
`Inventory.Total_Waste_Percent / Inventory.Shelf_Life`

Rule sets group multiple rules together. You can associate a rule set with both:

- An attribute group or item catalog category (ICC)
- A list of valid business entities (such as an item, item revision, item supplier, item supplier site, item organization, or item supplier site organization)

This enables the system to validate that the rules include only attributes from the associated attribute group or ICC. A rule set associated with an attribute group can only use the attributes in that group within its expressions. A rule set associated with an ICC can only use the attribute groups valid for that ICC.

Apply a rule set to each business entity separately - rule expressions cannot default from one business entity to another. For example, you cannot check an item attribute against an item organization attribute. Also, a multi-row attribute group cannot use attributes from any other attribute groups in its expressions.

You can create composite rule sets to aggregate rule sets that operate using different attribute groups and ICCs. Activate a rule set by assigning it to a business entity. The composite rule set assigned to a business entity must include all of the attribute group and ICC-specific rule sets relevant to that business entity because you can only assign one rule set to each business entity.

## Validation and Assignment Expressions

Create validation and assignment expressions after defining rules and rule sets. The expressions that make up the integrity validation and assignments follow a spreadsheet-like syntax. Each attribute is referenced by its attribute group name, followed by the attribute name:

`PhysicalAttributes.UNIT_WEIGHT`

In the above example, PhysicalAttributes is the internal name of the attribute group, and "UNIT\_WEIGHT" is the internal name of the attribute.

Combine attributes into more complex expressions using build-in operators and functions such as:

---

|                                                  |                                                                                                      |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Comparison Operators                             | == != < <= > >= in                                                                                   |
| Logical Operators                                | and or not (you can also use && and   )                                                              |
| String Functions                                 | compare contains endsWith match startsWith<br>+ indexOf length lowercase substring trim<br>uppercase |
| Math Functions                                   | + - * / sum abs amount min max round<br>roundup rounddown                                            |
| Date Functions                                   | + (Date + days) - (Date - days)                                                                      |
| Comparison to current Production Value Functions | changed delta percent previous                                                                       |

---

For example, the following expression checks if the item name's 3rd letter is "A" or "E":  
`substring(Item_Primary.Name,2,3) == "A" or substring(Item_Primary.Name,2,3) == "E"`

## Example

The following integrity constraints are typical of what you might expect for an item, in this case, a computer motherboard:

### Assignments

- Lead Percent equals Total Lead Mass divided by Unit Weight
- Sellable Date (date when item can be sold) equals 10 days after the Availability Date

### Validations

- The Minimum CPU Speed (in MHz) must be less than the Maximum CPU Speed.
- If the Purchasable Flag equals Yes, then the List Price cannot be null.
- Unit Height cannot change by more than 3% without requiring approval.

To implement the above rules, create the appropriate rule sets and then assign them to the item entity:

1. Create assignment rule set.
2. Create validation rule set.
3. Create composite rule set.
4. Assign composite rule set to the business entity.

**Note:** There is no limit to the number of rule sets that you can define for a business entity. Typically, define one rule set for each attribute group and ICC. If you create multiple rule sets for a business entity, you must create one composite rule set that contains all of these rule sets, then assign the composite rule set to the business entity. You can only assign one rule set to a business entity.

## Related Topics

Administering Rule Sets, page 6-4

Assigning Rule Sets to Entities, page 6-40

Building Rule Expressions, page 6-16

Viewing Attributes Used in Rule Sets, page 6-37

Including other Rule Sets, page 6-38

Viewing Rule Set Dependencies, page 6-39

Using On-Demand Rule Set Execution, page 6-42

Copying Item Rules Data to Other Instances, page 6-43

## Administering Rule Sets

When you create the rule sets to address your business needs, first create the rule sets for all necessary attribute groups and item catalog categories (ICCs). Typically, assign one rule set for each attribute group and ICC. Next, determine if you can consolidate rule sets together into composite rule sets. There is no limit to the number of rule sets that you can define for a business entity, but, ideally, assign only one composite rule set to each business entity.

Generally, rule sets that perform assignments should execute before validation rule sets, since validation rule sets ensure that the results of the assignment rule sets are valid.

### Computer Motherboard Rule Set Example

The following integrity constraints are typical of an item, in this case a computer motherboard:

### Assignments

- Lead Percent equals Total Lead Mass divided by Unit Weight.
- Sellable Date (date when you can sell the item) equals 10 days after the Availability Date.

### Validations

- Minimum CPU Speed (in MHz) must be less than the Maximum CPU Speed.
- If Purchasable Flag is set to Yes, then List Price has to be set.
- Unit Height cannot change more than 3% without requiring approval.

To implement the above rules, create the appropriate rule sets and then assign the sets to the Item entity following these steps:

1. Create Assignment Rule Set  
See steps below.
2. Create Validation Rule Set  
See steps below.
3. Create Composite Rule Set  
See: Including Other Rule Sets, page 6-38
4. Assign Composite Rule Set to business entity  
See: Assigning Rule Sets to Entities, page 6-40

The procedures below refer to this example while explaining how to enter rules.

### To create a new rule set:

1. Navigate to the Setup Workbench. In the Items tab, click the Rules subtab.
2. In the **Update Rule Set Assignment** page, select Item Rules from the left side menu.
3. In the **Item Rule Sets** page, click Create.
4. In the **Item Rules: Create Rule Set** page, enter Name, Description and Internal Name for the rule set.

**Caution:** You cannot change the Internal Name after the rule set is created.

5. In the Composite field, select NO.

For rule sets containing rules, set the Composite field to NO. For rule sets that contain other rule sets, set the Composite field to YES.

### **To create a new assignment rule**

The process for creating assignment and validation rules is the same up to this point. Follow the next set of steps to create assignment rules. Otherwise, skip to "To create a new validation rule:".

6. In the Type field, select Assignments.
7. Select the Item Catalog Category radio button.
8. Enter or search for and select an ICC.
9. Select all appropriate business entities for this rule set.
10. Click Continue.

This creates an empty rule set with the appropriate ICC context. All of the attribute groups within the selected ICC are available for creating assignment rules. The following steps create the actual rules.

11. Within the **Item Rules: Rule List** page, select the Rules tab (if not already selected).
12. Click Add Above.

You can also click Add Below. When you have more than one rule, these buttons enable you to create rules in a particular sequence.

13. In the **Item Rules: Create Attribute Assignment Rule** page, create an assignment rule by entering the following information:
  - Name - assignment rule name
  - Description
  - Set Value of Attribute - the name of the attribute where the computed value of the assignment rule is stored.

**Additional Information:** You can use all item primary attributes as read-only values for rule evaluation at all business entity levels (for example: Item, Item Revision, Item Supplier, and so on) while creating rules. Only Item business entity level assignment rules can set the value of the following item primary attributes: Item, Description, and Status. Modifying item primary attributes only triggers Item business entity rules.



- If Expression - enter the If part of an if/then expression, if required. If the attribute assignment rule has no requirements, then leave this field blank.  
This If expression must be blank or equal True for the assignment to take place.
- If Expressions in the Then Expression region - You can add multiple If expressions in the Then Expression region. They are nested within the previous If expression located above the Then Expression region. If there is no If expression to enter here, then leave this field blank. It defaults to true and is always selected.

### Example of Nested If/Then Expressions

The following example explains how the system reads multiple nested if/then expressions entered in the **Item Rules: Create Attribute Assignment Rule** page.

```
if (main rule expression) = true
  if (If Expression #1 in the Then Expressions region) = true,
  then (Then Expression #1)
  else if (If Expression #2 in the Then Expressions region) =
  true, then (Then Expression #2)
  else if (If Expression #3 in the Then Expressions region) =
  true, then (Then Expression #3)
end if
```

- Then Expression - enter an expression that assigns an attribute value
- Explanation Message - Optional. Use this field to inform the user about the assignment that took place.

**Additional Information:** You do not need to verify that attributes do not equal null, or, in the case of division, zero. The system ignores the assignment in this case.

### Computer Motherboard Rule Set Example

To define the first assignment rule for the Computer Motherboard Rule Set Example described above, the field values are entered as follows:

#### **Create Assignment Rule: Lead Percent is Total Lead Mass divided by Unit Weight**

| Field Name  | Value                                                   |
|-------------|---------------------------------------------------------|
| Name        | Lead_Percent                                            |
| Description | Lead Percent is Total Lead Mass divided by Unit Weight. |

| Field Name             | Value                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Set Value of Attribute | Hazard.Lead_Percent                                                                                   |
| If Expression          |                                                                                                       |
| Select If Expression   |                                                                                                       |
| Then Expression        | round((Hazard.Lead_Mass /<br>PhysicalAttributes.UNIT_WEIGHT) * 100,<br>2)                             |
| Explanation Message    | Percent is Mass (\$Hazard.Lead_Mass\$)<br>divided by Weight (\$PhysicalAttributes.<br>UNIT_WEIGHT\$). |

The screenshot shows the 'Item Rules: Create Attribute Assignment Rule' configuration window. The left sidebar has 'Item Rules' selected. The main area contains the following sections:

- Name:** Lead\_Percent
- Description:** Lead Percent is Total Lead Mass divided by Unit Weight.
- Set Value of Attribute:** Hazard.Lead\_Percent
- If Expression:** ☒ TIP If Expression has to evaluate to true for assignment to occur.
- Then Expressions:**
  - ☒ TIP Evaluation stops after the first 'If Expression' is selected.
  - ☒ TIP An empty 'If Expression' will always be selected.
  - ☒ TIP An empty 'Then Expression' will cause the value to be cleared.
- Select:** Add Above, Add Below, Delete
- Select All | Select None**
- Select If Expression:** ☒
  - Then Expression:** round((Hazard.Lead\_Mass / Physical\_Attributes.Unit\_Weight) \* 100, 2)
- Explanation Message:**
  - ☒ TIP Enter Attribute Values by surrounding name with \$. Example: \$PhysicalAttributes.Ler
  - Percent is Mass (\$Hazard.Lead\_Mass\$) divided by Weight (\$Physical\_Attributes.Unit\_Weight\$).

- Click Validate to ensure that you typed the attribute names correctly.

**Important:** You must know the internal names of the attribute groups and attributes that you want to use in expressions. You

cannot look up valid attribute groups and attributes from this page.

15. Click Apply.

This enters the first assignment rule into the rule set and returns you to the **Rule Set: Rule List** page. From this page, you can continue to add assignment rules to the rule set using the Add Above or Add Below buttons and repeating the above steps. Rules execute in the order listed within the rule set. Use the Sequence field and the Update Sequence button to change the rule order. If an attribute's expression depends on a previously calculated value, you must ensure that the previous value calculation occurs before it is needed in a later rule.

**Computer Motherboard Rule Set Example**

The following if/else if expression describes the second assignment rule for the Computer Motherboard Rule Set Example described above.

```
if (Item_Primary.Sellable_Flag == "Yes")  
  
    if (assignedTo("VeryFastDelivery"))  
    then Marketing.Sellable_Date = Planning.Availability_Date + 3  
  
    else if (assignedTo("FastDelivery"))  
    then Marketing.Sellable_Date = Planning.Availability_Date + 6  
  
    else Marketing.Sellable_Date = Planning.Availability_Date + 10
```

The "assignedTo" function verifies that an item is assigned to a particular Alternate Catalog Category.

From the **Rule Set: Rule List** page, click Add Below to add this second assignment rule. The table below shows the field values to enter in the **Item Rules: Create Attribute Assignment Rule** page.

**Create Assignment Rule: Calculate Sellable Date based on the Availability Date**

| Field Name             | Value                                                   |
|------------------------|---------------------------------------------------------|
| Name                   | Sellable_Date_Calculation                               |
| Description            | Calculate Sellable date based on the Availability Date. |
| Set Value of Attribute | Marketing.Sellable_Date                                 |
| If Expression          | Item_Primary.Sellable_Flag == "Yes"                     |
| Select If Expression   | assignedTo("Delivery", "VeryFastDelivery")              |

| Field Name           | Value                                  |
|----------------------|----------------------------------------|
| Then Expression      | Planning.Availability_Date + 3         |
| Select If Expression | assignedTo("Delivery", "FastDelivery") |
| Then Expression      | Planning.Availability_Date + 6         |
| Select If Expression | (field value is Null)                  |
| Then Expression      | Planning.Availability_Date + 10        |
| Explanation Message  | (field value is Null)                  |

#### To create a new validation rule

16. In the **Item Rule Sets** page, click Create.
17. In the **Item Rules: Create Rule Set** page, enter Name, Description and Internal Name for the rule set.

**Caution:** You cannot change the Internal Name after the rule set is created.

18. In the Composite field, select NO.  
For rule sets containing rules, set the Composite field to NO. For rule sets that contain other rule sets, set the Composite field to YES.
19. In the Type field, select Validations.
20. Select the Item Catalog Category radio button.
21. Enter or search for and select an ICC.
22. Select all appropriate business entities for this rule set.
23. Click Continue.  
This creates an empty rule set with the appropriate ICC context. All of the attribute groups within the selected ICC are available for creating validation rules. The following steps create the actual rules.
24. Within the **Item Rules: Rule List** page, select the Rules tab (if not already selected).

25. Click Add Above.

You can also click Add Below. When you have more than one rule, these buttons enable you to create rules in a particular sequence.

26. In the **Item Rules: Create Validation Rule** page, create a validation rule by entering the following information:

- Name - validation rule name
- Description
- Severity - determines the action taken if the validation fails. Severity actions include:
  - Reject - You cannot save the business entity until the validation passes.
  - Needs Approval - requires a change order.
  - Warning - The Explanation Message displays, but you can still save the entity.
- If Expression - enter the If part of an if/then expression, if required. If the attribute validation rule has no requirements, then leave this field blank.
- Validation Condition - enter an expression that determines if a condition is true or false.
- Explanation Message - Optional. Use this field to inform the user about the validation that took place.

**Additional Information:** You do not need to verify that attributes do not equal null, or, in the case of division, zero. The system ignores the validation in this case. If you want to explicitly test for a null value, use the "isnull" function.

**Computer Motherboard Rule Set Example**

To define the first validation rule for the Computer Motherboard Rule Set Example described above, the field values are entered as follows:

**Create Validation Rule: Minimum CPU Speed (in MHz) must be less than the Maximum CPU Speed.**

| Field Name           | Value                                                                    |
|----------------------|--------------------------------------------------------------------------|
| Name                 | Max_Min_CPU_Check                                                        |
| Description          | Check that Minimum CPU Clock Speed is less than Maximum CPU Clock Speed. |
| Severity             | Reject                                                                   |
| If Expression        |                                                                          |
| Validation Condition | Motherboard_Spec.Min_CPU_Speed <= Motherboard_Spec.Max_CPU_Speed         |
| Explanation Message  | Minimum CPU Speed must be less than or equal to Maximum CPU Speed.       |

Items Documents Structures Change Management Security Functions Catalogs Rules

Item Catalog Categories Cross Reference Types Lifecycle Phases Attribute Groups Temp Rules

Rule Set Assignment

**Item Rules**

**Item Rules: Create Validation Rule** Apply

\* Name

Description

\* Severity

**If Expression**

☒ **TIP** If Expression has to evaluate to true for validation to occur.

**Validation Condition**

☒ **TIP** Validation fails when Condition evaluates to False, passes if it evaluates to True.

\*

**Explanation Message**

☒ **TIP** Enter Attribute Values by surrounding name with \$. Example: \$PhysicalName

27. Click Validate to ensure that you typed the attribute names correctly.

**Important:** You must know the internal names of the attribute groups and attributes that you want to use in expressions. You cannot look up valid attribute groups and attributes from this page.

28. Click Apply.

This enters the first validation rule into the rule set and returns you to the **Rule Set: Rule List** page. From this page, you can continue to add validation rules to the rule set by selecting the checkbox next to the first rule, using the Add Above or Add Below buttons, and repeating the above steps. Rules execute in the order listed within the rule set. Use the Sequence field and the Update Sequence button to change the rule order. If an attribute's expression depends on a previously calculated value, you must ensure that the previous value calculation occurs before it is needed in a later rule.

### Computer Motherboard Rule Set Example

From the **Rule Set: Rule List** page, click Add Below to add this second validation rule. The table below shows the field values to enter in the **Item Rules: Create Attribute Validation Rule** page.

**Create Validation Rule: If the Purchasable Flag equals Yes, then the List Price cannot be null.**

| Field Name           | Value                                                        |
|----------------------|--------------------------------------------------------------|
| Name                 | Require_List_Price                                           |
| Description          | List Price must be set if Purchasable Flag is set to "Yes".  |
| Severity             | Reject                                                       |
| If Expression        | Purchasing.Purchasable == "Yes"                              |
| Validation Condition | (!isnull(Purchasing.List_Price))                             |
| Explanation Message  | If Purchasable is set to Yes then List Price cannot be null. |

**Tip:** The "!" in "!isnull" is used for "not".

Now, create the third validation rule in the example:

**Validation Rule: Unit Height cannot change by more than 3% without requiring approval.**

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| Name                 | Height_Change                                               |
| Description          | Unit Height cannot change by more than 3%.                  |
| Severity             | Needs Approval                                              |
| If Expression        |                                                             |
| Validation Condition | abs(percent(PhysicalAttributes.<br>Unit_Height)) > 3.0      |
| Explanation Message  | A Unit Height change of 3% or more requires a Change Order. |



**Note:** The percent function returns the percentage change of the new value to the value in Production. This change can be positive or negative. The abs function converts it to an absolute change.

**To include other rule sets within a rule set (create a composite rule set)**

29. See: Including Other Rule Sets, page 6-38

**To delete a rule set:**

1. In the **Item Rule Sets** page, search for and select the rule sets to delete.
2. Click Delete.

**To copy a rule set:**

1. In the **Item Rule Sets** page, search for and locate the rule set to copy.
2. In the row containing the rule set, click the Copy icon.  
The **PIM Rules: Copy Rule Set** page appears.
3. Enter a unique Internal Name for the rule set.  
The Name, Description, and Business Entities default from the original rule set, but you can change them.
4. Click Continue.  
This takes you to the **Rule Set: Rule List** page. All of the rules from the original rule set are copied into this new rule set, too. You can now update the attached rules and rule set as needed.

**To update a rule set:**

1. In the **Item Rule Sets** page, search for and locate the rule set to update.
2. In the row containing the rule set, click the Update icon.  
The **Item Rules: Rule List** page appears. From here, you can update the rule set description and the attached rules, as well as add or delete rules and update the rule sequence.

## Related Topics

Overview of Item Business Rules, page 6-1

Assigning Rule Sets to Entities, page 6-40

Building Rule Expressions, page 6-16  
Viewing Attributes Used in Rule Sets, page 6-37  
Including other Rule Sets, page 6-38  
Viewing Rule Set Dependencies, page 6-39  
Using On-Demand Rule Set Execution, page 6-42  
Copying Item Rules Data to Other Instances, page 6-43

## Building Rule Expressions

The rule administrator writes assignment and validation rules based on the components of the logical data model. This topic describes the logical data model components and the syntax used to write business rules.

You can use the following data model components when building rule expressions:

- attributes
- attribute groups
- records, when using multi-row attribute groups
- alternate catalog categories
- business entities
- current production values

## Item Business Rule Expression Reference

Use the syntax described in this section to build rule expressions.

### Constants

All values used in rule expressions consist of the following simple types:

| Type   | Description                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------|
| String | All strings and characters.<br><br>Strings are delimited by double or single quotes. The escape character is "\". |

| Type     | Description                                                                                                                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number   | <p>All decimals, integers, and so on.</p> <p>Depending on the user's Oracle E-Business Suite preferences, numbers can have a "." or "," as a decimal point. If "," is used as a decimal point, then function arguments use ";" as the argument delimiter.</p>                                                  |
| Date     | <p>Date only.</p> <p>Enter dates in the format specified in your Preference Option. Date constants automatically reformat if a user logs in with a different preference.</p> <p>You can add days to the date using numbers. For example, compute the sell date by adding 30 days to the availability date.</p> |
| DateTime | Combination of date and time                                                                                                                                                                                                                                                                                   |
| Time     | Time only                                                                                                                                                                                                                                                                                                      |

## Attributes and Catalogs

- Attributes

Indicate a user defined or operational attribute by appending it to its attribute group (separated by a period). For example:

`<Attribute Group Name>.<Attribute Name>`

Use the internal name of the attribute group and attribute. The internal name must not contain any spaces. You can find the internal names of attribute groups and attributes from the Attribute Groups page (see: Defining Item Attributes and Attribute Groups, page 4-21) or from the attribute groups associated with a particular item catalog category (see: Associating Attribute Groups with an Item Catalog Category, page 4-48).

**Important:** Attribute names are case sensitive. For example, Item\_Primary.Secondary\_UOM is not the same as Item\_Primary.Secondary\_uom.

- Attribute Units of Measure

For attributes with a defined unit of measure (UOM), add the UOM to the end of the expression. For example:

<Attribute Group Name>.<Attribute Name>.UOM

**Tip:** Note that all comparisons between amounts are automatically adjusted to take into account different UOMs, so accessing the UOM is not necessary for comparison purposes.

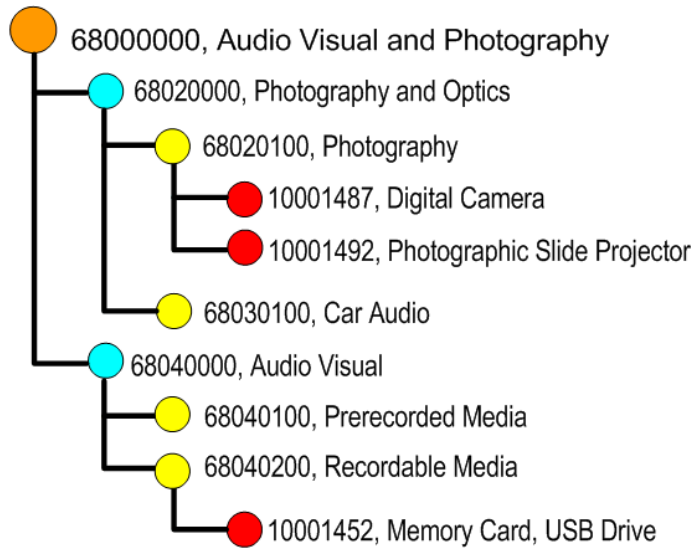
- Item Catalog Categories

You cannot use item catalog categories in expressions. To make a rule set specific to an item catalog category (ICC), assign it to that ICC during rule set creation. That rule set then applies to each entity with the ICC attached as a parent or as an ancestor.

- Alternate Catalogs

An administrator can define many alternate catalogs. For example, GS1, the supply chain standards organization, defines a categorization system named GPC. GPC contains four levels: Segment, Family, Class and Brick. The figure below illustrates some levels within the Audio Visual/Photography segment:

### GPC Categorization System Example



You can access alternate catalog information two ways with item rules:

- Determine if an entity is assigned to an alternate catalog or category.
- Access the values of an alternate catalog category.

If an item is assigned to an alternate category, all of the parent categories are also searchable.

**Note:** You can not use item business rules to assign an item to an alternate category.

Item rules referring to ICCs and Alternate Catalogs can use levels to identify the level in a catalog hierarchy tree. The root of the tree equals 1 and level 0 designates the node that the entity is assigned to, no matter what level in the hierarchy that this node actually appears. The following table shows the levels of the catalog hierarchy from the above GPC Categorization System example:

| Level  | Code     | Description              |
|--------|----------|--------------------------|
| 1      | 68000000 | Audio Visual/Photography |
| 2      | 68020000 | Photography/Optics       |
| 3      | 68020100 | Photography              |
| 4 or 0 | 10001487 | Digital Camera           |

### AssignedTo Function

The AssignedTo function can use levels:

#### Syntax

- `assignedTo(catalog)`
- `assignedTo(catalog, category)`
- `assignedTo(catalog, category, level)`

You can use assignedTo with one, two, or three parameters:

| Parameter | Type              | Explanation                                                                                                                                                                          |
|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| catalog   | String (constant) | Name of Alternate Catalog. Use only constant names (no expressions). If this is the only parameter provided, then determine if the item is assigned to any category in this catalog. |
| category  | String (constant) | Name of Category. If no other parameter is provided, then determine if the category matches any ancestor category of the item.                                                       |
| level     | Number            | Determine if the category matches at a specific level. Numbers count from the root category through the branches in the hierarchy. Level 0 is always the parent node.                |

### assignedTo Examples

- Determine if an item is assigned to the Spring Collection alternate catalog:  
`assignedTo("Spring Collection")`

- Determine if an item is assigned to a category starting with 68:  
`assignedTo ("GPC", "68000000")`
- Determine if Level 2 of a Geographic Categorization is Washington:  
`assignedTo ("Location", "Washington", 2)`

The `assignedTo` functions work for any item-related business entity. For example, if you are checking an Item Supplier entity, then the function runs for the relevant item belonging to that Item Supplier entity.

### Accessing Alternate Catalog Attributes

To write expressions that go beyond determining assignment to a category, you can access the categories and its attributes directly. If an expression referencing the alternate catalog occurs, the system loops through all of the selected categories and evaluates the expression by instantiating each one. Use the following syntax:

`<Catalog Specifier>.<Category Specifier>.<Attribute Specifier>`

For example:

`Catalog[Retail Hierarchy].AncestorCategory.Name == "USWest"`

**Caution:** You can use only one alternate catalog per expression.

You can use multiple `ParentCategory` and `AncestryCategory` specifiers, as long as they all refer to the same alternate catalog. The following list provides more information about each specifier:

- `<Catalog Specifier>`

The catalog is specified by the keyword "Catalog" followed by the name of the catalog in square brackets.

- `<Category Specifier>`

We support 2 category specifiers, `ParentCategory` and `AncestorCategory`. `ParentCategory` references the immediate category(s) assigned to the item. `AncestorCategory` references the immediate category, its parents, its parent's parents, and so on, to the root category.

- `<Attribute Specifier>`

The Attribute Specifier is the internal name of the attribute. This also retrieves the Name and Description of the attribute. Other information about the attribute is represented and stored as flexfields and is accessed using the Flexfield keyword. Examples include:

- `Catalog[Medical].AncestorCategory.Name == "Pediatrics"`
- `Catalog[RMS].ParentCategory.Name == "Fashion"`

- `[Electronics].ParentCategory.FlexField[HDTV] == "Yes"`

### Multi-Row Attribute Groups

Multi-row attribute group expressions can only include attributes from a single multi-row attribute group. The system only processes the modified rows of a multi-row attribute group.

### Operators and Functions

The following section describes how to use operators and functions when building rule expressions.

### Null Values

Any attribute can have an empty (null) value. In a typical business process, not all attribute values are entered at the same time. Multiple people enter data for an item over a period of time. For example, the Production Engineer enters an item's weight, while the Marketing Manager enters the description. To facilitate this form of processing, rules referencing attributes that have no (null) value are ignored. This means that the expression is neither true or false; expressions that evaluate to null are ignored.

Therefore, there is no need to write a validation rule with the following logic:

```
if (not isnull(PhysicalAttributes.UNIT_WEIGHT)) then PhysicalAttributes.UNIT_WEIGHT <= 10
```

The "if" part of the above rule is unnecessary. If Weight equaled null, the validation would be ignored. If you want to verify that an attribute has a non null value, use the "isnull" function. "isnull" verifies that an attribute contains a value. "isnull" is the only way to test for empty values.

### Boolean Expressions

Boolean expressions return TRUE, FALSE or null. Use them in "If" expressions or in validations. Consider the following when writing Boolean expressions:

**General Comparison Operators:** `== != < <= > >=`

Use the following syntax:

```
expression1 == expression2
expression1 != expression2
expression1 < expression2
expression1 <= expression2
expression1 > expression2
expression1 >= expression2
expression1 in (expression2, expression3, ?)
```

If one or both expressions are null, then the result is null. To check for null values, you must use the "isnull" function. A string comparison is case insensitive. For case sensitive



comparisons, use the string "compare" function. The "in" function is equivalent to (expression1 == expression2) or (expression1 == expression3) or ...(and so on).

#### **Null Comparison: isnull**

Use the following syntax:

```
isnull(expression)
```

The function "isnull" returns TRUE if its argument is a NULL value. Otherwise, it returns FALSE. This function lets you explicitly test whether a value is null. Other functions are ignored if the value of the argument is null. Use !isnull to check that an attribute is not null.

#### **String Search: compare contains endsWith match startsWith**

Use the following syntax:

```
compare(string1, string2)
contains(look_for_string, look_in_string)
endsWith(look_for_string, look_in_string)
match(pattern, look_in_string)
startsWith(look_for_string, look_in_string)
```

All string search functions are case sensitive. If you want to do a case insensitive comparison, use the "=" comparison operator. "compare" returns 0 when string1 is exactly equal to string2, -1 if string1 is lexicographically less than string2, and +1 if string1 is lexicographically greater than string2. "match" performs a regular expression search on a string. For a description of regular expression construct behavior, see Friedl, Jeffrey E. F. (2002). *Mastering Regular Expressions*. 2nd Edition. O'Reilly and Associates.

#### **Logical Operators: and or not**

Use the following syntax:

```
expression1 and expression2
expression1 or expression2
not expression1
```

The logical "and" function returns the following results based on whether or not expressions A and B are true:

| A     | B     | A and B |
|-------|-------|---------|
| False | False | False   |
| False | True  | False   |
| True  | False | False   |
| True  | True  | True    |

| <b>A</b> | <b>B</b> | <b>A and B</b> |
|----------|----------|----------------|
| False    | null     | False          |
| null     | False*   | null           |

\* The processor stops after it finds the first False. Therefore, there is a difference between "False and null" and "null and False".

The logical "or" function returns the following results based on whether or not expressions A and B are true:

| <b>A</b> | <b>B</b> | <b>A or B</b> |
|----------|----------|---------------|
| False    | False    | False         |
| False    | True     | True          |
| True     | False    | True          |
| True     | True     | True          |
| True     | null     | True          |
| null     | True*    | null          |

\* The processor stops after it finds the first True. Therefore, there is a difference between "True and null" and "null and True".

The logical "not" function returns the following results based on whether or not expression A is true:

| <b>A</b> | <b>not(A)</b> |
|----------|---------------|
| False    | True          |
| True     | False         |
| null     | null          |

#### **Comparison to Current Production: changed delta percent previous**

When using tolerance rules, you can compare the new value of an attribute to the

current production value.

Use the following syntax:

`changed(attribute)`

`changed(attributeGroup)`

`delta(attribute)`

`percent(attribute)`

`previous(attribute)`

"changed" returns TRUE if the new value of an attribute differs from the current production value. This function also works with null values. If you specify only the Attribute Group, then the function returns TRUE if any attribute in that Attribute Group changes.

"delta" returns the difference between new and existing attributes. String comparisons are case-insensitive. For Booleans expressions, TRUE is considered greater than FALSE. The following table shows the result of different scenarios:

| Operator                                                       | Number, Date, or DateTime      |
|----------------------------------------------------------------|--------------------------------|
| <code>new &lt; current production</code>                       | new - current production value |
| <code>new &gt; current production</code>                       | new - current production value |
| <code>new == current production</code>                         | 0                              |
| current production value does not exist                        | null                           |
| both new and current production value are null                 | 0                              |
| either new or current production value are null (but not both) | null                           |

"percent" is only valid for numbers. It returns:

`(delta(attribute)/"current production value")*100`

"previous" accesses the actual previous value.

## Number Operators

Use the following operators in numerical expressions:

**Math: + - \* / sum**

Use the following syntax:

`expression1 + expression2`

`expression1 - expression2`

```
expression1 * expression2
expression1 / expression2
sum(expression1, expression2, ?)
```

All of the above mathematical expressions return null if any argument is null and division by zero returns null. The number of decimal digits returned by division is the maximum number of digits from expression1 and expression2. Use "sum" to add up a series of values.

#### **abs**

Use the following syntax to return the absolute value of an expression.

```
abs(expression)
```

#### **Example**

For example, the percentage weight change must be less than 10.

```
abs(percent(PhysicalAttributes.UNIT_WEIGHT))<=10
```

#### **amount**

Use the following syntax to return an amount in a given unit of measure. This ensures that comparisons or calculations occur in the desired unit of measure.

```
amount(expression, target UOM)
```

#### **Example**

The weight must be less than or equal to 10 kg:

```
PhysicalAttributes.UNIT_WEIGHT<=amount(10, 'kg')
```

#### **min, max**

Use the following syntax to return the minimum or maximum value in a series of values. You can also use this function for arrays or in query expressions.

```
min(expression1, expression2, ?)
```

#### **Example**

In the following example, the result is "2006-11-30".

```
max("2006-10-12", "2006-11-30")
```

#### **round, roundup, rounddown**

Use the following syntax to round a number to the specified decimal places. "round" rounds to the nearest value. "roundup" rounds away from zero and "rounddown" rounds toward zero.

```
round(expression, decimals)
```

```
roundup(expression, decimals)
```

```
rounddown(expression, decimals)
```

#### **Example**

| Example Expression | Result |
|--------------------|--------|
| round(1.5758, 2)   | 1.58   |

| Example Expression                | Result |
|-----------------------------------|--------|
| <code>roundup(1.5758, 2)</code>   | 1.58   |
| <code>rounddown(1.5758, 2)</code> | 1.57   |

## String Functions

Use the following functions with string data types.

### **+**

Use the following syntax to concatenate two expressions and return the resulting string. This syntax also returns a valid string if the expressions consist of other data types.

`expression1 + expression2`

### **indexOf**

Use this syntax to return a position of "look\_for\_string" in "look\_in\_string". The string position starts at zero. If "look\_for\_string" is not found, then the processor returns a value of -1. The search is case sensitive. If either expression is null, the resultant value is null.

`indexOf(look_for_string, look_in_string)`

### **length**

Use this syntax to return the length of the given string. If the expression is null, then the resultant value is null.

`length(expression)`

### **lowercase**

Use this syntax to return the lowercase equivalent of the given expression. If the expression is null, then the resultant value is null.

`lowercase(expression)`

### **substring**

Use this syntax to return a substring of "string" starting at "start" and ending before "end". If "end" is omitted, then return to the end of the string. The string position begins at zero. If "start" is less than zero, then start at the beginning of the string. If "end" is greater than the length of the string, then return up to the end of the string. If any expression is null, then the resultant value is null.

`substring(string, start)`  
`substring(string, start, end)`

### **trim**

Use this syntax to remove all leading and trailing (but not middle) whitespace characters from a string. If the expression is null, then the resultant value is null.

`trim(expression)`

### **uppercase**

Use this syntax to return the uppercase equivalent of the given expression. If the

expression is null, then the resultant value is null.

`uppercase(expression)`

## Date Functions

Use the following functions with date data types.

**+ -**

Use the following syntax to add or subtract a certain number of days from a date. A single number is interpreted as days. Otherwise, use the ISO 8601 format.

`expression1 + expression2`

`expression1 - expression2`

For example, `Item.Logistics.LeadTime + 3` equals 3 days after the Item Lead Time.

## Supported Attributes

You can access all attributes for the following business entities using the `<Attribute Group Name>.<Attribute>` syntax.

- item
- item revision
- item organization
- item supplier
- item supplier site
- item supplier site organization

Attributes in the supplier, supplier site, and organization business entities do not use attribute group names; access the attributes directly using the `<Business Entity>.<Attribute>` syntax (for example, `Supplier.Supplier_Name`).

See below for more information about using the following types of attributes.

- Rule Set Context Attributes
- Item Primary Attributes
- Item Supplier Attributes
- Supplier Attributes
- SupplierSite Attributes
- Organization Attributes

### Rule Set Context Attributes

Rule set context attributes are a special set of seeded attributes that give some context to the rule execution.

| Context Attributes        | Description                               |
|---------------------------|-------------------------------------------|
| Context.RuleSetVersion    | Always set to "1.0"                       |
| Context.ExecutionDate     | Date that rules are invoked.              |
| Context.ExecutionDateTime | Date and time that rules are invoked.     |
| Context.BatchID           | Enter the Batch ID when running Import.   |
| Context.BatchName         | Enter the Batch Name when running Import. |

#### Example

```
Context.ExecutionDate>="7/6/2007"
```

### Item Primary Attributes

You can use all item primary attributes as read-only values for rule evaluation (in If/Then expressions) at all business entity levels (for example: Item, Item Revision, Item Supplier, and so on) while creating rules. Changes to an item's primary attributes only trigger Item business entity rules. The item primary attributes available for rule evaluation include:

- Item\_Primary.Name
- Item\_Primary.Description
- Item\_Primary.Status
- Item\_Primary.Primary\_UOM
- Item\_Primary.Secondary\_UOM
- Item\_Primary.Engineering\_Item
- Item\_Primary.Approval\_Status
- Item\_Primary.Defaulting

Only Item business entity assignment rules can set the value of the following item primary attributes:

- Name
- Description
- Status

You cannot change the value of any other item primary attributes using a rule.

**Example**

```
Item_Primary.Sellable_Flag == "Yes"
```

**Item Supplier Attributes**

Use the following two attributes for the item supplier, item supplier site, and item supplier site organization business entities when creating rules. The attribute group is `Intersection_Primary`.

- `Primary_Flag`
- `Status`

**Item Supplier Attribute Example**

```
Intersection_Primary.Primary_Flag
Intersection_Primary.Status=="Production"
```

**Supplier Attributes**

Use the following supplier attributes when creating rules. Access the supplier attributes using the attribute group `"Supplier"`.

- `Supplier_Name`
- `Supplier_Number`
- `DUNS`
- `Taxpayer_ID`
- `Tax_Registration_Number`

**Supplier Attribute Example**

```
Supplier.Supplier_Name == "Acme"
```

**SupplierSite Attributes**

Use the following supplier site attributes when creating rules. Access the supplier site attributes using the attribute group `"SupplierSite"`.

- `Supplier_Site_Name`
- `City`



- State
- Country

**Supplier Site Attribute Example**  
`SupplierSite.State=="CA"`

**Organization Attributes**

Use the organization attribute "Code" when creating rules. Access the organization attribute using the attribute group "Organization".

**Organization Attribute Example**  
`Organization.Code=="9289"`

**Execution Behavior**

Rules only run during item creation, when attributes change, or when assigning an alternate catalog and the rule contains an alternate catalog reference.

**Additional Information:** Rules run before the item number and description generating functions, so the Item Number and Description fields do not contain values when the rules run. However, the item number and description generating functions can use attribute values assigned by rules.

**Rules and Change Management**

An attribute requires a change order if one of the following conditions are met:

- an attribute's status equals "Change Order Required"
- A validation rule (using an attribute) with severity equal to "Needs Approval" runs

If an attribute requires a change order, any related attributes must also require change orders. Related attributes include any attribute used in a validation or assignment rule that contains another attribute requiring a change order.

**Related Attribute Example**

If Condition: `General.Shippable == "Yes"`

Assignment: `Logistics.PostalLength = Physical.Length + Physical.Girth`

If `General.Shippable`, `Logistics.PostalLength`, `Physical.Length` or `Physical.Girth` requires a change order, then all other attributes require a change order, also.

**Note:** For validation rules, if any attribute requires a change order, then all updated attributes in the validation rule must be included in the same change order.

If the attributes computed in assignment rules are used in subsequent rules, then they

can form a chain of dependencies. In order to ensure that the data remains consistent, the system propagates the change order requirement along the dependency chain. For example, consider the following two assignment rules:

- `Physical.Girth = Physical.Width + Physical.Depth`
- `Logistics.PostalLength = Physical.Length + Physical.Girth`

If "Physical.Depth" is updated and requires a change order, then the change order requirement propagates to `Logistics.PostalLength`. Note that the change order only propagates along updated attributes. If an attribute is not updated, then it does not require a change order (for example, `Physical.Length` above).

If an assignment rule assigns a value to an attribute in an attribute group with a change policy of "Not Allowed", then the application treats this item like a failed validation and rejects it. For example, consider the following assignment:

`Logistics.PostalLength = Physical.Length + Physical.Girth`

If the "Logistics.PostalLength" change policy is "Not Allowed" and "Physical.Length" changes, then the item is rejected.

This prevents inconsistent data from entering the database. If the system ignores the assignment rule, then the relationship between `PostalLength`, `Length` and `Girth` no longer holds. To avoid an error, ensure that all attributes dependent upon each other have change policies of "Not Allowed". In the above example, the "Logistics" and "Physical" attribute group change policies should be the same (see: Defining Change Policies, page 3-16). If setting "Physical" to "Not Allowed" is not feasible, then ensure that `Length` and `Girth` do not change by other means, because the changed entity will be rejected.

## Regular Expression Syntax

Use the following syntax when building rule expressions.

### Characters

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <code>x</code>     | The character <code>x</code>                                                            |
| <code>\\</code>    | The backslash character                                                                 |
| <code>\0n</code>   | The character with octal value <code>0n</code> ( $0 \leq n \leq 7$ )                    |
| <code>\0nn</code>  | The character with octal value <code>0nn</code> ( $0 \leq n \leq 7$ )                   |
| <code>\0mnn</code> | The character with octal value <code>0mnn</code> ( $0 \leq m \leq 3, 0 \leq n \leq 7$ ) |

---

|                     |                                                             |
|---------------------|-------------------------------------------------------------|
| <code>\xhh</code>   | The character with hexadecimal value 0xhh                   |
| <code>\uhhhh</code> | The character with hexadecimal value 0xhhhh                 |
| <code>\t</code>     | The tab character (' <code>\u0009</code> ')                 |
| <code>\n</code>     | The newline (line feed) character (' <code>\u000A</code> ') |
| <code>\r</code>     | The carriage-return character (' <code>\u000D</code> ')     |
| <code>\f</code>     | The form-feed character (' <code>\u000C</code> ')           |
| <code>\a</code>     | The alert (bell) character (' <code>\u0007</code> ')        |
| <code>\e</code>     | The escape character (' <code>\u001B</code> ')              |
| <code>\cx</code>    | The control character corresponding to x                    |

---

### ***Character classes***

---

|                                    |                                                                       |
|------------------------------------|-----------------------------------------------------------------------|
| <code>[abc]</code>                 | a, b, or c (simple class)                                             |
| <code>[^abc]</code>                | Any character except a, b, or c (negation)                            |
| <code>[a-zA-Z]</code>              | a through z or A through Z, inclusive (range)                         |
| <code>[a-d[m-p]]</code>            | a through d, or m through p: <code>[a-dm-p]</code> (union)            |
| <code>[a-z&amp;&amp;[def]]</code>  | d, e, or f (intersection)                                             |
| <code>[a-z&amp;&amp;[^bc]]</code>  | a through z, except for b and c: <code>[a-dz]</code> (subtraction)    |
| <code>[a-z&amp;&amp;[^m-p]]</code> | a through z, and not m through p: <code>[a-lq-z]</code> (subtraction) |

---

### ***Predefined character classes***

---

|    |                                                       |
|----|-------------------------------------------------------|
| .  | Any character (may or may not match line terminators) |
| \d | A digit: [0-9]                                        |
| \D | A non-digit: [^0-9]                                   |
| \s | A whitespace character: [ \t\n\x0B\f\r]               |
| \S | A non-whitespace character: [^\s]                     |
| \w | A word character: [a-zA-Z_0-9]                        |
| \W | A non-word character: [^\w]                           |

---

### ***Boundary matchers***

---

|    |                                                           |
|----|-----------------------------------------------------------|
| ^  | The beginning of a line                                   |
| \$ | The end of a line                                         |
| \b | A word boundary                                           |
| \B | A non-word boundary                                       |
| \A | The beginning of the input                                |
| \G | The end of the previous match                             |
| \Z | The end of the input but for the final terminator, if any |
| \z | The end of the input                                      |

---

**Greedy quantifiers**

---

|            |                                                |
|------------|------------------------------------------------|
| $X?$       | $X$ , once or not at all                       |
| $X^*$      | $X$ , zero or more times                       |
| $X^+$      | $X$ , one or more times                        |
| $X\{n\}$   | $X$ , exactly $n$ times                        |
| $X\{n,\}$  | $X$ , at least $n$ times                       |
| $X\{n,m\}$ | $X$ , at least $n$ but not more than $m$ times |

---

**Reluctant quantifiers**

---

|             |                                                |
|-------------|------------------------------------------------|
| $X??$       | $X$ , once or not at all                       |
| $X^*?$      | $X$ , zero or more times                       |
| $X^+?$      | $X$ , one or more times                        |
| $X\{n\}?$   | $X$ , exactly $n$ times                        |
| $X\{n,\}?$  | $X$ , at least $n$ times                       |
| $X\{n,m\}?$ | $X$ , at least $n$ but not more than $m$ times |

---

**Possessive quantifiers**

---

|            |                          |
|------------|--------------------------|
| $X?^+$     | $X$ , once or not at all |
| $X^*^+$    | $X$ , zero or more times |
| $X^+^+$    | $X$ , one or more times  |
| $X\{n\}^+$ | $X$ , exactly $n$ times  |

---

|             |                                         |
|-------------|-----------------------------------------|
| $X\{n,\}+$  | X, at least n times                     |
| $X\{n,m\}+$ | X, at least n but not more than m times |

---

### **Logical operators**

---

|       |                         |
|-------|-------------------------|
| $XY$  | X followed by Y         |
| $X Y$ | Either X or Y           |
| $(X)$ | X, as a capturing group |

---

## **Precedence**

The precedence of character-class operators is as follows, from highest to lowest:

1. Literal escape `\x`
2. Grouping `[...]`
3. Range `a-z`
4. Union `[a-e][i-u]`
5. Intersection `[a-z&&[aeiou]]`

Note that a different set of metacharacters are in effect inside a character class versus outside a character class. For instance, the regular expression `"` loses its special meaning inside a character class, while the expression `-` becomes a range forming metacharacter.

For a more precise description of the behavior of regular expression constructs, refer to [Mastering Regular Expressions](#), 2nd Edition, Jeffrey E. F. Friedl, O'Reilly and Associates, 2002.

## **Related Topics**

Overview of Item Business Rules, page 6-1

Administering Rule Sets, page 6-4

Assigning Rule Sets to Entities, page 6-40

Viewing Attributes Used in Rule Sets, page 6-37

Including other Rule Sets, page 6-38

Viewing Rule Set Dependencies, page 6-39

Using On-Demand Rule Set Execution, page 6-42

Copying Item Rules Data to Other Instances, page 6-43

## Viewing Attributes Used in Rule Sets

### To view attributes used in rule sets:

1. Navigate to the Setup Workbench. In the Items tab, click the Rules subtab.
2. In the **Update Rule Set Assignment** page, select Item Rules from the left side menu.
3. In the **Item Rule Sets** page, search for and select the name of a rule set.
4. In the **Item Rules: Rule List** page, select the Attributes tab.

You can search for the attributes included in the rule set using the following parameters:

- Business Entity
  - Attribute Group
  - Attribute
  - Data Type
5. Click Go once you have entered your search parameters.  
The search results display below the search parameters.

## Related Topics

Overview of Item Business Rules, page 6-1

Administering Rule Sets, page 6-4

Assigning Rule Sets to Entities, page 6-40

Building Rule Expressions, page 6-16

Including other Rule Sets, page 6-38

Viewing Rule Set Dependencies, page 6-39

Using On-Demand Rule Set Execution, page 6-42

Copying Item Rules Data to Other Instances, page 6-43

## Including other Rule Sets

Since you can only assign one rule set to each business entity, create composite rule sets to bundle rule sets together. You can create a hierarchy of rule sets. For example, create a composite assignment rule set, a composite validations rule set, then a master composite mixed rule set that contains the above two rule sets. Assign the master composite mixed rule set to the business entity.

### To include other rule sets within a rule set:

1. Navigate to the Setup Workbench. In the Items tab, click the Rules subtab.
2. In the **Update Rule Set Assignment** page, select Item Rules from the left side menu.
3. In the **Item Rule Sets** page, click Create.
4. In the **Item Rules: Create Rule Set** page, enter Name, Description and Internal Name for the rule set.

**Caution:** You cannot change the Internal Name after the rule set is created.

5. In the Composite field, select YES.  
For rule sets containing rules, set the Composite field to NO. For rule sets that contain other rule sets, set the Composite field to YES.
6. In the Type field, select the type of rules you plan to include in the composite rule set. Choose from:
  - Assignments
  - Validations
  - Mixed - select Mixed if you plan to include assignment and validation rule sets in the composite rule set.

**Note:** You have the option to create multiple layers of composite rule sets, some of which might only contain assignment or validation rules. Use a mixed composite rule set to include both types of rule sets together in one composite rule set.

7. Click Continue.  
The **Item Rules: Rule List** page appears.



8. In the Included Rule Sets tab of the **Item Rules: Rule List** page, click Add Above or Add Below to add a rule set.

**Note:** The rule sets execute in the order listed in the Included Rule Sets tab. List validations after assignments if you want to validate some of the assignments.

9. In the **Rule Set Search** page, search for and select the rule sets to add to your composite rule set.

The system returns you to the **Item Rules: Rule List** page.

10. Enter the sequence for the rules to run. Click Apply.

#### **To remove a rule set from a composite rule set:**

Select the composite rule set. In the **Item Rules: Rule List** page, select the rule to remove, then click Remove.

#### **To update the rule set sequence in a composite rule set:**

Select the composite rule set. In the **Item Rules: Rule List** page, enter a new sequence number in the Sequence field for each rule set. Click Apply.

## **Related Topics**

Overview of Item Business Rules, page 6-1

Administering Rule Sets, page 6-4

Assigning Rule Sets to Entities, page 6-40

Building Rule Expressions, page 6-16

Viewing Attributes Used in Rule Sets, page 6-37

Viewing Rule Set Dependencies, page 6-39

Using On-Demand Rule Set Execution, page 6-42

Copying Item Rules Data to Other Instances, page 6-43

## **Viewing Rule Set Dependencies**

From the **Item Rules: Rule List** page for a rule set, you can view the attributes used in the rule set, if the rule set is included in other rule sets (part of a composite rule set), and what business entities use the rule set.

**To view the attributes used in a rule set:**

Navigate to the **Item Rules: Rule List** page for a rule set. In the lower region of the page, select the Attributes tab.

The search results region within the Attributes tab lists all attributes used in the rule set. Use the search parameters to filter the attributes listed in the search results.

**To view which rule sets include the selected rule set:**

Navigate to the **Item Rules: Rule List** page for a rule set. In the lower region of the page, select the Rule Sets Where Included tab.

The search results region within the Rule Sets Where Included tab lists all composite rule sets that include the rule set appearing in the **Item Rules: Rule List** page. Use the search parameters to filter the composite rule sets listed in the search results.

**To view which entities use the rule set:**

Once a rule set (typically a composite rule set) is assigned to an entity, you can view which entity uses the rule set.

Navigate to the **Item Rules: Rule List** page for a rule set. In the lower region of the page, select the Where Used tab. The search results region lists the business entity that the rule set validates.

## Related Topics

Overview of Item Business Rules, page 6-1

Administering Rule Sets, page 6-4

Assigning Rule Sets to Entities, page 6-40

Building Rule Expressions, page 6-16

Viewing Attributes Used in Rule Sets, page 6-37

Including other Rule Sets, page 6-38

Using On-Demand Rule Set Execution, page 6-42

Copying Item Rules Data to Other Instances, page 6-43

## Assigning Rule Sets to Entities

After creating rule sets, you must activate them by assigning them to the appropriate business entity. You can activate them directly by assigning the rule set to a business entity, but you can only assign one rule set to each business entity. Activate a rule set indirectly by using composite rule sets to group multiple rule sets together (see: Including other Rule Sets, page 6-38). You can then assign one composite rule set to a business entity. Once a rule set is activated, the rules are applied the next time a user

updates the business entity data.

### To assign a rule set to an entity:

1. Navigate to the Setup Workbench. In the Items tab, click the Rules subtab.
2. In the **Update Rule Set Assignment** page, click Add Another Row.

The screenshot displays the 'Update Rule Set Assignment' page. The page has a navigation bar at the top with tabs: Items, Documents, Structures, Change Management, Security, Functions, Catalogs, and Value Sets. Below the navigation bar, there are sub-tabs: Item Catalog Categories, Cross Reference Types, Lifecycle Phases, Attribute Groups, Templates, GTIN, Item Type, and Attribute Groups. The main content area is titled 'Update Rule Set Assignment' and includes a sidebar on the left with 'Rule Set Assignment' and 'Item Rules'. The main area contains an 'Assignment Table' with a 'Delete' button and an 'Add Another Row' button. Below the table, there are links for 'Select All' and 'Select None'. The table has two columns: 'Select Business Entity' and 'Rule Set'. The 'Select Business Entity' column has a dropdown menu open, showing options: 'Item', 'Item Organization/Store', 'Item Supplier', 'Item Supplier Site', and 'Item Supplier Site Store'. The 'Rule Set' column lists corresponding rule sets: 'Master Item Supplier Rule Set', 'Master Item Supplier Site Org Rule Set', 'Master Item Rule Set', and 'Master Item Supplier Site Rule Set'. The page includes 'Cancel' and 'Apply' buttons at the top right and bottom right. The bottom of the page shows a navigation bar with tabs: Items, Documents, Structures, Change Management, Security, Functions, Catalogs, Value Sets, Home, Logout, Preferences, Personalize Page, and Diagnostic. The footer text reads 'Copyright (c) 2008, Oracle. All rights reserved.'

3. Select a business entity and a rule set to validate it.
  - You can only create one assignment per business entity.
  - If you need to update an assignment, select the existing assignment, click Delete, then click Apply. Finally, click Add Another Row.
  - Assign a rule set for each business entity that you want validated.
4. Click Apply.

Test the rules in the item specification pages by updating attribute values in the appropriate attribute groups. The rules should validate any updated values and any defined error messages should appear.

## Related Topics

Overview of Item Business Rules, page 6-1

Administering Rule Sets, page 6-4

Building Rule Expressions, page 6-16  
Viewing Attributes Used in Rule Sets, page 6-37  
Including other Rule Sets, page 6-38  
Viewing Rule Set Dependencies, page 6-39  
Using On-Demand Rule Set Execution, page 6-42  
Copying Item Rules Data to Other Instances, page 6-43

## Using On-Demand Rule Set Execution

You can use On-Demand Rule Set Execution to quickly select one or more items, then apply item business rules to them. Search for items using the same criteria you can use when conducting a simple or advanced item search. The search results displays the items as well as the rule sets to apply to those items before giving you the opportunity to submit the rule set execution concurrent program. The execution results display whether each item passed or failed any validation rules applied, as well as any updates made to each item as a result of assignment rules applied.

### To execute rule sets across a group of items:

1. Navigate to the **Update Rule Set Assignment** page. Click the Item Rules link.
2. In the **Item Rules** page, search for an item rule to execute on-demand. Select the name link for the rule that you want to execute.
3. In the **Item Rule Set Details** page, in the Selected Business Entity field, select the business entity for which to apply the rule set.  
  
If the rule set selected applies to more than one business entity, then you must choose one business entity over which to execute the rule set.
4. Click On-demand Execution.
5. In the **On-demand Rule Execution : Select Item** page, search for items over which to apply the rule set. See: Searching for Items, *Oracle Product Hub User's Guide*.
6. After specifying the item search criteria, click Continue.
7. In the **Search Results** page, a message appears, informing you of the rule set that will be applied to the items in the search results if you submit the results.
8. Click Submit.

You are returned to the **Item Rule Set Details** page. A message appears, providing the On-demand Rule Execution concurrent request Batch ID and Concurrent Request ID.

**To review rule set execution results:**

1. Navigate to the **Requests** page. Search for and select the details for the Request ID submitted earlier.
2. In the **Request: (Request ID)** page, verify that the request completed normally. If so, click View Log.

The log lists the items to which the rule set was applied.

**Additional Resources:** If a running Ondemand Rule Execution Java Concurrent Program terminates unexpectedly, then refer to Instructions for Restarting Concurrent Programs with a Manual Restart Mode, page H-1.

## Related Topics

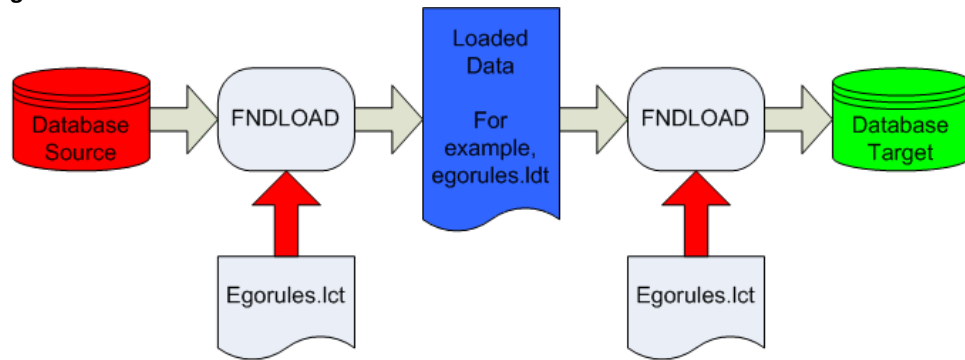
Overview of Item Business Rules, page 6-1  
Administering Rule Sets, page 6-4  
Assigning Rule Sets to Entities, page 6-40  
Building Rule Expressions, page 6-16  
Viewing Attributes Used in Rule Sets, page 6-37  
Including other Rule Sets, page 6-38  
Viewing Rule Set Dependencies, page 6-39  
Copying Item Rules Data to Other Instances, page 6-43

## Copying Item Rules Data to Other Instances

Oracle provides a method for copying item rules data from one instance to another. This method is useful when you want to define rules in a test environment to ensure the rules function as expected before defining the rules in your production environment.

To copy item rules data from one instance to another, use FNDLOAD, a generic loader used to export and import many types of data between instances. Oracle provides a configuration file, `egorules.lct`, that specifies the conversion between the database store and file records, and is read by FNDLOAD. When exporting, FNDLOAD uses `egorules.lct` to generate an output file, `egorules.ldt`, that contains the item rules data. When importing, use these same files with FNDLOAD to create or update item rules data in the target instance.

### Copying Item Rules Data Between Instances



### Item Rules Data Copied Between Instances

| Data Description         | Database Table Name                    |
|--------------------------|----------------------------------------|
| Rule Set Information     | EGO_RULE_SETS_B, EGO_RULE_SETS_TL      |
| Included Rule Set        | EGO_INCLUDED_RULESETS                  |
| Business Entity          | EGO_INCLUDED_BUSINESS_ENTITIES         |
| User Rules               | EGO_USER_RULES_B,<br>EGO_USER_RULES_TL |
| Attributes Used in Rules | EGO_RULE_ATTRIBUTE_USAGES              |

### To download the rule set data from the source environment into a data file:

First, use the a command to download the egorules.lct file. Next, use the FNDLOAD download command to download the rule set data from the source environment into a data file. If you want to download a composite rule set, comprised of other rule sets, then FNDLOAD downloads all of the rule sets related to the selected rule set.

1. Log on to the application server and initialize the application environment variable.
2. Run the following command to download the egorules.lct file.

```
FNDLOAD apps 0 Y DOWNLOAD $EGO_TOP/patch/115/import/egorules.lct .  
/out.ldt EGO_RULE_SETS [RULESET_NAME=<ruleset name>]  
[BUSINESS_ENTITY=<business entity>]  
ORACLE Password:
```

### To upload the rule set data from a data file into the target environment:

First, upload the item rules data syntax, then upload the rule set data from a data file into the target environment.

1. Log on to the application server and initialize the application environment variable.
2. Run the following command to upload the item rules data syntax.

```
FNDLOAD apps 0 Y UPLOAD $EGO_TOP/patch/115/import/egorules.lct .  
/out.ldt  
[- CUSTOM_MODE=FORCE] [LOG_NAME=<log file name>]  
ORACLE Password:
```

**Important:** Before FNDLOAD applies the data to the target instance, some item rule validation checks occur (the upload scripts of the egorules.lct file contain the validation logic). If the rule cannot pass the validation checks, then the upload process skips the rule and logs the information in the upload log file. The following errors are generated when a rule fails the validation checks:

- ICC is not defined: The rule-related ICC is not defined.
- Attribute group is not defined: The attribute group used in the rule is not defined.
- Attribute group is not associated with the ICC: The attribute group used in the rule is not associated with the ICC.
- Attribute has the wrong data type: The attribute has a different data type in the target environment.

**Caution:** FNDLOAD uses the following strategies to avoid writing over rules data in target environments:

- If the last update time of the rule set in the target environment is later than the last update time in the egorules.ldt file, then the rule set in the target environment is not updated.
- If the last update time of the rule set in the target environment is equal to or earlier than the last update time in the egorules.ldt file, then the rule set in the target environment is updated.
- You have the option to use the FORCE parameter to skip the check on the rule set last update time.

### To view the download/upload log file:

The validation and import results appear in the FNDLOAD log file after the upload process completes. The log file contains:

- The last update time comparison results.
- Validation results for the rules.
- Other exceptions that occurred during the FNDLOAD execution.

If you specified the LOG\_NAME when uploading the item rules data syntax, the log was created under the directory specified in the target database's utl\_file\_dir parameter. Determine the value of this parameter (and, therefore, determine the directory) using the following command.

```
select substr(value,1,instr(value||',' ','))-1) from v$parameter where  
name = 'utl_file_dir'
```

### To validate the rule set in a target instance:

Oracle recommends that you validate rule sets imported to the target instance. For rule sets associated with an ICC, the import process is unable to validate that the attribute groups associated with the ICC in the target instance are used at the same business entity level (for example, item vs. item organization level) as they are in the source instance. Update rules in either the **Update Assignment Rule** page or the **Update Validation Rule** page.

1. In the target instance, navigate to the **Update Rule Set Assignment** page. Select Item Rules from the left side menu.
2. Search for the item rule set that you want to validate.
3. In the search results, click the Update icon for the item rule set.
4. In the **Item Rules: Rule List** page, Rules tab, search for a rule that you want to validate.
5. Click the Update icon for the rule.
6. In the **Update Assignment Rule** or **Update Validation Rule** page, click Validate.  
If you receive validation errors, then fix the attributes used in the expressions as needed.

## Related Topics

Overview of Item Business Rules, page 6-1

Administering Rule Sets, page 6-4



Assigning Rule Sets to Entities, page 6-40  
Building Rule Expressions, page 6-16  
Viewing Attributes Used in Rule Sets, page 6-37  
Including other Rule Sets, page 6-38  
Viewing Rule Set Dependencies, page 6-39  
Using On-Demand Rule Set Execution, page 6-42



---

## Setting Up Change Categories and Types

This chapter covers the following topics:

- Overview of Change Categories and Types
- Defining Change Categories
- Enabling Change Category Function Security
- Creating Priority, Reason, and Classification Codes
- Creating Statuses
- Defining Workflow Templates
- Creating Task Templates
- Defining Header/Line Type Attributes and Attribute Groups
- Defining Header Types
- Defining Line Types
- Optimizing the Change Management Text Index
- Defining Change Category Search Criteria and Display Formats
- Defining Change Category Reports
- Subscribing to a Change Management Business Event
- Customizing Change Management Workflows
- Implementing User Defined Functions for Change Type Attributes

### Overview of Change Categories and Types

You must perform the following tasks to define Change categories and their associated types.

| Task                                                           | Required? |
|----------------------------------------------------------------|-----------|
| Defining Change Categories                                     |           |
| Defining Change Types                                          | Yes       |
| Defining Line Types                                            |           |
| Defining Change Type Attribute Groups                          |           |
| Defining Change Line Attribute Groups                          |           |
| Setting Up Change Attribute Group Security                     |           |
| Implementing User-Defined Functions for Change Type Attributes |           |
| Associating Change Type Attributes                             |           |
| Customizing Change Management Workflows                        |           |
| Defining Approval Routing Templates                            | Yes       |
| Creating User-Defined Priority Codes                           | Yes       |
| Creating User-Defined Reason Codes                             |           |
| Creating User-Defined Statuses                                 |           |
| Defining Change Category Criteria Templates                    |           |
| Defining Change Category Results Formats                       |           |
| Defining Change Reports                                        |           |

## Defining Change Categories

You can use *change categories* to define and manage changes required by your enterprise. In addition to the seeded change categories (ideas, issues, change requests, change notifications, change orders, new item requests, file approval, file review) you can create change categories specific to your business needs. For example, you can create the change category Enhancement Requests to track your customers' product

enhancement requests. You cannot delete seeded change categories; however, you can disable them.

You can configure each change category to have revised items or request lines based on the business purpose of the category. For example, a Change Order can have revised items as they allow a change order to implement item related changes. A request line enables you to request a change or specify a task related to an item and assign the line to a person or group. You can associate criteria templates and result formats to a change category for frequently executed search criteria. See: Managing Search Criteria, page 3-29 and Managing Display Formats, page 3-35.

The system provides five base change categories that can be used to create other categories:

- Ideas - To capture suggestions, innovations, improvements etc. from customers and internal users.
- Issues - to track, manage and resolve various product/process related issues.
- Change Request - to request changes and get approval for requested changes.
- Change Order - to implement requested changes and revise items.
- Change Notification - to distribute documents and files to multiple recipients.

The system also provides the following three change categories, but does not allow you to duplicate them:

- File Review - to allow for an informal process of review and feedback on attachments.
- File Approval - to allow for a formal process of review and approval of attachments.
- New Item Request (NIR) - to provide a formal process of definition and approval of a new item.

### **To create a new change category:**

1. On the **Categories** page in the Setup Workbench, select one of the following change categories and click Duplicate.
  - Idea
  - Issue
  - Change Request
  - Change Order

- Change Notification

**Note:** Creating new categories by duplicating existing categories enables your enterprise to better control the general change categories in use. It also makes it easier to introduce new categories that are similar to the basic issue, change request and change order categories. Note, however, that duplication is not allowed for the following change categories:

- File Review
- File Approval
- New Item Request (NIR)

2. On the **Duplicate Category** page, change the data in the fields as necessary to create your new category.

**Important:** When duplicating a change category, all information from the Basic Information subtab is copied. For new change categories based on existing change order categories, the following seeded line types are copied:

- Item AML changes
- Item attachment changes
- Item attribute changes

Note the following fields:

- **Plural Name**

Enter alphanumeric text to specify the plural of the name that appears in the user interface.

- **Sort Sequence**

Enter a number to specify the tab sequence in which the categories will appear in the user interface.

- **Autonumbering**

If you select the Number Generation method Override at Type Level, then autonumbering must be specified at the Type level, and therefore does not inherit the numbering schema used for the parent change category.

3. Click Apply to complete creation of the new change category.

If, after creating a new change category you wish for the new category to be displayed as a tab in the item workbench, you must create a form function passing the change category name as a parameter, then assign the form function to the appropriate Menu in the user interface, and finally assign the Menu to a Responsibility, which in turn is granted to a user. For more information, see: *Enabling Change Category Function Security*, page 7-5.

All new change categories created will be available to the user for search and reporting under the general Change Management Menu.

After creating a new change category, you can configure the Header Types, Line Types and Reports. For more details see:

- Defining Header Types, page 7-28
- Defining Line Types, page 7-35
- Defining Change Category Reports, page 7-42

## Enabling Change Category Function Security

You can specify form functions to control who can view and create change category objects. A form function must be added to the change management security menu (EGO\_CHGMGMT\_USER\_SECURITY), which in turn is referenced by a user responsibility (such as Development Manager or Development Engineer). You can also enable (in the Item Catalog workbench) a tab to show all instances of a change category for the item in context. For example, create an item enhancement request tab function that enables you to view enhancement requests from the item workbench by clicking on an Enhancement Requests tab.

### To create a subtab for a change category function within an item's Change Management tab:

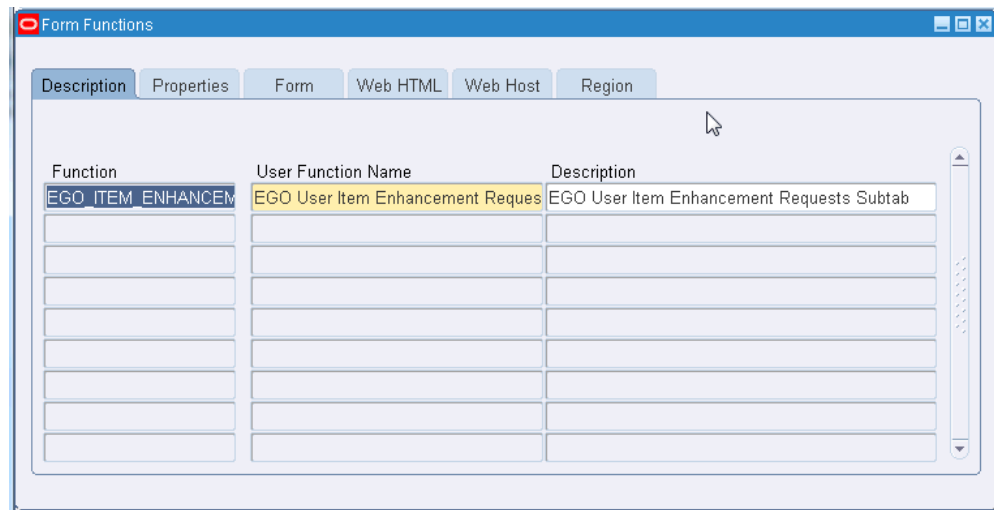
1. Using the System Administrator responsibility, navigate to the **Form Functions** window.
2. In the Function field, enter the name of the function for a particular change category. Do not include any spaces when naming the function.

For example, when creating the function for the Item Enhancement Request tab, use the name EGO\_ITEM\_ENHANCEMENT\_REQUESTS.

3. In the User Function Name field, enter a descriptive name of the function.

For example, a description appropriate for the EGO\_ITEM\_ENHANCEMENT\_REQUESTS function is EGO User Item Enhancement Requests Subtab.

4. Optionally, in the Description field, enter a description of the function.



The screenshot shows a window titled "Form Functions" with several tabs: Description, Properties, Form, Web HTML, Web Host, and Region. The "Description" tab is active. It displays a table with three columns: "Function", "User Function Name", and "Description". The first row of the table is highlighted in yellow and contains the following text: "EGO\_ITEM\_ENHANCEM" under "Function", "EGO User Item Enhancement Reques" under "User Function Name", and "EGO User Item Enhancement Requests Subtab" under "Description". There are several empty rows below the first one. A mouse cursor is visible over the "Description" tab.

| Function          | User Function Name               | Description                               |
|-------------------|----------------------------------|-------------------------------------------|
| EGO_ITEM_ENHANCEM | EGO User Item Enhancement Reques | EGO User Item Enhancement Requests Subtab |
|                   |                                  |                                           |
|                   |                                  |                                           |
|                   |                                  |                                           |
|                   |                                  |                                           |
|                   |                                  |                                           |
|                   |                                  |                                           |
|                   |                                  |                                           |
|                   |                                  |                                           |

5. Select the Properties tab.
6. In the Type field, select the Function Type Meaning "SSWA jsp function".
7. Select the Web HTML tab.
8. In the HTML Call field, enter OA.jsp? page=/oracle/apps/ego/item/eu/webui/EGOITEMCHANGEMGMTLISTPGL&akRegionApplicationId=431&addBreadCrumb=RP&OAHP=EGO\_USER\_WORKBENCH\_HOMEPAGE=EGO\_ITEM\_ENHANCEMENT\_REQUESTS= ENHANCEMENT\_REQUEST.

**Note:** Boldfaced characters represent the form function and the internal name of the change category, respectively.

Accept the default values for all other fields.

9. Click Save.

### To create a form function that lists all instances of a change category for an item:

1. Using the System Administrator responsibility, navigate to the **Form Functions** window.
2. In the Function field, enter the name of the function for all instances of a change category for an item. Do not include any spaces when naming the function.

For example, when creating the function for the instances of enhancement requests,



use the name ENG\_CREATE\_ENHANCEMENT\_REQUEST.

3. In the User Function Name field, enter a descriptive name of the function.  
For example, a description appropriate for the ENG\_CREATE\_ENHANCEMENT\_REQUEST function is Engineering Create Enhancement Request.
4. Optionally, in the Description field, enter a description of the function.
5. Select the Properties tab.
6. In the Type field, select the Function Type Meaning "SSWA jsp function".
7. Select the Web HTML tab.
8. In the HTML Call field, enter OA.jsp?  
page=/oracle/apps/eng/changemgmt/webui/SelectChgTypePG&changeMgmtTypeC  
ode=**ENHANCEMENT\_REQUEST**

**Note:** Boldfaced characters represent the internal name of the change category.

Accept the default values for all other fields.

9. Click Save.

### **To secure who can view and create a change category object:**

To control who can view and create change category objects, add the form function that shows all instances of a change category for an item to the change management security menu (EGO\_CHGMGMT\_USER\_SECURITY), then reference this menu by a user responsibility (such as Development Manager or Development Engineer).

In order to secure who can view and create change category objects, consider performing the following subtasks:

- Add a form function to the change management navigation menu, EGO\_CHG\_MGMT\_MENU.
  - Add a form function to the change management security menu, EGO\_CHGMGMT\_USER\_SECURITY.
  - Add a form function to the item change management tab menu.
1. Using the System Administrator responsibility, navigate to the **Menu** window.

**To add a form function to the change management navigation menu**

2. In the Menus window, find and select the change management navigation menu, EGO\_CHG\_MGMT\_MENU.

The menu displays a list of change management functions in sequential order.

3. In the Seq field, select a row where you would like to insert your form function.

For example, to insert your form function between rows 5 and 6, select the Seq = 5 row.

4. Click New.

An empty row appears between rows 5 and 6.

5. Select the empty Seq field. Enter the sequence in which this form function will appear in the change management security menu.

For example, enter Seq = 5.5.

6. In the Prompt field, enter the name of the form function that will appear in the change management navigation menu.

For example, enter Create Enhancement Request.

7. In the Function field, search for and select the function that shows all instances of a change category for an item.

For example, select Engineering Create Enhancement Request.

The screenshot shows the 'Menus' application window with the following fields and table:

Menu:  View Tree...

User Menu Name:

Menu Type:

Description:

| Seq | Prompt           | Submenu                | Function                  | Description | Grant                               |
|-----|------------------|------------------------|---------------------------|-------------|-------------------------------------|
| 1   | Change Simple S  |                        | Eng Menu Change Simple    |             | <input checked="" type="checkbox"/> |
| 2   | Change Advanced  |                        | Eng Menu Change Advanced  |             | <input checked="" type="checkbox"/> |
| 3   | Create Change O  |                        | Engineering Create Change |             | <input checked="" type="checkbox"/> |
| 4   | Create Change R  |                        | Engineering Create Change |             | <input checked="" type="checkbox"/> |
| 5   | Create Change N  |                        | Engineering Create Change |             | <input checked="" type="checkbox"/> |
| 5.5 | Create Enhancerr |                        | Create Enhancement Req    |             | <input checked="" type="checkbox"/> |
| 6   | Change Reports   |                        | Engineering Menu Change   |             | <input checked="" type="checkbox"/> |
| 7   |                  | Collab Dev Change Mgmt |                           |             | <input checked="" type="checkbox"/> |
| 8   |                  | Collab Dev Change Mgmt |                           |             | <input checked="" type="checkbox"/> |
| 9   |                  | Collab Dev Change Mgmt |                           |             | <input checked="" type="checkbox"/> |

8. Click Save.

A note informs you that your request to recompile your menus in the database has been submitted. Click OK.

### To add a function to the change management security menu

Next, add the function for all instances of a change category for an item to the change management security menu EGO\_CHGMMGMT\_USER\_SECURITY.

9. In the Menu field, find and select the EGO\_CHGMMGMT\_USER\_SECURITY menu.

10. In the Seq field, select a row where you would like to insert your form function.

For example, to insert your form function between rows 5 and 6, select the Seq = 5 row.

11. Click New.

An empty row appears between rows 5 and 6.

12. Select the empty Seq field. Enter the sequence in which this form function will appear in the change management security menu.

For example, enter Seq = 5.5.

13. In the Function field, search for and select the function that shows all instances of a change category for an item.

For example, select Engineering Create Enhancement Request.

14. Click Save.

A note informs you that your request to recompile your menus in the database has been submitted. Click OK.

**To add a function to the item change management tab menu**

15. In the Menu field, find and select the EGO\_USER\_CHANGE\_MANAGEMENT\_TAB menu.
16. In the Seq field, select a row where you would like to insert your form function.  
For example, to insert your form function between rows 5 and 6, select the Seq = 5 row.
17. Click New.  
An empty row appears between rows 5 and 6.
18. Select the empty Seq field. Enter the sequence in which this form function will appear in the change management security menu.  
For example, enter Seq = 5.5.
19. In the Prompt field, enter the name of the change category object.  
For example, enter Enhancement Requests.
20. In the Function field, search for and select the function for the change category.  
For example, select EGO User Item Enhancement Requests Subtab.
21. Click Save.  
A note informs you that your request to recompile your menus in the database has been submitted. Click OK.

Menu

EGO\_USER\_CHANGE\_MANAGEMENT

View Tree...

User Menu Name

EGO User Change Management Tab

Menu Type

HTML Tab

Description

EGO User Change Management Tab

| Seq | Prompt          | Submenu | Function                | Description | Grant                               |
|-----|-----------------|---------|-------------------------|-------------|-------------------------------------|
| 1   | Issues          |         | EGO User Item Issues Su |             | <input checked="" type="checkbox"/> |
| 2   | Change Requests |         | EGO User Item Change R  |             | <input checked="" type="checkbox"/> |
| 3   | Change Orders   |         | EGO User Item Change C  |             | <input checked="" type="checkbox"/> |
| 4   | Ideas           |         | EGO User Item Ideas Sub |             | <input checked="" type="checkbox"/> |
| 5   | Requirements    |         | EGO User Item Requirem  |             | <input checked="" type="checkbox"/> |
| 6   | Enhancement Re  |         | EGO User Item Enhancer  |             | <input checked="" type="checkbox"/> |
|     |                 |         |                         |             |                                     |
|     |                 |         |                         |             |                                     |
|     |                 |         |                         |             |                                     |
|     |                 |         |                         |             |                                     |
|     |                 |         |                         |             |                                     |

Note

Request ID 9927891 has been submitted to recompile your menus in the database.

OK

+ AK

**What's Next**

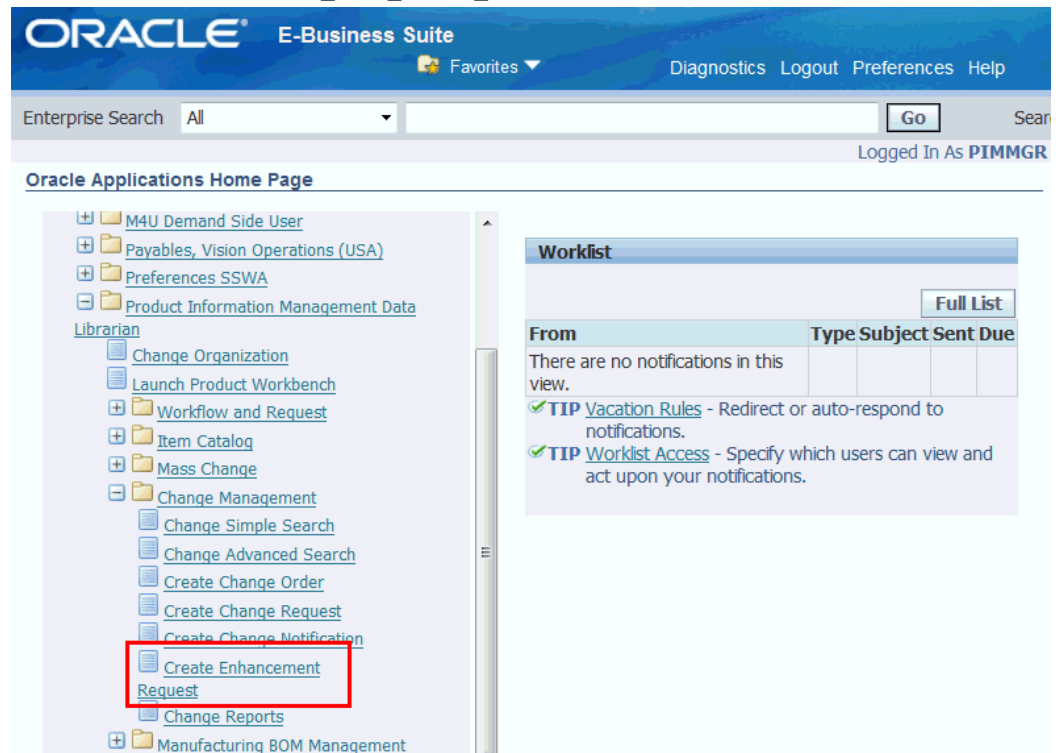
After enabling a subtab for a change category function within an item's Change Management tab, navigate to an item's **Overview** page, then select the Change Management tab. A subtab for the change category function is now available.

**New Subtab in EGO\_USER\_CHANGE\_MANAGEMENT\_TAB Menu**

The screenshot displays the Oracle Item Catalog interface. At the top, the header includes the Oracle logo, item details (Item: CN92777, Revision: A, Description: Sentinel Custom Desktop, Organization: Vision Operations (V1)), and navigation links (Navigator, Favorites). Below the header, a navigation bar contains tabs for Item, Lifecycle, Change Management, Configuration, and Transactions. Under the Change Management tab, a subtab labeled 'Enhancement Requests' is highlighted with a red rectangle. Below the navigation bar, there is a search section with a search bar, a 'Go' button, and links for 'Advanced Search' and 'Shortcuts'. To the right of the search bar is a 'Create Document' dropdown and another 'Go' button. Below the search section, there are options for 'Show Search Criteria', 'Search Criteria' (with a dropdown and 'Go' button), 'Personalize', 'Display Format' (with a dropdown and 'Go' button), 'Personalize', 'Item Revision' (set to A-A with a dropdown and 'Go' button), 'Create Report', and 'Export'. Below these options, there is a 'Create' section with a dropdown set to 'Doc ER' and a 'Go' button. Below the 'Create' section, there is a table with the following columns: Select, Details Name, Assigned To, Type, Priority, Status, Approval Status, and Need By Date. The table currently displays 'No results found.' Below the table, there are 'Create Report' and 'Export' buttons. At the bottom of the interface, there is a footer bar with links for Item, Lifecycle, Change Management, Configuration, Transactions, Recent, Home, Logout, Preferences, Help, Diagnostics, About this Page, and Privacy Statement. The footer also includes the copyright notice: Copyright (c) 2006, Oracle. All rights reserved.

Adding a form function to the Change Management navigation menu enables you to perform a new change management function directly from the Oracle Applications Home Page.

### New Form Function in EGO\_CHG\_MGMT\_MENU



## Creating Priority, Reason, and Classification Codes

Change priority, reason, and classification codes are set up similarly, but are used for different purposes as described below.

Change priorities enable you to define the degree of urgency of change requests and change orders in a way that accommodates your business processes. You can create priority codes to capture different priorities (such as High, Medium, or Low). Priority codes are applicable to all change categories and their types.

Use change reasons to categorize and identify causes for changes. The system uses reasons for reference only. Reason codes enable you to track the reason for which the change has been created. Create reason codes to capture reasons for the change (such as quality improvement, design improvement, cost reduction, test failure and nonconformance). Reason codes are applicable to all change categories and their types.

Classifications provide a mechanism for companies to categorize change orders and to also indicate to users exactly how the change order will impact their production. The user selects a valid classification code from a list of values. Specify a valid classification code in the change header type. Classifications are available only to change categories whose base category is Change Order.

You cannot delete seeded priority, reason, or classification codes. You can delete user-

defined priority, reason, or classification codes before they have been used. Instead of deleting, disable a code and define a new one specific to your use. You can disable a code on a given date by specifying a date in the Inactive On field.

#### **To create a change priority:**

1. On the **Search: Item Catalog Categories** page in the Setup Workbench, click the Change Management tab.
2. On the **Categories** page, click the Codes tab.
3. On the **Priorities** page, click Add Another Row.
4. In the new row, enter the following:

##### **Priority**

Enter alphanumeric text to specify the name of the priority.

##### **Priority Sequence**

Enter a number between 0 and 9. Use 0 to denote the highest priority and 9 the lowest. Priority sequence is used to determine the sort sequence.

##### **Description**

Enter a description for the priority.

**Note:** You can delete a change priority as long as it has not yet been used--once in use, a priority cannot be deleted.

#### **To create a change reason:**

1. On the **Search: Item Catalog Categories** page in the Setup Workbench, click the Change Management tab.
2. On the **Categories** page, click the Codes tab.
3. On the **Priorities** page, click the Reasons link.
4. On the **Reasons** page, click Add Another Row. In the new row, enter the following:

##### **Reason**

Enter alphanumeric text to describe a unique reason for initiating a change. For example, you could define a reason named OBSOLETE to indicate an obsolete part, or WAIVER to indicate a component change on a bill of material.

##### **Description**

Enter a description of the reason.



**Note:** You can delete a change reason as long as it has not yet been used--once in use, a reason cannot be deleted.

**To create classifications:**

1. On the **Search: Item Catalog Categories** page in the Setup Workbench, click the Change Management tab.
2. On the **Categories** page, click the Codes tab.
3. On the **Priorities** page, click the Classifications link.
4. On the **Classifications** page, click Add Another Row.
5. In the new row, provide the name of the Classification Code and a Description.

**Note:** You can delete a classification code as long as it has not yet been used. Once in use, a classification code cannot be deleted.

## Creating Statuses

Statuses enable you to manage an issue/change through its lifecycle. You define statuses to indicate various states of an issue/change (for example, Open, On Hold, Complete and Cancelled).

Statuses are applicable to all change categories and their types. You cannot delete or disable seeded statuses; however, you can define new statuses specific to your business processes. You can disable user-defined statuses on a given date by specifying a date in the Inactive On field.

Change categories are managed through their statuses and an approval workflow. Each status can have one workflow associated with it, and that workflow is automatically launched when changes enter into the status.

**Note:** You can change the name of any status, even those provided by the system. If a status is already in use you cannot delete it. Also, you cannot delete any of the system provided status.

**To create a status:**

1. On the **Search: Item Catalog Categories** page in the Setup Workbench, click the Change Management tab.
2. On the **Categories** page, click the Workflow tab.

3. On the **Statuses** page, click Add Another Row.

4. In the new row, enter the following:

- **Status**

Provide a name for the new status.

- **Status Type**

Select the type that applies to the status:

- **Open**

When a change order has a status of Open, users can update it. However, when the workflow is running and Update Allowed is not selected, users cannot update it. Note that Update Allowed specifies that even if a workflow is running, the change object can still be updated. Otherwise, if a workflow is running, it cannot be updated, except for the posting of comments.

- **Scheduled**

Once in use, the change cannot be updated. Scheduled change orders are picked up and implemented by auto-implement manager (when auto-implement is set up).

- **Released**

Cannot be updated.

- **Approval**

A workflow is mandatory for this status. Once the Approval is granted, the status cannot be updated. You can associate a workflow template containing an approval request step for Approval status types only.

- **Implemented**

For each change header type, only one status can use Implemented. Implemented change orders cannot be reopened and canceled. Implemented must always be the final status.

**Note:** Once in use, you cannot delete or update Status Types (except for the Display Name).

- **Description**

Enter a description of the status.

- **Inactive On**

Optionally, enter a date on which the status becomes inactive. As of this date, you can no longer assign the status.

## Defining Workflow Templates

Workflow templates enable you to predefine a business approval process. Change categories such as issues, change requests and change orders can only be approved via the successful completion of an approval routing. Workflows are supported at change header and line levels. You can create and maintain header and line workflow templates for each change type using workflow templates listed in the Setup Workbench under the Change Management Workflow tab.

### Creation of a new Workflow Template

#### Create Workflow Template

• Indicates required field

Cancel Apply

• Name

Description

Design Issue Approval

• Start Date

End Date

Type

Approval

Add Step

Show All Details

Details Step

Approval

Definition

Definition and Approval

Generic

| Step            | Response Required | Type | Assigned To | Days to Respond | Update | Delete |
|-----------------|-------------------|------|-------------|-----------------|--------|--------|
| No data exists. |                   |      |             |                 |        |        |

Cancel Apply

Workflow templates enable you to define a workflow for a change object; you can use workflows for a variety of purposes, some of which include informational messages (FYI), request for comment, or approval requests. Create workflow templates for change header and line level usage separately and associate them at the change header and change line level, respectively.

Workflow templates are made up of steps--each step describes a workflow process and specifies the assignees. For example, you can create steps to request approval, request comment, or send an FYI notification.

When creating a workflow template, you must specify the type of template.

**Important:** You cannot change a workflow template's type after creation.

The following workflow template types are supported:

## Approval

The Approval workflow template type is valid only for workflows with a status type of Approval.

## Definition

The Definition workflow template type is used primarily for workflows in New Item Requests with a status type of Open.

## Definition and Approval

The Definition and Approval workflow template type is used primarily for New Item Requests of status type Approval.

## Generic

The Generic workflow template type is used for all other status types.

You can only associate certain workflow types with specific statuses, as shown in the following table:

### ***Valid Workflow Types for Change Categories (except New Item Request)***

| Status Type     | Valid Workflow Type |
|-----------------|---------------------|
| Approval/Review | Approval            |
| Others          | Generic             |

### ***Valid Workflow Types for New Item Requests***

| Status Type                                           | Valid Workflow Type |
|-------------------------------------------------------|---------------------|
| Open                                                  | Definition          |
| Approval/Review, Definition and Approval, or Approval | Approval            |
| Others                                                | Generic             |

Workflow templates are comprised of approval steps; each approval step describes a workflow process and specifies the assignees. For example, you can create steps to request approval, request comment, or send an FYI notification.

You can use any of the following seeded workflow processes in a workflow template:

## Request Approval

The Request Approval workflow enables you to request approvals from a person or group.

### **FYI**

The FYI workflow enables you to send an FYI notification to a person or group.

### **Request Comment**

The Request Comment workflow enables you to request comments from a person or group.

### **Definition**

The New Item Request process uses a Definition workflow to enable the step assignee to associate item attribute groups to an item.

### **Definition and Approval**

The Definition and Approval workflow is similar to the Definition workflow, but also requires an approval by the step assignee. For more details, see: Defining New Item Request Workflows, page 3-21.

Plan and document the approval processes needed for specific types of changes in your enterprise. Pre-planning of these processes enables you to define workflow templates that adhere to your business processes.

You can manage parallel and serial approvals by assigning individual people, a specific role, multiple people, or groups to a routing step. Item roles need to be mapped to the change roles assigned on an approval routing step to ensure a person or group is assigned when the change is created (see Implementing Change Management Role Based Security, page 11-24 for details on mapping item roles to change roles).

### **To create a header workflow template:**

1. In the Applications tree menu, click the Setup Workbench link.
2. On the **Search: Item Catalog Categories** page, click the Change Management tab.
3. On the **Categories** page, click the Workflow tab.
4. On the **Workflow Templates** page, select Header Templates and click Create.
5. On the **Create Header Workflow Template** page, note the following fields:

#### **Start Date**

Defaults to the current date. Start date is the date from which the workflow template is available.

#### **End Date**

The date upon which this workflow template is no longer unavailable.

#### **Type**

Identifies the type of workflow template that you are creating. Available workflow template types are:

- Approval

The Approval workflow template type is valid only for workflows with status type Approval.

- Definition

The Definition workflow template type is used primarily for workflows in New Item Requests with status type Open.

- Definition and Approval

The Definition and Approval workflow template type is used primarily for New Item Requests of status type Approval.

- Generic

The Generic workflow template type is used for all other status types.

Certain workflow types can be associated with specific statuses. The following table shows the association between the header status type and the workflow type for all change categories:

***Header Workflow Types and their Usage in Change Categories and Statuses***

| <b>Change Category/<br/>Workflow Type</b> | <b>Definition</b>         | <b>Definition and Approval</b> | <b>Approval</b>               | <b>Generic</b>             | <b>Notification</b> |
|-------------------------------------------|---------------------------|--------------------------------|-------------------------------|----------------------------|---------------------|
| NIR                                       | Available for Open Status | Available for Approval Status  | Available for Approval Status | Available for all Statuses | -                   |
| Idea                                      | -                         | -                              | Available for Approval Status | Available for all Statuses | -                   |
| Issue                                     | -                         | -                              | Available for Approval Status | Available for all Statuses | -                   |
| Change Request                            | -                         | -                              | Available for Approval Status | Available for all Statuses | -                   |

| Change Category/<br>Workflow Type | Definition | Definition and Approval | Approval                      | Generic                    | Notification |
|-----------------------------------|------------|-------------------------|-------------------------------|----------------------------|--------------|
| Change Notification               | -          | -                       | Available for Approval Status | Available for all Statuses | -            |
| Change Order                      | -          | --                      | Available for Approval Status | Available for all Statuses | -            |
| File Review                       | -          | -                       | -                             | Use in All Statuses        | -            |
| File Approval                     | -          | -                       | Available for Approval Status | Available for all Statuses | -            |

6. Click Add Step.

7. On the **Add Step** page, provide the following information:

#### Step

Enter the number of the step. This determines the order in which the steps are executed.

#### Workflow Process

Select the workflow process that describes this step. After selecting a workflow process, click Go. Selecting FYI changes the fields that follow; only the Instructions field appears. You can create custom workflow processes using Oracle Workflow Builder.

**Important:** Users receive a role based on the workflow process that is chosen here. If Request Comment is selected, the person from whom you are requesting the comment receives the Change Reviewer role. If Request Approval is selected, the person from whom you are requesting the approval receives the Change Approver role. If FYI is selected, the person to whom you are sending the FYI receives the Change Reviewer role. These roles are assigned in addition to any other roles these users may already have on the change object. The administrator can edit the privileges for these roles.

## Assignee

Following are the choices for Assignee:

- User Entered

You must select a particular user who is restricted to the Roles or Groups specified in the Assignees list.

- Derived

The assignee is derived based on user's roles. Hence, the list of available assignees (in the Assignees section) all have the same role.

## Response Required

Following are the rule choices for responses:

- All Assignees

All assignees must respond (or approve).

- One Assignee

Only one of the assignees must respond (or approve).

- Mandatory Assignees

When you add assignees, you specify that they are Optional or Mandatory. You must always have at least one Mandatory responder/approver.

## Days to Respond

Enter the number of days--from the time this step is executed--in which you will need a response. Note that this is not the amount of time that transpires from the time the approval routing is submitted, but that it is the number of days from the time this particular step begins.

## Instructions

Optionally, enter instructions for the assignees to follow in their responses.

## Assignees

You can add assignees based on roles, membership in groups, or by selecting a specific person.

**Note:** When you select a role, you are selecting either a change management or item role. If you assign someone by item role, then all users with this role on the item become Reviewers. If you assign someone by change management role, then all users with this role on the change become Approvers.



8. Click Apply to save this step.
9. On the **Update Workflow Template** page, verify that all steps required for the template have been added, and then click Apply.

### **To create a line workflow template:**

Create line workflow templates using the same steps described in To create a header workflow template, page 7-19.

- Workflows are not associated to a particular status for change lines.
- Approval workflows are not supported at the line level, so the only two seeded workflow processes that you can select are Request Comment and FYI.

The following workflow types are supported at the line level:

- Notification

The notification workflow template type is available at the line level only. The workflow process associated to this type enables users to track assignee responses and manage the change line. You can assign notification workflows to Change Notification lines only.

- Generic

Use the generic workflow template type for change lines across the categories that support lines (the Change Request and Change Notification categories).

## **Creating Task Templates**

*Task templates* define the various tasks that must be completed for a change order. You can create task templates only for categories whose base change category is Change Order. After creating task templates, you can use them to define the organization policies for change order header types.

## Creation of a Task Template

### Create Task Template

• Indicates required field

Cancel Apply

\* Name

Description

\* Organization

\* Start Date   
(example: 17-Mar-2005)

End Date   
(example: 17-Mar-2005)

### Change Tasks

Select a Task(s) and... Remove

Select All | Select None

| Select                   | Sequence | Task                      | Assignee Type | Assignee      | Description                | Mandatory                           |
|--------------------------|----------|---------------------------|---------------|---------------|----------------------------|-------------------------------------|
| <input type="checkbox"/> | 10       | Unit cost analysis        | Person        | Mary Robinson | Perform unit cost analysis | <input checked="" type="checkbox"/> |
| <input type="checkbox"/> | 20       | Implementation cost analy | Person        | Ravi Patel    | Perform implementation ci  | <input checked="" type="checkbox"/> |
| <input type="checkbox"/> | 30       | Affected products/assemb  | Person        | Mary Robinson | ected products/assemblies  | <input checked="" type="checkbox"/> |

Add Another Row

Cancel Apply

Create task templates for specific organizations. This enables you to associate multiple task templates with a change order type. The organization in which you create the change order determines the task template used. Specifying a task as Mandatory ensures that the task must be completed before the change order can proceed to the next status.

After creating task templates, you can use them to define the organization policies for change order header types. Refer to the topic Defining Header Types, page 7-28 for more information about this.

### To create a task template:

1. On the **Categories** page in the Setup Workbench, click the Task Templates tab.
2. On the **Task Templates** page, click Create.
3. On the **Create Task Template** page, provide data in the required fields and click Add Another Row in the Change Tasks table to specify the tasks associated with the change order header type. Note the following fields:
  - **Sequence**  
Determines the order in which the tasks appear on the page.
  - **Task**  
The name of the task.
  - **Assignee Type**

Specify whether you wish to assign the task to a person or group.

- **Assignee**

The person (or group) to whom the task is assigned.

- **Mandatory**

Specifies that the task must be completed before the change order can proceed to the next status.

## Defining Header/Line Type Attributes and Attribute Groups

Defining change header and line attributes and attributes groups enables you to capture additional information related to issues, change requests, and change orders. You can create user-defined attributes with validation logic and associate them to change types as a collection of attributes within an attribute group.

Attributes are defined by their names and values, and are saved within attribute groups. You can associate attribute groups to a change header type or change line type. You must define separate attribute groups for header types and line types. You can reuse the same attribute group across different change categories and their change types. Users enter the values for the attributes on the pages that you create for each change type. You can also define how attributes are displayed for your change objects to improve usability.

Prior to creating user-defined attributes, do the following:

- Group related attributes within the same attribute group.
- Create a change object role with custom privileges to define attribute group security, if necessary. You can secure the attribute group by setting Edit/View privileges. Later on, only users with certain roles can view or edit those attributes.
- Determine the data type of the attributes (for example, number, char, date).

**Important:** You cannot edit the data type after you save the attribute.

- Set up the Unit of Measure Classes (for example, Currency) and Units of Measure (for example, US Dollars).
- Establish the validation rules for each attribute and create the corresponding value sets. For more details, see *Defining Value Sets for User-Defined Attributes*, page 4-1.
- Decide how you want the attribute group to be displayed - single-row or multi-row.

The figure below shows how the display options are set for the different attributes. The

figure on Defining Change Attribute Types shows the results of these display option settings. For example, you can see that the attribute Customer is set to display as a text field, and indeed it does. Note that the attribute Customer uses the value set Customer, which is really a table that is displayed as a list of values (LOV) text field. Also notice that the attribute Customer is maintained as part of a unique key. You can add to/edit the unique key as long as doing so does not destroy the uniqueness (creating duplicates) of existing records.

Also note in the figure below that Enabled parameter is set to No for the Company Name attribute, indicating that this attribute will not be displayed on the change page. You can always disable attributes; however, you cannot delete attributes if the attribute group has already been associated with a change type.

### Defining change attributes and attribute groups

**Attribute Group Details** Update

Internal Name: **cust\_priorities**  
 Display Name: **Customer Priorities**  
 Description:  
 Multi-Row: **Yes**

**Data Security**  
 View Privilege:  
 Edit Privilege:

**Attributes** Maintain Unique Key Add Attribute

Select Object: Delete

Select All | Select None

| Select                   | Sequence Name | Internal Name | Display Name | Data Type | Display As | Value Set Name | Part of Unique Key | Enabled | Required | Indexed | Column      | Edit |
|--------------------------|---------------|---------------|--------------|-----------|------------|----------------|--------------------|---------|----------|---------|-------------|------|
| <input type="checkbox"/> | 10            | Customer      | Customer     | Char      | Text Field | Customer       | Yes                | Yes     | No       | No      | C_EXT_ATTR1 |      |
| <input type="checkbox"/> | 20            | Priority      | Priority     | Char      | Text Field | Priority       | No                 | Yes     | No       | No      | C_EXT_ATTR2 |      |

**Where Used**

| Name       | Data Level | Enabled | Display Pages |
|------------|------------|---------|---------------|
| Feature ER | Change     | Yes     | Priorities    |

### Defining change type attributes (setting attribute values)

**Customer Priorities**

\* Indicates required field

Customer  Priority ☐ High ☐ Medium ☐ Low

Review ☒

Date    
 (DD-MON-RRRR)

Cancel Apply Add Another

### To create a change attribute and attribute group:

1. From the **Oracle Applications Home Page**, select an appropriate responsibility, then Setup Workbench. Select the Change Management tab.
2. Select the Header Attributes link.

3. In the **Header Attributes Search** page, find an existing change header attribute group or create a new one. To create a new one, click Create.
4. On the **Create Attribute Group** page, enter the following:
  - **Internal Name** - Required. The internal name of the attribute group. This name must be unique.
  - **Display Name** - Required. The display name of the attribute group. This name is not required to be unique.
  - **Description** - Optional. Describes the attribute group.
  - **Behavior** - Select whether to create a Single-Row or Multi-Row attribute group. The default behavior is single-row.
5. If you select Single-Row, then enter the **Number of columns in the page layout**. Required.

This determines the number of columns used to display the attributes within the attribute group.
6. If you select Multi-Row, then enter the **Number of columns shown in the table** and the **Number of rows shown in the table**. Required.

These two fields determine how the multi-row attributes within the attribute group are displayed.
7. In the **View** field, optionally select a view privilege.

If you leave this field blank, then anyone can view attribute values within this attribute group when this attribute group is associated with a change object. If you select a view privilege, then only those users with the privilege can view the associated attribute values.
8. In the **Edit** field, optionally select an edit privilege.

If you leave this field blank, then anyone can add or change attribute values within this attribute group when this attribute group is associated with a change object. If you select an edit privilege, then only those users with the privilege can edit the associated attribute values.

## Related Topics

Defining Item Attributes and Attribute Groups, page 4-21

## Defining Header Types

Change header types provide enterprises a way to accommodate business processes and classify types of changes within a change category. For example, you can define different change header types such as Quality Change, Product Change, and Performance Change to capture the various types of changes in your enterprise.

After creating and saving the basic information for a change header type, you can associate the following to the type:

- Attribute Groups and Pages
- Codes
- Workflow
- Configuration
- Organization Policies

### To create a change header type:

1. On the Change Management **Categories** page in the Setup Workbench, select a change category, then select the Types tab in the lower half of the page.
2. If you already know of a similar change header type, select it and click Duplicate. Otherwise, click Create.
3. On the **Create <Change Category Name>Type** page, enter information in the following fields:
  - **Name**- Required. Enter the name of the new change header type.
  - **Description** - Optional.
  - **Start Date and End Date** - Required. Use the Start Date and End Date fields to designate the date range in which you can use the change header type.
  - **Number Generation**  
Required. Select a number generation method.

**Note:** To enable autonumbering, you must enter both a prefix and the next available number.

- Sequence Generated

If the change header type numbering method is sequence generated, then all change headers will be automatically generated based on the sequence generation rules specified here. You can specify a prefix, starting number, and increment.

- **User Entered**

If you specify user entered, then the user is required to manually enter a number for new headers of this change header type.

**Note:** The User Entered number generation method is not supported for the New Item Request change category.

- **Function Generated**

Enables you to associate user-defined functions based on change type attributes. These functions will automatically create change numbers in the sequenced defined in the function.

See: Creating User-defined Functions, page 4-38.

- **Inherit From Parent**

If the change header type numbering method is inherited from the parent, then the change header type number generation method will be the same as that defined for the parent change category.

**Note:** Inherit From Parent only appears as a Number Generation selection choice for the change header type when the associated change category's (the parent) Number Generation is set to Override at Type Level.

**Note:** When importing new item requests using a spreadsheet, the associated change header type must have Sequence Generated or Function Generated autonumbering. User Entered autonumbering results in an import error.

- **Subject - Required.** You can specify that the subject of the change header type is an item, item revision, item and component, or nothing. Add Subject Selection Criteria to optionally restrict the subject by other related attributes such as catalog category, item lifecycle phase or item type. For example, you can set up a change order header type named Production Motherboard Change where the subject of the change is restricted to items in the Motherboard catalog category with an item lifecycle phase of Production.

- **Assignee Type** - Optional. You can set up the change header type to assign change objects by Person, Group, or Item Role. A user may have more than one role for an item because you can assign item roles at the organization, item catalog category, and item level. If you choose to assign change objects by item role, then you need to choose an Assignee Type that specifies the level in which to look up the assignee by item role:
  - Group - Assigns the change header to members of the specified group.
  - Item Role at Catalog Category Level - Assigns the change header to users with a role on the subject item that is inherited from the item catalog category.
  - Item Role at Item Level - Assigns the change header to users with a role on the subject item.
  - Item Role at Organization Level - Assigns the change header to users with a role on the subject item that is inherited within the organization.
  - Person - Assigns the change header to the person specified.
- **Assignee** - If you selected an assignee type, then select an item role or group for the assignee type.

4. Click Apply.

After creating and saving the basic information for the change header type, you can complete the change header type setup as described in the following tasks.

**To associate attribute groups and pages with a change header type:**

Associate attribute groups defined for headers to a change type. These attribute groups enable you to capture additional information regarding the type of change created, or business process-specific attributes required to process the change. You can create pages for a change header type and then organize the attribute groups associated with the change header type in a logical manner.

1. On the **Basic Information** page of the change header type to which you wish to associate attributes groups, click Attribute Groups.
2. Click Add Attribute Groups.
3. Search for and select the attribute groups you wish to associate and click Apply.

**To associate pages with a change header type**

4. On the **Basic Information** page of the change header type to which you wish to associate attributes groups, click Pages.



5. On the **Pages** page, click Create Page.
6. Enter the basic information in the required fields. Note that the number entered in Sequence determines the order in which the page links appear on the change object's detail page.
7. Click Add Another Row in the Attribute Groups section. Enter the Sequence number to specify the order in which the attribute groups will appear on the page.
8. Click Apply.

### **To associate codes with a change header type:**

Specify the valid codes for Priority, Reason and Classifications for the change header type (Classification codes are only available to change header types based on the Change Order category). This enables you to limit the available values for a user to only the selected values for each of these codes. To create these codes, refer to Creating Priority, Reason, and Classification Codes, page 7-13.

1. On the **Basic Information** page of the change header type to which you wish to associate attributes groups, click Codes.
2. On the **Codes** page, select the valid priorities and reasons codes for the change header type.
3. Click Apply.

### **To configure a change header type:**

Depending on your business process, you can elect to enable only certain primary attributes and sections for a change header type. For example, for a Supplier Requested Issue, you can disable the People section, as you may not want the Supplier to have visibility on who has what role in your organization.

1. On the **Basic Information** page of the change header type, click Configuration.
2. On the **Configuration** page, select the sections and primary attributes to enable for the change header type.
3. Click Apply.

### **To associate workflows with a change header type:**

Define the various statuses for the change header type under the workflow tab. You can specify the valid statuses for promotion and demotion as well as associate one or more workflows with each status. For an Approval status, you can only select an Approval type workflow.

You can associate multiple workflows to a change status and specify one of them as a

default. Use any one of the associated workflows at run time.

**Note:** If a mandatory workflow step is unassigned in an approval workflow, the workflow does not launch until all mandatory steps have valid assignees. If a workflow is associated to the first status (Open) on a change and a mandatory step in the workflow is unassigned, the change can only be saved as a draft. The change can move to the Open status after the step is assigned.

When you create a status you specify a status type. These status types determine some of the operational characteristics of the change header during its workflow:

- Open

When a change header has a status of Open, users can update the change object. However, there are two exceptions:

- If the workflow is running and Update Allowed is not selected, you cannot update the change object.
- If an Open status follows an Approval status within a workflow, you cannot update the change object. You cannot update a change object once it has gone through an approval process.

- Scheduled

Users cannot update the change object (unless reopened). Scheduled change objects are picked up and implemented by the auto-implement manager. Change headers must be Scheduled before they can enter the Implemented status.

- Released

Ready to be scheduled or implemented. Cannot be updated (unless the change is reopened).

- Approval

A workflow is mandatory for this status. Once the Approval is granted, the change object cannot be updated. You can associate a workflow template containing an approval request step for Approval status types only.

- Implemented

The changes specified in the change object take effect. Implemented change objects cannot be reopened and canceled. Once implemented, no further changes can be made. For each change header type, only one status can use Implemented. Implemented must always be the final status.

Certain workflow types can be associated with the statuses defined here. Following are the associations between status type and workflow type for non-new item requests:

| Status Type | Valid Workflow Type |
|-------------|---------------------|
| Approval    | Approval            |
| Others      | Generic             |

Following are the associations between status types and workflow types for new item requests:

| Status Type | Valid Workflow Type                |
|-------------|------------------------------------|
| Open        | Definition                         |
| Approval    | Definition & Approval, or Approval |
| Others      | Generic                            |

1. On the **Basic Information** page of the change header type, click Workflow to define the workflow for the change header type.
2. On the **Workflow** page, click Add Another Row in the Statuses section.

**Note:** The statuses of Open and Completed were automatically associated with the header type when the header type was created.

3. Specify a number that indicates the order in which the statuses are executed, and select a status.
4. Optionally, select a default workflow for the status.
5. Click Apply.
6. After creating a workflow for a change header type, you can click the Update Properties icon to specify for each status the valid promotion and demotion statuses and assign one or more workflows. For workflows that contain an Approval status, you must assign a workflow template that contains at least one Request Approval step. You can also select the following for each Approval workflow:
  - Enable Digital Signature

The Enable Digital Signature list only appears when electronic records and signatures are enabled for Oracle Engineering.

- Allow Updates

Selecting this option enables the user to cancel a change line during the Approval status.

- Auto Promote

Selecting this option and specifying a valid promotion status enables the system to automatically promote the change to this status after successful completion of the Approval status.

- Auto Demote

Selecting this option and specifying a valid demotion status enables the system to automatically demote the change to this status after failing the Approval process.

## 7. Click Apply.

### To specify organization policies for change headers:

For change orders only, you can specify organization policies for change header types such that task templates and propagation rules are applied to every change order created in a specific organization. Task templates identify the specific tasks that must be performed before a status is considered complete (for details, see *Creating Task Templates*, page 7-23). If a business is comprised of hierarchically related organizations, the changes specified in a change order may need to be propagated to several other organizations in your company. For example, you may have an organization, Vision Operations, which has two manufacturing organizations in the VisionMfg hierarchy named Seattle Manufacturing and Chicago Manufacturing. You initially create change orders in the Vision Operations organization. Once the change orders reach a certain status in the workflow (for example, Scheduled), the change orders can be propagated (for example, copied to the destination organization with the Status initially set to Open) to the two manufacturing organizations in the hierarchy.

You can set up the organization policy such that a change order is automatically propagated downward to the other organizations in the hierarchy whenever the change order reaches a particular status. However, the changes only propagate to the organizations within the hierarchy to which the user entering the change order has access. For example, if another user who only has security access to the Vision Operations and Seattle Manufacturing organizations enters change orders under this organization policy, the change orders only propagate to Vision Operations and Seattle Manufacturing and not to Chicago Manufacturing.

1. On the **Basic Information** page of the change order header type for which you wish to create organization policies, click Organization Policies.
2. On the **Organization Policies** page, add the organization to which you want to add the organization policy.

Click Add Another Row.

3. Select the newly added organization, then click the Task Template tab.
4. Click Add Another Row to specify the task template you wish to use. Note that you can assign more than one task template. You can specify the status in which to start the task and the status in which to complete the task.
5. Click the Propagation Rules tab. Before specifying the organization hierarchy for propagation, you have the option of selecting a particular status that, when entered, will automatically propagate the change order to a default organization hierarchy. To do so, select Auto Propagate and specify the Status for Propagation. If you wish to manually propagate change orders, do not select Auto Propagate.
6. In the Organization Hierarchy for Propagation table, click Add Another Row, then select the Organization Hierarchy to which the change order will be propagated. While you can add more than one organization hierarchy to the propagation rules, the change order can only be propagated to one. Also, you can select one organization hierarchy as the default to which all change orders are propagated.

**Note:** *Both* an organization and the task templates/propagation rules must be specified before you click Apply. If you select an organization only and click Apply, no organization policy has been defined, thus nothing is saved.

## Defining Line Types

Change orders employ tasks and revised items to capture their various elements or "to do's." Change orders do not support ad-hoc line types. Issues, Change Notifications, Ideas, and Change Requests, however, use *change lines*, which serve a similar role. NIR and File Approval and File Review do not support line types. Change lines capture all the details required for a change.

Change line types enable you to capture specific changes to an item or tasks related to a change. For example, you can define different line types to capture item related changes (such as Item Attribute changes, Component redesign changes, Attachment changes and Part Obsolescence).

## Change Line Types

### Categories

| Select Category: <input type="button" value="Duplicate"/> <input type="button" value="Update"/> |                     | Previous 1-5 Next 5 |             |          |                 |                |
|-------------------------------------------------------------------------------------------------|---------------------|---------------------|-------------|----------|-----------------|----------------|
| Select Name                                                                                     | Description         | Created By          | Start Date  | End Date | Search Criteria | Display Format |
| <input type="radio"/> Change Order                                                              | Change Order        | SYSADMIN SYSADMIN   | 01-Jul-2004 |          |                 |                |
| <input type="radio"/> New Item Request                                                          | New Item Request    | SYSADMIN SYSADMIN   | 01-Jul-2004 |          |                 |                |
| <input checked="" type="radio"/> Change Request                                                 | Change Request      | SYSADMIN SYSADMIN   | 01-Jul-2004 |          |                 |                |
| <input type="radio"/> Issue                                                                     | Issue               | SYSADMIN SYSADMIN   | 01-Jul-2004 |          |                 |                |
| <input type="radio"/> Enhancement Request                                                       | Enhancement Request | Steve Williams      | 04-Aug-2003 |          |                 |                |

### Category: Change Request

| Basic Information   <b>Types</b>   Line Types   Reports                                                                                |                              |               |                |             |          |  |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------|----------------|-------------|----------|--|
| Select Line Type: <input type="button" value="Duplicate"/> <input type="button" value="Update"/> <input type="button" value="Create"/> |                              |               |                |             |          |  |
| Select Name                                                                                                                            | Description                  | Subject Type  | Created By     | Start Date  | End Date |  |
| <input checked="" type="radio"/> A Part                                                                                                | Supplier / Manufacturer Part | None          | Steve Williams | 04-Jan-2005 |          |  |
| <input type="radio"/> Comp Redsg                                                                                                       | Component Redesign           | Item Revision | Steve Williams | 31-Jul-2003 |          |  |
| <input type="radio"/> Dsgn Chng                                                                                                        | Design Change                | Item Revision | Steve Williams | 31-Jul-2003 |          |  |
| <input type="radio"/> Dwg Change                                                                                                       | Drawing Change               | Miscellaneous | Steve Williams | 31-Jul-2003 |          |  |
| <input type="radio"/> Sfter Chng                                                                                                       | Software Change              | Item Revision | Steve Williams | 31-Jul-2003 |          |  |
| <input type="radio"/> Spec Chng                                                                                                        | Specification Change         | Item Revision | Steve Williams | 31-Jul-2003 |          |  |

Basic Information | Types | **Line Types** | Reports

Creation of a Line type is similar to that of the Change Header type. Each Line type could have a default assignee as well as an associated Subject.

## Defining a Change Line Type

### Create Change Request Line Type

\* Indicates required field

\* Name

Description

\* Start Date

(example: 17-Mar-2005)

End Date

(example: 17-Mar-2005)

**Subject Selection Criteria**

\* Subject

| Select                                         | Attribute       | Value | Allowed |
|------------------------------------------------|-----------------|-------|---------|
|                                                | No data exists. |       |         |
| <input type="button" value="Add Another Row"/> |                 |       |         |

**Default Assigned To**

Assignee Type  Assignee

You can associate workflows at the line level. Generic and Notification workflow types are supported for lines—approval workflows are not supported for lines. You have the option to associate multiple workflows to a line type, with one of the workflows set as the default.

### To create a change line type:

1. On the **Categories** page in the Setup Workbench, select any change category that supports line types, and click the Line Types tab.
2. If you already know of a similar line type, select it and click Duplicate. Otherwise, click Create.
3. On the **Create <Change Category Name> Line Type** page, note the following fields:

- **Subject**

You can specify an object upon which to base the line type. Restrict the subject object further by item catalog category, lifecycle phase, or item type. For example, you can create an issue line type to restrict the subject to all items in a particular catalog category whose value you specify. Then, when users file issues, they select from a list of values. If you do not restrict the subject, then all items are available for the line type.

- **Assignee Type**

Typically users have different roles at different levels within an enterprise. To ensure users are assigned the correct role on line types, you can choose from the following Assignee Types:

- **Group**

The change line type is assigned to members of the group that is specified.

- **Item Role at Catalog Category Level**

The change line type is assigned to users who have a role on the subject item that is inherited from the item catalog category.

- **Item Role at Item Level**

The change line type is assigned to users who have a role on the subject item.

- **Item Role at Organization Level**

The change line type is assigned to users who have a role on the subject item that is inherited within the organization.

- **Person**

The change line type is assigned to the person specified.

### To associate workflows with a change line type

You can associate multiple workflow templates to a change line type, setting one of

them as the default workflow.

**Note:** Only Generic and Notification workflow types are supported for change line types. The Approval workflow type is not supported.

4. In the Workflows section of the **Create <Change Category Name> Line Type** page, click Add Another Row.
5. Search for and select a workflow template.
6. Optionally, with the workflow template selected, click Set as Default to select this template as the default workflow for the change line type.
7. Click Apply.

### **To associate attributes, attribute groups, and pages with a change line type:**

You can associate attribute groups to a change line type, then associate the attribute groups to a change line type page. These attribute groups enable you to capture additional information regarding the type of change being created, or business process-specific attributes that are required to process the change. Use pages for a line type to organize the attribute groups associated with the line type in a logical manner. For example, you may have created and specified the attribute groups Cost Information and Inventory Impact for a particular change line type. Now you wish to make them available via the user interface. Set up a new page (for example, Related Information) on which to make available the interface for the attributes Cost Information and Inventory Impact.

1. On the **Basic Information** page of the change line to which you wish to associate attributes groups, click the Attribute Groups link.
2. On the **Attribute Groups** page, search for and select the attribute groups you wish to associate.
3. Click Apply.

### **To associate pages with a change line type**

Associate the item change type's attribute groups to a page.

4. On the **Basic Information** page of the change line to which you wish to associate pages, click the Pages link.
5. On the **Pages** page, click Create Page.
6. Enter the basic information in the required fields. Note that the number entered in Sequence determines the order in which the page links appear on the change line's



detail page.

7. Click Add Another Row in the Attribute Groups section. Enter the Sequence number to specify the order in which the attribute groups will appear on the page.

**Note:** You cannot associate pages with the following change order line types:

- AML changes
- Attribute changes
- Attachment changes

8. Once you have added all of the attribute groups that you want to appear on the page, click Apply.

## Optimizing the Change Management Text Index

To take advantage of the Oracle text search features such as keyword search, stemming search, and fuzzy search— run the concurrent program Optimize Change Management Text Index. Any user with the Development Manager responsibility can submit this concurrent program.

As change management data changes over time due to ongoing additions, deletions, or update operations, the query response time may gradually decrease. Optimizing the change management text index using the Optimize Change Management Text Index concurrent program removes old data and minimizes index fragmentation, and therefore can improve query response time.

Optimize the Change Management Text Index:

- After performing many change actions.
- After updating many change actions.
- On a regular basis (optimize the text index at regular intervals, for example, twice a week.).
- When you notice slow performance for change simple search.

**Note:** This program optimizes at most 16,000 change objects per single run. To continue optimizing more changes, re-run the program.

**To optimize the Change Management Text Index:**

1. In the Development Manager responsibility, click the "View Concurrent Requests" link.
2. In the **Requests** page, click Submit Request.
3. In the **Schedule Request: Name** page, select New Request and then search for and select the program name Optimize Change Management Text Index.
4. Optionally, enter a request name to use later when searching for the request.
5. Select one of the following language and numeric display options:
  - American English
  - Arabic
  - Korean
6. Click Next to continue setting parameters or Submit to use the default parameters.
7. In the **Schedule Request: Parameters** page, click Next.
8. In the **Schedule Request: Schedule** page, specify whether to use a new or saved schedule.

When using a new schedule, specify the following:

- Start Date
- Recurrence  
Specify whether to repeat the request or not, how often to repeat, and when to end the recurrence.
- Apply the Interval  
Specify whether to optimize the index from the completion of the prior index run or from the start of the prior index run.

Optionally, click Advanced Schedule. This provides additional scheduling options.

When using a saved schedule, search for and select the name of the saved schedule. Click Next.

9. On the **Schedule Request: Notifications** page, specify the recipients to receive any of the following notifications:
  - Normal - The request completed normally.

- Warning - Lists any warnings received upon completion.
- Error - Lists any errors received upon completion.

Click Next.

10. On the **Schedule Request: Delivery** page, select printer, email fax, or FTP options for delivery of program results.

Click Next.

11. On the Schedule Request: Review page, review the options selected.

If you are satisfied with the options selected, click Submit. Otherwise, click Back to choose different options.

Once you click Submit, an **Information** page appears, listing the request ID.

## Defining Change Category Search Criteria and Display Formats

Criteria templates offer you a way to save frequently used search criteria—essentially, acting as a saved list of specific attributes and attribute values. Criteria templates that an administrator creates are available to all users and serve to expedite searches for issues, ideas, change requests and change orders, thereby saving users the time it would take to specify search criteria and execute frequent change management searches.

Users also have the option of creating criteria templates for their own private use. When both a user and administrator define a default criteria template for a particular change category, the user-defined criteria template takes precedence. Because criteria templates contain user-defined attributes, you must always define them in the context of a change category. You can define as many criteria templates as you wish, and you can also denote the most commonly used as the default criteria template. For example, you may define criteria templates to quickly find Open Change Requests, My Issues, or Scheduled Change Orders.

Display formats enable you to predefine search results views for each change category. You can use these views to look at different sets of attributes of the change objects (for example, issues, change requests, change orders) that are returned by a search. Both administrators and users can create display formats. Administrator-created display formats are available to all users. User-created display formats are available only to the users who created them. You can define as many display formats as you wish, and you can also denote the most commonly used as the default display format. You can include any primary, operational, or user-defined attributes in your display format definitions. You can include display sections in your display formats that provide links directly to the section from your search results. This enables you to display links in your search results to any user-defined change type page (such as Cost Information) or any of the standard display sections: Lines, Attachments, Action Log, Approval, Approval History, Dependencies, Revisions, and People.

Change category search criteria templates and display formats are set up and function similarly to item catalog category search criteria templates and display formats. The topics *Managing Search Criteria*, page 3-29 and *Managing Display Formats*, page 3-35 in the *Setting Up the Item Catalog* chapter describe in detail how to set up and use search criteria templates and display formats (for change objects as well as items).

## Defining Change Category Reports

Change management reports are basically search criteria and display formats that you can save, browse, e-mail, or print. You can create reports for any change management category. You can also browse a report sequentially or using a summary view. Browsing a report sequentially enables you to step through each change object in a report. A summary view displays the report in tabular column format. You can send a report to other users including registered suppliers and customers.

An administrator can create a report for the most common or frequent searches that users of the system perform. These administrator-defined reports save an ordinary user the time of selecting a change category, search criteria, and display format, and repeating the same search over and over again. You can give reports meaningful names as well. For example, a search for all open high priority issues may yield dozens of issues, each identified by issue name and number. You can name the report for these search results *High Priority Open Issues Report* as shown in the figure below. Report security is consistent with search security: you can only access a change object on which you have the required role. Also, administrator-created reports are available to all users, although only the administrator can edit these reports.

## High Priority Issue Report

### Create Report

For any duplicated attribute the search will be conducted on both attribute values.

Cancel Apply

Indicates required field

Change Category Issue

Name High Priority Open Issues Report

Description A report of current high priority Open Issues.

### Search Criteria

Search Criteria High Priority Issues Co

Select Criteria: Clear Delete Duplicate Add Criteria

Select All Select None

| Select Attribute Group           | Attribute         | Operator | Value |
|----------------------------------|-------------------|----------|-------|
| <input type="checkbox"/> Primary | ^ Number          | is       |       |
| <input type="checkbox"/> Primary | ^ Name            | is       |       |
| <input type="checkbox"/> Primary | ^ Assigned To     |          |       |
| <input type="checkbox"/> Primary | ^ Status          | is       | Open  |
| <input type="checkbox"/> Primary | ^ Approval Status | is       |       |
| <input type="checkbox"/> Primary | ^ Priority        | is       | High  |
| <input type="checkbox"/> Primary | Reason            | is       |       |

TIP ^ denotes the Attribute is indexed. You must enter value for at least one indexed attribute to perform search.

### Display Format

Display Format Assignment Information Co

| Type | Number | Name | Status | Approval Status | Priority | Assigned To | Need By Date | Action Log |
|------|--------|------|--------|-----------------|----------|-------------|--------------|------------|
|------|--------|------|--------|-----------------|----------|-------------|--------------|------------|

## To create a report:

1. In the Setup Workbench, on the **Categories** page, click the Reports tab. When the page refreshes, click Create Report.

2. When the **Create Report** page refreshes, provide the following:

### Name

Provide a name for the Report.

### Description

Optionally, provide a description of the Report.

### Criteria

Select the criteria to be used in the search. You can use an existing search criteria template (if one exists for the change category selected), or add criteria here by clicking Add Criteria.

### Format

Select a format for the report. You can use an existing display format by selecting it from the Display Format pulldown and clicking Go.

3. In the **Export Format** section, you can optionally select an export template and output format, such as .pdf , .rtf , .html , .xls , and .xml , in which you want to

generate the report.

4. Click Apply.

## Related Topics

Defining Change Category Search Criteria and Display Formats, page 7-41

Managing Search Criteria, page 3-29

Managing Display Formats, page 3-35

## Subscribing to a Change Management Business Event

Change management business events represent an activity or task that can be extended or customized. For example, creation of items and engineering change orders (ECO's) are examples of business events.

Change Management business events enable you to extend and customize your environment. You subscribe to Change Management business events to:

- Execute custom code on the event information
- Send event information to a workflow process
- Send event information to other queues or systems

Business events are represented within workflow processes by event activities. Business events can be synchronous or asynchronous. That is, when a local event occurs, the subscribing event can be executed during the same time as the transaction (synchronous), or the subscribing event can be deferred (asynchronous).

You use Oracle Workflow to set up business events.

You can subscribe to the following business events:

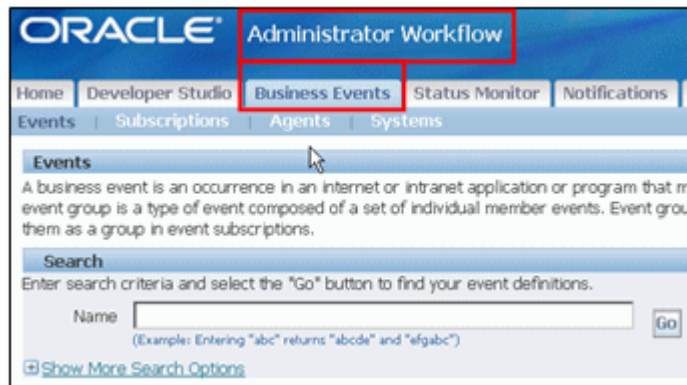
| Business Event                                               | Event Type                | Description                                                 |
|--------------------------------------------------------------|---------------------------|-------------------------------------------------------------|
| oracle.apps.eng.cm.<br>changeObject.<br>changeApprovalStatus | CM Approval Status Change | Business event for change<br>object approval status change. |
| oracle.apps.eng.cm.<br>changeObject.changePriority           | CM Priority Change        | Business event for change<br>object priority changes.       |
| oracle.apps.eng.cm.<br>changeObject.changeStatus             | CM Status Change          | Business event for change<br>object status changes.         |

| Business Event                                               | Event Type                           | Description                                               |
|--------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------|
| oracle.apps.eng.cm.<br>changeObject.<br>changeWorkflowStatus | CM Workflow Status Change            | Business event for change object Workflow Status changes. |
| oracle.apps.eng.cm.<br>changeObject.create                   | CM Create                            | Business event for the creation of a change object.       |
| oracle.apps.eng.cm.<br>changeObject.postComment              | CM Post Comment                      | Business event for a posted comment on a change object.   |
| oracle.apps.eng.cm.<br>changeObject.reassign                 | CM Reassign                          | Business event for the reassignment of a change object.   |
| oracle.apps.eng.cm.<br>changeObject.<br>requestComment       | CM Request Comment                   | Business event for a comment request of a change object.  |
| oracle.apps.eng.cm.<br>changeObject.submit                   | CM Submit                            | Business event for the submission of a change object.     |
| oracle.apps.eng.cm.<br>changeObject.update                   | CM Update                            | Business event for the update of a change object.         |
| oracle.apps.eng.cm.<br>changeOrder.<br>changeScheduleDate    | CM Schedule Date Change              | Business event for change order schedule date change.     |
| oracle.apps.eng.cm.import.<br>complete                       | CM Import Complete                   | Change Management Import is completed.                    |
| oracle.apps.eng.cm.<br>revisedItem.<br>changeScheduleDate    | CM Revised Item Schedule Date Change | Business event for revised item schedule date change.     |
| oracle.apps.eng.cm.<br>revisedItem.changeStatu               | CM Revised Item Status Change        | Business event for revised item status changes.           |

### To enable Change Management business events:

1. Log into Oracle Applications as Workflow Administrator.

2. Select the Business Events tab.

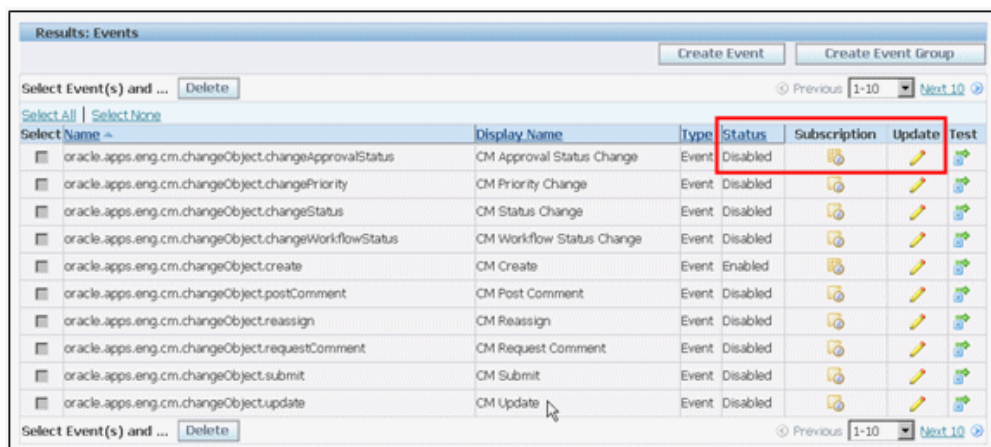


3. Search for the business event to which you want to subscribe.

**Note:** If you do not know the name of the event that you want to search for, use the following search string: Oracle.apps.eng.cm.



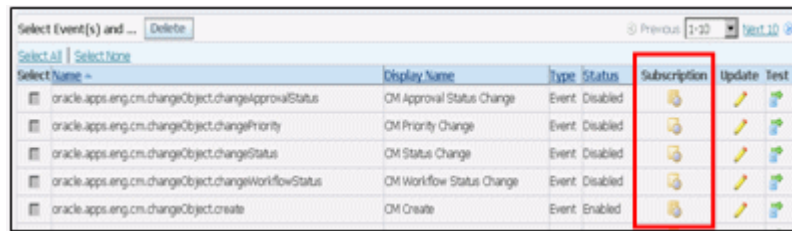
The search results include a list of change management business events. By default, all business events are disabled.



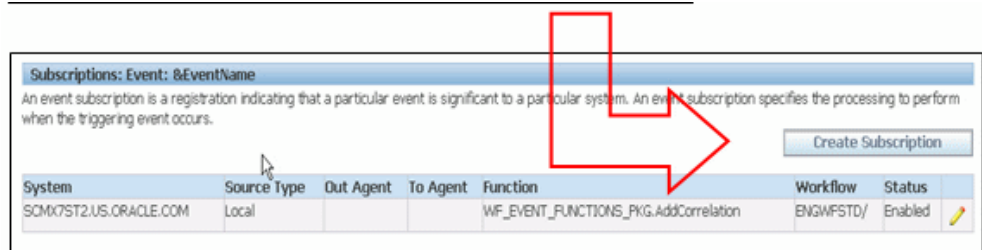
4. To enable a business event, click the Update icon, which opens a page for the selected event where you can enable the event.
5. Click the Subscription icon to define the subscription code (also known as the Event



trigger).



| Select Name                                                                   | Display Name              | Type  | Status   | Subscription | Update | Test |
|-------------------------------------------------------------------------------|---------------------------|-------|----------|--------------|--------|------|
| <input type="checkbox"/> oracle.apps.eng.cm.changeObject.changeApprovalStatus | OM Approval Status Change | Event | Disabled |              |        |      |
| <input type="checkbox"/> oracle.apps.eng.cm.changeObject.changePriority       | OM Priority Change        | Event | Disabled |              |        |      |
| <input type="checkbox"/> oracle.apps.eng.cm.changeObject.changeStatus         | OM Status Change          | Event | Disabled |              |        |      |
| <input type="checkbox"/> oracle.apps.eng.cm.changeObject.changeWorkflowStatus | OM Workflow Status Change | Event | Disabled |              |        |      |
| <input type="checkbox"/> oracle.apps.eng.cm.changeObject.create               | OM Create                 | Event | Enabled  |              |        |      |



Subscriptions: Event: &EventName

An event subscription is a registration indicating that a particular event is significant to a particular system. An event subscription specifies the processing to perform when the triggering event occurs.

Create Subscription

| System                | Source Type | Out Agent | To Agent | Function                              | Workflow  | Status  |
|-----------------------|-------------|-----------|----------|---------------------------------------|-----------|---------|
| SCM7ST2.US.ORACLE.COM | Local       |           |          | WF_EVENT_FUNCTIONS_PKG.AddCorrelation | ENGWFSTD/ | Enabled |

## Related Topics

Setting Up the Business Event System, *Oracle Workflow Administrator's Guide*

Change Management Business Events, page E-1

## Customizing Change Management Workflows

Oracle supports the extension of workflows. Extensions include using existing seeded processes to build new workflow processes and modifying parameters of an activity without changing process logic (for example, adding custom logic in an Abort Approval Routing subprocess).

If the Change Management seeded workflows do not meet your business processing needs, you can customize or extend the seeded Change Management Workflow processes by creating your own processes, modifying the seeded ones, adding a new activity to a seeded process, and/or modifying custom hook PL/SQL procedures.

## Suggestions for Customizing Workflows

- If you decide to create a new workflow process for Priority Change rather than modifying the seeded workflow, you must write your own custom logic to start your custom workflow in the custom hook called in Start Workflow.
- If you add a new workflow process in the Change Approval Routing Step (ENGSTEP), then the process will be enabled as an Approval Routing Step Workflow in the Approval Routing pages. The workflow process is automatically

called from the Approval Routing workflow.

- The activity will be shared by more than one object's process, so create the activity in the ENG: Standard (ENGWFSTD) item type. Otherwise, create the activity in the appropriate object's item type.
- Instead of modifying workflow notifications, you should create new messages. Oracle Workflow Builder does not maintain version information for objects such as item types, item type attributes, messages, and lookup types. For these objects, the latest definition always applies, so you must consider whether a change to any of these objects is backward compatible. If the modification affects existing processes, you should create a new object rather than edit the existing object.

**Important:** Oracle provides support only for its seeded activities, processes, and the types of extensions described in this guide. Oracle does not provide support for your custom activities and processes.

## Related Topics

Workflows for Change Management, page A-1

## Implementing User Defined Functions for Change Type Attributes

You can define your own custom logic by adding user-defined functions and actions to existing change pages. In such cases you do not need to customize the entire page.

You can register user-defined functions for executing customer-specific business rules and calculations. These functions can be written in Java or PL/SQL. URL functions can also pass the values of certain parameters to the URL string and redirect users to a secure page. For each function, you need to register the list of parameters, their data types, and how they get their values by specifying parameter types.

Actions are trigger points for functions and can be displayed as buttons or links. You can determine the conditional visibility of the button/links and also prompt the user based on his or her input.

### **Example: Implement User-Defined Functions for Change Type Attributes**

The following example shows how you would calculate Total Cost by using the attribute group Implementation Cost.

| Sequence | Attribute Group: Implement Cost | Attribute Group: Implement Cost | Attribute Group: Implement Cost | PL/SQL Function: Calculate Cost | PL/SQL Function: Calculate Cost | PL/SQL Function: Calculate Cost |
|----------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
| Sequence | Attribute Name                  | Data Type                       | Mapping Attribute & Parameters  | Parameter Name                  | Data Type                       | Parameter Type                  |
| 1        | Suppliers Charges               | Number                          | --a` ---a` ---a`                | Cost 1                          | Number                          | Input                           |
| 2        | Manufacturing                   | Number                          | --a` ---a` ---a`                | Cost 2                          | Number                          | Input                           |
| 3        | Engineering                     | Number                          | --a` ---a` ---a`                | Cost 3                          | Number                          | Input                           |
| 4        | Field/Repair                    | Number                          | --a` ---a` ---a`                | Cost 4                          | Number                          | Input                           |
| 5        | Total Implement Cost            | Number                          | --a` ---a` ---a`                | Cost Result                     | Number                          | Return Value                    |

1. Register the PL/SQL function "Calculate Cost" (based on the PL/SQL procedure that resides in the PL/SQL package) with all required parameters.
2. Navigate to the "ECO" change order type and add the action to the attribute group "Implement Cost" by clicking Update Action. The action executes the function on the change pages. After creating the action, map the function parameters to the corresponding attributes.

The mapping section on the **Action Detail** page provides the mapping information for function parameters. You can also map parameters of functions to the object's primary key value (for example, CHANGE\_ID for change objects).

#### When setting up an action:

1. Specify a role-based privilege to secure the action.  
See: Creating Custom Privileges, page 11-27.
2. Add the Dynamic Visibility Function (written in Java only) to your action if there is a certain condition that needs to be satisfied before users can view the action button

or link. For example, if one or all of the fields are empty, then the user cannot see the button/link to execute the action.

- 3. Add the Dynamic Prompt Function (written in Java only) to your action if the label for the button/link needs to be changed depending on a certain condition. For example, if the Total Implement Cost attribute is empty, then the button label is Apply, otherwise the button label is Apply Changes.

**Important:** Dynamic Prompt and Dynamic Visibility function parameters must be mapped to the corresponding attributes as well.

**Associate action with attribute group to execute a function**

Action Details for Calculate Cost

Basic Information

Object Name: Change  
Classification: ECO  
Attribute Group: Implementation Costs  
Sequence: 1  
Action Name: Calculate Cost  
Description: Calculate Cost  
Function: Calculate Cost  
Security Privilege:  
Execution Method: User Action  
Display Style: Button  
Prompt Application: Oracle Engineering Online  
Prompt Message Name: Calculate Cost  
Visible: Yes  
Dynamic Prompt Function:  
Dynamic Visibility Function:

Mappings for Function Calculate Cost

| Parameter Name | Mapped To       | Mapped Attribute     |
|----------------|-----------------|----------------------|
| Cost 1         | Attribute Group | Suppliers Charges    |
| Cost 2         | Attribute Group | Manufacturing        |
| Cost 3         | Attribute Group | Engineering          |
| Cost 4         | Attribute Group | Field / Repair       |
| Cost Result    | Attribute Group | Total Implement Cost |

**Change type attribute page displays action button**

Implementation Costs

Indicates required field

Material: 137.5 Dollars  
Suppliers Charges: 175 Dollars  
Manufacturing: 150.85 Dollars  
Engineering: 120.87 Dollars  
Field / Repair: 250.45 Dollars  
Quality Assurance: 160 Dollars  
Total Implement Cost: 994.47 Dollars  
Result

Calculate Cost

Cancel Apply

**Related Topics**

Creating User-Defined Functions for Item Attributes, page 4-38

# Setting Up Item Structures

This chapter covers the following topics:

- Overview of Item Structures
- Defining Structure Types
- Associating a Structure Name with a Structure Type
- Defining Structure Attributes
- Defining Component Attributes
- Defining Display Formats and Search Criteria for Structures
- Creating an Item Catalog Category Structure

## Overview of Item Structures

You can define item structures in the Product Workbench (see: *Creating Structures, Oracle Product Hub User’s Guide*) or in Oracle Bills of Material (see: *Creating a Bill of Material, Oracle Bills of Material User’s Guide*). However, before defining structures, you must complete the following tasks:

| Task                                                                                      | Required? |
|-------------------------------------------------------------------------------------------|-----------|
| Create structure types. See: Defining Structure Types, page 8-2                           | Yes       |
| Create structure names. See: Associating a Structure Name with a Structure Type, page 8-4 | Yes       |

| Task                                                                                                                                   | Required? |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Associate structure names with structure types. See: Associating a Structure Name with a Structure Type, page 8-4                      | Yes       |
| Define structure attributes. See: Defining Structure Attributes, page 8-6                                                              | No        |
| Define component attributes. Defining Component Attributes, page 8-8.                                                                  | No        |
| Define search criteria templates for a structure type. See: Defining Display Formats and Search Criteria for Structures, page 8-9      | No        |
| Define search results display formats for a structure type. See: Defining Display Formats and Search Criteria for Structures, page 8-9 | No        |
| Create an item catalog category structure. See: Creating an Item Catalog Category Structure, page 8-6                                  | No        |

## Related Topics

Product Workbench Overview, *Oracle Product Hub User's Guide*

Searching for Items in a Structure, *Oracle Product Hub User's Guide*

## Defining Structure Types

A structure type enables you to classify and characterize item structures throughout the lifecycle of an item. For example, you can have structure types for Engineering, Manufacturing, and Service. You can define an unlimited number of structure names for a structure type, enabling you to create several different structures of the same type that you can use for comparison. For example, you can create several different engineering BOMs (EBOM1, EBOM2, EBOM3) using the structure type Engineering BOM with different component costs and BOM grading attributes. Comparing these BOMs enables you to make better decisions earlier in the lifecycle of your products.

You can associate item attachment categories, search criteria templates and search results formats with a structure type. Structure types are hierarchical, with structure names, item attachments categories, search criteria templates and search results formats inherited throughout the hierarchy.

### To create a structure type:

To help classify your existing structures, you can use the following seeded structure types, or create your own structure types.

| Internal Name       | Display Name      | Allow Subtypes |
|---------------------|-------------------|----------------|
| All-Structure Types | Root              | Y              |
| Packaging Hierarchy | Packaging BOM     | N              |
| Asset BOM           | Asset BOM         | Y              |
| Cad BOM             | Cad BOM           | Y              |
| Design Structure    | Design Structure  | Y              |
| MBOM                | Manufacturing BOM | Y              |

**Important:** When you upgrade to a new software release, ensure that you use the newest seeded Packaging Hierarchy instead of an older version. Either delete or rename any older versions of a Packaging Hierarchy structure type.

1. On the **Search: Item Catalog Categories** page in the Setup Workbench, click the Structures tab.
2. On the **Search: Structure Types** page, click the "Create Structure Type" link.
3. On the **Create Structure Type** page, enter the following information:

**Internal Name**

The structure type's internal name.

**Display Name**

The structure type's display name.

**Description**

A brief description of the structure type.

**Parent Structure Type**

Select a parent structure type. The parent structure type specifies the immediate parent of the structure type being created, thereby determining the structure type

hierarchy.

If you leave this field blank, the parent structure type displays as Root. If you use a root structure type as a parent, then the following information defaults to the descendants of the parent:

- Structure Header Attributes
- Item Attachment Categories
- Display Formats
- Import Formats
- Search Criteria
- Component Attribute Groups

See: Viewing Item Structures, *Oracle Product Hub User's Guide*

#### **Inactive On**

Optionally, you can specify a date on which the structure type will become inactive. Making a type inactive has the following implications:

- You cannot create structures of that type or any of its descendants.
- You cannot use that type or any of its descendants as a parent structure type upon creation of a structure type.

#### **Allow Subtypes**

Select Allow Subtypes if you want this structure type to be a parent structure type or a member of a structure type hierarchy.

4. Click Apply.

## **Related Topics**

Creating a Bill of Material , *Oracle Bills of Material User's Guide*

Managing the Packaging Hierarchy, *Oracle Product Hub User's Guide*

## **Associating a Structure Name with a Structure Type**

You can define multiple structure names for each structure type and enable them in different organizations. For example, you can have three engineering BOMs (EBOM1, EBOM2, EBOM3) that use the structure type Engineering BOM; you can use these to compare designs. Manufacturing BOMs might be slightly different in each global region, because similar parts are procured from different suppliers. You can have a



structure type Manufacturing BOM with different structure names (MBOM\_US, MBOM\_EMEA, MBOM\_APAC) for each of your manufacturing regions around the globe. Structures within a structure type share the same common characteristics.

**To create a structure name:**

1. On the **Search: Item Catalog Categories** page in the Setup Workbench, click the Structures tab.
2. On the **Search: Structure Types** page, click the Structure Names tab.
3. On the **Search: Structure Names** page, click the "Create Structure Names" link.
4. On the **Create Structure Name** page, provide the following information:

**Display Name**

The display name of the structure.

**Description**

The description of the structure name.

**Structure Type**

The associated structure type.

**Preferred for Structure Type**

Specifies this structure name to be the default name for all structures created within this structure type.

**Organization Assignment**

A structure name can be assigned to one or more organizations with different inactive dates. This enables you to use the structure name within the assigned organizations.

**Inactive Date**

The date on which the organization assignment is no longer valid. One or more organizations may have different inactive dates.

5. Click Apply.

**To associate a structure name with a structure type:**

1. On the **Search: Item Catalog Categories** page in the Setup Workbench, click the Structures tab.
2. On the **Search: Structure Types** page, find the structure type you wish to associate and click its corresponding name link.

3. On the **Basic Information** page, click the "Structure Names" link.
4. On the **Structure Names** page, click Create.
5. On the **Create Structure Name** page, enter the following information:
  - Internal Name
  - Display Name
  - Description
  - Preferred for Structure Type

Select this box if you want this structure name to be the default name for all structures created within this structure type.
6. On the **Create Structure Name** page, assign the structure name to one or more organizations by selecting the Assign box in an organization row(s). This enables you to use the structure name within the assigned organizations.

Optionally, click Assign All to assign the structure name to all organizations.
7. For each assigned organization, you can optionally specify an Inactive On date. This makes the structure name no longer valid as of the Inactive On date.
8. Click Apply to create and save the structure name.

Optionally, click Add Another to create another new structure name.

## Related Topics

Defining Structure Types, page 8-2

## Defining Structure Attributes

Structure attributes capture user-defined information about a structure. Structure attributes are defined similarly to item attributes. First, define a structure attribute group, then the structure attributes within the group. Instead of associating the structure attribute group with an item catalog category as you would for item attributes, associate the structure attribute group with your structure type. Next, create structure pages (similar to item pages) and associate the structure attribute group with the page and the page with the structure type.

### To define a structure attribute group and its attributes

1. On the **Search: Structure types** page, click the Attributes link.

2. On the **Search: Attribute Groups** page, click Create.
3. On the **Create Attribute Group for Bom Structure User Attributes** page, create an attribute group that can apply to a particular structure name.

The fields on this page function the same as they do for item user defined attribute groups. For information about each field, refer to *Defining Item Attributes and Attribute Groups*, page 4-21.
4. Click Apply and Add Attributes.
5. In the **Create Attribute** page, create an attribute for the structure attribute group.

The fields on this page function the same as they do for item user defined attributes. For information about each field, refer to *Defining Item Attributes and Attribute Groups*, page 4-21.
6. Click Apply if you are finished creating attributes or click Apply and Add Another to continue adding attributes to the structure attribute group.

If you click Apply, then you are returned to the **Attribute Group Details** page.

#### **To associate a structure attribute group with a structure type:**

1. On the **Attribute Group Details** page, click the Structure Types link.
2. On the **Search: Structure Types** page, select a structure type.
3. On the **Basic Information** page, click the Structure Header Attribute Groups link.
4. On the **Attribute Groups** page, click Add Attribute Groups.
5. on the **Add Attribute Groups** page, search for and select the attribute groups to add to the structure type.
6. Click Apply.

You are returned to the **Attribute Groups** page.

#### **To create a structure attribute page:**

Create a structure page that enables you to define attribute values for a structure.

1. On the **Attribute Groups** page, click the Structure Pages link.
2. On the **Structure Pages** page, click Create Page.
3. On the **Create/Edit Structure Pages** page, enter values in the following required fields:

- Display Name - the display name of the structure attribute page.
  - Internal Name - the internal name of the structure attribute page.
  - Sequence - the order in which this page is listed compared to other pages.
4. In the Attribute Groups region of the page, click Add Another Row.
  5. In the Sequence field, enter the order in which this attribute group appears on the structure attribute page.
  6. In the Display Name field, search for and select an attribute group.
  7. Once you have added all of the structure attribute groups you want to appear on the page, click Apply.

## Related Topics

Defining Structure Types, page 8-2

## Defining Component Attributes

You can define user defined attributes for components in a structure to capture additional information at the component level on a BOM. Component attributes are defined similarly to item attributes. First, define a component attribute group, then the component attributes within the group. Instead of associating the component attribute group with an item catalog category as you would for item attributes, associate the component attribute group with your structure type. You do not need to create a page to display component attribute values; component attribute values are only viewable and maintainable from the Product Workbench and the Product Workbench does not use pages.

**Note:** Child structure types can inherit component attribute groups from a parent structure type. You cannot delete component attribute groups inherited from a parent structure type at the child structure type level.

### To define a component attribute group and its attributes:

1. On the **Search: Structure Types** page, click Attributes.
2. On the **Search: Attribute Groups** page, click Component Attribute Groups.
3. Click Create to create a new component attribute group.

4. On the **Create Attribute for BOM Component User Attributes** page, create an attribute group that can apply to a particular structure.

The fields on this page function the same as they do for item user defined attribute groups. For information about each field, refer to *Defining Item Attributes and Attribute Groups*, page 4-21.

5. Click **Apply** and **Add Attributes**.
6. In the **Create Attribute** page, create an attribute for the structure attribute group.  
The fields on this page function the same as they do for item user defined attributes. For information about each field, refer to *Defining Item Attributes and Attribute Groups*, page 4-21.
7. Click **Apply** if you are finished creating attributes or click **Apply** and **Add Another** to continue adding attributes to the structure attribute group.

If you click **Apply**, then you are returned to the **Attribute Group Details** page.

#### **To associate a component attribute group with a structure type:**

1. On the **Attribute Group Details** page, click the **Structure Types** link.
2. On the **Search: Structure Types** page, select a structure type.
3. On the **Basic Information** page, click the **Component Attribute Groups** link.
4. On the **Component Attribute Groups** page, click **Add Attribute Groups**.
5. on the **Add Component Attribute Groups** page, search for and select the component attribute groups to add to the structure type.
6. Click **Apply**.

You are returned to the **Component Attribute Groups** page.

The page now shows the selected component attribute groups associated with the structure type.

## **Related Topics**

*Defining Structure Types*, page 8-2

## **Defining Display Formats and Search Criteria for Structures**

### **Search Criteria**

You can search for structure names based on different search criteria such as display

name, internal name, description or parent structure type. Criteria templates enable you to save frequently performed search criteria based on a list of attributes. Criteria templates for structures enable you to search components within a structure. You can associate with a structure type an existing criteria template for a given item catalog category, or create a new criteria template to search items within a structure. You can specify a default criteria template for each structure type.

Administrators and users can define search criteria for a given structure type. All search criteria defined by administrators are available to all users viewing a structure of the given structure type to which the search criteria is associated. All search criteria defined by a user are only available to that user. On selecting a search criteria, only those components that satisfy the criteria will be listed in the hierarchy and the components that do not match the criteria will be filtered out. Intermediate nodes will be listed even though they might not satisfy the criteria to maintain the context for a lower level component. Primarily search criteria are useful to filter only relevant components in a bill having a large number of components. Search criteria templates can be built using item and component primary, operational, and user defined attributes.

For instructions on how to create and maintain search criteria for structures, see: [Managing Search Criteria](#), page 3-29.

### Display Formats

Administrators and users can define display formats for a given structure type. All display formats defined by administrators are available to all users viewing a structure of the given structure type to which the display format is associated. User display formats are only available to that user. Display formats enable users to view attribute information across an entire structure. The display formats can include item and component primary, operational, and user defined attributes. You can define as many results formats as you wish, and you can also denote the most commonly used as the default results format.

For instructions on how to create and maintain display formats for structures, see: [Managing Display Formats](#), page 3-35.

Oracle provides the following system defined display formats:

| Display Format Name   | Fields Included in the Display Format                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| Component Information | Component Name, Description, Revision, Category                                                                      |
| Component Details     | Component Name, Description, Revision, Item Sequence, Operation Sequence, UOM, Quantity, Planning%, Yield, Item type |
| Change Order          | Component Name, Description, Revision, Implemented, Change Order                                                     |

---

|                       |                                                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Comments              | Component Name, Description, Revision, Comments                                                                                            |
| Effectivity           | Component Name, Description, Revision, Effectivity Control, Effective From, Effective To, Change Order                                     |
| Lifecycle Status      | Component Name, Description, Revision, Lifecycle Phase, Approval Status, Item Status, Percent Complete, Schedule End Date, Progress Status |
| Long Description      | Component Name, Description, Revision, Long Description                                                                                    |
| Material Control      | Component Name, Description, Revision, Supply Type, Subinventory, Locator, Auto Request Material, Enforce Integer Requirements             |
| Order Management      | Component Name, Description, Revision, Check ATP, Optional, Mutually Exclusive, Minimum Quantity, Maximum Quantity, Basis                  |
| Reference Designators | Component Name, Description, Revision, Quantity, Quantity Related, Reference Designator                                                    |
| Shipping              | Component Name, Description, Revision, Shippable, Include on Shipping documents, Required to Ship, Required for Revenue                    |

---

## Creating an Item Catalog Category Structure

You can define a default item catalog category (ICC) structure for use by items created within the ICC. To use the default structure, you must define items as follows:

- Engineering box checked
- Revision effectivity enabled

When creating a structure for the item, use the same structure type and structure name as the ICC structure. The ICC structure components then automatically default to the item's new structure.

Child ICC structures can inherit components from parent ICC structures as long as a valid parent ICC structure version falls within the child ICC version's effective date range.

## Prerequisites

- ☐ Enable the use of item catalog category versions by setting the profile option "Enable PIM for Telco Features" to Yes.
- ☐ Create an item catalog category draft version. See: Defining Item Catalog Categories, page 3-2.

### To create an item catalog category structure:

1. In the **Item Catalog Categories** page, search for and select an item catalog category (ICC).

**Additional Information:** You can only associate a structure to a root (parent) item catalog category.

2. In the **Basic Information** page, click Draft.
3. In the **Transaction Attributes** page, click Structure.
4. In the **Structure** page, select a structure type and structure name.

Optionally, enter a structure description.

**Note:** You cannot select a structure type and structure name if the parent ICC has a structure. The child ICC inherits the parent's structure type and structure name along with the parent ICC structure components. However, you can add components to the inherited structure at the child ICC level.

5. Click Create Structure.

The Structure page refreshes with the structure header information and the First Level Components region.

6. Click Update to add components to the structure.

In the **Update Structure** page, you can update the structure header information and add components.

**Additional Information:** Any inherited components display, along



with the name of the category from which they were inherited. If a parent ICC has a defined structure header, then the structure header information (structure type, structure name) defaults the child ICC structure.

7. Click Search and Add Components.

You can perform a simple search in the **Search and Add Components: Simple Search** page or click Advanced Search.

Alternatively, you can click Add Another Row.

8. In the **Search and Add Components: Search Results** page, select the items that you want to add as components to the structure.

9. Click Apply.

This returns you to the **Update Structure: (structure name)** page.

10. In the **Update Structure: (structure name)** page, change the quantity of each component if desired.

11. Click Apply.

#### **To remove a component from the structure**

12. Optionally, in the **Update Structure** page, select the component to remove, then click Remove.

**Note:** You cannot remove components inherited from a parent ICC structure.

13. Click Apply.

#### **To add attribute values to the structure components**

14. In the **Structure** page, you can add attributes to any component. Click the Additional Attributes icon for a particular component.

15. In the **View Additional Attributes: (component)** page, select the attribute group that contains the attributes you want to define.

**Note:** You cannot update attributes for inherited components.

16. Click Update.

In the **Update Additional Attributes: (component)** page, enter the attribute values.

17. Click Apply.

This returns you to the **View Additional Attributes: (component)** page.

You can add attribute values to as many attributes and attribute groups as necessary. Once you define all of the component's attributes, click Return to Structure Page.

Once you have finished defining the item catalog category's structure and transaction attributes, release the ICC to create a new version. See: "To release an item catalog category draft version" in Defining Item Catalog Categories, page 3-2.

### **To update an item catalog category structure:**

To update an item catalog category structure, you must lock the Draft version of the ICC.

1. In the item catalog category's **Basic Information** page, click Lock.  
If only the Unlock button is visible, then the ICC is already locked.
2. Click Draft.
3. In the **Transaction Attributes** page, click Structure.
4. In the **Structure** page, click Update.

**Additional Information:** You cannot update inherited components or the structure type and structure name if they are inherited. If you change a structure type or structure name at the root ICC level, a warning appears, saying that this change will cause the deletion of any associated structure components and attribute values. If you accept the change, you can add new components and attribute values.

### **To update the components within a structure**

5. In the **Update Structure: (ICC Name)** page:
  - Add or remove components as described above in "To create an item catalog category structure".
  - Change the quantity of components as necessary.
  - Click Apply.

### **To validate the structure**

When updating the components within a parent ICC's structure, it is possible to create duplicate components further down the hierarchy. You can use the Validate Structure

button to warn you when this problem occurs. The following example explains how the duplicate components problem can occur.

### **Duplicate Components Example**

In this example, you create the following initial ICC structures.

#### **Parent (Root) ICC Structure**

- Component 1
- Component 2

#### **Child ICC Structure**

- Component 1 (inherited from Parent ICC Structure)
- Component 2 (inherited from Parent ICC Structure)
- Component 3

Later, you decide that Component 3 must exist at the Parent ICC structure level. When you add Component 3 to the Parent ICC structure and release this draft ICC version, you receive a warning that, by adding Component 3, Component 3 becomes a duplicate component in the Child ICC structure.

The draft version of the Child ICC structure now contains a duplicate component. You cannot release a draft version in this state. You have three choices:

1. Release a new version of the Parent ICC with Component 3 removed from the ICC structure.
2. Remove the initial Component 3 added at the Child ICC structure level.
3. Change the effective version date of either the Parent ICC or Child ICC so Component 3 is no longer inherited in this version of the Child ICC.

Users can avoid duplicate components by using the Validate Structure button before releasing an updated draft ICC.

6. After updating the components within a structure, in the **Structure** page, click Validate Structure.

The **Structure Validation Results** page opens. It displays any duplicate components in relevant versions of other item catalog categories within the hierarchy.

You can release the parent ICC version even though it causes duplicate components in child ICC structures, however you cannot release any draft child ICC versions that contain duplicate components.

### **To update structure component attributes**

7. In the **Structure** page, you can change the attributes of any component. Click the

Additional Attributes icon for a particular component.

8. In the **View Additional Attributes: (component)** page, select the attribute group that contains the attributes you want to change.
9. Click Update.  
In the **Update Additional Attributes: (component)** page, enter the new attribute values.
10. Click Apply.

### To update the structure header:

You can change the structure type and structure name of an ICC structure, but doing so deletes all components and their attributes.

1. In the **Update Structure: (ICC Name)** page, select a different structure type or structure name.

The following warning message appears:

Changing the structure type and/or the structure name will delete all components and their attributes. Do you still want to continue?

If you click Yes, then the **Update Structure: (ICC Name)** page appears again, with all components removed.

2. Select the new structure type and structure name you want to use for the updated ICC structure.
3. Add components and attributes as described previously in "To create an item catalog category structure:".

## Related Topics

To associate an ICC structure with an item, *Managing Item Structures, Oracle Product Hub User's Guide*

---

# Setting Up Content Management Repositories

This chapter covers the following topics:

- Integrating Oracle E-Business Suite Applications with Content Management Repositories
- Setting Up a Content Management Repository

## Integrating Oracle E-Business Suite Applications with Content Management Repositories

You can attach unstructured data to many different business objects or entities in Oracle E-Business Suite (EBS). For example, attach a product data sheet in PDF format (the unstructured data) to an item (the business object). The following table lists other examples of objects and unstructured data:

| Examples of Objects                                     | Types of Unstructured Data                                                            |
|---------------------------------------------------------|---------------------------------------------------------------------------------------|
| Change objects such as a change order or change request | Files such as word processing documents, spreadsheets, videos, images, and Web pages. |
| Items                                                   | Folders containing related item information such as material safety data sheets.      |
| Structures                                              | URL                                                                                   |

Store unstructured data in one or more content management repositories and set up EBS to integrate with the repositories. Integrate EBS with a repository using one of the following technologies:

- WebDAV (Web-based Distributed Authoring and Versioning) - a set of extensions to the HTTP protocol, enabling users to collaboratively manage files on remote Web servers. For more information, refer to [www.webdav.org](http://www.webdav.org).

The following content management repositories are certified for use in three EBS applications - Oracle Projects, Oracle Internal Controls Manager, and Oracle Product Hub:

- Oracle Files, releases 9.0.3 and 9.04, using WebDAV, is certified for use with EBS release 11i10
- Oracle Content Services, release 10.1.2, using WebDAV, is certified for use with EBS release 11i10
- Oracle Universal Content Management (UCM), using WebDAV, is certified for use with EBS release 12 and higher.

Theoretically, EBS integrates with any WebDAV-enabled, non-Oracle content management repository (for example, Documentum), but it is only certified with the Oracle content management repositories for the Oracle applications listed above.

Integrating Oracle Universal Content Management with EBS using WebDAV gives you the most robust set of content management features. The following table lists examples of features available when using UCM

| Feature                                                                                                          |
|------------------------------------------------------------------------------------------------------------------|
| Upload a new file to a folder in the content repository and attach the file to an EBS object.                    |
| Attach an existing file/folder in the content repository to an EBS object.                                       |
| Render content of an attached content repository file from EBS.                                                  |
| Detach files/folders from an EBS object.                                                                         |
| Use the Where Attached feature to find all EBS objects to which a content repository file/folder is attached to. |
| Browse content repository folders within the EBS.                                                                |
| Single sign-on integration between the EBS and content repository.                                               |
| Attach a specific version of a file in the content repository to an EBS object.                                  |
| Submit a review/approval from the EBS for specific versions of files in the content repository.                  |

## Related Topics

Setting Up a Content Management Repository, page 9-3

E-Business Suite Attachments chapter, *Oracle Product Hub User's Guide*

## Setting Up a Content Management Repository

Oracle recommends using Oracle Universal Content Management (UCM) as the repository to store files. UCM can integrate with EBS using WebDAV.

The following instructions also apply to setting up repositories other than UCM, although Oracle only certifies the use of the Oracle content management repositories for the Oracle applications listed in Integrating Oracle E-Business Suite Applications with Content Management Repositories, page 9-1.

**Caution:** Consider restricting access to the Repository Setup page. By default, any user assigned a seeded responsibility has access to this page and can register any URL as an external content repository . To restrict access, see: Creating Responsibilities, page 2-1.

## Prerequisites

- ☐ Set up attachment categories. See the Attachments chapter in the *Oracle E-Business Suite Developer's Guide*.

### To set up the repository:

1. Navigate to the **Repository Setup** page.

Repository Setup

Select Object:

Test Connection

Remove

|

Update

| Select                           | Short Name | Name                       | Description                | Integration Mode | Service URL                                                                                                 | WebDAV Connection URL                                                                                           |
|----------------------------------|------------|----------------------------|----------------------------|------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <input type="radio"/>            | EBS        | E-Business Suite           |                            |                  |                                                                                                             |                                                                                                                 |
| <input type="radio"/>            | OFO        | OFO                        | OFO                        | WebDAV           | <a href="http://files.oraclecorp.com/app">http://files.oraclecorp.com/app</a>                               | <a href="http://files.oraclecorp.com/content">http://files.oraclecorp.com/content</a>                           |
| <input type="radio"/>            | OFNA       | Oracle Files North America | Oracle Files North America | WebDAV           | <a href="http://tdslnx531.oracleads.com:7778/files/app">http://tdslnx531.oracleads.com:7778/files/app</a>   | <a href="http://tdslnx531.oracleads.com:7778/files/app">http://tdslnx531.oracleads.com:7778/files/app</a>       |
| <input type="radio"/>            | ADS10G     | ADS OCS10G Repository      | ADS OCS10G Repository      | WebDAV           | <a href="http://sa0040.oracleads.com/content">http://sa0040.oracleads.com/content</a>                       | <a href="http://sa0040.oracleads.com/content">http://sa0040.oracleads.com/content</a>                           |
| <input checked="" type="radio"/> | CDB        | CDB                        | CDB                        | WebService       | <a href="http://ap6001tch.us.oracle.com:7780/content/ws">http://ap6001tch.us.oracle.com:7780/content/ws</a> | <a href="https://ap6001tch.us.oracle.com:8000/content/dav">https://ap6001tch.us.oracle.com:8000/content/dav</a> |

2. Click on Update to add a repository

3. Enter data in the following fields:

- Short Name - the short name of the repository.
- Name - the descriptive name of the repository.
- Description - optional. The description of the repository.
- Integration Mode - select either WebDAV or Web Service.
- Service URL - URL pointing to the Web Service servlet of the Repository instance.

For example: [http\(s\)://<host>:<port>/content/ws](http(s)://<host>:<port>/content/ws) See the content repository installation guide (for example, Oracle Content Database Installation Guide) for the correct URL.

- WebDAV Connection URL - URL pointing to the DAV servlet of the Content Repository instance.

For example: [http\(s\)://<host>:<port>/content/dav](http(s)://<host>:<port>/content/dav) or [http\(s\)://<host>:<port>/files](http(s)://<host>:<port>/files). See the content repository installation guide (for example, Oracle Content Database Installation Guide) for the correct URL.

- Sequence - If you have more than one repository, the sequence determines the order in which the repositories display during user selection.
- Certificate path - If the repository is SSL enabled, then this field contains the location to the repository digital certificate stored in the EBS mid-tier. The certificate must be accessible and readable.



**Update Repositories**  
\* Indicates required field

| *Short Name | *Name                      | Description                | *Integration Mode | *Service URL          | *WebDAV Connection URL |
|-------------|----------------------------|----------------------------|-------------------|-----------------------|------------------------|
| EBS         | E-Business Suite           | E-Business Suite           |                   |                       |                        |
| OFNA        | Oracle Files North America | Oracle Files North America | WebDAV            | http://tdslhx531.orac | http://tdslhx531.orac  |
| ADS10G      | ADS OCS10G Repository      | ADS OCS10G Repository      | WebDAV            |                       |                        |
| CDBWS       | CDBWS                      | CDBWS                      | WebService        |                       |                        |
| WebDAV      | WebDAV                     | WebDAV                     | WebDAV            | https://contentub.us  | https://contentub.us   |
| UCM         | UCM                        | UCM                        | WebDAV            | http://ucmserver:port | http://ucmserver:port  |

**Enter the connection details for the UCM server**

[Add Another Row](#)

4. Click Apply.

### To test the repository connection:

1. In the **Repository Setup** page, select a repository and click Test Connection.
2. In the **Repository Confirmation** page, enter the user ID and password for the repository and click Test.

If the user ID and password for the repository are correct and the repository is operational, a message appears notifying you that the connection with the repository is successful.

### To remove a repository:

Before removing a repository, you must first detach all of the attachments stored in the repository from EBS objects.

1. In the **Repository Setup** page, select the repository to remove.
2. Click Remove.

This deletes the repository definition from the EBS. The installed repository instance remains intact.

## Related Topics

Integrating Oracle E-Business Suite Applications with Content Management Repositories, page 9-1

E-Business Suite Attachments, *Oracle Product Hub User's Guide*

My Oracle Support (<https://support.oracle.com>), Document ID 1061947.1



---

# Setting Up E-Business Suite Attachments

## Overview of E-Business Suite Attachments

Oracle Product Hub provides integration to Oracle Files, Oracle Content Services, and Oracle Universal Content Management (UCM). These content repositories provide users with a wide range of file management capabilities. UCM, however, provides the most flexibility for organizing unstructured data and Oracle recommends using UCM over other repositories. The content of this chapter explains how to set up E-Business Suite attachments with UCM. Some of the set up steps, features, and processes do not apply to other repositories.

## Structuring Folders

Oracle Universal Content Management (UCM) provides a great deal of flexibility in the structuring of data. The following security considerations should factor in to implementation decisions about the organization of unstructured data (files/folders):

- Examine what content is internal to the enterprise and what may be shared externally as part of the collaboration process or on a website.
- Access privileges by organizations modeled as Organizations in EBS.
- Access controlled by item catalog categories or types of items.
- Access controlled by business objects in EBS such as Items, Change Management, and Purchase Orders.
- Access controlled by the lifecycle state of the item.

Following are some examples for structuring folders/libraries.

**Example 1: Model Libraries as Item Catalog Categories**

|                           |            |                       |
|---------------------------|------------|-----------------------|
| /Engines                  | Library    | Item Catalog Category |
| /M10000                   | Folder     | Item                  |
| /M10000 Rev 1.0           | Sub Folder | Item Revision         |
| /Functional Specification | Sub Folder | Document Type         |
| / CAD View                | Sub Folder | Document Type         |
| /M10000 Rev 2.0           | Sub Folder | Item Revision         |
| /M10000 CO                | Folder     | Item specific CR/CO   |
| /Manufacturer 1           | Library    | Manufacturer Name1    |
| /Manufacturer 2           | Library    | Manufacturer Name2    |
| /Change Management        | Library    | Generic CR/CO         |
| /Intranet                 | Library    | Public to Enterprise  |
| /Internet                 | Library    | Public to www         |

**Example 2: Model Libraries as Organizations or Divisions in a Company**

|                           |            |                      |
|---------------------------|------------|----------------------|
| /Org 1                    | Library    | Organization 1       |
| /M10000                   | Folder     | Item                 |
| /M10000 Rev 1.0           | Sub Folder | Item Revision        |
| /Functional Specification | Sub Folder | Document Type        |
| / CADView                 | Sub Folder | Document Type        |
| /M10000 Rev 2.0           | Sub Folder | Item Revision        |
| /M10000 CO                | Sub Folder | Item specific CR/CO  |
| /Org 2                    | Library    | Organization 2       |
| /Manufacturer 1           | I Library  | Manufacturer Name1   |
| /Manufacturer 2           | Library    | Manufacturer Name2   |
| /Change Management        | Library    | Generic CR/CO        |
| /Intranet                 | Library    | Public to Enterprise |
| /Internet                 | Library    | Public to www        |

### Example 3: Model Libraries as Business Objects in EBS

|                           |            |                       |
|---------------------------|------------|-----------------------|
| /Items                    | Library    | Business Object       |
| /Catalog Group            | Folder     | Item Catalog Category |
| /M10000                   | Folder     | Item                  |
| /M10000 Rev 1.0           | Sub Folder | Item Revision         |
| /Functional Specification | Sub Folder | Document Type         |
| / CADView                 | Sub Folder | Document Type         |
| /M10000 Rev 2.0           | Sub Folder | Item Revision         |
| /M10000 CO                | Sub Folder | Item specific CR/CO   |
| /Change Management        | Library    | Generic CR/CO         |
| /Project Management       | Library    | Business Object       |
| /Manufacturer 1           | Library    | Manufacturer Name1    |
| /Manufacturer 2           | Library    | Manufacturer Name2    |
| /Intranet                 | Library    | Public to Enterprise  |
| /Internet                 | Library    | Public to www         |

### Example 4: Model Libraries as Lifecycle Phases

|                           |            |                       |
|---------------------------|------------|-----------------------|
| /Concept                  | Library    | Business Object       |
| /Catalog Group            | Folder     | Item Catalog Category |
| /M10000                   | Folder     | Item                  |
| /M10000 Ver A             | Sub Folder | Item Revision         |
| /Functional Specification | Sub Folder | Document Type         |
| / CADView                 | Sub Folder | Document Type         |
| /M10000 Ver B             | Sub Folder | Item Revision         |
| /Design                   | Library    | Generic CR/CO         |
| /Test                     | Library    |                       |
| /Release to Mfg           | Library    |                       |
| /Catalog Group            | Folder     | Item Catalog Category |
| /M10000                   | Folder     | Item                  |
| /M10000 Rev 1.0           | Sub Folder | Item Revision         |
| /Functional Specification | Sub Folder | Document Type         |
| / CADView                 | Sub Folder | Document Type         |
| /M10000 Rev 2.0           | Sub Folder | Item Revision         |
| /M10000 CO                | Sub Folder | Item specific CR/CO   |
| /Obsolete                 | Library    |                       |
| /Manufacturer 1           | Library    | Manufacturer Name1    |
| /Manufacturer 2           | Library    | Manufacturer Name2    |
| /Intranet                 | Library    | Public to Enterprise  |
| /Internet                 | Library    | Public to www         |

## Attachment Review and Approval

Attachment Review or Approval provides:

- Flexible and scalable way of defining different types of reviews/approvals;
- Capture of user-defined attributes for them;
- Secure collaboration
- Flexible and advanced approval routing definition.

Complex parallel and serial approval routes can be created as predefined templates to enforce strict approval processes, or ad hoc routings to obtain additional approvals within the enterprise can be specified.

Attachments can be selected and submitted for Review or Approval. The Review and Approval workflows can be independent of each other. The Approval and Review process use predefined approval templates with support for ad-hoc addition of new steps. You can add steps to the process only if a workflow routing is associated with the particular status. You cannot delete predefined (in the item catalog category) steps. You can add additional assignees to a step, but you cannot delete predefined assignees. Based on the Approval or Review type chosen, the workflow/approval routing is determined. Appropriate notifications are sent at the various stages of the approval process including a final notification of approval. Approvers can be derived from a user's role on the business object.

The system indicates the current status of the attachment. For files residing in a Oracle Universal Content Management (UCM) repository, the status refers to the file itself. When you submit a file for review and approval from within UCM, you are asking for review and approval of the file content.

An attachment could have one of the following statuses of Approved/ Reviewed / Submitted for Approval/Submitted for Review or Rejected. If the file status is not one of these, then it means that the attachment has not been submitted through a formal review and approval process and is in a status of Draft or Unapproved. As the attachment progresses through the Approval process the attachment status should be automatically updated and the approval or review itself should be appropriately routed to different users for their review or approval.

The Attachment Review and Approval process is built using the Change Management Framework with Approval and Review being modeled as Change Categories. Thus, creating the different Approval and Review types is similar to Defining the Header Types for a Change Category, with a few minor differences. For example Lines and associated Line types cannot be defined for the Approval and Review types. Similarly the References, Dependencies and Attachments sections are not available.

Once an Attachment Approval or Review type has been defined, then with each status a workflow could be associated. An Approval type must have an Approval status with an approval workflow associated with it. For more details about creating workflows, see Defining Workflow Templates, page 7-17.

## Related Topics

Managing Attachments, *Oracle Product Hub User's Guide*

## Change Management for Attachments

Change Management for attachments provides change control functionality for all types of attachments, a crucial and important piece of non-structured information about the product.

Types of attachment changes supported using a change order include:

- adding a specific version of an attachment at the item or item revision level
- creating a new version for an existing attachment using by checking out the current version and checking in a new version.
- changing the attached version for an existing attachment.
- changing attachment attributes. Supported attributes include Name, Description, and Attachment Category.
- deleting attachments.

Change control is achieved by providing a mechanism to author, track and manage changes to attachments through a change order approval and implementation process. You can:

- Specify items and attachments that are revised or effected in a change order.
- Author and specify the exact changes that are desired or required to resolve an issue or change request.
- Route the change order for approval to the appropriate people responsible for the revised item based on item roles, change desired and lifecycle of item.

Once the change order is approved, implement the changes manually or automatically on the specified implementation date.

Change Management for attachments is supported through the Change Management functionality. User can create a change order and specify changes to the attachments of a reviewed item. There are no additional steps required in the setting up of a change order to be able to do attachment changes for a revised item.

## Related Topics

Defining Change Categories and Types, *Oracle Product Hub Implementation Guide*





---

# Implementing Roles and Role Based Security

This chapter covers the following topics:

- Overview of Roles and Role Based Security
- Administering Roles
- Administering People, Groups, and Companies
- Implementing Role Based Security
- Implementing Change Management Role Based Security
- Creating Custom Privileges

## Overview of Roles and Role Based Security

Role-based data security enables users to secure individual data objects and user-defined attribute groups. Your ability to view, edit and perform certain actions on an object or attribute group is determined by your role on it. Users with the privilege to add people can assign users to object roles. You can assign roles for the following objects:

- Catalog
- Catalog Category
- Change
- Group
- Item
- Item Catalog Category

*Roles*

A role is a collection of privileges. Roles are a convenient way to group privileges into a bundle that can later on be assigned to users, groups, or companies. For example, a user with the Design Engineer role on an item is granted the privilege to View Item, Edit Item, Add/Delete Item Attachments and more. However, the Design Engineer is not granted the privilege to Promote Item/Item Revision Lifecycle Phase.

You can assign roles to a person, group, company, or all users. To simplify item security maintenance, you can assign item roles at the catalog category, catalog, organization, item catalog category, or item level. You can specify default roles for all items in an organization. For example, you can assign the Item Reviewer role to the Engineering group so that all engineers can search for and view all items in the item catalog. For each item catalog category, you can specify which people can create items by assigning them the Catalog Category User role. For example, you can assign just the New Product Introduction engineers the Catalog Category User on the item catalog categories for which they are responsible for maintaining. Next, you can assign default role assignments to people by organization for all items in the item catalog category. For example, you can assign your Motherboard engineers a Design Engineer role for the Motherboard item catalog category. You can assign the buyer for Motherboards a Buyer role. At the item level, you can assign direct roles to enable access to specific items. For example, you can assign the role of Supplier Engineer to suppliers with whom you are collaborating on specific items.

#### *Privileges*

A privilege defines a user's access to an object. For example, a user's ability to view an item and edit it is determined by his or her privileges on that item. You can define additional privileges to secure user-defined attribute groups. See: To implement attribute group security, page 11-19.

### Example of an Item Role

|                                             |                                             |        |              |
|---------------------------------------------|---------------------------------------------|--------|--------------|
| People                                      | Company                                     | Groups | <b>Roles</b> |
| Roles > Role Search Results >               |                                             |        |              |
| <b>Role: Product Information Manager</b>    |                                             |        |              |
| Name                                        | Product Information Manager                 |        |              |
| Description                                 | Product Information Manager                 |        |              |
| Object Name                                 | Item                                        |        |              |
| <b>Privileges</b>                           |                                             |        |              |
| <div>Update</div>                           |                                             |        |              |
| <div>Previous 1-25 Next 25</div>            |                                             |        |              |
| <b>Name</b>                                 | <b>Description</b>                          |        |              |
| View Target Markets                         | View Target Markets                         |        |              |
| Add/Delete Approved Manufacturer Parts Item | Add/Delete Approved Manufacturer Parts Item |        |              |
| Add/Delete Customer Item Cross References   | Add/Delete Customer Item Cross References   |        |              |
| Add/Delete Item Attachment                  | Add/Delete Item Attachment                  |        |              |
| Add/Delete Item Cross References            | Add/Delete Item Cross References            |        |              |
| <b>Role Mappings</b>                        |                                             |        |              |
| <div>Update</div>                           |                                             |        |              |
| <b>Type</b>                                 | <b>Role Name</b>                            |        |              |
| Change Order                                | Change Product Manager                      |        |              |
| Issue                                       | Change Product Manager                      |        |              |
| File Approval                               | Approver                                    |        |              |
| New Item Request                            | Change Product Manager                      |        |              |

To implement roles and role-based security, perform the following tasks:

| Task                                                           | Required? |
|----------------------------------------------------------------|-----------|
| Administering Roles, page 11-3                                 | Yes       |
| Administering People, Groups, and Companies, page 11-6         | No        |
| Implementing Role-based Security, page 11-11                   | Yes       |
| Implementing Change Management Role-based Security, page 11-24 | Yes       |
| Creating Custom Privileges, page 11-27                         | No        |

## Administering Roles

You can create roles for the following objects:

- **Catalog**  
A catalog role assigned to a user determines which actions that user can perform on items in a catalog.
- **Catalog Category**  
A catalog category role assigned to a user determines which actions that user can perform on items in a catalog category.
- **Change**  
The change role assigned to a user for a change object (for example, issue, change request, change order) determines which actions that user can perform on the change object. For example, a user with an Approver role on a change request is granted the View Basic Change Information and Edit/Delete Change privileges.
- **Group**  
A group role assigned to a user for a group determines which actions that user can perform on the group.
- **Item**  
The item role assigned to a user for an item determines which actions that user can perform on the item. You can also inherit item privileges assigned at the catalog, catalog category, and item catalog category levels. For example, a user with the seeded Catalog Category User role on the Motherboard catalog category is granted the "Create Item of this Catalog Category" privilege, which enables him or her to create Motherboard items.
- **Item Catalog Category**  
The item catalog category role assigned to a user determines which actions that user can perform on items in a specific item catalog category.

**Additional Information:** In order to administer roles, a system administrator must grant you a role at the Oracle Applications user level. This role is granted by default in seeded responsibilities that include the Role submenu. The Role submenu, which includes the Roles link, is available from the Development Manager and Development Engineer responsibilities, as well as others.

If you cannot access the Roles link from your responsibility or if you cannot view, create, or update roles, then contact your system administrator. For more information, see:

- *Creating and Updating Roles, Oracle E-Business Suite Security Guide.*

- Excluding functions in the Menu Exclusions tab, Responsibilities Window, *Oracle E-Business Suite Security Guide*.

### To view roles:

You can view roles to examine the privileges that are associated with them.

1. Click the Roles link in the Application tree menu.
2. On the **Roles** page, enter the role name and click Go. If you know the specific object that the role is associated with, select it.
3. On the **Roles Search Results** page, click the name link of the role for which you were searching.
4. On the **Role Detail** page, you can view all the privileges associated with the role. You can also view the role mappings, which identify item roles and their corresponding change object roles.

### To create a role:

1. From the applications tree menu, click Roles.
2. On the **Roles** page, click Create Role.
3. On the **Create Role** page, select the object for which you are creating a role and click Next.
4. Provide a name and description for the new role. Select the object privileges you want to associate with this new role and click Apply. Be aware that selecting the Edit privilege does not implicitly grant the view privilege. So, if you want someone to be able to edit an item, they must be able to view it first. Every role must have at least one privilege associated with it.

**Note:** Click Save and Map Roles if you wish to map an item role to a change object role.

### To update a role:

1. In the applications tree menu, click Roles.
2. On the **Roles** page, enter the role name and click Go. If you know the specific object that the role is associated with, select it.
3. On the **Role Search Results** page, click the name link of the role for which you were

searching.

4. On the **Role: (Role Name)** page, click Update in the Privileges region.
5. On the **Update Role: (Role Name)** page, you can edit the role name, description, or select/unselect the privileges associated with it.
6. Click Apply.

#### **To update role mappings**

7. On the **Role: (Role Name)** page, click Update in the Role Mappings region.
8. On the **Role Mappings** page, you can select a different role name or no role name for each change object.
9. Click Apply.

#### **To delete a role:**

On the **Role Search Results** page, select the role you want to delete and click Delete.

**Note:** Once a role has been granted it cannot be deleted.

## **Related Topics**

Overview of Roles and Role Based Security, page 11-1

Administering People, Groups, and Companies, page 11-6

Implementing Role Based Security, page 11-11

Implementing Change Management Role Based Security, page 11-24

Creating Custom Privileges, page 11-27

## **Administering People, Groups, and Companies**

You can assign a person, group, company, or all users a role on a particular object, such as the Product Catalog (a catalog object) or a mobile phone (an item object). Groups are useful for managing role assignments for items, catalogs, and change management. Groups are also useful in change management approval routings, where one or all people in a group must approve a change, or all people in a group should be notified of a change.

You can create groups of people that include internal users, suppliers and customers. A group consists of one or more members. You can create groups of people for different teams or departments that serve similar business functions. The person that creates the

group is by default a member and the Group Administrator. Each member in a group is assigned a group role that determines which privileges are granted to the member of that group. There are three privileges a group role may grant a member: View Group Header, View Group Members, and Manage Group. For example, you can create a Supplier group made up of all your suppliers; however, you do not want any of the members to see who else is in the group. In this case, each of the members would be assigned the Group Reviewer role, which only allows them to see the group header information, but not the other members. In another scenario, you may wish to create an Engineering group where you want members to be aware of who else is in the group. In this case, you can grant each member the View Group Members role. Only a person or group with the Group Administrator role can also manage the group. The Manage Group privilege enables a user to add/delete members and change their roles.

### Group details

**ORACLE** Home Logout Help Preferences

People Company **Groups** Roles

Groups > My Own Groups > Group Details

**Group Detail**

Name: Vision Engineering  
Description: Vision Engineering  
Group Email Address: steve.williams@oracleads.com

Send Email to Members Update

**Members** Add

Select Users and... Delete

Select All Select None

| Select Name                              | Company           |
|------------------------------------------|-------------------|
| <input type="checkbox"/> Steve Williams  | Vision Enterprise |
| <input type="checkbox"/> Mary Robinson   | Vision Enterprise |
| <input type="checkbox"/> Ravi Patel      | Vision Enterprise |
| <input type="checkbox"/> Rebecca Swanson | Vision Enterprise |

**Group People** Update

Select Users and... Delete

Select All Select None

| Select Role                                  | Type    | Name               | Company           | Start Date  | End Date |
|----------------------------------------------|---------|--------------------|-------------------|-------------|----------|
| <input type="checkbox"/> View Group Members  | Group   | Vision Engineering |                   | 31-Jul-2003 |          |
| <input type="checkbox"/> Group Administrator | Person  | Steve Williams     | Vision Enterprise | 31-Jul-2003 |          |
| <input type="checkbox"/> Group Reviewer      | Company | Vision Enterprise  | Vision Enterprise | 31-Jul-2003 |          |

Copyright 2003 Oracle Corporation. All rights reserved. Home Logout Help Preferences Privacy Statement

You can add three types of people to a group or assign them to a role on an object directly. The Type attribute on the **People Search** and **Person: (Person Name)** pages corresponds to the three basic types of people in the system:

- Internal

When you create a user using the E-Business Suite Define User form, you define an employee in the People field. This person, now identified by the username entered, is classified as Internal in the system. You can search for and view details about this person.

- Customer

When you create a user using the E-Business Suite Define User form, you define a customer in the Customer field. This person, now identified by the username entered, is classified as a Customer in the system. You can search for and view details about this person.

- Vendor

When you create a user using the E-Business Suite Define User form, you define a supplier in Supplier field. This person, now identified by the username entered, is classified as a Vendor in the system. You can search for and view details about this person.

Customer and Vendor are *external* people.

**Important:** You may have defined people, such as contractors and temporary employees, in Oracle HRMS. You may also have defined people as customers in Oracle Customer Online. These people are not recognized as valid people in the system despite the fact that they are in HRMS and Customer Online. You cannot search for or view details about these people.

## Prerequisites

- ☐ Before adding people to a group, you must run the concurrent program "EGO Setup Default Organization for User".

### To search for people:

1. In the Applications tree menu, click People, Groups and Companies.
2. Click the People tab.
3. On the **People** page, enter the name of the person for whom you are searching in the Search field.

You can search by first name, last name or user name. You can enter a partial name and search on this. For example, enter "wil". A list of all people with a name beginning with "wil" is returned.

4. Click Go.

The name you entered is returned in the People Search Results region. If the name you entered is not listed in the People Search Results, check the spelling of the name and try searching again. If you are unable to locate the name, the person is not registered in E-Business Suite, or is not valid in the system. Try registering the person in the E-Business Suite first, or make sure the person is valid in the system.



### **To create a group:**

Groups are communities of people you can define in order to collaborate. You can represent all your teams as groups in the system and give your group access to the system's objects. You can send an e-mail to everyone in a group with a click of a button. To view and create a group, you must have the following list of privileges, such as in the seeded Group Administrator role:

- View Group Header
- View Group Members
- Manage Group

1. In the Applications tree menu, click People, Groups and Companies.
2. Click the Groups tab.
3. On the **Groups** page, click Create Group.
4. On the **Create Group** page, provide the following:

#### **Group**

Provide a name that identifies the group.

#### **Description**

Provide a brief description of the group.

#### **Group Email Address**

Specify the group mailing list, if you have one, as your group's Email address.

5. Click Apply to save the group, or click Add Another Row to create another group. After clicking Apply, the **Group Details** page appears.

6. On the **Group Details** page, you can perform the following actions:
  - Click Update in the Group Detail region to edit group information.
  - Click Add to add additional members to the group.
  - If you wish to delete a member, select that group member and click Delete.
  - Click Update in the Group People region to assign a group object role to a specific person, group or company.

The system automatically assigns the following seeded roles:

- Group Administrator to the person who created the group

- View Group Members to all members of the group
- Group Reviewer to any member of the group creator's company

**Tip:** Assign the Group Administrator role to an additional person as a backup.

### **To add/remove members to a group:**

1. In the Applications tree menu, click People, Groups and Companies.
2. Click the Groups tab.
3. On the **Groups** page, click My Own Groups.
4. Select a group and click Edit Membership.
5. On the **Group Detail** page, click Add, and select one or more persons from the list of values.
6. To remove members, select each member to delete and click Delete.

### **To search for a group:**

1. In the Applications tree menu, click People, Groups and Companies.
2. Click the Groups tab.
3. On the **Groups** page, enter the full or partial name of the group. Click Go.
  - To locate all groups in which you are a member, click My Group Memberships.
  - To locate all groups that you own, click My Own Groups.
4. If the group name you entered is not listed in the Group Search Results, check the spelling of the name and try searching again. If you are unable to locate the group, it has not yet been created. Create the group using the "Create Group" link on the Groups page.
5. On the **Group Search Results** page, select a group and click Request Membership to join a group. The owner is sent an Email notification and can add you as a member of that group.
6. Click a group name to see all the members of a group (only if you are the owner or a member of that group or have the View Group Members privilege assigned to

you).

### **To find groups that you own:**

1. In the Applications tree menu, click the "People, Groups and Companies" link.
2. Click the Groups tab.
3. On the **Groups** page, click the "My Own Groups" link to locate all groups that you own.

See "To create a group:" above for more details about updating groups.

### **To search for a company:**

1. In the Applications tree menu, click the "People, Groups and Companies" link.
2. Click the Companies tab.
3. On the **Companies** page, enter the name of the company for which you are searching, or enter a letter from the alphabet. If you only know part of the name of the company for which you are searching, enter that name. For example, if you are searching for a company named "AAA Supplier," type "AAA" in the search field. You will then see a list of all companies with "AAA" in their name.
4. Click Go. The name you entered is displayed in the Company Search Results field. If the company name you entered is not listed in the Company Search Results, check the spelling of the name and try searching again. If you are unable to locate the company, it is likely not registered in E-Business Suite. Try registering the customer company using Oracle Customer Online or Oracle Order Management. Try registering the Vendor/Supplier Company using Oracle Purchasing.

## **Related Topics**

Overview of Roles and Role Based Security, page 11-1

Administering Roles, page 11-3

Implementing Role Based Security, page 11-11

Implementing Change Management Role Based Security, page 11-24

Creating Custom Privileges, page 11-27

## **Implementing Role Based Security**

To implement role based security on a particular object, attach a role to the object and a person, group, company, or all users to the role on the object. For example, for each

item catalog category, you can specify which people can create items by assigning them the Catalog Category User role on the item catalog category.

To implement role based security on a user-defined attribute group, assign people to an item role containing custom privileges for the attribute group.

## Prerequisites

- ☐ Create roles. See: *Administering Roles*, page 11-3
- ☐ Optionally, create groups. See: *Administering People, Groups, and Companies*, page 11-6

### To assign people to a catalog:

Depending on their assigned role, people can assign items to a catalog or view the catalog hierarchy. You can assign either of the following seeded roles to people in a catalog:

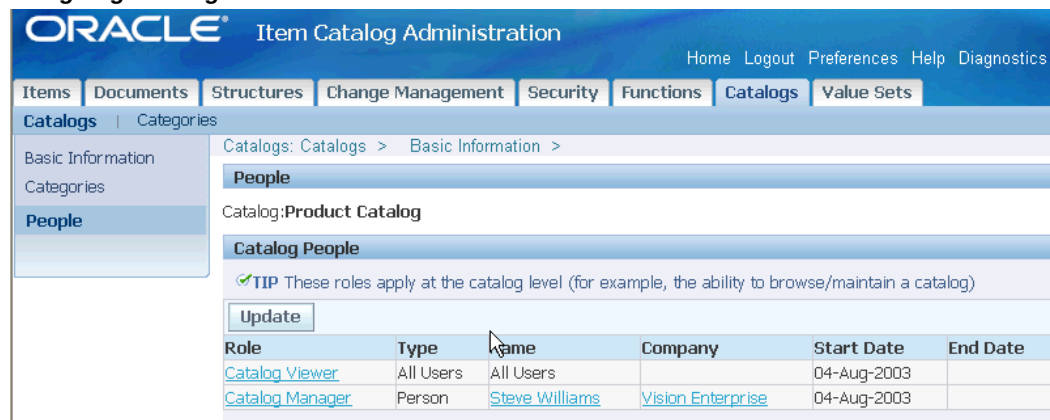
#### Catalog Manager

The Catalog Manager role enables users to view the catalog hierarchy and also enables them to assign items to this catalog.

#### Catalog Viewer

The Catalog Viewer role enables users to view the catalog hierarchy, but they cannot assign items to it.

### Assigning Catalog Roles



The screenshot shows the Oracle Item Catalog Administration web interface. The top navigation bar includes links for Home, Logout, Preferences, Help, and Diagnostics. The main navigation menu on the left includes Items, Documents, Structures, Change Management, Security, Functions, Catalogs, and Value Sets. The 'Catalogs' tab is selected, and the 'People' sub-tab is active. The breadcrumb trail shows 'Catalogs: Catalogs > Basic Information >'. The main content area displays 'Catalog: Product Catalog' and 'Catalog People'. A tip states: 'TIP These roles apply at the catalog level (for example, the ability to browse/maintain a catalog)'. Below the tip is an 'Update' button and a table listing assigned roles.

| Role                            | Type      | Name                           | Company                           | Start Date  | End Date |
|---------------------------------|-----------|--------------------------------|-----------------------------------|-------------|----------|
| <a href="#">Catalog Viewer</a>  | All Users | All Users                      |                                   | 04-Aug-2003 |          |
| <a href="#">Catalog Manager</a> | Person    | <a href="#">Steve Williams</a> | <a href="#">Vision Enterprise</a> | 04-Aug-2003 |          |

1. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
2. On the **Search: Catalogs** page, select the catalog to which you wish to assign people.

3. On the **Basic Information** page for the catalog, click the "People" link.
4. On the **People** page, click Update. After the page refreshes, click Add Another Row.
5. After the **Update People** page refreshes, select a role, the type of people to use the role, and the name of the company, group, or person given the role on the catalog. Specify a start date and, optionally, an end date.
6. Click Apply.

#### **To delete people from a catalog**

7. In the **Update People** page, select a row and click Remove.
8. Click Apply.

#### **To assign people to a catalog category:**

**Note:** There are no seeded roles for the catalog category object. You must create a role for the catalog category before you can assign people.

1. On the **Search: Item Catalog Categories** page, click the Catalogs tab.
2. On the **Search: Catalogs** page, select the catalog containing the category to which you wish to assign people.
3. On the catalog's **Basic Information** page, click the Categories link on the left side of the page.  
The **Categories** page displays the valid categories for the selected catalog.
4. Click on the name link of the valid category to which you wish to assign people.
5. On the **Category Details** page, click the People link.
6. On the **People** page, under the Category People heading, click Update. After the page refreshes, click Add Another Row.
7. After the **Category People** page refreshes, select a role, the type of people to use the role, and the name of the company, group, or person given the role on the catalog. Specify a start date and, optionally, an end date.
8. Click Apply.

#### **To delete people from a catalog category**

9. In the **Category People** page, select a row and click Remove.

10. Click Apply.

### **To grant group roles to people:**

You can grant a person, group or company a specific role for a group. You must have the Manage Groups privilege for the group in order to grant others a role on the group.

1. In the Applications tree menu, click People, Groups and Companies.
2. Click the Groups tab.
3. On the **Groups** page, click My Own Groups or search for a group for which you have the Manage Groups privilege.
4. Click on a group name.
5. In the **Group Details** page, under the Group People heading, click Update.
6. In the **Update Group People** page, click Add Another Row.
7. Select a role, the type of people to use the role, and the name of the company, group, or person given the role on the group. Specify a start date and, optionally, an end date.
8. Click Apply.

### **To assign people to an item:**

You can assign roles for certain users directly to a single item or assign roles indirectly using roles inherited from the following objects that contain the item:

- an alternate catalog
- an alternate catalog category
- an item catalog category
- an organization

### Item-specific role assignments

The screenshot shows the Oracle Item Catalog interface. At the top, it says 'ORACLE Item Catalog' and 'Item: VI40001 Revision: B'. Below this, it says 'Organization: Vision Operations (V1)'. There are several tabs: 'Item', 'Lifecycle', 'Change Management', 'Configuration', 'Transactions', 'Specifications', 'Revisions', 'Attachments', 'Organizations', 'People', 'Associations', and 'Syndication'. The 'People' tab is selected. Below the tabs, there is a search bar with 'Item' in the dropdown, a 'Go' button, and links for 'Advanced Search', 'Shortcuts', and 'Create Document'. Below the search bar, there is a section titled 'Item People'. Under this, there is a 'Direct Item Roles' section with an 'Update' button. Below that is a table with columns: Role, Type, Name, Company, Start Date, and End Date. The table contains two rows: 'Supplier Engineer' (Person, James Indus, Industrial Dressler, 06-Aug-2003) and 'Item Author' (Person, Steve Williams, Vision Enterprise, 31-Jul-2003). Below the 'Direct Item Roles' section is an 'Inherited Item Roles' section with a 'Previous' button, a '1-10' dropdown, and a 'Next 10' button. Below that is another table with the same columns: Role, Type, Name, Company, Start Date, and End Date. The table contains four rows: 'Design Engineer' (Person, Mary Robinson, Vision Enterprise, 04-Aug-2003), 'Design Engineer' (Person, Mary Robinson, Vision Enterprise, 06-Aug-2003), 'Design Reviewer' (Company, Vision Enterprise, 18-Sep-2002), and 'Design Reviewer' (Group, Vision Engineering, 06-Aug-2003).

| Role              | Type   | Name           | Company             | Start Date  | End Date |
|-------------------|--------|----------------|---------------------|-------------|----------|
| Supplier Engineer | Person | James Indus    | Industrial Dressler | 06-Aug-2003 |          |
| Item Author       | Person | Steve Williams | Vision Enterprise   | 31-Jul-2003 |          |

| Role            | Type    | Name               | Company           | Start Date  | End Date |
|-----------------|---------|--------------------|-------------------|-------------|----------|
| Design Engineer | Person  | Mary Robinson      | Vision Enterprise | 04-Aug-2003 |          |
| Design Engineer | Person  | Mary Robinson      | Vision Enterprise | 06-Aug-2003 |          |
| Design Reviewer | Company | Vision Enterprise  |                   | 18-Sep-2002 |          |
| Design Reviewer | Group   | Vision Engineering |                   | 06-Aug-2003 |          |

1. In the Applications tree menu, use either the "Simple Search" or "Advanced Search" links to search for the item to which you wish to add people.
2. Click the item's link in the search results.
3. On the item's **Overview** page, click People.
4. On the **Item People** page, click Update.

**Note:** You can only update the People page for an item from within the master organization.

5. On the **Update Item People** page, click Add Another Row. Provide the following information:

#### Role

Specify the role of the person you are adding.

#### Type

Specify the type of person you are adding (All Users, Company, Person, Group).

#### Name

Enter the name of the company, person, or group.

#### Start Date

Specify the date on which the person/group gains access to the item.

### End Date

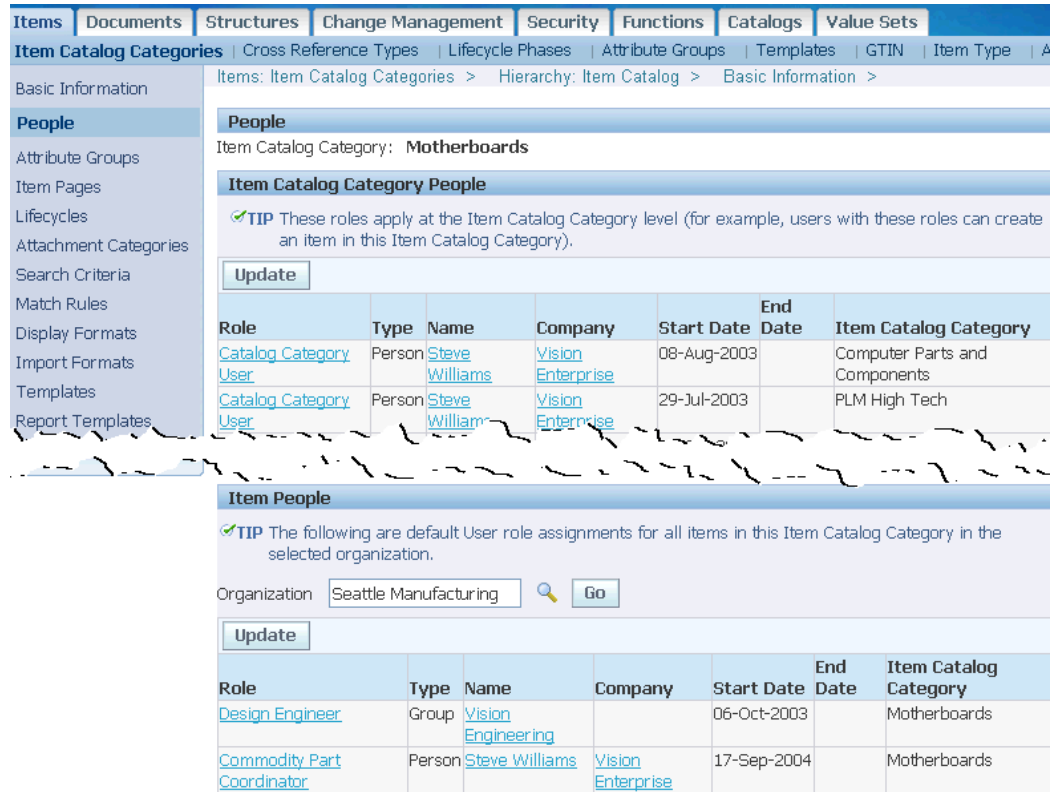
Optionally, specify the date on which the person/group no longer has access to the item.

6. Click Apply.

## To assign people to an item catalog category:

You can grant roles to people by item catalog category for all organizations or by item catalog category in a particular organization. Item catalog categories at a lower level in the catalog hierarchy inherit roles granted to people at a higher level item catalog category. You cannot remove or edit inherited roles.

### Item Catalog Category People



| Role                                  | Type   | Name                           | Company                           | Start Date  | End Date | Item Catalog Category         |
|---------------------------------------|--------|--------------------------------|-----------------------------------|-------------|----------|-------------------------------|
| <a href="#">Catalog Category User</a> | Person | <a href="#">Steve Williams</a> | <a href="#">Vision Enterprise</a> | 08-Aug-2003 |          | Computer Parts and Components |
| <a href="#">Catalog Category User</a> | Person | <a href="#">Steve Williams</a> | <a href="#">Vision Enterprise</a> | 29-Jul-2003 |          | PLM High Tech                 |

| Role                                       | Type   | Name                               | Company                           | Start Date  | End Date | Item Catalog Category |
|--------------------------------------------|--------|------------------------------------|-----------------------------------|-------------|----------|-----------------------|
| <a href="#">Design Engineer</a>            | Group  | <a href="#">Vision Engineering</a> |                                   | 06-Oct-2003 |          | Motherboards          |
| <a href="#">Commodity Part Coordinator</a> | Person | <a href="#">Steve Williams</a>     | <a href="#">Vision Enterprise</a> | 17-Sep-2004 |          | Motherboards          |

1. Navigate to the Setup Workbench. On the **Item Catalog Categories** page, search for and select an item catalog category.
2. On the **Basic Information** page, click the People link.



### **To assign people to an item catalog category for all organizations**

3. On the **People** page, under the Item Catalog Category People heading, click Update.

This updates roles for the item catalog category across all organizations.

4. In the **Edit People** page, click Add Another Row. Provide the following information:

#### **Role**

Specify the role of the person you are adding.

#### **Type**

Specify the type of person you are adding (All Users, Company, Person, Group).

#### **Name**

Enter the name of the company, person, or group.

#### **Start Date**

Specify the date on which the person/group gains access to the item catalog category.

#### **End Date**

Optionally, specify the date on which the person/group no longer has access to the item catalog category.

5. Click Apply.

### **To assign people to an item catalog category for a particular organization**

6. On the **People** page, under the Item People heading, search for and select an organization.

7. Under the Organization field, click Update.

This updates roles for the item catalog category only in the selected organization.

8. In the **Edit Item People** page, click Add Another Row. Provide the following information:

#### **Role**

Specify the role of the person you are adding.

#### **Type**

Specify the type of person you are adding (All Users, Company, Person, Group).

#### **Name**

Enter the name of the company, person, or group.

### Start Date

Specify the date on which the person/group gains access to the item catalog category.

### End Date

Optionally, specify the date on which the person/group no longer has access to the item catalog category.

9. Click Apply.

## To assign people to all items in an organization:

You can grant a specific role to a certain person, group, company, or all users that applies to all items in an organization. This is useful when you want to grant one person access to many items.

**Note:** You can grant organization-level roles if you are assigned the Item Administration function through one of your responsibilities.

If you cannot grant organization-level roles, then contact your system administrator. For more information, see:

- Creating and Updating Roles, *Oracle E-Business Suite Security Guide*.
- Excluding functions in the Menu Exclusions tab, Responsibilities Window, *Oracle E-Business Suite Security Guide*.

### Organization Level Item Role Assignments

Items Documents Structures Change Management Security Functions Catalogs Value Sets

Security >

**Organization Roles**

✓TIP Default roles will be assigned to all objects in the selected organization.

Organization

Object Type **Items**

| Role                                   | Type   | Name                           | Company                           | Start Date  | End Date |
|----------------------------------------|--------|--------------------------------|-----------------------------------|-------------|----------|
| <a href="#">Product Manager</a>        | Person | <a href="#">Steve Williams</a> | <a href="#">Vision Enterprise</a> | 06-Jun-2005 |          |
| <a href="#">Design Engineer</a>        | Person | <a href="#">Mary Robinson</a>  | <a href="#">Vision Enterprise</a> | 06-Jun-2005 |          |
| <a href="#">Manufacturing Engineer</a> | Person | <a href="#">Ravi Patel</a>     | <a href="#">Vision Enterprise</a> | 06-Jun-2005 |          |

1. In the Applications tree menu, click Setup Workbench.
2. On the **Item Catalog Categories** page, click the Security tab.
3. On the **Organization Roles** page, select an Organization to which to grant a role and click Go. The table returns all people, groups and companies who have been

granted roles in the selected organization.

4. Click Update.
5. On the **Update Organization Roles** page, click Add Another Row. A new row appears in the table. Enter the following information:
  - **Role**  
Select a role for this grant.
  - **Type**  
Select the type of grant: a grant to a single person, a group, a company, or all users.
  - **Name**  
Enter the name of the person, group, or company to which you are making this grant.
  - **Start Date**  
Select a start date from which this grant is effective.
  - **End Date**  
Select an end date on which the grant will no longer be effective.
6. Click Apply.

**To revoke a specific role from a person for all items in the organization**

7. On the **Update Organization Roles** page, select the person(s) or group(s) whose role you wish to delete, and click Remove.
8. Click Apply.

**To edit a role grant of a person for all items in an organization**

9. On the **Update Organization Roles** page, select the person or group whose role you wish to edit, and provide a new start date or end date.
10. Click Apply.

**To implement attribute group security:**

When implementing role-based item security, you can create custom privileges to control the view and edit permissions for specific item attribute groups. You can control which users can view and/or edit certain attribute groups for an item by assigning a role granting those specific privileges. By default, an item role's View Item and Edit Item

privileges control whether or not you can view or edit item attributes that are not controlled specifically at the item attribute group level. In other words, when implementing item security you do not have to specify a view or edit privilege for each item attribute group.

### Example: Attribute Group Security

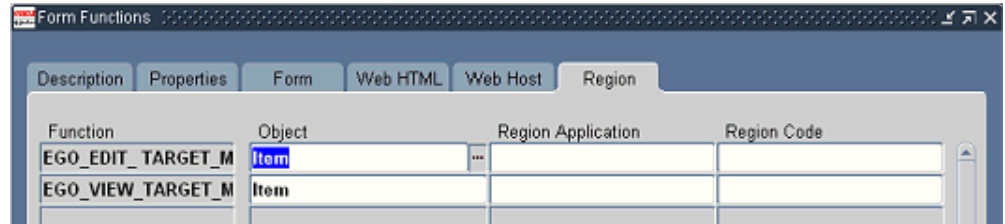
Suppose your company is designing, along with your supplier, a new motherboard for its next generation of desktop computers. To improve design collaboration you would like to securely share item information about the motherboard—both internally between departments and externally with your suppliers and contract manufacturers. The Supplier Engineer should only be able to view specific item attribute groups such as the Technical Specifications. The Supplier Engineer should not be able to view the Market Research attributes. The Engineering Manager and Marketing Manager should be able to view and edit the Market Research attributes, while a Design Engineer should only view the Market Research attributes. There are three sets of Market Research attributes (attribute groups): Key Metrics, Target Markets, and Competitors.

1. Select the Application Developer responsibility, navigate to the Form Functions form, and create Form Functions for each privilege that controls view and edit permissions for the Market Research attribute groups. See: Creating Custom Privileges, page 11-27.

#### Defining form functions to create user-defined item role privileges

| Function          | User Function Name  | Description         |
|-------------------|---------------------|---------------------|
| EGO_EDIT_TARGET_M | Edit Target Markets | Edit Target Markets |
| EGO_VIEW_TARGET_M | View Target Markets | View Target Markets |

| Function          | Type | Maintenance Mode Support | Context Dependence |
|-------------------|------|--------------------------|--------------------|
| EGO_EDIT_TARGET_M |      | None                     | Responsibility     |
| EGO_VIEW_TARGET_M |      | None                     | Responsibility     |



2. Select the Development Manager responsibility and navigate to the Setup Workbench. In the **Attribute Group Details** page for each Market Research-related attribute group (for example, Target Markets) specify the View Privilege (for example, View Target Markets) and Edit Privilege (for example, Edit Target Markets) in the Business Entities region.

### Defining Item Attribute group with view and edit privileges

#### Attribute Group Details

|                                      |                |
|--------------------------------------|----------------|
| Internal Name                        | Target_Markets |
| Display Name                         | Target Markets |
| Description                          |                |
| Behavior                             | Multi-Row      |
| Number of columns in the page layout |                |
| Number of columns shown in the table |                |
| Number of rows shown in the table    |                |

| Business Entities |              |                     |                     |
|-------------------|--------------|---------------------|---------------------|
| Business Entities |              | Privilege           |                     |
| Name ▲            | Style to SKU | View                | Edit                |
| Item              | Defaulting   | View Target Markets | Edit Target Markets |

| Attributes                                               |          |                        |                             |           |            |                |      |
|----------------------------------------------------------|----------|------------------------|-----------------------------|-----------|------------|----------------|------|
| Select Object: <a href="#">Delete</a>                    |          |                        |                             |           |            |                |      |
| <a href="#">Select All</a>   <a href="#">Select None</a> |          |                        |                             |           |            |                |      |
| Select                                                   | Sequence | Internal Name          | Display Name                | Data Type | Display As | Value Set Name | Part |
| <input type="checkbox"/>                                 | 10       | Target_Mkt_Name        | <a href="#">Name</a>        | Char      | Text Field |                | Yes  |
| <input type="checkbox"/>                                 | 20       | Target_Mkt_Description | <a href="#">Description</a> | Char      | Text Field |                | No   |

| Where Used                    |                 |
|-------------------------------|-----------------|
| Name                          | Business Entity |
| Computer Parts and Components | Item            |
| Computer Systems              | Item            |

- On the **Item Role Detail** page for the Marketing Manager and Engineering Manager roles, grant the following privileges:
  - View Target Markets
    - View Key Metrics
    - View Competitors
  - Edit Target Markets
    - Edit Key Metrics
    - Edit Competitors

### Defining the Marketing Manager role with specific item attribute group privileges

| Role: Marketing Manager |                   |
|-------------------------|-------------------|
| Name                    | Marketing Manager |
| Description             | Marketing Manager |
| Object Name             | <b>Item</b>       |

| Privileges                             |                                        |
|----------------------------------------|----------------------------------------|
| <input type="button" value="Update"/>  |                                        |
| Name                                   | Description                            |
| <b>View Target Markets</b>             | View Target Markets                    |
| Add/Delete Item Attachment             | Add/Delete Item Attachment             |
| <b>Edit Competitors</b>                | Edit Competitors                       |
| Edit Item                              | Edit Item                              |
| <b>Edit Key Metrics</b>                | Edit Key Metrics                       |
| Edit Privileges on A&D Item Attributes | Edit Privileges on A&D Item Attributes |
| <b>Edit Target Markets</b>             | Edit Target Markets                    |
| Grant/Revoke Role on the Item          | Grant/Revoke Role on the Item          |
| <b>View Competitors</b>                | View Competitors                       |
| View Item                              | View Item                              |
| View Item Attachments List             | View Item Attachments List             |
| View Item Organization Assignments     | View Item Organization Assignments     |
| View Item People List                  | View Item People List                  |
| View Item Project/Task Associations    | View Item Project/Task Associations    |
| View Item Revision List                | View Item Revision List                |
| View Item / Item Revision Lifecycle    | View Item / Item Revision Lifecycle    |
| <b>View Key Metrics</b>                | View Key Metrics                       |
| View Privileges on A&D Item Attributes | View Privileges on A&D Item Attributes |

| Role Mappings                         |
|---------------------------------------|
| <input type="button" value="Update"/> |

For the Design Engineer role, grant the following privileges:

- View Target Markets
- View Key Metrics
- View Competitors

Do not grant any of the Market Research privileges to the Supplier Engineer role.

### Defining the Supplier Engineer role with no specific item attribute group privileges

| Role: Supplier Engineer |                   |
|-------------------------|-------------------|
| Name                    | Supplier Engineer |
| Description             | Supplier Engineer |
| Object Name             | Item              |

| Privileges                             |                                        |
|----------------------------------------|----------------------------------------|
| <div>Update</div>                      |                                        |
| Name                                   | Description                            |
| View Privileges on A&D Item Attributes | View Privileges on A&D Item Attributes |
| Add/Delete Item Attachment             | Add/Delete Item Attachment             |
| View Item                              | View Item                              |
| View Item Attachments List             | View Item Attachments List             |
| View Item Revision List                | View Item Revision List                |

| Role Mappings     |                          |
|-------------------|--------------------------|
| <div>Update</div> |                          |
| Type              | Role Name                |
| Requirements      | Change Supplier Engineer |
| Issue             | Change Supplier Engineer |

[Return to Role Search Results](#)

## Related Topics

Overview of Roles and Role Based Security, page 11-1

Administering Roles, page 11-3

Administering People, Groups, and Companies, page 11-6

Implementing Change Management Role Based Security, page 11-24

Creating Custom Privileges, page 11-27

## Implementing Change Management Role Based Security

You can assign change roles to a person, group, company, or all users. To simplify maintaining change management security, you can assign change roles directly to the change object or inherit them through item role mapping from the subject item of the change object. For example, you can map the Design Engineer item role to the Change Design Engineer role for issues, change requests, and change orders. So users with the Design Engineer role on the subject item of the change request header will inherit the Change Design Engineer role as well. You can also assign a default role to all internal users with the site level profile ENG: Internal User Default Role for Changes. A role that is explicitly granted to a user for a change object is a direct role assignment. Roles inherited from an item are inherited role assignments.

All seeded item roles are mapped to seeded change object roles. User-created item roles



do not have to be mapped to change roles. If you do not want a seeded item role mapped to a change role, edit that item role (such as Item Author) explicitly.

The change role assigned to a user for a change object (for example, issue, change request, change order) determines which actions that user can perform on the change object. For example, a user with an Approver role on a change request is granted the View Basic Change Information and Edit/Delete Change privileges. You can also specify which user-defined attribute groups a user can view and/or edit when granted a change role.

Following are the seeded change roles:

- Approver
- Assignee
- Creator
- Requestor
- Reviewer

### **To assign change roles directly to the change object:**

Change roles are assigned directly to the change object by the change type. When you defined your change types in Defining Header Types, page 7-28, you specified a workflow. Each workflow step with an Approval status requires at least one associated workflow template (see: Defining Workflow Templates, page 7-17). Each person, role, or group specified in the workflow template receives a role on the change object based on the approval routing activity. All Request Approval assignees get an Approver role. Request Comment and FYI assignees get a Reviewer role on the change object. You can also specify a Default Assigned To role for a person/group when you define a change type. The Assigned To person for every change object gets an Assignee role.

### **To inherit change roles through item role mapping:**

1. In the applications tree menu, click Roles.
2. On the **Roles** page, select the item object, enter the role name, and then click Go.
3. On the **Role Search Results** page, click the name link of the role for which you were searching.
4. On the **Role: (Role Name)** page, click Update in the Role Mappings region.
5. On the **Role Mappings** page, you can map each change object type to a specific change role.
6. Click Apply.

### Mapping an Item Role to Change Roles

| People                                      | Company                                     | Groups | Roles |
|---------------------------------------------|---------------------------------------------|--------|-------|
| Roles > Role Search Results >               |                                             |        |       |
| Role: Design Engineer                       |                                             |        |       |
| Name: Design Engineer                       |                                             |        |       |
| Description: PLM Design Engineer            |                                             |        |       |
| Object Name: Item                           |                                             |        |       |
| Privileges                                  |                                             |        |       |
| Update                                      |                                             |        |       |
| Previous 1-25 Next 25                       |                                             |        |       |
| Name                                        | Description                                 |        |       |
| View Target Markets                         | View Target Markets                         |        |       |
| Add/Delete Approved Manufacturer Parts Item | Add/Delete Approved Manufacturer Parts Item |        |       |
| Add/Delete Customer Item Cross References   | Add/Delete Customer Item Cross References   |        |       |
| Add/Delete Item Attachment                  | Add/Delete Item Attachment                  |        |       |
| Role Mappings                               |                                             |        |       |
| Update                                      |                                             |        |       |
| Type                                        | Role Name                                   |        |       |
| Change Notification                         | Change Design Engineer                      |        |       |
| Change Request                              | Change Design Engineer                      |        |       |
| Issue                                       | Change Design Engineer                      |        |       |
| Enhancement Request                         | Change Design Engineer                      |        |       |

### To assign a default change role to all internal users:

You can assign a default role to all internal users with the site level profile option ENG: Internal User Default Role for Changes.

1. Within the System Administrator responsibility, click System.
2. In the **System Profile Values** window, search for and select the site level profile option ENG: Internal User Default Role for Changes.
3. In the Site field, select the default change role for the site from the list of values.
4. Save your work.

### To implement attribute group security for change management:

When implementing role-based change management security you can set up privileges to control the view and edit permissions for specific change management attribute groups. You can control which users can view and/or edit certain attribute groups for a change object by assigning a role granting those specific privileges. By default, a change role's View Basic Change Information and Edit/Delete Change privileges control whether you can view or edit attributes that are not controlled specifically at the attribute group level. In other words, when implementing change management security you do not have to specify a view or edit privilege for each attribute group.

Suppose your company is co-designing with your supplier a new motherboard for its next generation of desktop computers. To improve communication with your supplier

on design changes to the motherboard, you would like to securely share change request information externally with your suppliers and contract manufacturers. The Supplier Engineer should only be able to view specific attribute groups.

1. Select the Application Developer responsibility, navigate to the Form Functions form, and create Form Functions for each privilege that controls view and edit permissions for the attribute groups. Specify "Change" or "Change Line" in the Object field for each Form Function.

See: Creating Custom Privileges, page 11-27

2. Select the Development Manager responsibility and navigate to the Setup Workbench. On the **Attribute Group Details** page for each attribute group, specify the View Privilege and Edit Privilege in the Business Entities region.
3. On the **Change Role Detail** page, grant the privileges that you defined as Form Functions in the previous steps.

## Related Topics

Overview of Roles and Role Based Security, page 11-1

Administering Roles, page 11-3

Administering People, Groups, and Companies, page 11-6

Implementing Role Based Security, page 11-11

Creating Custom Privileges, page 11-27

## Creating Custom Privileges

The system includes a number of seeded privileges. However, you have the option to create custom privileges to meet the needs of your business.

### To create a custom privilege:

1. Log into Oracle Forms using the System Administrator responsibility.
2. In the Application menu, select Function.
3. In the Form Functions window, provide the following information:

#### Function

Enter a unique function name. You may use this name when calling your function programmatically.

#### User Function Name

Enter a unique name that describes your function. You see this name when

assigning functions to roles.

**Description**

Provide a description of the function.

4. Click the Region tab, and select the Object for which you are creating this privilege.
5. Click Save.

**Related Topics**

Overview of Roles and Role Based Security, page 11-1

Administering Roles, page 11-3

Administering People, Groups, and Companies, page 11-6

Implementing Role Based Security, page 11-11

Implementing Change Management Role Based Security, page 11-24

---

# Setting Up Inbound Product Data Synchronization and Data Quality Management

This chapter covers the following topics:

- Setup Overview of Inbound Product Data Synchronization
- Creating Match Rules
- Defining Source Systems

## Setup Overview of Inbound Product Data Synchronization

Oracle Product Hub uses the Import Workbench to bring product data from disparate systems into a master product information repository, known as the Product Information Management Data Hub (PIMDH).

While importing data into a centralized data model, the Import Workbench identifies and resolves duplicates and errors while at the same time enriching existing data with external information. This process creates a blended record, known as the single source of truth. You can then create business reports and other documents related to the updates.

In order to import data from an external source system, you must define a source system for each data source and create match rules to help match imported data to existing data in the product information repository.

## Related Topics

Creating Match Rules, page 12-2

Defining Source Systems, page 12-5

Overview of Inbound Product Data Synchronization and Data Quality Management,  
*Oracle Product Hub User's Guide*

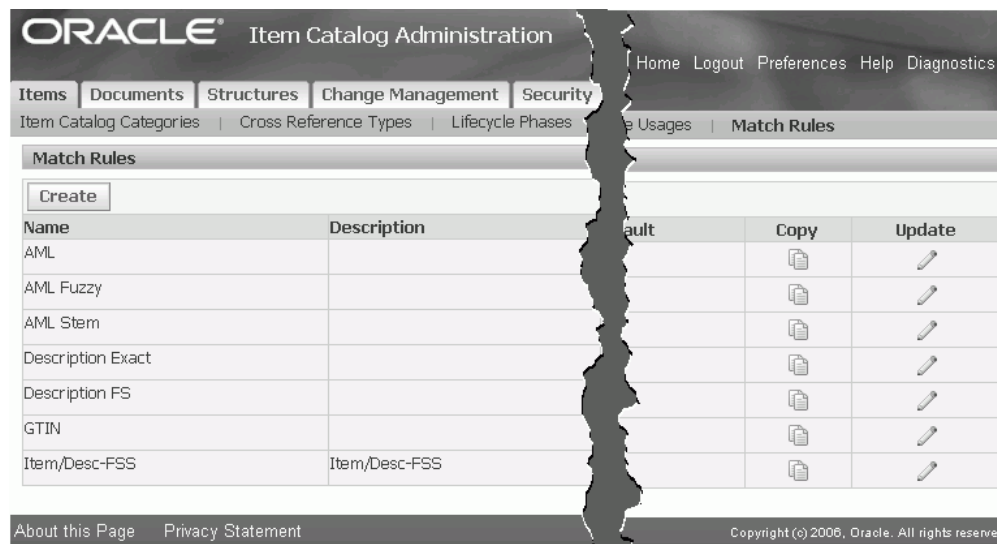
## Creating Match Rules

A match rule defines matching criteria in order to find matches between the incoming external (source system) items and Oracle Product Information Management Data Hub (PIMDH) items. Create match rules at the item catalog level.

After uploading the records into the interface table, PIMDH determines if cross-references for items exist in the production table. If the source system item matches a PIMDH item name, the system confirms the match. At the source system batch level, users can define the default matching rule and choose whether to run the matching rule automatically upon data upload. If you choose to run the matching rule automatically, then a concurrent request launches to find a match using the default matching rule for all unconfirmed records. Alternatively, you can select a match rule and launch the concurrent request to find a match for all unconfirmed items. You can run different matching rules against each batch one at a time while searching to match batch items to the appropriate PIMDH items. The matching process only applies to unconfirmed items. You cannot make changes to the batch while the matching concurrent request executes. The system searches to match source system items against all PIMDH items, including unapproved items.

### To create a match rule for an item catalog:

1. Navigate to the Setup Workbench. From the Items tab, click the **Match Rules** sub-tab. Click **Create**.



2. **Name** the match rule and provide a **Description**. Check **Set as Default** to have this match rule automatically appear as the default item catalog match rule for source systems. Optionally, you can select another default match rule when defining a source system.

3. Click **Add Criteria**.

4. Find and select the item attributes for which you want to add criteria using either the **Attribute Name**, **Business Entity**, or **Attribute Group** field. Click **Apply**.

Choose from the following business entities:

- Item
- Item Organization/Store
- Item Revision
- Item Supplier

**Tip:** You can match the following attributes by keywords, using fuzzy, stemming, and synonym searches:

- item
- item description
- item long description
- item catalog category
- manufacturer
- manufacturer part number
- supplier
- supplier part number

All other attributes must match exactly. To use synonym searches, you must load a thesaurus into the database and set the site level EGO: Thesaurus for Matching profile option to Yes.

Items Documents Structures Change Management Security Functions Catalogs Value Sets

Item Catalog Categories Cross Reference Types Lifecycle Phases Attribute Groups Templates GTIN Item Type Attrib

Items: Match Rules >

**Create Match Rule**

\* Indicates required field

Item Catalog Category **Item Catalog**

\* Name LAD Match

Description LAD Match

Revision **Latest**

☐ Set as Default

Cancel Apply

---

**Match Criteria**

✓TIP Keyword Search using Oracle Text Index is available for Item Number, Item Short and Long Descriptions, Item Catalog Category, Manufacturer and Manufacturer's Part Number. All other attributes are equivalent matches.

Select Criteria: Delete Add Criteria

Select All Select None

| Select Entity                                    | Attribute Group  | Attribute       | Search  | Keyword Match Condition | Additional Match Operators |                          |                          |
|--------------------------------------------------|------------------|-----------------|---------|-------------------------|----------------------------|--------------------------|--------------------------|
|                                                  |                  |                 |         |                         | Fuzzy                      | Stemming                 | Synonyms                 |
| <input type="checkbox"/> Item                    | Primary          | ^ Description   | Exact   |                         |                            |                          |                          |
| <input type="checkbox"/> Item                    | Primary          | ^ Item          | Keyword | All                     | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| <input type="checkbox"/> Item                    | Primary          | Approval Status | Exact   | Any                     |                            |                          |                          |
| <input type="checkbox"/> Item Supplier           | Primary          | ^ Supplier Name | Exact   | All                     |                            |                          |                          |
| <input type="checkbox"/> Item Supplier           | Primary          | Supplier Status | Exact   |                         |                            |                          |                          |
| <input type="checkbox"/> Item Organization/Store | Asset Management | Asset Item Type | Exact   |                         |                            |                          |                          |

- Once you have added all of the criteria, click **Apply**.

### To optionally create a match rule for an item catalog category:

After defining match rules at the item catalog level, you can optionally create additional match rules for individual item catalog categories.

- Navigate to the Setup Workbench. Click the **Item Catalog Categories** tab. Search for the item catalog category, then click an item catalog category name link.
- Select **Match Rules** from the list of options under the Item Catalog Categories sub-tab. From here, follow the same procedure to create match rules for the item catalog category as you did to create match rules for an item catalog.

**Note:** The item catalog match rules are listed along with the item catalog category match rules, but you cannot update them from within the item catalog category.



ORACLE®

Item Catalog Administration

[Home](#)
[Logout](#)
[Preferences](#)
[Help](#)
[Diagnostics](#)

[Items](#)
[Documents](#)
[Structures](#)
[Change Management](#)
[Security](#)
[Functions](#)
[Catalogs](#)
[Value Sets](#)

[Item Catalog Categories](#)
[Cross Reference Types](#)
[Lifecycle Phases](#)
[Attributes](#)
[Templates](#)
[GTIN](#)
[Item Type](#)

[Basic Information](#)
[People](#)
[Attribute Groups](#)
[Item Pages](#)
[Lifecycles](#)
[Attachment Categories](#)
[Search Criteria](#)
[Match Rules](#)
[Display Formats](#)
[Import Formats](#)
[Templates](#)
[Report Templates](#)
[Item Creation](#)

- [New Item Request](#)
- [Number Generation](#)
- [Description Generation](#)

Items: Item Catalog Categories > Basic Information >

Match Rules

Item Catalog Category: Motherboards

Create

| Name              | Description   | Item Catalog Category | Default | Copy | Update |
|-------------------|---------------|-----------------------|---------|------|--------|
| AML               |               | Item Catalog          | No      |      |        |
| AML Fuzzy         |               | Item Catalog          | No      |      |        |
| AML Stem          |               | Item Catalog          | No      |      |        |
| Description Exact |               | Item Catalog          | Yes     |      |        |
| Description FS    |               | Item Catalog          | No      |      |        |
| GTIN              |               | Item Catalog          | No      |      |        |
| Item/Desc-FSS     | Item/Desc-FSS | Item Catalog          | No      |      |        |
| LAD match         | LAD match     | Item Catalog          | No      |      |        |

[About this Page](#)
[Privacy Statement](#)

Copyright (c) 2006, Oracle. All rights reserved.

## Related Topics

Overview of Inbound Product Data Synchronization and Data Quality Management, *Oracle Product Hub User's Guide*

Defining Source Systems, page 12-5

## Defining Source Systems

When importing data from a source other than Oracle Product Information Management Data Hub (PIMDH), you must create a source system within Import Workbench.

### To define a source system:

1. Navigate to the Import Workbench and select the Source System Setup tab.
2. Click **Define Source System** to create a new source system.

Import | Publish History | **System Setup**

System Setup >

**Define Source System**

\* Indicates required field

\* Name

Type **Spoke**

Description

\* Code

Status **Active**

☒ Enable For Items

☐ Enable for Data Pool

**Data Load Options** | **Import Options** | **Mapping**

**Matching** | **Data Load Import**

Automatically Match on Data Load ☐ No ☒ Yes

Default Item Catalog Match Rule **Default Match Rule**

☐ Apply Default Match Rule To All Records

Automatically Confirm Single Matches ☒ No ☐ Yes

Automatically Confirm New Item on No Match ☒ No ☐ Yes

Automatically Import On Data Load ☐ No ☒ Yes

Default Revision Import Policy **Default**

3. Alternatively, you can search for and view any existing source system item that is enabled only for customer items. Enable it for items.
4. On the **Define Source System** page, enter the following for the source system:
  - a unique name
  - a unique code

**Note:** The code can be alphanumeric, but must not contain any spaces. Ensure that the code matches the System Code defined on the System Setup page within the corresponding Oracle Application Integration Architecture (AIA).

Select the **Enable For Items** check box if you plan to use this source system for item-related data.

**Additional Information:** Oracle Product Hub uses the same architecture as Oracle Customer Data Hub. The Enable For Items check box communicates that you plan to enter item-related data. You can always uncheck this box at a later time.

Select the **Enable for Data Pool** check box if the source system is a GDSN data pool. The system can perform certain tasks for GDSN batches, such as:

- using a group ID to process an entire item hierarchy
- automatically sending a message to the data pool once data is imported
- enabling you to send messages to the data pool as needed

**Additional Information:** You can specify a GDSN-enabled source system for both GDSN and non-GDSN batches. However, you cannot specify a non-GDSN source system for a GDSN batch.

5. On the **Define Source System** page, Click the Data Load Options sub-tab.

6. Select Yes by **Automatically Match on Data Load** if you want to run match rules.

**Tip:** If you want to load data from a source system and manually inspect it before running match rules, then select No.

7. If you are using match rules, then select a **Default Item Catalog Match Rule** from the list.

8. Determine whether to check or uncheck **Apply Default Match Rule To All Records**.

- **Uncheck** - The system applies the default item catalog category-level match rule to unconfirmed items with an assigned item catalog category. The system assigns the item catalog level match rule for items with no item catalog category.

**Check** - The system applies the default item catalog-level match rule to all unconfirmed items.

9. Select Yes if you want to **Automatically Confirm Single Matches** that you find when running match rules for the entire batch. If you select No, then the system tries to match all unconfirmed records using the default match rule.

10. Select Yes if you want to **Automatically Confirm New Item on No Match**. When you run match rules for the entire batch, this option enables you to automatically confirm an item with no matches as a new item. If you select No, then you must manually move any unmatched items from the Unconfirmed tab to the Confirmed tab, where they are marked with the match status of New.

**Note:** If the Item Number and Description are not provided, the Source System Item Reference and Description will be used as the Item Number and Description by default.

11. Select the appropriate **Import** option:
  - Immediately on data load - automatically updates the Oracle Item Master table with matched data.
  - Schedule Data and Time - select the date and time to import the confirmed items or structures in the batch.
  - None - leaves the data in the Import Workbench. You can manually import the data at a later time.
12. Select the appropriate **Default Revision Import Policy**:
  - Create New - creates a new version of the item.
  - Update Latest - updates the latest item revision upon import.
13. Click the **Import Options** sub-tab.

Define default import options here. You can change the import options for a particular batch later, if necessary.

**ORACLE® Item Import Management**  
Organization: Vision Operations (V1) Home Logout Pr

Import | **Source System Setup**  
Source System Setup >

**Define Source System**  
\* Indicates required field

\* Name  \* Code   
Type **Spoke** Status **Active**   
Description   
☒ Enable For Items  
☐ Enable for Data Pool

**Data Load Options** **Import Options** **Mapping**

**Change Order**  
Change Order  None

**Structure**  
Structure Type  Root  
Structure Name  Primary  
Effectivity Control Type  Date  
Structure Contains  Changed Components Only

**New Item Request Option**  
New Item Request  None

**Data Import**  
Data Import  Imp

14. Select one of the following **Change Order** options:

- Add To Existing - associates the imported data with an existing change order.
- Create New - creates a new change order for the imported data.

**Additional Information:** When creating a new change order, you have the opportunity to enter or select the following:

- Change Order Category
- Change Order Type
- Change Order Name
- Change Order Description

Once you select the change order type, depending on the setup, the Change Order Number field is automated or you must manually enter a change order number.

- None - imports the data into PIMDH without going through a change approval process.

15. Select one of the following **New Item Request** options:

- None - you cannot create any new item requests from this source system. Instead, the system creates draft items. Later, you can navigate to the Item Workbench, further define the draft items, and add it to a new item request. See:
- One per Item Catalog Category - create one new item request per item catalog category within an imported batch. This new item request can contain multiple line items.
- One per Item - create one new item request per item.

For more information about new item requests, see: *Creating New Item Requests, Oracle Product Hub User's Guide*.

16. If you are importing structures, select the following default options:

- Structure Type

**Caution:** Oracle Product Hub recommends avoiding the use of the Root structure type. User-defined attributes and import formats are not supported for the Root structure type.

- Structure Name
- Effectivity Control Type - choose Date or Model/Unit Number
- Structure Contains - choose All Components or Changed Components Only

17. Select one of the following **Data Import** options:

- Import All Data
- Create Cross References Only - Select this option if you want to create cross-references only. It will not import the data.

18. Click the **Mapping** sub-tab. Select one of the following **Mapping Process** options:

- Define in PIMDH - Enables you to define the mapping of attributes between an external source system and PIM Data Hub within PIM Data Hub.
- External - Enables you to upload an XSLT file containing mapping information between an external source system and PIM Data Hub.
- None - No mapping occurs.

ORACLE®

Item Import Management

Organization: Vision Operations (V1)

Import

Source System Setup

Source System Setup >

Define Source System

\* Indicates required field

\* Name

\* Code

Type

Spoke

Description

Status

Active

☒ Enable For Items

Data Load Options

Import Options

Mapping

Mapping Process

None

Define in PDH

External

None

About this Page

Privacy Statement

## Related Topics

Overview of Inbound Product Data Synchronization and Data Quality Management,  
*Oracle Product Hub User's Guide*

Creating Match Rules, page 12-2





---

## Workflows for Change Management

This appendix covers the following topics:

- Workflows for Change Management

### Workflows for Change Management

The system uses Oracle Workflow technology to automate business processes in Change Management and Group Member registration. You must set up Oracle Workflow to implement Oracle Product Hub.

**Additional Information:** Run wfver.sql under \$FND\_TOP/sql for detailed information about workflow versions.

### Workflow Setup

Refer to the Oracle Workflow Implementation and User Documentation required for setup.

### Overview of Change Management Workflows

Change Management uses Oracle Workflow technology to execute workflow routing for change objects and events occurring in a change object (such as Submit, Reassign, Priority Change, and Status Change actions). Change Management enables you to model your business processes in terms of generic order processes. You can model your business processes by customizing or extending seeded workflow processes and/or modifying custom hook PL/SQL procedures.

For more information about customizing workflow processes, refer to Customizing Change Management Workflows, page 7-47.

## Viewing Processes in Oracle Workflow Builder

From within the Oracle Workflow Builder you can view the different workflow processes and their associated functions, messages, subprocesses, notifications, and properties.

To view processes in Oracle Workflow Builder:

1. Within Oracle Workflow Builder, select Open from the File menu and connect to the database. Alternatively, you can connect to the workflow definitions file engchgmt.wft located in the product directory tree of your Oracle Applications server.
2. Expand the data source, and then select the item type branch within that data source.
3. Expand the processes branch within your item type and select a process activity to display the diagram of the process in a Process window.

| Display Name (Internal Name) | Purpose                                                                                                                                                                                                                                                                                                              | Location of Flat File Store on Server (Location of other Flat File Stores with Item Type embedded) |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| ENG: Standard(ENGWFSTD)      | Contains a collection of commonly used (by other Eng Workflow Item Types) Function Activities and Lookup Types.If the activity will be shared by more than one object's process, create the activity in the ENG: Standard (ENGWFSTD) item type. Otherwise, create the activity in the appropriate object's item type | \$ENG_TOP/patch/115/import //engchgmt.wft                                                          |
| Change (ENGCACT)             | Contains activities, attributes, notifications, and processes used by Change Action Process                                                                                                                                                                                                                          | \$ENG_TOP/patch/115/import //engchgmt.wft                                                          |
| Change Line (ENGCLACT)       | Contains activities, attributes, notifications, and processes used by Change Line Action Process                                                                                                                                                                                                                     | \$ENG_TOP/patch/115/import //engchgmt.wft                                                          |

| Display Name (Internal Name)            | Purpose                                                                                                    | Location of Flat File Store on Server (Location of other Flat File Stores with Item Type embedded) |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Change Workflow Routing (ENGCRRT)       | Contains activities, attributes, notifications, and processes used by Change Approval Routing Process      | \$ENG_TOP/patch/115/import//engchmgt.wft                                                           |
| Change Workflow Routing Step(ENGCSSTEP) | Contains activities, attributes, notifications, and processes used by Change Approval Routing Step Process | \$ENG_TOP/patch/115/import//engchmgt.wft                                                           |

## Pre-Seeded Workflow Processes

### *Change (ENGCACT)*

| Process Internal Name | Process Display Name | Description                                                                                                                                                               |
|-----------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INITIATE_CHANGE       | Initiate Change      | Sends an assignment notification to assignee and starts "Initiate Change Line" workflow if Change has a line. This workflow is started when users submit a Change Object. |
| PRIORITY_CHANGE       | Priority Change      | Sends a priority change notification to Requestor, Owner, and Assignee for changes. This workflow is started when users change the priority for a Change Object.          |
| REASSIGN_CHANGE       | Reassign Change      | Sends an assignment notification to assignee. This workflow is started when users reassign a Change Object.                                                               |

| Process Internal Name | Process Display Name | Description                                                                                                                                                                                                                                                                                |
|-----------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQUEST_COMMENT       | Request Comment      | Sends a notification to request comment to specified people on the Request Comment Action page. When the recipient responds to the request notification, Workflow sends a response notification to the requestor. This workflow is started when users request comment for a Change Object. |
| RESPONSE_FYI          | Send Response FYI    | This process is called when the recipient responds to the request notification in Request Comment workflow process.                                                                                                                                                                        |
| STATUS_CHANGE         | Status Change        | Sends a status change notification to Requestor, Owner, and Assignee for a change. This workflow is started when users change the status for a Change Object.                                                                                                                              |

#### ***Change Line (ENGCLACT)***

| Process Internal Name | Process Display Name | Description                                                                                                                                                                             |
|-----------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INITIATE_CHANGE       | Initiate Change      | Sends an assignment notification to assignee. This workflow is started from the Change Header's Initiate Change Workflow or when a user creates a Change Line if Change Header is Open. |
| REASSIGN_CHANGE       | Reassign Change      | Sends an assignment notification to the assignee. This workflow is started when a user reassigns a Change Line.                                                                         |

| Process Internal Name | Process Display Name | Description                                                                                                                                                                                 |
|-----------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STATUS_CHANGE         | Status Change        | Sends a status change notification to Header Requestor, Header Owner, Header Assignee, Line Owner and Line Assignee. This workflow is started when a user changes status for a Change Line. |

#### ***Change Workflow Routing (ENGCR)***

| Process Internal Name | Process Display Name   | Description                                                                                                                                                                                                                                                                       |
|-----------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROUTE_AGENT           | Workflow Routing Agent | Controls Workflow Routing and sends Approval Status change notification to Requestor, Owner, and Assignee for a Change. This workflow is started when a user submits an Workflow Routing for a Change Object. Also this workflow is aborted when user aborts an Workflow Routing. |

### ***Change Workflow Routing Step (ENGCSSTEP)***

| <b>Process Internal Name</b>    | <b>Process Display Name</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQUEST_DEFINITION              | Definition                  | Sends a definition request notification to specified assignees on Workflow Routing page and controls definition request for New Item Request. If the assignees do not respond by the specified period, workflow will also send a reminder notification. Once the step is completed, workflow will return control parent Workflow Routing Agent workflow. This workflow is started from Workflow Routing Agent based on the definition of Workflow Routing.                           |
| REQUEST_DEFINITION_A<br>ND_APPR | Definition and Approval     | Sends a definition and approval request notification to specified assignees on Workflow Routing page and controls definition and approval request for New Item Request. If the assignees do not respond by the specified period, workflow will also send a reminder notification. Once the step is completed, workflow will return control parent Workflow Routing Agent workflow. This workflow is started from Workflow Routing Agent based on the definition of Workflow Routing. |

| Process Internal Name | Process Display Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQUEST_APPROVAL      | Request Approval     | Sends an approval request notification to specified assignees on Workflow Routing page and controls approval request. If the assignees do not respond by the specified period, workflow will also send a reminder notification. Once the step is approved, workflow will return control parent Workflow Routing Agent workflow. This workflow is started from Workflow Routing Agent based on the definition of Workflow Routing. |
| REQUEST_COMMENT       | Request Comment      | Sends a comment request notification to specified assignees on Workflow Routing page and controls comment requests. If the assignees do not respond by the specified period, workflow will also send a reminder notification. Once the step is completed, workflow will return control parent Workflow Routing Agent workflow. This workflow is started from Workflow Routing Agent based on the definition of Workflow Routing.  |

| Process Internal Name | Process Display Name | Description                                                                                                                                                                                                                                                                   |
|-----------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEND_FYI              | FYI                  | Sends an FYI notification to specified assignees on Workflow Routing page. Once the step is completed, workflow will return control parent Workflow Routing Agent workflow. This workflow is started from Workflow Routing Agent based on the definition of Workflow Routing. |

Following are two Process Attributes for the Workflow Routing Step workflow process:

**1. Default Change Role Instance Set (DEFAULT\_CHANGE\_ROLE)**

Assigned on current or last Workflow Routing. Implicit grants securely based on the steps workflow process.

ENG\_CHANGE\_WF\_APPROVERS -> Granted Role: Approver  
(ENG\_CHANGE\_APPROVER)

ENG\_CHANGE\_WF\_REVIWERS -> Granted Role: Reviewer  
(ENG\_CHANGE\_REVIEWER).

Implicit Role (Instance Set) to be granted is defined in the Step workflow process attributes

DEFAULT\_CHANGE\_ROLE

In case of seeded Step Workflow Process

Request Approval (DEFAULT\_CHANGE\_ROLE:  
"ENG\_CHANGE\_WF\_APPROVERS")

The step assignees get "Approver" implicit role from instance set  
ENG\_CHANGE\_WF\_APPROVERS.

Request Comment & FYU (DEFAULT\_CHANGE\_ROLE:  
"ENG\_CHANGE\_WF\_REVIEWERS")

The step assignees get "Reviewer" implicit role from instance set  
ENG\_CHANGE\_WF\_REVIEWERS. Oracle Workflow does not support other roles  
by default.

**2. Activity Condition Code (ACTIVITY\_CONDITION\_CODE)**

Attribute will indicate whether or not the Condition radio box is displayed on the Workflow Routing Create/Update step page. Acceptable values are Y or N.



**3. Enable Flag in Type: Definition (DEFINITION)**

Attribute will indicate whether or not the this process is displayed in the Workflow Process pull-down list based on the Definition Type of Workflow Routing (Template) on the Workflow Routing Create/Update step page. Acceptable values are Y or N.

**4. Enable Flag in Type: Approval (APPROVAL)**

Attribute will indicate whether or not the this process is displayed in the Workflow Process pull-down list based on the Approval Type of Workflow Routing (Template) on the Workflow Routing Create/Update step page. Acceptable values are Y or N.

**5. Enable Flag in Type: Definition and Approval (DEFINITION\_APPROVAL)**

Attribute will indicate whether or not the this process is displayed in the Workflow Process pull-down list based on the Definition and Approval Type of Workflow Routing (Template) on the Workflow Routing Create/Update step page. Acceptable values are Y or N.

**6. Enable Flag in Type: Generic (GENERIC)**

Attribute will indicate whether or not the this process is displayed in the Workflow Process pull-down list based on the Generic Type of Workflow Routing (Template) on the Workflow Routing Create/Update step page. Acceptable values are Y or N.

**7. Association Enabled Flag (ASSOC\_ENABLED)**

Attribute will indicate whether or not the the association for the assignee is available in the Workflow Routing Step for New Item Request on the Create/Update step page. Acceptable values are Y or N. This attribute is only applicable for "Definition" and "Definition and Approval" Workflow.

## **Seeded Workflow Item Attributes**

The following table describes the common item attributes for Change Management Workflow. Each Workflow Item Type should have these attributes.

**All Item Type**

| <b>Internal Name</b>   | <b>Display Name</b>    | <b>Description</b>                   | <b>Type</b> | <b>Length</b> | <b>Misc info (used for etc)</b> |
|------------------------|------------------------|--------------------------------------|-------------|---------------|---------------------------------|
| CHANGE_ID              | Change Id              | Engineering Change Unique Identifier | Number      | -             | Primary Key Values.             |
| CHANGE_NOTICE          | Change Number          | Engineering Change Number            | Text        | 30            | Old Primary Key Values.         |
| CHANGE_NAME            | Change Name            | Engineering Change Name              | Text        | 240           | Token replacement for messages. |
| ORGANIZATION_ID        | Organization Id        | Organization Id                      | Number      | -             | Old Primary Key Values.         |
| ORGANIZATION_CODE      | Organization Code      | Organization Code                    | Text        | 3             | Token replacement for messages. |
| CHANGE_MANAGEMENT_TYPE | Change Management Type | Change Management Type               | Text        | 40            | Token replacement for messages. |
| DESCRIPTION            | ECO Description        | ECO Description                      | Text        | 2000          | Token replacement for messages. |
| STATUS                 | Status                 | Change Status                        | Text        | 80            | Token replacement for messages. |
| APPROVAL_STATUS        | Approval Status        | ECO Approval Status                  | Text        | 80            | Token replacement for messages. |
| PRIORITY               | Priority               | Priority                             | Text        | 50            | Token replacement for messages. |

| Internal Name          | Display Name           | Description                                | Type   | Length | Misc info (used for etc)                                                      |
|------------------------|------------------------|--------------------------------------------|--------|--------|-------------------------------------------------------------------------------|
| REASON                 | Reason                 | Reason                                     | Text   | 50     | Token replacement for messages.                                               |
| ASSIGNEE_NAME          | Assignee Name          | Assignee Name                              | Text   | 360    | Token replacement for messages.                                               |
| ASSIGNEE_COMPANY       | Assignee Company       | Assignee Company                           | Text   | 360    | Token replacement for messages.                                               |
| WF_USER_ROLE           | WF Owner User Role     | Role of Person Who Started the Workflow    | Role   | -      | Runtime information for the work item.                                        |
| WF_USER_ID             | WF Owner User Id       | User Id of Person Who Started the Workflow | NUMBER | -      | Runtime information for the work item.                                        |
| HOST_URL               | Host URL and port      | Host URL and port                          | Text   | -      | Runtime information for the work item.                                        |
| DEFAULT_STYLE_SHEET    | Default Style Sheet    | Default Style Sheet                        | Text   | 100    | /OA_HTML/cabo/styles/bla f. cssTemporary placeholders for HTML notifications. |
| CHANGE_DETAIL_PAGE_URL | Change Detail Page URL | Change Detail Page URL                     | URL    | -      | -                                                                             |

| Internal Name     | Display Name                  | Description                        | Type     | Length | Misc info (used for etc)                                                                                    |
|-------------------|-------------------------------|------------------------------------|----------|--------|-------------------------------------------------------------------------------------------------------------|
| MESSAGE_TEXT_BODY | Message Text Body             | Message Text Body                  | Document | -      | Temporary placeholders for notifications.                                                                   |
| MESSAGE_HTML_BODY | Message HTML Body             | Message HTML Body                  | Document | -      | Temporary placeholders for notifications.                                                                   |
| REVIEWERS_ROLE    | Change Standard Reviewer Role | Change Standard Reviewer Role      | Role     | -      | Temporary placeholders for notifications.                                                                   |
| ASSIGNEE_ROLE     | Assignee Role                 | Assignee Role                      | Role     | -      | Temporary placeholders for notifications.                                                                   |
| ACTION_ID         | Change Action Id              | Change Action Id                   | Number   | -      | Primary Key for Change Actions (ENG_CHANGE_ACTIONS.ACTION_ID)<br><br>Runtime information for the work item. |
| ADHOC_PARTY_LIST  | Adhoc Party List              | Adhoc Party List                   | Text     | -      | Runtime information for the work item.                                                                      |
| ADHOC_PARTY_ROLE  | Adhoc Party Role              | Adhoc Party Role for Change Action | Role     | -      | Temporary placeholders for notifications.                                                                   |

| Internal Name    | Display Name       | Description                                                             | Type     | Length | Misc info (used for etc)                                                                        |
|------------------|--------------------|-------------------------------------------------------------------------|----------|--------|-------------------------------------------------------------------------------------------------|
| RESPONSE_TIMEOUT | Response Timeout   | Response Timeout in Minutes for Action calculated from Response_By_Date | Number   | -      | -                                                                                               |
| RESPONSE_COMMENT | Response Comment   | Response Comment                                                        | Text     | -      | Temporary placeholders for notifications.                                                       |
| ATTACHMENTS      | Attachments        | Attachments                                                             | Document | -      | Temporary placeholders for notifications: Notification Detail Page Attachment LinkFND: entity== |
| WF_USER_ID       | WF Owner User Id   | User Id of Person Who Started/Aborted the Workflow                      | NUMBER   | -      | Runtime information for the work item.                                                          |
| WF_USER_ROLE     | WF Owner User Role | Role of Person Who started/aborted the Workflow                         | Role     | -      | Runtime information for the work item.                                                          |

| Internal Name    | Display Name           | Description           | Type   | Length | Misc info (used for etc)                                               |
|------------------|------------------------|-----------------------|--------|--------|------------------------------------------------------------------------|
| WF_ADMINISTRATOR | WF Administrator       | WF Administrator      | Role   | -      | Constant: SYSADMINF or WFERROR: DEFAULT_ERROR (Default Error Process.) |
| WF_SIG_POLICY    | Signature Policy       | Signature Policy      | Text   | -      | To use Signature Policy defaulting in future.                          |
| ERROR_TIMEOUT    | WF Error Timeout Value | Timeout Value in Days | Number | -      | Constant: 0For WFERROR: DEFAULT_ERROR (Default Error Process.)         |
| RCS              | RCS version            | RCS version           | Text   | 240    | RCS version                                                            |
| FROM_ROLE        | From Role              | From Role             | Role   | -      | Item Attribute which holds the From Role. (Message #FROM_ROLE)         |

**ENGCLACT Specific Item Attributes**

| <b>Internal Name</b>  | <b>Display Name</b>          | <b>Description</b>                        | <b>Type</b> | <b>Length</b> | <b>Misc info (used for etc)</b>           |
|-----------------------|------------------------------|-------------------------------------------|-------------|---------------|-------------------------------------------|
| CHANGE_LINE_ID        | Change Line Id               | Engineering Change Line Unique Identifier | Number      | -             | Primary Key Values.                       |
| LINE_NAME             | Change Line Name             | Engineering Change Line Name              | Text        | 240           | Token replacement for messages.           |
| LINE_DESCRIPTION      | Change Line Description      | Change Line Description                   | Text        | 4000          | Token replacement for messages.           |
| LINE_SEQUENCE_NUMBER  | Change Line Sequence Number  | Change Line Sequence Number               | Number      |               | Token replacement for messages.           |
| LINE_STATUSES         | Change Line Status           | Change Line Status                        | Text        | 80            | Token replacement for messages.           |
| LINE_ASSIGNEE_NAME    | Change Line Assignee Name    | Change Line Assignee Name                 | Text        | 360           | Token replacement for messages.           |
| LINE_ASSIGNEE_COMPANY | Change Line Assignee Company | Change Line Assignee Company              | Text        | 360           | Token replacement for messages.           |
| LINE_REVIEWERS_ROLE   | Change Line Reviewer Role    | Change Line Reviewer Role                 | Role        | -             | Temporary placeholders for notifications. |
| LINE_ASSIGNEE_ROLE    | Change Line Assignee Role    | Change Line Assignee Role                 | Role        | -             | Temporary placeholders for notifications. |

| Internal Name    | Display Name | Description | Type     | Length | Misc info (used for etc)                                                                        |
|------------------|--------------|-------------|----------|--------|-------------------------------------------------------------------------------------------------|
| LINE_ATTACHMENTS | Attachments  | Attachments | Document | -      | Temporary placeholders for notifications: Notification Detail Page Attachment LinkFND: entity== |

**ENGCRIT Specific Item Attributes**

| Internal Name    | Display Name                               | Description                                                                                                       | Type   | Length | Misc info (used for etc)                  |
|------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|-------------------------------------------|
| ROUTE_ID         | Route Id (Workflow Routing Id)             | Route Id                                                                                                          | Number | -      | Primary Key Values for Change Route.      |
| STEP_ID          | Current Step Id (Workflow Routing Step Id) | Current Step Id                                                                                                   | Number | -      | Primary Key Values for Current Step Id.   |
| ROUTE_POPLE_ROLE | Route (Workflow Routing) People Role       | All people in the Workflow Routing workflow that have already been notified (i.e. completed or in process steps). | Role   | -      | Temporary placeholders for notifications. |



**ENGSTEP Specific Item Attributes**

| <b>Internal Name</b>  | <b>Display Name</b>                      | <b>Description</b>                                                                                                | <b>Type</b> | <b>Length</b> | <b>Misc info (used for etc)</b>           |
|-----------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------|---------------|-------------------------------------------|
| ROUTE_ID              | Workflow Routing Id                      | Workflow Routing Id                                                                                               | Number      | -             | Primary Key Values for Change Route.      |
| STEP_ID               | Workflow Routing Step Id                 | Current Step Id                                                                                                   | Number      | -             | Primary Key Values for Step Id.           |
| ROUTE_PEO<br>PLE_ROLE | Workflow Routing People Role             | All people in the Workflow Routing workflow that have already been notified (i.e. completed or in process steps). | Role        | -             | Temporary placeholders for notifications. |
| STEP_SEQ_N<br>UM      | Workflow Routing Step Sequence Number    | Workflow Routing Step Sequence Number                                                                             | Number      | -             | Temporary placeholders for notifications. |
| STEP_PEOPL<br>E_ROLE  | Workflow Routing Step People Role        | Workflow Routing Step People Role                                                                                 | Role        | -             | Temporary placeholders for notifications. |
| STEP_COND<br>ITION    | Workflow Routing Step Activity Condition | Workflow Routing Step Activity Condition                                                                          | Text        | 30            | Temporary placeholders for notifications. |

| Internal Name              | Display Name                                               | Description                                                | Type   | Length | Misc info (used for etc)                 |
|----------------------------|------------------------------------------------------------|------------------------------------------------------------|--------|--------|------------------------------------------|
| STEP_VOTING_OPTION         | Workflow Routing Step Voting Option for Activity Condition | Workflow Routing Step Voting Option for Activity Condition | Lookup | -      | Standard Voting Option.                  |
| YES_RESPONSE_PERCENTAGE_NT | Yes Response Percentage of Votes                           | Yes Response Percentage of Votes                           | Number | -      | Temporary placeholders for notifications |
| NO_RESPONSE_PERCENTAGE_T   | No Response Percentage of Votes                            | No Response Percentage of Votes                            | Number | -      | Temporary placeholders for notifications |

## Change Management Workflow PL/SQL Packages

Change Management Workflow Private APIs will contain the following list of packages. These are private APIs for reference purposes only for your customization.

| Package Name      | File Name                    | Description                                                                                                                             | Location of Flat File Store on Server |
|-------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| ENG_WORKFLOW_PUB  | ENGBWKFS.pls<br>ENGBWKFB.pls | Private APIs: for developers or consultants at Oracle:<br><br>This package contains All PL/SQL Function for Workflow Function Activity. | \$ENG_TOP/patch/115/sql               |
| ENG_WORKFLOW_UTIL | ENGUWKFS.pls<br>ENGUWKFS.pls | Private APIs: This package contains utility functions and procedures for Eng Workflows.                                                 | \$ENG_TOP/patch/115/sql               |

| Package Name              | File Name                    | Description                                                                            | Location of Flat File Store on Server |
|---------------------------|------------------------------|----------------------------------------------------------------------------------------|---------------------------------------|
| ENG_WORKFLOW_<br>NTF_UTIL | ENGUNTFS.pls<br>ENGUNTFS.pls | This package contains utility functions and procedures for Eng Workflow Notifications. | \$ENG_TOP/patch/115<br>/sql           |

| Package Name         | File Name                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Location of Flat File Store on Server |
|----------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| ENG_WORKFLOW_<br>EXT | ENGXWKFS.pls<br>ENGXWKFB.pls | <p>This package contains procedures that serve as "hooks" into the Eng Workflows for customization of the workflows.</p> <p>The "StartCustomWorkflow " and the "AbortCustomWorkflow" are called from ENG_WORKFLOW_UTIL package when a workflow process is started and aborted, respectively. Users with customized workflows can add their custom logic in these two procedures to be executed at the appropriate times.</p> <p>Also GetCustomMessageBody is called from the seeded PL/SQL Document APIs when system generates notification. Users can customize the PL/SQL procedure to modify notification contents.</p> <p>Additionally, users can create new procedures in this package for custom workflow activities.</p> | \$ENG_TOP/patch/115<br>/sql           |

---

## Item and Catalog Business Events

This appendix covers the following topics:

- Business Events

### Business Events

Using Oracle Workflow, you can automatically trigger actions based on the following business events. Examples of actions you can trigger include issuing notifications and launching a workflow.

The following table lists business events, as well as a description and the event parameters of each event.

#### *Item Business Events*

| Event Name                              | Event Description                                                                                                     | Event Parameter Names and Descriptions                                                                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>postAML.Change | Item AML is changed (such as a manufacturer part number is added, updated, or deleted)<br><br>Applicable: SSWA, FORMS | DML_TYPE: CREATE, UPDATE, DELETE<br><br>INVENTORY_ITEM_ID: Item ID<br><br>ORGANIZATION_ID: Organization ID<br><br>MANUFACTURER_ID: Manufacturer ID<br><br>MFG_PART_NUM: Manufacturer Part Number |

| Event Name                                       | Event Description                                                                  | Event Parameter Names and Descriptions                                                                                                                                                 |
|--------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>postItemCreate          | This event is raised after an item is created.<br><br>Applicable: SSWA, FORMS, API | INVENTORY_ITEM_ID: Item ID<br><br>ITEM_NUMBER: Item Name<br><br>ORGANIZATION_ID: Organization ID<br><br>ORGANIZATION_CODE: Organization Code<br><br>ITEM_DESCRIPTION: Item Description |
| oracle.apps.ego.item.<br>postItemUpdate          | This event is raised after an item is updated.<br><br>Applicable: SSWA, FORMS, API | INVENTORY_ITEM_ID: Item ID<br><br>ITEM_NUMBER: Item Name<br><br>ORGANIZATION_ID: Organization ID<br><br>ORGANIZATION_CODE: Organization Code<br><br>ITEM_DESCRIPTION: Item Description |
| oracle.apps.ego.item.<br>postGTINAttributeChange | This event is obsolete.                                                            | -                                                                                                                                                                                      |
| oracle.apps.ego.item.<br>postItemApproved        | On Item Approval<br><br>Applicable: SSWA                                           | INVENTORY_ITEM_ID: Item ID<br><br>ORGANIZATION_ID: Organization ID                                                                                                                     |

| Event Name                                  | Event Description                                                                                                                                                              | Event Parameter Names and Descriptions                                                                                                                                                                                                                                                                                      |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>postItemBulkload   | <p>Raised during bulk item processing that is create or update. This event is applicable to spreadsheet import, and Item Open Interface.</p> <p>Applicable: XL import, IOI</p> | <p>REQUEST_ID: Concurrent Request ID that processed the bulk item import.</p> <p><b>Additional Resources:</b><br/>If a running Bulkload Program For Spread Sheet Upload concurrent program terminates unexpectedly, then refer to Instructions for Restarting Concurrent Programs with a Manual Restart Mode, page H-1.</p> |
| oracle.apps.ego.item.<br>postXrefChange     | <p>Item cross-reference creation, update, or deletion</p> <p>Applicable : SSWA, FORMS</p>                                                                                      | <p>DML_TYPE: CREATE, UPDATE, DELETE</p> <p>INVENTORY_ITEM_ID: Item ID</p> <p>ORGANIZATION_ID: Organization ID</p> <p>CROSS_REFERENCE_TYPE: Cross-reference Type</p> <p>CROSS_REFERENCE: Cross-reference</p>                                                                                                                 |
| oracle.apps.ego.item.<br>postRevisionChange | <p>This event is raised after an item revision is created or updated.</p> <p>Applicable: SSWA, FORMS, IOI, XL import, API</p>                                                  | <p>DML_TYPE: CREATE, UPDATE, DELETE</p> <p>INVENTORY_ITEM_ID: Item ID</p> <p>ORGANIZATION_ID: Organization ID</p> <p>REVISION_ID: Revision ID</p> <p>REQUEST_ID: Concurrent Request ID</p>                                                                                                                                  |

| Event Name                                  | Event Description                                                                                                                                                              | Event Parameter Names and Descriptions                                                                                                                                                                                                                                                 |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>postItemRoleChange | An item role grant is changed (added, modified, or removed)<br><br>Applicable: SSWA                                                                                            | DML_TYPE: CREATE, UPDATE, DELETE<br><br>INVENTORY_ITEM_ID: Item ID<br><br>ORGANIZATION_ID: Organization ID<br><br>ROLE_ID: Role ID<br><br>PARTY_TYPE: Party Type (Person, Group or Company)<br><br>PARTY_ID: Party ID<br><br>START_DATE: Start Date                                    |
| oracle.apps.ego.batch.<br>postbatchprocess  | Post Batch Import of Item and Item Related Entities<br><br>This event is raised when a batch is processed.                                                                     | REQUEST_ID: Concurrent Request ID that processes the batch.                                                                                                                                                                                                                            |
| oracle.apps.ego.item.<br>preAttributeChange | Item Attribute group row is created, updated, or deleted.<br><br>This event is raised before the change is committed.<br><br>Applicable: SSWA, Concurrent Program, WebADI, API | DML_TYPE: CREATE, UPDATE, DELETE<br><br>ATTR_GRP_NAME: Attribute Group Name<br><br>EXTENSION_ID: Extension ID that uniquely identifies the record in the UDA extension table<br><br>INVENTORY_ITEM_ID: Item ID<br><br>ORGANIZATION_ID: Organization ID<br><br>REVISION_ID: Revision ID |



| Event Name                                   | Event Description                                                                                                                                                | Event Parameter Names and Descriptions                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>postAttributeChange | <p>Item Attribute group row is created, updated, or deleted.</p> <p>This event is raised after the change is committed.</p> <p>Applicable: SSWA</p>              | <p>DML_TYPE: Indicates the type of operation. Values can be CREATE, UPDATE, or DELETE.</p> <p>ATTR_GRP_NAME: Attribute Group Name</p> <p>EXTENSION_ID: Extension ID that uniquely identifies the record in the UDA extension table</p> <p>INVENTORY_ITEM_ID: Item ID</p> <p>ORGANIZATION_ID: Organization ID</p> <p>REVISION_ID: Revision ID</p> <p>PK1_VALUE: Supplier ID</p> <p>PK2_VALUE: SupplierSite ID</p> <p>DATA_LEVEL_ID: Data Level ID</p> |
| oracle.apps.ego.item.<br>associations.create | <p>Item Association Creation</p> <p>This event is raised after an item's association with a supplier, suppliersite, or suppliersite organization is created.</p> | <p>INVENTORY_ITEM_ID: Item ID</p> <p>ORGANIZATION_ID: Organization ID</p> <p>ASSOCIATION_ID: Association Id</p> <p>ASSOCIATION_TYPE_CODE : Indicates the type of association (supplier, suppliersite, or suppliersite organization).</p>                                                                                                                                                                                                             |

| Event Name                               | Event Description                                                                                                                                                | Event Parameter Names and Descriptions                                                                                                                                                                                                  |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.associations.delete | <p>Item Association Deletion</p> <p>This event is raised after an item's association with a supplier, suppliersite, or suppliersite organization is deleted.</p> | <p>INVENTORY_ITEM_ID: Item ID</p> <p>ORGANIZATION_ID: Org ID</p> <p>SUPPLIER_ID: Supplier ID</p> <p>SUPPLIER_SITE_ID: Supplier Site ID</p>                                                                                              |
| oracle.apps.ego.item.associations.update | <p>Item Association Update</p> <p>This event is raised after an item's association with a supplier, suppliersite, or suppliersite organization is updated.</p>   | <p>INVENTORY_ITEM_ID: Item ID</p> <p>ORGANIZATION_ID: Organization ID</p> <p>ASSOCIATION_ID: Association Id</p> <p>ASSOCIATION_TYPE_CODE: Indicates the type of association (supplier, suppliersite, or suppliersite organization).</p> |

#### ***Catalog Business Events***

| Event Name                                     | Event Description                                                                                                         | Event Parameter Names and Descriptions                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.postCatalogCategoryChange | <p>Catalog category is changed such as when categories are created, updated or deleted</p> <p>Applicable: SSWA, FORMS</p> | <p>DML_TYPE: Indicates the type of operation. Values can be CREATE, UPDATE, or DELETE.</p> <p>CATEGORY_NAME: Category Name</p> <p>CATEGORY_ID: Category ID</p> |

| Event Name                                           | Event Description                                                                                                                                                                                                                      | Event Parameter Names and Descriptions                                                                                                                                                                                                                                      |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>postCatalogAssignmentChange | <p>Post Item Catalog Assignment Change Event</p> <p>This event is raised when an item's assignment to a category (within a catalog) is created, updated or deleted.</p> <p>Applicable: SSWA, FORMS, IOI</p>                            | <p>DML_TYPE: Indicates the type of operation. Values can be CREATE, UPDATE, or DELETE.</p> <p>INVENTORY_ITEM_ID: Item ID</p> <p>ORGANIZATION_ID: Organization ID</p> <p>CATALOG_ID: Catalog ID</p> <p>CATEGORY_ID: Category ID</p> <p>REQUEST_ID: Concurrent Request ID</p> |
| oracle.apps.ego.item.<br>postValidCategoryChange     | <p>Post Valid Category Change Event</p> <p>This event is raised when a category is added to or removed from a catalog.</p> <p>This event is also raised when the attributes of catalog are changed.</p> <p>Applicable: SSWA, FORMS</p> | <p>DML_TYPE: Indicates the type of operation. Values can be CREATE, UPDATE, or DELETE.</p> <p>CATEGORY_SET_ID: Category Set ID</p> <p>CATEGORY_ID: Category ID</p> <p>PARENT_CATEGORY_ID: Parent Category ID</p>                                                            |

### ***Publishing Business Events***

| <b>Event Name</b>                                | <b>Event Description</b>                                                                                                                                                                                                                                                             | <b>Event Parameter Names and Descriptions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>syncItemCatalogCategory | A seeded event used when publishing Item Catalog Categories to other Oracle Product Hub systems. The seeded subscription invokes the Item Catalog Category Web Service, page G-18 for publishing through the Service Invocation Framework of SOA Gateway and should not be modified. | <p>Refer to Implementing Service Invocation Framework, <i>Oracle E-Business Suite Integrated SOA Gateway Implementation Guide</i> for details about using the SIF framework.</p> <p>Event Data: Input Payload for the webservice.</p> <p>WFBES_SOAP_USERNAME: SOAP user name to invoke webservice.</p> <p>WFBES_SOAP_PASSWORD_MOD: Module Name to retrieve the password stored using the FND vault.</p> <p>WFBES_SOAP_PASSWORD_KEY: Module Key to retrieve the password stored using the FND vault.</p> |

| Event Name                                    | Event Description                                                                                                                                                                                                                                                       | Event Parameter Names and Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.extfwk.<br>syncValueSet       | A seeded event used when publishing Value Sets to other Oracle Product Hub systems. The seeded subscription invokes the Value Set Web Service, page G-2 for publishing through the Service Invocation Framework of SOA Gateway and should not be modified.              | <p>Refer to Implementing Service Invocation Framework, <i>Oracle E-Business Suite Integrated SOA Gateway Implementation Guide</i> for details about using the SIF framework.</p> <p>Event Data: Input Payload for the webservice.</p> <p>WFBES_SOAP_USERNAME: SOAP user name to invoke webservice.</p> <p>WFBES_SOAP_PASSWORD_MOD: Module Name to retrieve the password stored using the FND vault.</p> <p>WFBES_SOAP_PASSWORD_KEY: Module Key to retrieve the password stored using the FND vault.</p> |
| oracle.apps.ego.extfwk.<br>syncAttributeGroup | A seeded event used when publishing Attribute Groups to other Oracle Product Hub systems. The seeded subscription invokes the Attribute Group Web Service, page G-12 for publishing through the Service Invocation Framework of SOA Gateway and should not be modified. | <p>Refer to Implementing Service Invocation Framework, <i>Oracle E-Business Suite Integrated SOA Gateway Implementation Guide</i> for details about using the SIF framework.</p> <p>Event Data: Input Payload for the webservice.</p> <p>WFBES_SOAP_USERNAME: SOAP user name to invoke webservice.</p> <p>WFBES_SOAP_PASSWORD_MOD: Module Name to retrieve the password stored using the FND vault.</p> <p>WFBES_SOAP_PASSWORD_KEY: Module Key to retrieve the password stored using the FND vault.</p> |

| Event Name                                          | Event Description                                                                                                                                                                                                                                                                                                         | Event Parameter Names and Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>publishItemCatalogCategory | A seeded event used when publishing Item Catalog Categories to external systems such as Siebel, Oracle Billing and Revenue Management (BRM), and Oracle Design Studio via Batch Mode. It is raised from the <b>Publish Item Catalog Categories Batch</b> page and indicates the submittal of an ICC batch for publishing. | <p>BATCH_ID: Unique ID for the ICC batch to publish.</p> <p>PUBLISH_PARENT_HIERARCHY: Indicates whether or not to publish the parent ICC hierarchy along with an ICC. Values include:</p> <ul style="list-style-type: none"> <li>Y - publish parent ICC hierarchy.</li> <li>N - do not publish parent ICC hierarchy.</li> </ul> <p>PUBLISH_CHILD_HIERARCHY: Indicates whether or not to publish the child ICC hierarchy along with an ICC. Values include:</p> <ul style="list-style-type: none"> <li>Y - publish child ICC hierarchy.</li> <li>N - do not publish child ICC hierarchy.</li> </ul> |
| oracle.apps.ego.extfwk.<br>publishValueSet          | A seeded event used when publishing Value Sets to external systems such as Siebel and BRM via Batch Mode. It is raised from the <b>Publish Value Sets Batch</b> page and indicates the submittal of a Value Set batch for publishing.                                                                                     | <p>BATCH_ID: Unique ID for the batch of ValueSets to publish.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Event Name                           | Event Description                                                                                                                                                                                                        | Event Parameter Names and Descriptions                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oracle.apps.ego.item.<br>publishItem | A seeded event used when publishing Items to external systems such as Siebel and BRM via Batch Mode. It is raised from the <b>Publish Items Batch</b> page and indicates the submittal of an Items batch for publishing. | <p>BATCH_ID: Unique ID for the batch of Items to publish.</p> <p>STRUCTURE_FLAG: Indicates whether or not to publish an item's structure along with the item. Values include:</p> <ul style="list-style-type: none"> <li>Y - publish structure.</li> <li>N - do not publish structure.</li> </ul> <p><b>Note:</b> oracle.apps.bom.structure.publishStructure is no longer raised when publishing structures.</p> |

#### *Other Business Events*

| Event Name                                           | Event Description                            | Event Parameter Names and Descriptions |
|------------------------------------------------------|----------------------------------------------|----------------------------------------|
| oracle.apps.ego.gtin.<br>uccnetEvent                 | UCCnet GDSN Event                            | -                                      |
| oracle.apps.ego.item.<br>postCustItemXrefChange      | Post Customer Item Cross Reference Change    | -                                      |
| oracle.apps.ego.item.<br>postRelatedItemChange       | Post Related Item Change                     | -                                      |
| oracle.apps.ego.orchestration.<br>postProcessMessage | Send Post Process Confirmation Message Event | -                                      |

### Common Event Parameters

| Event Parameter Name | Description                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event_Type           | This parameter identifies the type of transaction that raised the event. The values are Single or Bulk.                                                                                              |
| Last_Updated_By      | Identifier of the user.                                                                                                                                                                              |
| Last_Update_Date     | Identifies the date the transaction occurred.                                                                                                                                                        |
| Created_By           | Identifier of the user.<br><br><b>Note:</b> In bulk events that include both creation and update transactions, this parameter is does not apply and is not available.                                |
| Creation_Date        | Identifies the date the creation transaction occurred.<br><br><b>Note:</b> In bulk events that include both creation and update transactions, this parameter is does not apply and is not available. |
| Request_Identifier   | When Event_Type = Bulk, this identifies the concurrent request that processed the bulk data set.                                                                                                     |

## Related Topics

To view sample subscription functions, see *Sample Subscription Functions*, *Business Event Data for Oracle Bills of Material*, *Oracle Bills of Material User's Guide*.

Using Item Business Events, *Oracle Product Hub User's Guide*



---

## Reading Data for User-Defined Attributes

This appendix covers the following topics:

- Overview

### Overview

You can define an unlimited number of attribute groups for items without programming them. You can define attributes using the setup user interface; definitions are stored in metadata tables. However, you may also need to read attribute data to interface with third party or custom systems.

#### **Generate database views to read User-defined attributes data**

Using the attribute group setup screens, you can generate database views for reading user-defined attributes data. These views hide the complexity of dealing with the attribute meta-data.

#### **Using PL/SQL to Access Data for Attributes**

Another way to access user-defined attribute data for a particular item is to use PL/SQL. The EGO\_USER\_ATTRS\_DATA\_PUB procedure follows:

**Additional Information:** You can run EGO\_USER\_ATTRS\_DATA\_PUB.GET\_USER\_ATTRS\_DATA instead of EGO\_USER\_ATTRS\_DATA\_PUB.GET\_USER\_ATTRS\_EXT\_DATA, but EGO\_USER\_ATTRS\_DATA\_PUB.GET\_USER\_ATTRS\_DATA does not retrieve the following WHO column values:

- CREATED\_BY
- CREATION\_DATE
- LAST\_UPDATED\_BY

- LAST\_UPDATE\_DATE
- LAST\_UPDATE\_LOGIN

---

**Procedure GET\_USER\_ATTRS\_EXT\_DATA**

---

|                               |                                         |
|-------------------------------|-----------------------------------------|
| p_api_version                 | IN NUMBER                               |
| ,p_object_name                | IN VARCHAR2                             |
| ,p_pk_column_name_value_pairs | IN<br>EGO_COL_NAME_VALUE_PAIR_ARRAY     |
| ,p_attr_group_request_table   | IN EGO_ATTR_GROUP_REQUEST_TABLE         |
| ,p_user_privileges_on_object  | IN EGO_VARCHAR_TBL_TYPE DEFAULT<br>NULL |
| ,p_entity_id                  | IN VARCHAR2 DEFAULT NULL                |
| ,p_entity_index               | IN NUMBER DEFAULT NULL                  |
| ,p_entity_code                | IN VARCHAR2 DEFAULT NULL                |
| ,p_debug_level                | IN NUMBER DEFAULT 0                     |
| ,p_init_error_handler         | IN VARCHAR2 DEFAULT FND_API.<br>G_FALSE |
| ,p_init_fnd_msg_list          | IN VARCHAR2 DEFAULT FND_API.<br>G_FALSE |
| ,p_add_errors_to_fnd_stack    | IN VARCHAR2 DEFAULT FND_API.<br>G_FALSE |
| ,p_commit                     | IN VARCHAR2 DEFAULT FND_API.<br>G_FALSE |
| ,x_attributes_row_table       | OUT NOCOPY<br>EGO_USER_ATTR_ROW_TABLE   |

---

---

|                          |                                            |
|--------------------------|--------------------------------------------|
| ,x_attributes_data_table | OUT NOCOPY<br>EGO_USER_ATTR_DATA_EXT_TABLE |
|--------------------------|--------------------------------------------|

**Caution:** If you use procedure  
GET\_USER\_ATTRS\_DATA instead of  
GET\_USER\_ATTRS\_EXT\_DATA, then use  
OUT NOCOPY  
EGO\_USER\_ATTR\_DATA\_TABLE instead.

|                  |                     |
|------------------|---------------------|
| ,x_return_status | OUT NOCOPY VARCHAR2 |
| ,x_errorcode     | OUT NOCOPY NUMBER   |
| ,x_msg_count     | OUT NOCOPY NUMBER   |
| ,x_msg_data      | OUT NOCOPY VARCHAR2 |

---

#### **p\_api\_version**

Callers should pass a constant value of 1.0.

#### **p\_object\_name**

Callers should pass a constant value of 'EGO\_ITEM'

#### **p\_pk\_column\_name\_value\_pairs**

This is a table of EGO\_COL\_NAME\_VALUE\_PAIR\_OBJ objects specifying the Inventory Item ID and Organization ID for which the caller wants attribute data. Callers should create the table with code like the following:

```
l_pk_column_values EGO_COL_NAME_VALUE_PAIR_ARRAY;
l_pk_column_values := EGO_COL_NAME_VALUE_PAIR_ARRAY(
EGO_COL_NAME_VALUE_PAIR_OBJ('INVENTORY_ITEM_ID', <your inventory item
ID>), EGO_COL_NAME_VALUE_PAIR_OBJ('ORGANIZATION_ID', <your
organization ID>));
```

#### **p\_attr\_group\_request\_table**

This is a table of EGO\_ATTR\_GROUP\_REQUEST\_OBJ objects, one for each attribute group whose values the caller wants. The objects also support requests for only some of the attributes in the attribute group. Each EGO\_ATTR\_GROUP\_REQUEST\_OBJ object in the table will look like the following:

```
EGO_ATTR_GROUP_REQUEST_OBJ(
NULL,
431,
```

'EGO\_ITEMMGMT\_GROUP',

<your Attribute Group internal name>,

<your Revision ID if the Attribute Group is associated at the Revision level; otherwise NULL>,

NULL,

NULL

<an optional comma-delimited list of the Attribute internal names whose values you want, or NULL>;

#### **p\_user\_privileges\_on\_object**

This is an optional table of VARCHAR2s, one for each privilege the caller wants validated against the View privileges of the attribute groups being requested. For example, if a user invokes a procedure that calls `Get_User_Attrs_Data`, and the calling procedure wants to enforce attribute group data security, the calling procedure could pass a list of the user's privileges based on his/her role on the current item. If this parameter is non-null, `Get_User_Attrs_Data` will enforce data security for those attribute groups that have a View privilege defined.

#### **p\_entity\_id, p\_entity\_index, p\_entity\_code**

Parameters for use with `ERROR_HANDLER` package. Can safely be defaulted unless caller wants to distinguish among errors for multiple items; if so, please investigate the `ERROR_HANDLER` package for more details.

#### **p\_debug\_level**

Parameter to control debug logging; can safely be defaulted.

#### **p\_init\_error\_handler, p\_init\_fnd\_msg\_list, p\_add\_errors\_to\_fnd\_stack**

More parameters for use with `ERROR_HANDLER` package; can safely be defaulted.

#### **p\_commit**

Controls whether or not `Get_User_Attrs_Data` issues a COMMIT statement at the end of processing; present to comply with API standards, but since `Get_User_Attrs_Data` does not perform any DML, can safely be defaulted.

#### **x\_attributes\_row\_table**

This is a table of `EGO_USER_ATTR_ROW_OBJ` objects, each one representing a row of user-defined attributes data (for example, one database table row of data). It contains the Extension ID value from `EGO_MTL_SY_ITEMS_EXT_VL` (stored as `ROW_IDENTIFIER`) and serves mainly as a way to organize the objects in `x_attributes_data_table`.

#### **x\_attributes\_data\_table**

This is a table of `EGO_USER_ATTR_DATA_OBJ` objects, each one representing a single attribute value. All of the attribute values for a given row of data will share the same `ROW_IDENTIFIER` value; as with `x_attributes_row_table`, the `ROW_IDENTIFIER` value

is the Extension ID from EGO\_MTL\_SY\_ITEMS\_EXT\_VL. Each object also contains the attribute internal name (stored as ATTR\_NAME) and its value (stored as ATTR\_DISP\_VALUE).

**x\_return\_status, x\_errorcode, x\_msg\_count, x\_msg\_data**

Four standard OUT parameters indicating what happened in processing. The first, x\_return\_status, will equal FND\_API.G\_RET\_STS\_SUCCESS if processing succeeded, FND\_API.G\_RET\_STS\_ERROR if an expected error case was encountered, or FND\_API.G\_RET\_STS\_UNEXP\_ERROR if something went wrong during processing. X\_errorcode is not used. X\_msg\_count indicates how many error messages were logged with ERROR\_HANDLER; if x\_msg\_count is 1, then x\_msg\_data contains that message (to save the caller the inconvenience of interacting with ERROR\_HANDLER for only one message).



---

# Synchronizing Item User-Defined Attributes with Item Descriptive Elements

This appendix covers the following topics:

- Overview

## Overview

You can define an unlimited number of user-defined attributes for items without programming. These attributes can have different data types, value sets, and can be parametrically searched. This appendix describes how to synchronize user-defined attribute values with descriptive element values. Oracle provides a public API to update Item Descriptive Element values. It also raises a business event when any pre-selected user-defined attribute values change. You can subscribe to this event and then call the Item Descriptive Element API to synchronize the descriptive elements with the attribute values. A business event is raised when any pre-selected item user-defined attribute values change.

See: Event Manager, and Standard API for an Event Rule Function, *Oracle Workflow Developer's Guide*.

## User-Defined Attributes Business Event

You can configure user-defined attribute groups to raise a Business Event every time a row is added, altered, or deleted. The name of the Business Event raised is

`oracle.apps.ego.item.attributesChanged`

The parameters passed by this Business Event are

INVENTORY\_ITEM\_ID  
ORGANIZATION\_ID  
EXTENSION\_ID  
ATTR\_GROUP\_NAME

The Event is disabled by default and must be explicitly enabled for synchronization to work.

## Item Descriptive Elements API

Oracle provides the following API (with accompanying data type) to maintain Descriptive Elements for a given item.

The data type used by the API follows:

ITEM\_DESC\_ELEMENT\_TABLE: this is an associative array (also known as a PL/SQL table or an index-by table) of ITEM\_DESC\_ELEMENT records:  
ITEM\_DESC\_ELEMENT IS RECORD

```
(
    ELEMENT_NAME      VARCHAR2(30)
    ,ELEMENT_VALUE     VARCHAR2(30)
    ,DESCRIPTION_DEFAULT VARCHAR2(1)
);
```

ELEMENT\_NAME: Column ELEMENT\_NAME from  
MTL\_DESCRIPTIVE\_ELEMENTS

ELEMENT\_VALUE: Column ELEMENT\_VALUE from MTL\_DESCR\_ELEMENT\_VALUES

DESCRIPTION\_DEFAULT: Column DEFAULT\_ELEMENT\_FLAG from  
MTL\_DESCR\_ELEMENT\_VALUES (indicates whether this Descriptive Element  
will be used in the description generation)

The API signature is:

```
INV_ITEM_CATALOG_ELEM_PUB.Process_Item_Descr_Elements
(
    p_api_version          IN NUMBER
    ,p_init_msg_list       IN VARCHAR2 DEFAULT fnd_api.g_FALSE
    ,p_commit_flag         IN VARCHAR2 DEFAULT fnd_api.g_FALSE
    ,p_validation_level    IN NUMBER   DEFAULT
INV_ITEM_CATALOG_ELEM_PUB.g_VALIDATE_ALL
    ,p_inventory_item_id   IN NUMBER   DEFAULT -999
    ,p_item_number         IN VARCHAR2 DEFAULT NULL
    ,p_item_desc_element_table IN ITEM_DESC_ELEMENT_TABLE
    ,x_generated_descr     OUT VARCHAR2
    ,x_return_status       OUT VARCHAR2
    ,x_msg_count           OUT NUMBER
    ,x_msg_data            OUT VARCHAR2
);
```

### **p\_api\_version**

Pass the value 1.0 for this parameter.

### **p\_init\_msg\_list**

This parameter can safely be defaulted.

### **p\_commit\_flag**

Indicates whether or not the API commits the transaction.

### **p\_validation\_level**

Determines how much validation occurs; can safely be defaulted.

### **p\_inventory\_item\_id**



Pass the value 1.0 for this parameter.

**p\_item\_number**

This parameter can safely be defaulted.

**p\_item\_desc\_element\_table**

Data type defined above.

**x\_generated\_descr**

Returns the Item Description generated from the updated Descriptive Elements. If you want to update the Item Description with this value, use Item Open Interface.

**x\_return\_status**

Returns one of the following:

`FND_API.G_RET_STS_SUCCESS, FND_API.G_RET_STS_ERROR, FND_API.G_RET_STS_UNEXP_ERROR`

**x\_msg\_count**

Number of messages logged by the API.

**x\_msg\_data**

The message logged if exactly one message was logged.

## Subscribing to the Event and Calling the API

As mentioned earlier, the Business Event `oracle.apps.ego.item.attributesChanged` must be explicitly enabled through the graphical user interface in order for synchronization to work. After this step is complete, you must subscribe to the User-Defined Attributes Business Event. See: *Event Manager, Oracle Workflow Developer's Guide*.

`oracle.apps.ego.item.attributesChanged`

As part of Event Subscription, define an Event Subscription Rule Function that will be called when the Event is raised. See Standard API for an Event Rule Function, *Oracle Workflow Developer's Guide*.

This Rule Function should:

- Identify the attribute group that raised the Event and determine the correct Descriptive Elements for that attribute group (perhaps through customer-supplied mapping metadata).
- Fetch the data altered by the Event (see the following Sample Rule Function for more details) and process it as necessary.
- Call the Item Descriptive Elements API that is passing the processed data.

## Sample Rule Function

```

FUNCTION Synch_Attrs_With_Desc_Elems (
    p_subscription_guid IN RAW
    ,p_event              IN OUT NOCOPY WF_EVENT_T
)
RETURN VARCHAR2
IS

TYPE ATTR_TO_DESC_ELEM_MAPPING_REC IS RECORD
(
    ATTR_NAME           VARCHAR2(30)
    ,DESC_ELEM_NAME     VARCHAR2(30)
    ,DESCRIPTION_DEFAULT VARCHAR2(1)
);
TYPE ATTR_TO_DESC_ELEM_MAPPING_TBL IS TABLE OF
ATTR_TO_DESC_ELEM_MAPPING_REC
INDEX BY BINARY_INTEGER;

l_parameter_list      WF_PARAMETER_LIST_T;
l_next_parameter      WF_PARAMETER_T;
l_dml_type_param      WF_PARAMETER_T;
l_attr_group_name_param WF_PARAMETER_T;
l_inventory_item_id_param WF_PARAMETER_T;
l_organization_id_param WF_PARAMETER_T;
l_revision_id_param   WF_PARAMETER_T;
l_pk_column_values    EGO_COL_NAME_VALUE_PAIR_ARRAY;
l_request_table       EGO_ATTR_GROUP_REQUEST_TABLE;
x_attributes_row_table EGO_USER_ATTR_ROW_TABLE;
x_attributes_data_table EGO_USER_ATTR_DATA_TABLE;
x_return_status       VARCHAR2(1);
x_errorcode           NUMBER;
x_msg_count           NUMBER;
x_msg_data            VARCHAR2(1000);
l_current_mapping_rec ATTR_TO_DESC_ELEM_MAPPING_REC;
l_mapping_rec_table   ATTR_TO_DESC_ELEM_MAPPING_TBL;
l_attributes_data_index NUMBER;

l_current_row_obj      EGO_USER_ATTR_ROW_OBJ;
l_current_data_obj     EGO_USER_ATTR_DATA_OBJ;
l_current_item_desc_elem INV_ITEM_CATALOG_ELEM_PUB.
ITEM_DESC_ELEMENT;

l_item_desc_elem_table INV_ITEM_CATALOG_ELEM_PUB.
ITEM_DESC_ELEMENT_TABLE;

x_generated_descr      VARCHAR2(240);

l_return_status        VARCHAR2(30) := 'SUCCESS';

BEGIN

    l_parameter_list := p_event.getParameterList();

    -----
    -- Sort the parameters that the Business Event passed
    -----
    FOR i IN l_parameter_list.FIRST .. l_parameter_list.LAST
    LOOP

        l_next_parameter := l_parameter_list(i);
        IF (l_next_parameter.getName() = 'DML_TYPE') THEN
            l_dml_type_param := l_next_parameter;
        ELSIF (l_next_parameter.getName() = 'ATTR_GROUP_NAME') THEN

```

```

l_attr_group_name_param := l_next_parameter;
ELSIF (l_next_parameter.getName() = 'INVENTORY_ITEM_ID') THEN
    l_inventory_item_id_param := l_next_parameter;
ELSIF (l_next_parameter.getName() = 'ORGANIZATION_ID') THEN
    l_organization_id_param := l_next_parameter;
ELSIF (l_next_parameter.getName() = 'REVISION_ID') THEN
    l_revision_id_param := l_next_parameter;
END IF;

```

```

END LOOP;

```

```

-----
Copy the PK values into our EGO_COL_NAME_VALUE_PAIR_ARRAY object
-----

```

```

l_pk_column_values :=
    EGO_COL_NAME_VALUE_PAIR_ARRAY(
        EGO_COL_NAME_VALUE_PAIR_OBJ('INVENTORY_ITEM_ID',
            l_inventory_item_id_param.getValue())
        , EGO_COL_NAME_VALUE_PAIR_OBJ('ORGANIZATION_ID',
            l_organization_id_param.getValue())
    );

```

```

-----
Prepare an EGO_ATTR_GROUP_REQUEST_TABLE object with one element, an
EGO_ATTR_GROUP_REQUEST_OBJ for the Attr Group that raised the Event
-----

```

```

l_request_table := EGO_ATTR_GROUP_REQUEST_TABLE();
l_request_table.EXTEND();
l_request_table(l_request_table.LAST) :=
    EGO_ATTR_GROUP_REQUEST_OBJ(
        NULL      --ATTR_GROUP_ID
        , 431
        , 'EGO_ITEMMGMT_GROUP'
        , l_attr_group_name_param.getValue()
        , l_revision_id_param.getValue()
        , NULL     --DATA_LEVEL_2
        , NULL     --DATA_LEVEL_3
        , NULL     --ATTR_NAME_LIST
    );

```

```

-----
Get the data for this Event, if there is any
-----

```

```

IF (l_dml_type_param.getValue() <> 'DELETE') THEN

```

```

    EGO_USER_ATTRS_DATA_PUB.Get_User_Attrs_Data(
        p_api_version          => 1.0
        , p_object_name        => 'EGO_ITEM'
        , p_pk_column_name_value_pairs => l_pk_column_values
        , p_attr_group_request_table => l_request_table
        , p_user_privileges_on_object => NULL
        , p_entity_id          => NULL
        , p_entity_index       => NULL
        , p_entity_code        => NULL
        , p_debug_level        => 0
        , p_init_error_handler  => FND_API.G_FALSE
        , p_init_fnd_msg_list  => FND_API.G_FALSE
        , p_add_errors_to_fnd_stack => FND_API.G_FALSE
        , p_commit             => FND_API.G_FALSE
        , x_attributes_row_table => x_attributes_row_table
        , x_attributes_data_table => x_attributes_data_table
        , x_return_status      => x_return_status
        , x_errorcode          => x_errorcode
        , x_msg_count          => x_msg_count
    );

```

```

,x_msg_data          => x_msg_data
);

-----
If the call failed, we report that an error occurred
(in a production situation, we would retrieve and pass
on all errors on the ERROR_HANDLER message stack)
-----
IF (x_return_status IS NULL OR
    x_return_status <> FND_API.G_RET_STS_SUCCESS) THEN
    IF (x_return_status = FND_API.G_RET_STS_ERROR) THEN
        RAISE FND_API.G_EXC_ERROR;
    ELSE
        RAISE FND_API.G_EXC_UNEXPECTED_ERROR;
    END IF;
END IF;
END IF;

-----
Process the Descriptive Elements according to the
Attr Group that raised the Event and the DML type
-----
IF (l_attr_group_name_param.getValue() = 'Pella_Windows') THEN

-----
Find mappings from Attr Group Attrs to Desc Elems; in this example,
hard code the mapping from Attr Group Attrs to Desc Elems, but in a
production situation customers may have some stored mapping between
the Attrs in an Attr Group associated to a Catalog Category and the
Descriptive Elements for that Catalog Category (similarly, in this
example we always set DESCRIPTION_DEFAULT as 'Y' for simplicity)
-----
l_current_mapping_rec.ATTR_NAME := 'Style';
l_current_mapping_rec.DESC_ELEM_NAME := 'Style';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'GB_Options';
l_current_mapping_rec.DESC_ELEM_NAME := 'Grille Bar Options';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'Wood_Type';
l_current_mapping_rec.DESC_ELEM_NAME := 'Wood Type';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'Frame_Area';
l_current_mapping_rec.DESC_ELEM_NAME := 'Frame Area';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'GRADE';
l_current_mapping_rec.DESC_ELEM_NAME := 'Grade';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'PW_Exterior';
l_current_mapping_rec.DESC_ELEM_NAME := 'Primed Wood Exterior';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=

```

```

l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'DATE';
l_current_mapping_rec.DISC_ELEM_NAME := 'Date';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

ELSIF (l_attr_group_name_param.getValue() = 'WD_Attributes') THEN

l_current_mapping_rec.ATTR_NAME := 'STYLE';
l_current_mapping_rec.DISC_ELEM_NAME := 'WD Style';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'WD_Grade';
l_current_mapping_rec.DISC_ELEM_NAME := 'WD Grade';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'DATETIME';
l_current_mapping_rec.DISC_ELEM_NAME := 'Date';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

l_current_mapping_rec.ATTR_NAME := 'URL_link';
l_current_mapping_rec.DISC_ELEM_NAME := 'URL';
l_current_mapping_rec.DESCRPTION_DEFAULT := 'Y';
l_mapping_rec_table(l_mapping_rec_table.COUNT + 1) :=
l_current_mapping_rec;

END IF;

-----
For every Desc Elem, get its value (a NULL value, or a DML_TYPE of
DELETE,
will result in a NULL Elem value), set its DESCRIPTION_DEFAULT flag,
and
add it to the ITEM_DESC_ELEMENT_TABLE
-----
FOR i IN l_mapping_rec_table.FIRST .. l_mapping_rec_table.LAST
LOOP

l_current_mapping_rec := l_mapping_rec_table(i);
l_current_item_desc_elem.ELEMENT_NAME := l_current_mapping_rec.
DESC_ELEM_NAME;
l_current_item_desc_elem.DESCRPTION_DEFAULT := l_current_mapping_rec.
DESCRPTION_DEFAULT;

-----
This example uses single-row Attr Groups, so we don't need to make use
of the EGO_USER_ATTR_ROW_TABLE object; but for a multi-row Attr Group,
we would use the ROW_IDENTIFIER field of each EGO_USER_ATTR_ROW_OBJ to
find all the EGO_USER_ATTR_DATA_OBJ objects for that particular row
-----
IF (x_attributes_row_table IS NOT NULL AND
x_attributes_row_table.COUNT > 0 AND
x_attributes_data_table IS NOT NULL AND
x_attributes_data_table.COUNT > 0) THEN

l_attributes_data_index := x_attributes_data_table.FIRST;
WHILE l_attributes_data_index <= x_attributes_data_table.LAST
LOOP

```

```

EXIT WHEN l_current_item_desc_elem.ELEMENT_VALUE IS NOT NULL;
    l_current_data_obj := x_attributes_data_table
(l_attributes_data_index);
    IF (l_current_data_obj.ATTR_NAME = l_current_mapping_rec.
ATTR_NAME) THEN
        l_current_item_desc_elem.ELEMENT_VALUE := SUBSTRB
(l_current_data_obj.ATTR_DISP_VALUE, 1, 30);
        END IF;
        l_attributes_data_index := x_attributes_data_table.NEXT
(l_attributes_data_index);
        END LOOP;
    END IF;

-----
-- Now we add the ITEM_DESC_ELEMENT record into our table
-----
l_item_desc_elem_table(l_item_desc_elem_table.COUNT + 1) :=
l_current_item_desc_elem;

    END LOOP;
-----
Now that we have all the necessary data and metadata, we
call the API to update the Desc Elems for this Attr Group
(note that the API returns the Item Description generated
from the updated Descriptive Elements; if required, you
can update the Description through Item Open Interface)
-----
INV_ITEM_CATALOG_ELEM_PUB.Process_Item_Descr_Elements(
    p_api_version                => 1.0
    ,p_inventory_item_id          => l_inventory_item_id_param.
getValue()
    ,p_item_desc_element_table    => l_item_desc_elem_table
    ,x_generated_descr            => x_generated_descr
    ,x_return_status              => x_return_status
    ,x_msg_count                  => x_msg_count
    ,x_msg_data                   => x_msg_data
);
RETURN l_return_status;
EXCEPTION
    WHEN OTHERS THEN
        WF_CORE.CONTEXT('EGO_EF_QA_PUB', 'Synch_Attrs_With_Desc_Elems',
            p_event.getEventName(), p_subscription_guid);
        WF_EVENT.setErrorInfo(p_event, 'ERROR');
        RETURN 'ERROR';
END Synch_Attrs_With_Desc_Elems;

```

---

## Change Management Business Events

This appendix covers the following topics:

- Description of Change Management Business Events

### Description of Change Management Business Events

Change Management business events can be triggered for the following actions:

- Approval status change
- Priority changes
- Status changes
- Workflow status changes
- Creation of a change object

**Note:** To subscribe to any of the following business events, refer to Subscribing to a Change Management Business Event, page 7-44.

| Business Event Name                                  | Triggering Event        |
|------------------------------------------------------|-------------------------|
| oracle.apps.eng.cm.changeObject.changeApprovalStatus | Approval status changes |
| oracle.apps.eng.cm.changeObject.changePriority       | Priority changes        |
| oracle.apps.eng.cm.changeObject.changeStatus         | Status changes          |

| Business Event Name                                  | Triggering Event                  |
|------------------------------------------------------|-----------------------------------|
| oracle.apps.eng.cm.changeObject.changeWorkflowStatus | Workflow status changes           |
| oracle.apps.eng.cm.changeObject.create               | Creation of a change object       |
| oracle.apps.eng.cm.changeObject.postComment          | Posted comment on a change object |
| oracle.apps.eng.cm.changeObject.reassign             | Reassignment of a change object   |
| oracle.apps.eng.cm.changeObject.requestComment       | Common request of a change object |
| oracle.apps.eng.cm.changeObject.submit               | Submission of a change object     |
| oracle.apps.eng.cm.changeObject.update               | Update of a change object         |
| oracle.apps.eng.cm.changeOrder.changeScheduleDate    | Schedule date change              |
| oracle.apps.eng.cm.import.complete                   | Import is completed               |
| oracle.apps.eng.cm.revisedItem.changeScheduleDate    | Revised item schedule date change |
| oracle.apps.eng.cm.revisedItem.changeStatus          | Revised item status changes       |

## Change Management Dependencies and Interactions

Change Management business events, depend on the following systems and applications:

- Workflow
- Business events



- Business Intelligence Publisher
- Oracle TCA

Change Management business events can interact with the following:

- Items
- Multiple modules in Oracle Discrete Manufacturing, including: Engineering, Bills of Materials, Work In Process, and Planning
- Projects



---

## Seeded Item Metadata Libraries

This appendix covers the following topics:

- Sellable Product Information Library - Horizontal
- Product Management Library - Horizontal
- Communications Services Billing Library - Vertical
- Communications Product Details Library - Vertical

### Sellable Product Information Library - Horizontal

Oracle Product Hub provides the following seeded library containing attribute groups, attributes, and value sets. For more information about using data libraries, see: Using Seeded Attributes, page 4-36. The attribute groups in this library are assigned to the Item Revision business entity.

This library is not automatically installed. The system administrator must perform the following steps to upload the library after installing the Oracle Product Hub for Communications patch.

#### Installing the Library

1. Upload the value set library file. This file contains all of the value sets required by the Sellable Product Information Library, as well as those required by all other Oracle Product Hub for Communications libraries.

**Tip:** You only need to install this file once for all Oracle Product Hub for Communications libraries.

- **Filename**  
\$EGO\_TOP/patch/115/import/US/egomstrlibvs.ldt
- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefval.lct
$EGO_TOP/patch/115/import/US/egomstrlibvs.ldt
ORACLE Password:
```

2. Upload the Sellable Product Information Library file.

- **Filename**

\$EGO\_TOP/patch/115/import/US/egospilib.ldt

- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefag.lct
$EGO_TOP/patch/115/import/US/egospilib.ldt
ORACLE Password:
```

***Sellable Product Information Library - Horizontal***

| Attribute Group<br>Display Name<br>(Internal Name)                        | Attribute Display<br>Name (Internal<br>Name)       | Attribute Description                                                                                       | Value Set     |
|---------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------|
| <b>Behavior</b>                                                           |                                                    |                                                                                                             |               |
| Additional Billing<br>Information<br>(Additional_Billing_I<br>nformation) | Billing Entity Type<br>(Billing_Entity_Type)       | Identifies the type of<br>billing product. Use<br>the DVM for mapping<br>these values.                      | Billing_Type  |
| Single-Row                                                                |                                                    | <b>Note:</b> You can use<br>the billing type<br>instead and ignore<br>this attribute<br>during integration. |               |
| Assigned to the Item<br>business entity.                                  |                                                    |                                                                                                             |               |
| -                                                                         | Billing Service Type<br>(Billing_Service_Typ<br>e) | The service type<br>associated with the<br>product. It represents<br>the service the<br>product belongs to. | BRM_PERMITTED |

| Attribute Group<br>Display Name<br>(Internal Name)          | Attribute Display<br>Name (Internal<br>Name)            | Attribute Description                                                                                                                                                                                                                                                        | Value Set     |
|-------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Behavior</b>                                             |                                                         |                                                                                                                                                                                                                                                                              |               |
| Additional Entity<br>Details<br>(Additional_Entity_Details) | Effective Start Date<br>(Effective_Start_Date)          | Start date of the<br>product.                                                                                                                                                                                                                                                | EGO_Lib_NumVS |
| Single-Row                                                  |                                                         |                                                                                                                                                                                                                                                                              |               |
| Assigned to the Item<br>business entity.                    |                                                         |                                                                                                                                                                                                                                                                              |               |
| -                                                           | Effective End Date<br>(Effective_End_Date)              | End date of the<br>product.                                                                                                                                                                                                                                                  | EGO_Lib_NumVS |
| -                                                           | Internal Reference<br>Code<br>(Internal_Reference_Code) | Alphanumeric value<br>that identifies the<br>item. The value is<br>immutable (does not<br>change across<br>instances). No<br>validations or<br>constraints are<br>provided. Users must<br>define rules and<br>constraints to ensure<br>uniqueness and to<br>prevent updates. | -             |
| -                                                           | Entity Type<br>(Entity_Type)                            | -                                                                                                                                                                                                                                                                            | Entity_Type   |

| Attribute Group<br>Display Name<br>(Internal Name)                                  | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                                                                                                                                                                                                                                                                                                               | Value Set      |
|-------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Behavior</b>                                                                     |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |
| Product Details:<br>Definitions<br>(Prod_Details_Definit<br>ions)<br><br>Single-Row | Price Type<br>(Price_Type)                   | <p>Select the price type.<br/>The options are:</p> <ul style="list-style-type: none"> <li>One time. The customer pays once to buy the product.</li> <li>Recurring. The customer pays a fixed recurring fee to use the product. An example is a fixed monthly fee for local telephone service.</li> <li>Usage. The customer pays for the product based on usage. An example is the charge for electricity, based on how much you consume.</li> </ul> | Price_Type     |
| -                                                                                   | Structure Type<br>(Structure_Type)           | Indicates whether an item has children or a bundle. Sample values include Bundle, None, Customizable.                                                                                                                                                                                                                                                                                                                                               | Structure_Type |
| -                                                                                   | Sales Product (Sales_Product)                | Indicates that the item can be used for order type 'SalesOrder'.                                                                                                                                                                                                                                                                                                                                                                                    | EGO_Lib_Yes_No |
| -                                                                                   | Service Product<br>(Service_Product)         | Indicates whether the item is a service.                                                                                                                                                                                                                                                                                                                                                                                                            | EGO_Lib_Yes_No |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                           | Value Set           |
|----------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Behavior</b>                                    |                                              |                                                                                                                                                                 |                     |
| -                                                  | Product Level<br>(Product_Level)             | The numeric product level in the product hierarchy.                                                                                                             | EGO_Lib_NumVS       |
| -                                                  | Format (Format)                              | For training products, select a training format such as Instructor led or Web-based.                                                                            | Format              |
| -                                                  | Parent Product<br>(Parent_Product)           | Select the parent product. This field is for information only. It is not used for creating or managing products with components.                                | Parent_Product      |
| -                                                  | Quantity (Quantity)                          | The number of items in the unit of measure.                                                                                                                     | EGO_Lib_PosNumVS    |
| -                                                  | Pricing Code<br>(Pricing_Code)               | Identifies the type of pricing associated with the item. None = No pricing associated, Standard = Simple pricing model associated, Billing = BRM pricing model. | Pricing_Code        |
| -                                                  | Product Line<br>(Product_Line)               | The product belongs to this product line.                                                                                                                       | Object_Product_Line |
| -                                                  | Check Eligibility<br>(Check_Eligibility)     | System check customers' eligibility to buy the product. Used by Siebel.                                                                                         | EGO_Lib_Yes_No      |

| Attribute Group<br>Display Name<br>(Internal Name)                             | Attribute Display<br>Name (Internal<br>Name)     | Attribute Description                                                                                     | Value Set         |
|--------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------|
| <b>Behavior</b>                                                                |                                                  |                                                                                                           |                   |
| -                                                                              | Inclusive Eligibility<br>(Inclusive_Eligibility) | Used to ensure that the Eligibility matrix is validated when a quote or an order is created for the item. | EGO_Lib_Yes_No    |
| Product Details:<br>Marketing<br>(Product_Details_Marketing)<br><br>Single-Row | Targeted Industry<br>(Targeted_Industry)         | The industry to target with this product.                                                                 | Targeted_Industry |
| -                                                                              | Targeted Min Age<br>(Targeted_Min_Age)           | The minimum age of buyers for this product.                                                               | EGO_Lib_PosNumVS  |
| -                                                                              | Targeted Max Age<br>(Targeted_Max_Age)           | The maximum age of buyers for this product.                                                               | EGO_Lib_PosNumVS  |
| -                                                                              | Targeted Postal Code<br>(Targeted_Postal_Code)   | The postal code to target for sales of this product.                                                      | -                 |
| -                                                                              | Targeted Country<br>(Targeted_Country)           | The country where you want to sell this product.                                                          | Targeted_Country  |
| Product Details:<br>Service<br>(Product_Details_Service)<br><br>Single-Row     | MTBF (MTBF)                                      | The mean time between failure for the product.                                                            | EGO_Lib_PosNumVS  |
| -                                                                              | MTTR (MTTR)                                      | The mean time to repair the product.                                                                      | EGO_Lib_PosNumVS  |
| -                                                                              | Field Replacable<br>(Field_Replaceable)          | Indicates if this is a field-replaceable unit.                                                            | EGO_Lib_Yes_No    |



| Attribute Group<br>Display Name<br>(Internal Name)           | Attribute Display<br>Name (Internal<br>Name)        | Attribute Description                                                                              | Value Set          |
|--------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------|
| <b>Behavior</b>                                              |                                                     |                                                                                                    |                    |
| -                                                            | Return if Defective<br>(Return_if_Defective)        | Indicates if a defective product should be returned by the customer when a replacement is shipped. | EGO_Lib_Yes_No     |
| -                                                            | Tool (Tool)                                         | Indicates if this product is a tool, such as one used by field service engineers.                  | EGO_Lib_Yes_No     |
| Product Details:<br>Logistics<br>(Product_Details_Logistics) | Vendor Part Number<br>(Vendor_Part_Number)          | The vendor's part number for this product.                                                         | Vendor_Part_Number |
| <b>Single-Row</b>                                            |                                                     |                                                                                                    |                    |
| -                                                            | Lead Time<br>(Lead_Time)                            | The standard lead time for ordering the product, measured in weeks.                                | -                  |
| -                                                            | Ship Carrier<br>(Ship_Carrier)                      | The name of the shipping carrier for this product.                                                 | Ship_Carrier       |
| -                                                            | Ship Method<br>(Ship_Method)                        | The shipping mode. For example, air, ground, and so on.                                            | Ship_Method        |
| -                                                            | Allocate Below<br>Safety<br>(Allocate_Below_Safety) | Indicates if you allow allocation below the safe inventory level of this product.                  | EGO_Lib_Yes_No     |
| -                                                            | Auto Substitute<br>(Auto_Substitute)                | Indicates if you allow auto-substitutions.                                                         | EGO_Lib_Yes_No     |

| Attribute Group<br>Display Name<br>(Internal Name)                           | Attribute Display<br>Name (Internal<br>Name)                    | Attribute Description                                                                                                                            | Value Set      |
|------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Behavior</b>                                                              |                                                                 |                                                                                                                                                  |                |
| -                                                                            | Auto Allocate<br>(Auto_Allocate)                                | Indicates if you use automatic allocation during the fulfillment process.                                                                        | EGO_Lib_Yes_No |
| -                                                                            | Auto Explode<br>(Auto_Explode)                                  | Flag to indicate explosion of child components.                                                                                                  | EGO_Lib_Yes_No |
| -                                                                            | Shipping Required<br>(Shipping_Required)                        | Flag to indicate if shipping is required.                                                                                                        | EGO_Lib_Yes_No |
| Product Details:<br>Additional<br>Information<br>(Prod_Detail_More_I<br>nfo) | Division Code SAP<br>(Division_Code_SAP<br>)                    | User defined access code for the item.                                                                                                           | -              |
| Single-Row                                                                   |                                                                 |                                                                                                                                                  |                |
| -                                                                            | Global Product<br>Identifier<br>(Global_Product_Ide<br>ntifier) | A unique product identification string. This field is intended for use by integrators who need to move product information between applications. | -              |
| -                                                                            | Serialized<br>(Serialized)                                      | Indicates if instances of this product are tracked as serialized assets or simply as quantities of product                                       | EGO_Lib_Yes_No |
| -                                                                            | Model Product<br>(Model_Product)                                | This field is obsolete. It is provided as a reference for upgrade users of Configurator.                                                         | EGO_Lib_Yes_No |

| Attribute Group<br>Display Name<br>(Internal Name)              | Attribute Display<br>Name (Internal<br>Name)                | Attribute Description                                                                                | Value Set      |
|-----------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------|
| <b>Behavior</b>                                                 |                                                             |                                                                                                      |                |
| -                                                               | Tax Subcomponent<br>Flag<br>(Tax_Subcomponent<br>_Flag)     | Indicates whether to<br>compute the tax on a<br>bundle by adding up<br>the tax on its<br>components. | EGO_Lib_Yes_No |
| -                                                               | Project Resource<br>(Project_Resource)                      | Indicates if the<br>product is a service<br>for a project.                                           | EGO_Lib_Yes_No |
| -                                                               | Compensable<br>(Compensable)                                | Indicates if sales<br>personnel can receive<br>compensation for<br>selling the product.              | EGO_Lib_Yes_No |
| Additional Entity<br>Details<br>(Additional_Entity_D<br>etails) | Effective Start Date<br>(Effective_Start_Date<br>)          | Start date of the<br>product.                                                                        | -              |
| Single-Row                                                      |                                                             |                                                                                                      |                |
| Defined at the item<br>level.                                   |                                                             |                                                                                                      |                |
| -                                                               | Effective End Date<br>(Effective_End_Date)                  | End date of the<br>product.                                                                          | -              |
| -                                                               | Internal Reference<br>Code<br>(Internal_Reference_<br>Code) | Identifies the type of<br>product. This is a user<br>defined list.                                   | -              |
| Subject<br>Compatibility Rules<br>(Sub_Compatible_Rul<br>e)     | Serial Number<br>(Serial_Number)                            | Unique identifier of<br>the record in the<br>multi-row.                                              | -              |
| Multi-Row                                                       |                                                             |                                                                                                      |                |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)   | Attribute Description                                                                                                                                                                                                                                                                                                                            | Value Set           |
|----------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Behavior</b>                                    |                                                |                                                                                                                                                                                                                                                                                                                                                  |                     |
| -                                                  | Subject Product<br>(Subject_Product)           | <p>Select the subject product. The default is the product that is selected in the Products list, but you can select any product to replace it. NOTE: Make an entry in only one of the following fields:</p> <ul style="list-style-type: none"> <li>• Subject Product</li> <li>• Subject Product Line</li> <li>• Subject Product Class</li> </ul> | Product             |
| -                                                  | Rule Type<br>(Rule_Type)                       | Select the type of the rule.                                                                                                                                                                                                                                                                                                                     | Rule_Type           |
| -                                                  | Object Product<br>(Object_Product)             | Select the object product.                                                                                                                                                                                                                                                                                                                       | Product             |
| -                                                  | Object Product Line<br>(Object_Product_Line)   | Select the object product line.                                                                                                                                                                                                                                                                                                                  | Object_Product_Line |
| -                                                  | Object Product Class<br>(Object_Product_Class) | Select the object product class.                                                                                                                                                                                                                                                                                                                 | Product_Class       |
| -                                                  | Start Date<br>(Start_Date)                     | The date when the rule goes into effect.                                                                                                                                                                                                                                                                                                         | -                   |
| -                                                  | End Date (End_Date)                            | The date when the rule is no longer in effect.                                                                                                                                                                                                                                                                                                   | -                   |

| Attribute Group<br>Display Name<br>(Internal Name)  | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                                                 | Value Set            |
|-----------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Behavior</b>                                     |                                              |                                                                                                                                                                                       |                      |
| -                                                   | Matrix Name<br>(Matrix_Name)                 | The compatibility<br>group used for<br>grouping this rule.                                                                                                                            | Matrix_Name          |
| Pricing:Simple Price<br>List<br>(Simple_Price_List) | Price List Name<br>(Price_List_Name)         | Name of the price list.                                                                                                                                                               | Pricelist_Name       |
| Multi-Row                                           |                                              |                                                                                                                                                                                       |                      |
| -                                                   | Currency (Currency)                          | Currency associated<br>with the price list.                                                                                                                                           | Currency_Code_Select |
| -                                                   | Price Type<br>(Price_Type)                   | Charge type<br>associated with the<br>price.                                                                                                                                          | Price_Type           |
| -                                                   | Charge Period<br>(Charge_Period)             | Frequency of<br>applying the charge.                                                                                                                                                  | Charge_Period        |
|                                                     | List Price (List_Price)                      | The standard price<br>used for most<br>transactions and the<br>most commonly used<br>target price for price<br>adjustments.                                                           | -                    |
| -                                                   | Promotional Price<br>(Promotional_Price)     | If you enter a<br>promotional price, it<br>will be used instead of<br>the list price in all<br>functions that do not<br>explicitly identify the<br>list price as the target<br>price. | -                    |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)         | Attribute Description                                                                                                                                                                                        | Value Set               |
|----------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Behavior</b>                                    |                                                      |                                                                                                                                                                                                              |                         |
| -                                                  | Pricing Commit Type<br>(Pricing_Commit_Type)         | Field that identifies whether a price override (Committed) or a discount override (Dynamic) is defined for the charge.                                                                                       | Pricing_Commit_Type     |
| -                                                  | Dynamic Discount Method<br>(Dynamic_Discount_Method) | Whenever a discount override is applied to the charge, this field identifies the type of discount (either 'Percent' or 'Amount').                                                                            | Dynamic_Discount_Method |
| -                                                  | Service Price Percent<br>(Service_Price_Percent)     | A percentage that is used to mark up the price of the service product.                                                                                                                                       | -                       |
| -                                                  | Service Pricing Method<br>(Service_Pricing_Method)   | Specify the basis used for calculating the price markup of the service product from the covered product.                                                                                                     | Service_Pricing_Method  |
| -                                                  | Volume Discount<br>(Volume_Discount)                 | A volume discount is an adjustment to the price of a product based on the quantity of that product that is requested in the quote. For example, a 5% discount when the user requests five to 10 of the item. | Volume_Discount         |

| Attribute Group<br>Display Name<br>(Internal Name)            | Attribute Display<br>Name (Internal<br>Name)      | Attribute Description                                                                                                                                                                                                  | Value Set            |
|---------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Behavior</b>                                               |                                                   |                                                                                                                                                                                                                        |                      |
| -                                                             | Attribute<br>Adjustment<br>(Attribute_Adjustment) | Specifies the attribute pricing table name that provides attribute-based pricing adjustments for this line item product. This attribute pricing table uses attribute data from the Class to which the product belongs. | Attribute_Adjustment |
| -                                                             | Maximum Price<br>(Maximum_Price)                  | Maximum price that can be charged.                                                                                                                                                                                     | -                    |
| -                                                             | Minimum Price<br>(Minimum_Price)                  | Minimum price that can be charged.                                                                                                                                                                                     | -                    |
| -                                                             | Description<br>(Description)                      | Description about the charge.                                                                                                                                                                                          | -                    |
| -                                                             | Start Date<br>(Start_Date)                        | Start date of the price.                                                                                                                                                                                               | -                    |
| -                                                             | End Date (End_Date)                               | End date of the price.                                                                                                                                                                                                 | -                    |
| Promotion:More<br>Information<br>(Promotion_More_Information) | Promotion Type<br>(Promotion_Type)                | The type of the promotion. Options include coupon and bundled promotion.                                                                                                                                               | Promotion_Type       |
| Single-Row                                                    |                                                   |                                                                                                                                                                                                                        |                      |
| -                                                             | Score (Score)                                     | The ranking of promotion                                                                                                                                                                                               | Score                |
| -                                                             | Message (Message)                                 | The message for the promotion.                                                                                                                                                                                         | Message              |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                           | Value Set          |
|----------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Behavior</b>                                    |                                              |                                                                                                                                                 |                    |
| -                                                  | Instances (Instances)                        | The value that specifies how many times a customer is allowed to use this promotion. Options are One Per Customer, One Per Order, and No Limit. | Instances          |
| -                                                  | Show In Catalog (Show_In_Catalog)            | Indicates whether to display the promotion in the product catalogs.                                                                             | Show In Catalog    |
| -                                                  | Track As Agreement (Track_As_Agreement)      | Indicates whether to track the promotion as an agreement.                                                                                       | Track As Agreement |
| -                                                  | Period (Period)                              | Identifies the duration on the promotion after which the penalty is not applied if it is discontinued.                                          | Period             |
| -                                                  | Period UOM (Period_UOM)                      | Unit of measure for the period.                                                                                                                 | Period UOM         |
| -                                                  | Grace Period (Grace_Period)                  | Identifies the grace period before which the promotion can be discontinued without application of the penalty.                                  | Grace Period       |
| -                                                  | Grace UOM (Grace_UOM)                        | Unit of measure for the grace period.                                                                                                           | Grace UOM          |



| Attribute Group<br>Display Name<br>(Internal Name)                                                       | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                  | Value Set        |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Behavior</b>                                                                                          |                                              |                                                                                                                        |                  |
| Promotion:Charge<br>Plan:Non Recurring<br>Charge Details<br>(Charge_Plan_Non_R<br>ecurring)              | Non Recurring<br>Charge<br>(Non_Recurring)   | The total amount of<br>the nonrecurring<br>charge.                                                                     | EGO_Lib_NumVS    |
| -                                                                                                        | NRC Quantity<br>(NRC_Quantity)               | Non recurring<br>quantity.                                                                                             | EGO_Lib_PosNumVS |
| Promotion:Charge<br>Plan:Recurring<br>Charge Details<br>(Charge_Plan_Recurr<br>ing)                      | Recurring Charge<br>(Recurring)              | The schedule for the<br>recurring charge.                                                                              | EGO_Lib_NumVS    |
| Single-Row                                                                                               |                                              |                                                                                                                        |                  |
| -                                                                                                        | RC Adjust Amount<br>(RC_Adjust)              | The adjustment for a<br>recurring charge.                                                                              | EGO_Lib_NumVS    |
| -                                                                                                        | RC Adjust U/M<br>(RC_Adjust_UM)              | The time period to<br>which the recurring<br>charge adjustment<br>applies. Options<br>include Per Day and<br>Per Week. | RC_Adjust_UM     |
| Promotion:Charge<br>Plan: Charges,<br>Adjustment,Usage<br>Plan Details<br>(Charge_Plan_Adjust<br>_Usage) | Adjust Charge<br>(Adjust_Charge)             | A special adjustment<br>to the amount the<br>customer is charged.                                                      | EGO_Lib_NumVS    |
| Single-Row                                                                                               |                                              |                                                                                                                        |                  |
| -                                                                                                        | Adjust Reason<br>(Adjust_Reason)             | The reason for the<br>special adjustment.                                                                              | -                |

| Attribute Group<br>Display Name<br>(Internal Name)                      | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                                                                                                                                                                                                                                          | Value Set           |
|-------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Behavior</b>                                                         |                                              |                                                                                                                                                                                                                                                                                                                                                                                |                     |
|                                                                         | Commit (Commit)                              | Indicates if the user is finished making changes to the record.                                                                                                                                                                                                                                                                                                                | EGO_Lib_Yes_No      |
| -                                                                       | Charge Basis<br>(Charge_Basis)               | <p>Specify when the charge for a period is generated. Options include:</p> <ul style="list-style-type: none"> <li>• Schedule.<br/>Charges are generated after the usage period. For example, June charges are generated on July 1.</li> <li>• Advance.<br/>Charges are generated in advance of the usage period. For example, June charges are generated on June 1.</li> </ul> | Charge_Basis        |
| Promotion:<br>Commitment:<br>Charges Credits<br>(Commit_Charge_Credits) | Charge Credits Type<br>(Charge_Credits_Type) | Select the penalty.                                                                                                                                                                                                                                                                                                                                                            | Charge_Credits_Type |
| Multi-Row                                                               |                                              |                                                                                                                                                                                                                                                                                                                                                                                |                     |
| -                                                                       | Amount (Amount)                              | The amount of the penalty.                                                                                                                                                                                                                                                                                                                                                     | EGO_Lib_NumVS       |

| Attribute Group<br>Display Name<br>(Internal Name)                    | Attribute Display<br>Name (Internal<br>Name)            | Attribute Description                                                                                                                                                         | Value Set     |
|-----------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Behavior</b>                                                       |                                                         |                                                                                                                                                                               |               |
| -                                                                     | Plan (Plan)                                             | The name of the charge plan that is associated. The plan defines the charges that must be applied if the commitment for the promotion is terminated.                          | Plan          |
| Promotion:<br>Commitment:Terms<br>(Commitment_Terms<br>)<br>Multi-Row | Number<br>(Number_Value)                                | Enter a number to control what order this text will be displayed in within the list of terms.                                                                                 | EGO_Lib_NumVS |
| -                                                                     | Name (Name)                                             | Name of the commitment term.                                                                                                                                                  | -             |
| -                                                                     | Description<br>(Description)                            | Enter a brief description of the term.                                                                                                                                        | -             |
| -                                                                     | Commitment Terms<br>Type<br>(Commitment_Terms<br>_Type) | Select the type of the term. Options include Standard, Non-Standard, and Special. When you design screens and reports, you can decide to display only terms of certain types. | Term_Type     |
| Promotion:Upgrade<br>(Promotion_Upgrade<br>)<br>Multi-Row             | Original Promotion<br>(Original_Promotion<br>)          | The earlier promotion that customers can break without penalty.                                                                                                               | Promotion     |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                                                                                                                                                                                                                                                                                              | Value Set        |
|----------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Behavior</b>                                    |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| -                                                  | Commitment Start<br>(Commitment_Start)       | <p>Specify when the commitment starts for the new promotion.<br/>Options include:</p> <ul style="list-style-type: none"> <li>Original Start. The commitment starts at the same time the commitment for the earlier promotion started.</li> <li>Now. The commitment starts when the customer accepts the new promotion.</li> <li>Original End. The commitment starts when the commitment for the earlier promotion ends.</li> </ul> | Commitment_Start |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                                                                                                                                                                                                                                                                                            | Value Set     |
|----------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Behavior</b>                                    |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| -                                                  | Duration (Duration)                          | <p>Specify the duration of the new promotion. Options include:</p> <ul style="list-style-type: none"> <li>Original Duration. The duration of the new promotion is the same as the duration of the original promotion.</li> <li>New Duration. The duration of the new promotion is different from the duration of the original promotion. If you select this, you must enter the new duration in the Commitments view.</li> </ul> | Duration      |
| -                                                  | Penalty (Penalty)                            | The penalty for upgrading from the old to the new promotion, if any.                                                                                                                                                                                                                                                                                                                                                             | EGO_Lib_NumVS |
| -                                                  | Currency (Currency)                          | Currency in which the penalty is defined.                                                                                                                                                                                                                                                                                                                                                                                        | -             |

| Attribute Group<br>Display Name<br>(Internal Name)                    | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                                                                                      | Value Set        |
|-----------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Behavior</b>                                                       |                                              |                                                                                                                                                                                                                            |                  |
| Product Promotions:<br>Components<br>(Prod_Promotions_C<br>omponents) | Min (Min)                                    | Minimum cardinality<br>for the component of<br>the promotion.                                                                                                                                                              | EGO_Lib_PosNumVS |
| Single-Row                                                            |                                              |                                                                                                                                                                                                                            |                  |
| <b>Important:</b> This is<br>a component<br>attribute group.          |                                              |                                                                                                                                                                                                                            |                  |
| -                                                                     | Max (Max)                                    | Maximum cardinality<br>for the component of<br>the promotion.                                                                                                                                                              | EGO_Lib_PosNumVS |
| -                                                                     | Default<br>(Default_Value)                   | Default cardinality for<br>the component of the<br>promotion.                                                                                                                                                              | EGO_Lib_PosNumVS |
| -                                                                     | Recommendable<br>(Recommendable)             | Select this check box<br>to enable the system<br>to recommend this<br>product promotion to<br>end users when the<br>product exists in a<br>quote or order and is<br>not associated with<br>any other product<br>promotion. | EGO_Lib_Yes_No   |
| -                                                                     | Commitment<br>(Commitment)                   | Select this check box if<br>the product involves a<br>commitment.                                                                                                                                                          | EGO_Lib_Yes_No   |

| Attribute Group<br>Display Name<br>(Internal Name)                                                                              | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                    | Value Set            |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Behavior</b>                                                                                                                 |                                              |                                                                                                                                                          |                      |
| Component Pricing<br>(Component_Pricing)<br><br>Single-Row<br><br><b>Important:</b> This is<br>a component<br>attribute group.  | Adjustment Type<br>(Adjustment_Type)         | Select the type of price<br>adjustment applied to<br>the component<br>product of the<br>promotion.                                                       | Adjustment_Type      |
| -                                                                                                                               | Value (Value)                                | Value of the<br>adjustment.                                                                                                                              | EGO_Lib_NumVS        |
| -                                                                                                                               | Currency Code<br>(Currency_Code)             | Currency associated<br>with the value of the<br>adjustment.                                                                                              | Currency_Code_Select |
| -                                                                                                                               | Exchange Date<br>(Exchange_Date)             | Exchange date for the<br>currency.                                                                                                                       | -                    |
| Version: Structure<br>(Version_Structure)<br><br>Single-Row<br><br><b>Important:</b> This is<br>a component<br>attribute group. | Relationship Name<br>(Relationship_Name)     | Identifies the name of<br>the association of the<br>child to the parent. If<br>not specified then it<br>defaults to product<br>name.                     | -                    |
| -                                                                                                                               | Domain Type<br>(Domain_Type)                 | Identifies the type of<br>association of the<br>child to the parent. It's<br>a Domain Value Map<br>(DVM) and the values<br>are 'product' and<br>'class'. | Domain_Type          |

| Attribute Group<br>Display Name<br>(Internal Name)                                                                                                                                  | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                                                                        | Value Set        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Behavior</b>                                                                                                                                                                     |                                              |                                                                                                                                                              |                  |
| -                                                                                                                                                                                   | Product Class<br>(Product_Class)             | If the relationship name is 'Class', this refers to the Class name.                                                                                          | Product_Class    |
| -                                                                                                                                                                                   | Default Product<br>(Default_Product)         | If the domain type = 'class', then the domain for that child will have more than one product. This represents the default product among the set of products. | Product          |
| -                                                                                                                                                                                   | Max Cardinality<br>(Max_Cardinality)         | Maximum cardinality of the child in context of the parent.                                                                                                   | EGO_Lib_PosNumVS |
| -                                                                                                                                                                                   | Min Cardinality<br>(Min_Cardinality)         | Minimum cardinality of the child in context of the parent.                                                                                                   | EGO_Lib_PosNumVS |
| -                                                                                                                                                                                   | Default Cardinality<br>(Default_Cardinality) | Default cardinality of the child in context of the parent.                                                                                                   | EGO_Lib_PosNumVS |
| Product Promotions:<br>Pricing: Components:<br>Adjustments<br>(Product_Promotion_<br>Com_Adj)<br><br>Single-Row<br><br><b>Important:</b> This is<br>a component<br>attribute group. | Start Date<br>(Start_Date)                   | Start date for the adjustment defined for the components of a customizable item included in a promotion.                                                     | -                |



| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name) | Attribute Description                                                                                  | Value Set        |
|----------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------|
| <b>Behavior</b>                                    |                                              |                                                                                                        |                  |
| -                                                  | End Date (End_Date)                          | End date for the adjustment defined for the components of a customizable item included in a promotion. | -                |
| -                                                  | Adjustment Type (Adjustment_Type)            | Type of adjustment.                                                                                    | -                |
| -                                                  | Adjustment Value (Adjustment_Value)          | Value of the adjustment.                                                                               | EGO_Lib_NumVS    |
| -                                                  | Maximum Price (Maximum_Price)                | Maximum price that can be charged.                                                                     | EGO_Lib_PosNumVS |
| -                                                  | Minimum Price (Minimum_Price)                | Minimum price EGO_Lib_PosNumVS that can be charged.                                                    | -                |

## Product Management Library - Horizontal

Oracle Product Hub provides the following seeded library containing attribute groups, attributes, and value sets. For more information about using data libraries, see: Using Seeded Attributes, page 4-36. The attribute groups in this library are assigned to the Item Revision business entity.

This library is not automatically installed. The system administrator must perform the following steps to upload the library after installing the Oracle Product Hub for Communications patch.

### Installing the Library

1. Upload the value set library file. This file contains all of the value sets required by the Product Management Library, as well as those required by all other Oracle Product Hub for Communications libraries.

**Tip:** You only need to install this file once for all Oracle Product Hub for Communications libraries.

- **Filename**

\$EGO\_TOP/patch/115/import/US/egomstrlibvs.ldt

- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefval.lct
$EGO_TOP/patch/115/import/US/egomstrlibvs.ldt
ORACLE Password:
```

## 2. Upload the Product Management Library file.

- **Filename**

\$EGO\_TOP/patch/115/import/US/egopmtlib.ldt

- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefag.lct
$EGO_TOP/patch/115/import/US/egopmtlib.ldt
ORACLE Password:
```

### *Product Management Library - Horizontal*

| Attribute Group<br>Display Name<br>(Internal Name)                                          | Attribute Display<br>Name (Internal<br>Name) | Attribute<br>Description                                                                                                                                                                                                              | Value Set          |
|---------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Behavior</b>                                                                             |                                              |                                                                                                                                                                                                                                       |                    |
| Destination System<br>Specification<br>(Destination_Sys_Spe<br>cification)<br><br>Multi-Row | Destination System<br>(Destination_System)   | Users can add multiple destination locations. Either PIM ABCS or the Destination ABCS can look at this info and 'ignore' or 'accept' the product. The list of systems is driven by a value set that contains a list of spoke systems. | Destination_System |
| -                                                                                           | Sync Item<br>(Sync_Item)                     | This defines if the item should be synchronized or not.                                                                                                                                                                               | EGO_Lib_Yes_No     |

| Attribute Group<br>Display Name<br>(Internal Name)                             | Attribute Display<br>Name (Internal<br>Name) | Attribute<br>Description                                                                                                                                                                                                              | Value Set          |
|--------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Behavior</b>                                                                |                                              |                                                                                                                                                                                                                                       |                    |
| Destination Product<br>Role<br>(Destination_Product<br>_Role)<br><br>Multi-Row | Destination System<br>(Destination_System)   | Users can add multiple destination locations. Either PIM ABCS or the Destination ABCS can look at this info and 'ignore' or 'accept' the product. The list of systems is driven by a value set that contains a list of spoke systems. | Destination_System |
| -                                                                              | Role (Role)                                  | The Value Set is driven by Configurator, Billing, Fulfillment, Order Capture etc. (whatever role the product performs in that particular system).                                                                                     | Role               |
| -                                                                              | Selection (Selection)                        | -                                                                                                                                                                                                                                     | Selection          |
| -                                                                              | Selection Code<br>(Selection_Code)           | Customers use this as a discriminator to write code against this Code. NULL by default.                                                                                                                                               | Selection_Code     |

## Communications Services Billing Library - Vertical

Oracle Product Hub provides the following seeded library containing attribute groups, attributes, and value sets. For more information about using data libraries, see: Using Seeded Attributes, page 4-36. All seeded attribute groups in this library are assigned to the Item Revision business entity.

This library is not automatically installed. The system administrator must perform the following steps to upload the library after installing the Oracle Product Hub for Communications patch.

### Installing the Library

1. Upload the value set library file. This file contains all of the value sets required by the Communications Services Billing Library, as well as those required by all other Oracle Product Hub for Communications libraries.

**Tip:** You only need to install this file once for all Oracle Product Hub for Communications libraries.

- **Filename**

\$EGO\_TOP/patch/115/import/US/egomstrlibvs.ldt

- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefval.lct
$EGO_TOP/patch/115/import/US/egomstrlibvs.ldt
ORACLE Password:
```

2. Upload the Communications Services Billing Library file.

- **Filename**

\$EGO\_TOP/patch/115/import/US/egocsbib.ldt

- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefag.lct
$EGO_TOP/patch/115/import/US/egocsbib.ldt
ORACLE Password:
```

### *Communications Services Billing Library - Vertical*

| Attribute Group<br>Display Name<br>(Internal Name)                                          | Attribute Display<br>Name (Internal<br>Name) | Attribute<br>Description                                                                                                                                                                                                       | Value Set     |
|---------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Behavior</b>                                                                             |                                              |                                                                                                                                                                                                                                |               |
| Billing Products<br>Attributes General<br>(Billing_Prod_Attr_G<br>eneral)<br><br>Single-Row | Priority (Priority)                          | This priority sets the<br>order in which the<br>billing application<br>analyzes products<br>during the rating<br>process. The higher<br>the number, the<br>higher the priority;<br>zero is the lowest<br>priority you can set. | EGO_Lib_NumVS |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)       | Attribute<br>Description                                                                                                                                                                                                              | Value Set        |
|----------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Behavior</b>                                    |                                                    |                                                                                                                                                                                                                                       |                  |
| -                                                  | Multi-RUM Support<br>(Multi_RUM_Support<br>)       | Flag to indicate support of ratable usage metrics.                                                                                                                                                                                    | EGO_Lib_Yes_No   |
| -                                                  | Provisioning Tag<br>(Provisioning_Tag)             | This tag provides a framework to associate extended rating attributes that can enable rating or discounting to vary for a service.                                                                                                    | Provisioning_Tag |
| -                                                  | Supplier Tax ID<br>(Tax_Supplier)                  | This is the company or corporate division that sells the product and is responsible for collecting taxes.                                                                                                                             | Tax_Supplier     |
| -                                                  | Allow partial purchase<br>(Allow_Partial_Purchase) | Indicates whether customers are allowed to purchase a part of the product. For example, if a product gives customers 30 hours of internet connection for \$10, you can allow customers to purchase half the amount for half the cost. | EGO_Lib_NumVS    |
| -                                                  | Purchase Max Quantity<br>(Purchase_Max)            | Represents the maximum quantity of the product that can be purchased per transaction.                                                                                                                                                 | EGO_Lib_PosNumVS |

| Attribute Group<br>Display Name<br>(Internal Name)                                 | Attribute Display<br>Name (Internal<br>Name)                | Attribute<br>Description                                                                                                                     | Value Set                   |
|------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Behavior</b>                                                                    |                                                             |                                                                                                                                              |                             |
| -                                                                                  | Purchase Min<br>Quantity<br>(Purchase_Min)                  | Represents the<br>minimum quantity of<br>the product that can<br>be purchased per<br>transaction.                                            | EGO_Lib_PosNumVS            |
| -                                                                                  | Ownership Max Qty<br>(Own_Max)                              | Represents the<br>maximum quantity of<br>the product that can<br>be owned by the<br>customer at any given<br>point of time.                  | EGO_Lib_PosNumVS            |
| -                                                                                  | Ownership Min Qty<br>(Own_Min)                              | Represents the<br>minimum quantity of<br>the product that can<br>be owned by the<br>customer at any given<br>point of time.                  | EGO_Lib_PosNumVS            |
| Billing Products<br>Event Map<br>(Billing_Products_<br>Event_Map)<br><br>Multi-Row | Dynamic Discount<br>Method<br>(Dynamic_Discount_<br>Method) | Whenever a discount<br>override is applied to<br>the charge, the field<br>identifies the type of<br>discount, such as<br>'Percent' 'Amount'. | Dynamic_Discount_<br>Method |
| -                                                                                  | Event (Event)                                               | The name of the<br>billable event, or<br>charges modeled as<br>the event in the<br>billing application.                                      | Event                       |
| -                                                                                  | Measured By<br>(Measured_By)                                | Represents the ratable<br>usage metric (RUM).<br>RUM defines the<br>quantity measured<br>and the units used.                                 | Measured_By                 |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)     | Attribute<br>Description                                                                                                                                                                                                                                                                               | Value Set                  |
|----------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Behavior</b>                                    |                                                  |                                                                                                                                                                                                                                                                                                        |                            |
| -                                                  | Pricing Commit Type<br>(Pricing_Commit_Ty<br>pe) | Field that identifies<br>whether a discount<br>override or a price<br>override is defined<br>for the charge.                                                                                                                                                                                           | Pricing_Commit_Typ<br>e    |
|                                                    | Rate Plan Structure<br>(Rate_Plan_Structure)     | This identifies the<br>type of rate plan<br>associated with the<br>product. Single rate<br>plan - Only one rate<br>plan can be associated<br>with the event. Rate<br>Plan Selector – More<br>than one rate plan can<br>be selected from a<br>selector. Custom<br>event analysis –<br>custom rate plan. | BRM_Rate_Plan_Str<br>cture |
|                                                    | Rate Plan ID<br>(Rate_Plan_ID)                   | The identifier of the<br>rate plan associated<br>with the billable<br>event.                                                                                                                                                                                                                           | -                          |
| Rate Plan (Rate_Plan)<br>Multi-Row                 | Price List Name<br>(Pricelist_Name)              | Price list associated<br>with Rate plan of an<br>item.                                                                                                                                                                                                                                                 | Pricelist_Name             |
| -                                                  | Currency (Currency)                              | Currency associated<br>with rate plan.                                                                                                                                                                                                                                                                 | Currency_Code_Sele<br>cted |
| -                                                  | Rate Plan ID<br>(Rate_Plan_ID)                   | Rate plan unique<br>identifier.                                                                                                                                                                                                                                                                        | -                          |
| -                                                  | Rate Plan Name<br>(Rate_Plan_Name)               | Rate plan name                                                                                                                                                                                                                                                                                         | -                          |
| -                                                  | Rate Plan Description<br>(Rate_Plan_Desc)        | Rate Plan Description                                                                                                                                                                                                                                                                                  | -                          |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)                | Attribute<br>Description                                                                                                                | Value Set            |
|----------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Behavior</b>                                    |                                                             |                                                                                                                                         |                      |
| -                                                  | Rate Plan Selector ID<br>(Rate_Plan_Selector_ID)            | Rate plan selector unique identifier. The rate plan selector provides a framework to define attribute based pricing in billing systems. | -                    |
| -                                                  | Rate Plan Selector Name<br>(Rate_Plan_Selector_Name)        | Rate Plan selector name.                                                                                                                | -                    |
| -                                                  | Rate Plan Selector Description<br>(Rate_Plan_Selector_Desc) | Rate Plan Selector Description.                                                                                                         | -                    |
| -                                                  | Tax When<br>(Tax_When)                                      | Identifies when the tax must be applied.                                                                                                | Tax_When             |
| -                                                  | (Tax_Code)                                                  | This is user defined. Identifies the tax code that is applied.                                                                          | Tax_Code             |
| -                                                  | Bill in Advance<br>(Advance_Billing)                        | Flag to indicate whether advance billing is allowed.                                                                                    | EGO_Lib_Yes_No       |
| -                                                  | Charge Cycle Fees<br>(Charge_Cycle_Fees)                    | Field to indicate when the monthly charge is billed in advance of the billing cycle.                                                    | -                    |
| -                                                  | Charge Cycle Fees UOM<br>(Charge_Cycle_Fee_UOM)             | Unit of measure for the 'Charge cycle fees in advance of billing cycle'.                                                                | Charge_Cycle_Fee_UOM |



| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)     | Attribute<br>Description                                                                                                                                                                                                           | Value Set      |
|----------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Behavior</b>                                    |                                                  |                                                                                                                                                                                                                                    |                |
| -                                                  | Apply Cycle Fee<br>(Apply_Cycle_Fee)             | Flag to indicate that the monthly charge must be applied on the billing date.                                                                                                                                                      | EGO_Lib_Yes_No |
| Rate Data (Rate_Data)<br>Multi-Row                 | Rate Data ID<br>(Rate_Data_ID)                   | Rate data unique identifier.                                                                                                                                                                                                       | -              |
| -                                                  | Rate Data Name<br>(Rate_Data_Name)               | Rate data name.                                                                                                                                                                                                                    | -              |
| -                                                  | Override Credit Limit<br>(Override_Credit_Limit) | Flag to identify whether the credit limit is overridden. A credit limit is the maximum amount of a resource, such as currency or hours, that can accumulate in an account before the customer is prevented from using the service. | EGO_Lib_Yes_No |

| Attribute Group<br>Display Name<br>(Internal Name)  | Attribute Display<br>Name (Internal<br>Name)                                            | Attribute<br>Description                                                                                                                                                                                                                                                                                                                                                                                                          | Value Set                           |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Behavior</b>                                     |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |
| -                                                   | Quantity Discount<br>Bracket Basis<br>(Qty_Dis_Bracket_Ba<br>sis)                       | You can group<br>charges together into<br>quantity discount<br>brackets to charge<br>different amounts<br>based on quantity.<br>For example, for a<br>telephony service,<br>you could charge 10<br>cents a minute for the<br>first 120 minutes and<br>5 cents per minute<br>after 120 minutes.<br>Specifies how the<br>billing application<br>determines the<br>quantities it uses to<br>select the quantity<br>discount bracket. | Qty_Disc_Bracket_Ba<br>sis          |
| -                                                   | Quantity Discount<br>Bracket Basis<br>Resources<br>(Qty_Dis_Bracket_Ba<br>sis_Resource) | If the discount<br>bracket basis is<br>'resource balance', it<br>identifies the<br>resource.                                                                                                                                                                                                                                                                                                                                      | Qty_Disc_Bracket_Ba<br>sis_Resource |
| -                                                   | Purchase Proration<br>Information<br>(Pur_Proration)                                    | Specify how to charge<br>customers when they<br>purchase the product<br>in the middle of the<br>accounting cycle.                                                                                                                                                                                                                                                                                                                 | Proration                           |
| -                                                   | Cancel Proration<br>Information<br>(Cancel_Proration)                                   | Specify how to charge<br>customers when they<br>cancel the product in<br>the middle of the<br>accounting cycle.                                                                                                                                                                                                                                                                                                                   | Proration                           |
| Balance Impact<br>(Balance_Impact)<br><br>Multi-Row | Sequence Number<br>(Seq_Number)                                                         | Identifies the<br>sequence of the<br>balance impact.                                                                                                                                                                                                                                                                                                                                                                              | -                                   |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)                  | Attribute<br>Description                                                                                           | Value Set   |
|----------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Behavior</b>                                    |                                                               |                                                                                                                    |             |
| -                                                  | Rate Plan ID<br>(Rate_Plan_ID)                                | Rate plan unique<br>identifier.                                                                                    | -           |
| -                                                  | Rate Tier ID (Tier_ID)                                        | Rate tier unique<br>identifier.                                                                                    | -           |
| -                                                  | Day Time Range ID<br>(Day_Time_Range_ID<br>)                  | Day Time Range<br>unique identifier.                                                                               | -           |
| -                                                  | Day of the Week<br>Range ID<br>(Day_of_the_Week_R<br>ange_ID) | Day of the week<br>unique identifier.                                                                              | -           |
| -                                                  | Rate Data ID<br>(Rate_Data_ID)                                | Rate data unique<br>identifier                                                                                     | -           |
| -                                                  | Rate Data Minimum<br>(Rate_Data_Min)                          | Minimum quantity of<br>an event a customer<br>must use for the<br>balance impact or<br>charge to be valid.         | -           |
| -                                                  | Rate Data Maximum<br>(Rate_Data_Max)                          | Maximum quantity of<br>an event a customer<br>must use for the<br>balance impact or<br>charge to be valid.         | -           |
| -                                                  | Resource ID<br>(Resource_ID)                                  | Identifies the resource<br>that will be charged<br>for the product. For<br>example, dollars, free<br>minutes, etc. | Resource_ID |
| -                                                  | GLID (GLID)                                                   | An ID assigned to<br>events or charges to<br>record revenue.                                                       | GLID        |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name) | Attribute<br>Description                                                                                                                                                                                                        | Value Set       |
|----------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Behavior</b>                                    |                                              |                                                                                                                                                                                                                                 |                 |
| -                                                  | Impact Category<br>(Impact_Category)         | Defines the impact category associated with the rate.                                                                                                                                                                           | Impact_Category |
| -                                                  | Proratable<br>(Proratable)                   | Flag to identify that the resource is pro-rated.                                                                                                                                                                                | EGO_Lib_Yes_No  |
| -                                                  | Discountable<br>(Discountable)               | Flag to identify that the resource is discounted.                                                                                                                                                                               | EGO_Lib_Yes_No  |
| -                                                  | Sponsorable<br>(Sponsorable)                 | Enables the resource to be sponsored by another account as part of a sponsor group. In the billing application, you can create sponsor groups where a group owner is billed for certain charges for all of the group's members. | EGO_Lib_Yes_No  |
| -                                                  | Grantable (Grantable)                        | Flag to identify that the resource is granted by the balance impact. This will allow specifying the start date and end date on the balance impact.                                                                              | EGO_Lib_Yes_No  |
| -                                                  | Fixed Amount<br>(Fixed_Amount)               | Represents the fixed amount/price that is applied on the resource that is associated with the billing product.                                                                                                                  | -               |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)          | Attribute<br>Description                                                                                                                       | Value Set              |
|----------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Behavior</b>                                    |                                                       |                                                                                                                                                |                        |
| -                                                  | Scaled Amount<br>(Scaled_Amount)                      | Represents the amount that can be credited/debited based on rating of the resource. For example, usage of minutes or internet download limits. | -                      |
| -                                                  | Units (Units)                                         | Represents the units for the rating. Usually a picklist [Second, Minute, Kilobyte, Megabyte,...]                                               | Balance_Impact_Units   |
| -                                                  | Promotional Price<br>(Promotional_Price)              | Indicates a promotional or introductory price.                                                                                                 | -                      |
| -                                                  | Service Pricing<br>Method<br>(Service_Pricing_Method) | -                                                                                                                                              | Service_Pricing_Method |
| -                                                  | Service Price Percent<br>(Service_Price_Percent)      | A percentage that is used to mark up the price of the service product.                                                                         | -                      |
| -                                                  | Start Date<br>(Start_Date)                            | Start date of the balance impact.                                                                                                              | -                      |
| -                                                  | End Date (End_Date)                                   | End date of the balance impact.                                                                                                                | -                      |
| Tier Group<br>(Tier_Group)                         | Tier Name<br>(Tier_Name)                              | Name of the rate tier.                                                                                                                         | -                      |
| Multi-Row                                          |                                                       |                                                                                                                                                |                        |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name) | Attribute<br>Description                                                                                                                                                                                                                                                                                                     | Value Set              |
|----------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Behavior</b>                                    |                                              |                                                                                                                                                                                                                                                                                                                              |                        |
| -                                                  | Tier ID (Tier_ID)                            | Unique identifier of the rate tier.                                                                                                                                                                                                                                                                                          | -                      |
| -                                                  | Effectivity Mode (Effectivity_Mode)          | Identifies the effectivity mode associated with the date and time of the tier.<br><br>'Absolute' - sets the time period when the rate tier is in effect by specifying dates.<br><br>'Relative' -sets the time period when the rate tier is in effect by specifying the number of days relative to the product purchase date. | Effectivity_Mode       |
| -                                                  | Start Date Time (Start_Date_Time)            | Start date and time of the tier.                                                                                                                                                                                                                                                                                             | -                      |
| -                                                  | End Date Time (End_Date_Time)                | End date and time for the tier.                                                                                                                                                                                                                                                                                              | -                      |
| -                                                  | Relative Start (Relative_Start)              | Relative start from the product purchase date.                                                                                                                                                                                                                                                                               | -                      |
| -                                                  | Relative Start UOM (Relative_Start_UOM)      | UOM for the relative start date.                                                                                                                                                                                                                                                                                             | Relative_Start_End_UOM |
| -                                                  | Relative End (Relative_End)                  | Relative end when the rate tier is no longer active.                                                                                                                                                                                                                                                                         | -                      |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)           | Attribute<br>Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Value Set                  |
|----------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Behavior</b>                                    |                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |
| -                                                  | Relative End UOM<br><br>Relative_End_UOM               | UOM for the relative<br>end date.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Relative_Start_End_U<br>OM |
| -                                                  | Time Day Restrictions<br><br>Time_Day_Restrictio<br>ns | <p>Indicates whether<br/>Day Time restrictions<br/>are associated with<br/>the Rate Tier.</p> <ol style="list-style-type: none"> <li>1. No Restrictions:<br/>The rate tier has<br/>no day or time<br/>restrictions. If<br/>you have a day<br/>or time of day<br/>range in your<br/>rate structure,<br/>selecting this<br/>removes it.</li> <li>2. Day Restrictions:<br/>Set day of week<br/>and month of<br/>day restrictions<br/>for a rate.</li> <li>3. Time<br/>Restrictions: Set<br/>time of day<br/>restrictions for a<br/>rate.</li> </ol> | Time_Day_Restrictio<br>ns  |

| Attribute Group<br>Display Name<br>(Internal Name)                                                                                                                                     | Attribute Display<br>Name (Internal<br>Name)    | Attribute<br>Description                         | Value Set      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------|----------------|
| <b>Behavior</b>                                                                                                                                                                        |                                                 |                                                  |                |
| Day Time Range<br>(Day_Time_Range)<br><br>Multi-Row                                                                                                                                    | Day Time Range ID<br>(Day_Time_Range_ID<br>)    | Day and time range<br>unique identifier.         | -              |
| <b>Additional Information:</b> All date and time attributes within this group are time zone independent, meaning there is no need to change the time entered for different time zones. |                                                 |                                                  |                |
| -                                                                                                                                                                                      | Day Range Name<br>(Day_Range_Name)              | Day restrictions name.                           | -              |
| -                                                                                                                                                                                      | Day Range Start Date<br>(Day_Range_Start_Date2) | Starting day.                                    | -              |
| -                                                                                                                                                                                      | Day Range End Date<br>(Day_Range_End_Date2)     | Ending day.                                      | -              |
| -                                                                                                                                                                                      | Use Time of Day Ranges<br>(Use_Time_Day_Ranges) | Select whether or not to use time of day ranges. | EGO_Lib_Yes_No |
| -                                                                                                                                                                                      | Time of Day Range Name<br>(Time_Day_Range_Name) | Time range restrictions name.                    | -              |
| -                                                                                                                                                                                      | Start Time<br>(Start_Time2)                     | Start time.                                      | -              |



| Attribute Group<br>Display Name<br>(Internal Name)                                                                                                                                     | Attribute Display<br>Name (Internal<br>Name)                      | Attribute<br>Description              | Value Set        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------|------------------|
| <b>Behavior</b>                                                                                                                                                                        |                                                                   |                                       |                  |
| -                                                                                                                                                                                      | End Time<br>(End_Time2)                                           | End time.                             | -                |
| Days of the Week<br>Range<br>(Days_of_the_Week_<br>Range)                                                                                                                              | Day of the Week<br>Range ID<br>(Day_of_the_Week_R<br>ange_ID)     | Day of the week<br>unique identifier. | -                |
| Multi-Row                                                                                                                                                                              |                                                                   |                                       |                  |
| <b>Additional Information:</b> All date and time attributes within this group are time zone independent, meaning there is no need to change the time entered for different time zones. |                                                                   |                                       |                  |
| -                                                                                                                                                                                      | Day of the Week<br>Range Name<br>(Day_of_the_Week_R<br>ange_Name) | Day of the week<br>restrictions name. | -                |
| -                                                                                                                                                                                      | Starting Week Day<br>(Start_Day_of_the_W<br>eek)                  | Starting week day.                    | Days_of_the_Week |
| -                                                                                                                                                                                      | Ending Week Day<br>(End_Day_of_the_We<br>ek)                      | Ending week day.                      | Days_of_the_Week |
| -                                                                                                                                                                                      | Start Date for the<br>Week<br>(Start_Date_for_the_<br>Week2)      | Starting date for the<br>week.        | -                |

| Attribute Group<br>Display Name<br>(Internal Name)                                | Attribute Display<br>Name (Internal<br>Name)                          | Attribute<br>Description                                                                                                | Value Set      |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Behavior</b>                                                                   |                                                                       |                                                                                                                         |                |
| -                                                                                 | End Date for the<br>Week<br>(End_Date_for_the_Week2)                  | Ending date for the week.                                                                                               | -              |
| -                                                                                 | Use Time in Day of<br>the Week<br>(Use_Time_in_Day_of_the_Week)       | Flag to indicate whether time range is specified.                                                                       | EGO_Lib_Yes_No |
| -                                                                                 | Time in Day of week<br>range name<br>(Time_in_Day_of_week_range_name) | Time range restrictions name.                                                                                           | -              |
| -                                                                                 | Start Time in Day of<br>the Week<br>(Start_Time_in_Day_of_the_Week2)  | Start time.                                                                                                             | -              |
| -                                                                                 | End Time in Day of<br>the Week<br>(End_Time_in_Day_of_the_Week2)      | End time.                                                                                                               | -              |
| Billing Discount<br>Attributes<br>(Billing_Discount_Attributes)<br><br>Single-Row | Discount Type<br>(Discount_Type)                                      | Identifies the type of discount. It is equivalent to Product Type. Possible values include:<br>Subscription and System. | Discount_Type  |
| -                                                                                 | Priority (Priority)                                                   | Priority for applying the discount.                                                                                     | EGO_Lib_NumVS  |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)                       | Attribute<br>Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Value Set                     |
|----------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Behavior</b>                                    |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                               |
| -                                                  | Multiple Discounts<br>per Event<br>(Multi_Discounts_Per<br>_Event) | Identifies the way<br>discounts are applied,<br>if there are multiple<br>discounts associated<br>per event/charge.<br>Cascading discounts:<br>only apply to parts of<br>the charge packet that<br>have not already been<br>considered for<br>discount. Parallel<br>discounts: Evaluated<br>independently of<br>each other and are<br>always considered.<br>The entire charge<br>packet is discounted,<br>regardless of whether<br>it has been<br>discounted<br>previously.<br>Sequential discounts:<br>Applied as long as a<br>customer charge<br>remains. | Multi_Discounts_Per<br>_Event |
| -                                                  | Ownership<br>Maximum Quantity<br>(Ownership_Qty_Ma<br>ximum)       | Total quantity of<br>services/products<br>owned above which<br>the discount does not<br>apply. If set to 0<br>(default) then it is not<br>used.                                                                                                                                                                                                                                                                                                                                                                                                            | EGO_Lib_NumVS                 |
| -                                                  | Ownership Minimum<br>Quantity<br>(Ownership_Qty_Min<br>imum)       | Total Quantity of<br>services/products<br>owned below which<br>the discount does not<br>apply. If set to 0<br>(default) then it is not<br>used.                                                                                                                                                                                                                                                                                                                                                                                                            | EGO_Lib_NumVS                 |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)                                   | Attribute<br>Description                                                                                                                                                                                                                                                                                                                                                                         | Value Set            |
|----------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Behavior</b>                                    |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                  |                      |
| -                                                  | Provisioning Tag<br>(Provisioning_Tag)                                         | Provides a framework to associate extended rating attributes that can enable rating or discounting to vary for a service.                                                                                                                                                                                                                                                                        | Provisioning_Tag     |
| -                                                  | Purchase Maximum Quantity<br>(Purchase_Qty_Maximum)                            | Represents the maximum quantity of the discount that can be purchased per transaction.                                                                                                                                                                                                                                                                                                           | EGO_Lib_NumVS        |
| -                                                  | Purchase Minimum Quantity<br>(Purchase_Qty_Minimum)                            | Represents the minimum quantity of the discount that can be purchased per transaction.                                                                                                                                                                                                                                                                                                           | EGO_Lib_NumVS        |
| -                                                  | Usage Discount - Valid from middle of the cycle<br>(Usage_Discount_Valid_From) | The field defines validity rules for discounting a usage charge. Identifies the method of handling a discount purchased in the middle of an accounting cycle. Full Discount: Grants the discount for the whole accounting cycle. Prorated Discount: Grants the discount for the portion of the accounting cycle the discount is owned. No Discount: Grants no discount for the accounting cycle. | Usage_Cycle_Discount |

| Attribute Group<br>Display Name<br>(Internal Name)                | Attribute Display<br>Name (Internal<br>Name)                                             | Attribute<br>Description                                                                                                                                                           | Value Set                |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Behavior</b>                                                   |                                                                                          |                                                                                                                                                                                    |                          |
| -                                                                 | Usage Discount -<br>Valid to middle of the<br>cycle<br>(Usage_Discount_Val<br>id_To)     | Identifies the method<br>of handling a<br>discount canceled in<br>the middle of an<br>accounting cycle.                                                                            | Usage_Cycle_Discou<br>nt |
| -                                                                 | Usage Discount -<br>Valid only part of the<br>cycle<br>(Usage_Discount_Val<br>id_Only)   | Identifies the method<br>of handling a<br>discount purchased<br>and canceled in the<br>middle of an<br>accounting cycle.                                                           | Usage_Cycle_Discou<br>nt |
| -                                                                 | Cycle Discount -<br>Valid from middle of<br>the cycle<br>(Cycle_Discount_Vali<br>d_From) | The field defines<br>validity rules for<br>discounting a cycle<br>charge. Identifies the<br>method of handling a<br>discount purchased<br>in the middle of an<br>accounting cycle. | Usage_Cycle_Discou<br>nt |
| -                                                                 | Cycle Discount -<br>Valid to middle of the<br>cycle<br>(Cycle_Discount_Vali<br>d_To)     | Identifies the method<br>of handling a<br>discount canceled in<br>the middle of an<br>accounting cycle.                                                                            | Usage_Cycle_Discou<br>nt |
| -                                                                 | Cycle Discount -<br>Valid only part of the<br>cycle<br>(Cycle_Discount_Vali<br>d_Only)   | Identifies the method<br>of handling a<br>discount purchased<br>and canceled in the<br>middle of an<br>accounting cycle.                                                           | Usage_Cycle_Discou<br>nt |
| Billing Discount<br>Event Map<br>(Billing_Discount_Ev<br>ent_Map) | Event (Event)                                                                            | Represents the name<br>of a billable event or<br>charges modeled as<br>an event in the billing<br>application.                                                                     | Event                    |
| Multi-Row                                                         |                                                                                          |                                                                                                                                                                                    |                          |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)                | Attribute<br>Description                                                                                                                                                                                                             | Value Set               |
|----------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Behavior</b>                                    |                                                             |                                                                                                                                                                                                                                      |                         |
| -                                                  | Discount Structure<br>Type<br>(Discount_Structure_<br>Type) | Identifies the type of discount model associated with the product. Discount Model - Only one discount model can be associated with the event. Discount Model Selector- More than one discount model can be selected from a selector. | Discount_Structure_Type |
| -                                                  | Model / Model<br>Selector<br>(Model_Selector)               | The name of the discount model associated with the discount.                                                                                                                                                                         | -                       |
| -                                                  | Snowball (Snowball)                                         | If this discount/event combination is a snowball discount. A snowball discount enables distribution of group discounts to group members.                                                                                             | EGO_Lib_Yes_No          |
| -                                                  | Stop Discounting<br>(Stop_Discounting)                      | Identifies when to stop applying the discounts. [Never, When inactive, When canceled, When inactive or canceled]                                                                                                                     | Stop_Discounting        |

## Communications Product Details Library - Vertical

Oracle Product Hub provides the following seeded library containing attribute groups, attributes, and value sets. For more information about using data libraries, see: Using Seeded Attributes, page 4-36. All seeded attribute groups in this library are assigned to the Item Revision business entity.

This library is not automatically installed. The system administrator must perform the

following steps to upload the library after installing the Oracle Product Hub for Communications patch.

#### Installing the Library

1. Upload the value set library file. This file contains all of the value sets required by the Communications Product Details Library - Vertical, as well as those required by all other Oracle Product Hub for Communications libraries.

**Tip:** You only need to install this file once for all Oracle Product Hub for Communications libraries.

- **Filename**

\$EGO\_TOP/patch/115/import/US/egomstrlibvs.ldt

- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefval.lct
$EGO_TOP/patch/115/import/US/egomstrlibvs.ldt
ORACLE Password:
```

2. Upload the Communications Product Details Library - Vertical file.

- **Filename**

\$EGO\_TOP/patch/115/import/US/egocpdlib.ldt

- **Command**

```
FNDLOAD apps 0 Y UPLOAD
$EGO_TOP/patch/115/import/egoefag.lct
$EGO_TOP/patch/115/import/US/egocpdlib.ldt
ORACLE Password:
```

#### *Communications Product Details Library - Vertical*

| Attribute Group<br>Display Name<br>(Internal Name)          | Attribute Display<br>Name (Internal<br>Name)          | Attribute<br>Description                                                                  | Value Set       |
|-------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------|
| <b>Behavior</b>                                             |                                                       |                                                                                           |                 |
| Communications:<br>Product Info<br>(Comms_Product_Inf<br>o) | Network Element<br>Type<br>(Network_Element_T<br>ype) | Indicates whether the<br>network item<br>represents a node,<br>connection, or<br>network. | Network_Element |
| Single-Row                                                  |                                                       |                                                                                           |                 |

| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)              | Attribute<br>Description                                                                                                                                       | Value Set           |
|----------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Behavior</b>                                    |                                                           |                                                                                                                                                                |                     |
| -                                                  | Compound Product<br>(Compound_Product)                    | Specifies if this is a networking product which uses compound product validation rules.                                                                        | EGO_Lib_Yes_No      |
| -                                                  | Special Rating Max<br>Items<br>(Special_Rating_Max_Items) | Identifies the number of phone numbers allowed for the item when the Item represents a Special rating list. (Friends and Family list)                          | -                   |
| -                                                  | Special Rating Type<br>(Special_Rating_Type)              | Identifies the type of data in the special rating list.                                                                                                        | Special_Rating_Type |
| -                                                  | Composition Type<br>(Composition_Type)                    | This is fulfillment dependent information. It identifies what the item is composed of.<br>1. PartialItem 2. Wholeitem.                                         | Composition_Type    |
| -                                                  | Success Dependency<br>(Success_dependency)                | This field is used during order submission. It declares if all order items must fulfill successfully or else the whole order fails.<br>1. None 2. All or None. | Success_Dependency  |



| Attribute Group<br>Display Name<br>(Internal Name) | Attribute Display<br>Name (Internal<br>Name)           | Attribute<br>Description                                                                                         | Value Set      |
|----------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Behavior</b>                                    |                                                        |                                                                                                                  |                |
| -                                                  | Billable (Billable)                                    | This flag identifies the purchased products that will be sent to billing applications during order submission.   | EGO_Lib_Yes_No |
| -                                                  | Service Instance Enabled<br>(Service_Instance_Enabled) | This flag identifies a billing product which is also a bundle.                                                   | EGO_Lib_Yes_No |
| -                                                  | Fulfillment Item Code<br>(Fulfillment_Item_Code)       | This defaults to the ICC associated with the product, but there is a drop down to pick any ICC in the hierarchy. | -              |
| -                                                  | MSISDN Required<br>(MSISDN_Required)                   | MSISDN Required                                                                                                  | EGO_Lib_Yes_No |



---

## Web Services for Publishing

This appendix covers the following topics:

- Oracle Product Hub Web Services Overview
- Value Set Web Service
- Attribute Group Web Service
- Item Catalog Category Web Service
- Item Web Service
- Controlling the Output Payload
- Error Reporting

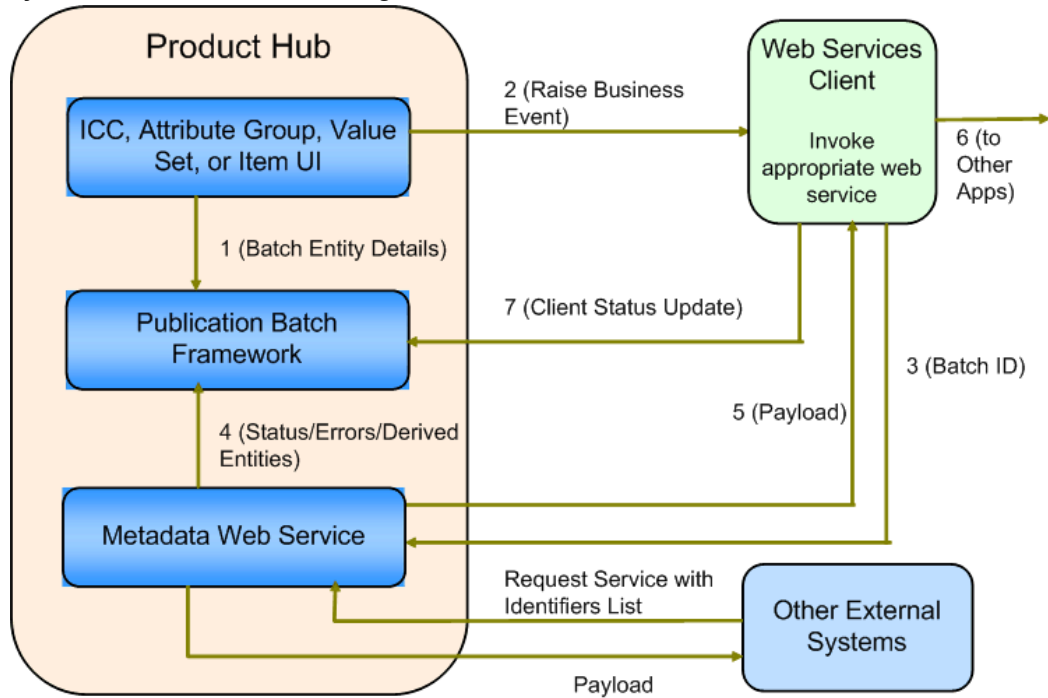
### Oracle Product Hub Web Services Overview

Users can publish item catalog categories, attribute groups, value sets, or items and extract more information about these entities by invoking entity specific Web Services for Oracle Product Hub. The following Web Services are available to Product Hub users:

- Item catalog category web service
- Attribute group web service
- Value set web service
- Item web service

The figure below shows a sample implementation that consumes one of these Web Services.

**System Flow and Interaction Diagram**



## Related Topics

Publishing Overview, *Oracle Product Hub User's Guide*

Item Catalog Category Web Service, page G-18

Attribute Group Web Service, page G-12

Value Set Web Service, page G-2

Item Web Service, page G-30

## Value Set Web Service

The value set Web Service includes the `getValueSetDetails` method, which returns the details of the value set according to the parameters requested in the output. Users can invoke the method on both value set versions and non-version value sets.

**Important:** Before using Publication Web Services, you must complete some manual post-installation steps. Complete the steps documented in My Oracle Support ID 1336146.1. This My Oracle Support ID also includes troubleshooting tips.

## Method: `getValueSetDetails`

This Web Service method provides an interface to extract the information about the value set and/or its hierarchy based on input.

### Modes

You can invoke the `getValueSetDetails` method using two modes:

#### Batch Mode

- Retrieves the details of the value set(s) published as part of a batch from the PIM Publication user interface.
- Use this mode when you can directly provide the batch ID to use the value set services. The parameters provided while creating the batch take precedence over the values provided for the parameters below while invoking the service.

#### List Mode

- Passes a list of value set identifiers directly as input to the `getValueSetDetails` method.
- Optionally provides any number of value set IDs for invoking the services.

While invoking the value set services in either mode, provide the following WS-Security and Parameter inputs:

### WS-Security

| Parameter Name | Type   | Optional | Description                          |
|----------------|--------|----------|--------------------------------------|
| User Name      | String | NO       | EBS "Applications User name"         |
| Password       | String | NO       | Password for EBS "Applications User" |

## Parameters

| Parameter Name | Type    | Optional                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BatchId        | Integer | No (batch mode)<br><br>Yes (non-batch mode)                                                   | The identifier of the Batch that was published from the PIM UI.                                                                                                                                                                                                                                                                                                                                                                                                |
| ValueSetId     | Integer | Yes (batch mode)<br><br>No (non-batch mode)                                                   | Value set ID for which the details need to be extracted/published.<br><br>In Batch Mode, the Value Set ID is derived for the selected value set, so there is no need to provide an ID.<br><br>In List Mode: <ul style="list-style-type: none"><li>• When publishing to external systems, you must provide the ValueSetId.</li><li>• When publishing to other Oracle Product Hub systems, you must provide either the ValueSetId or the ValueSetName.</li></ul> |
| ValueSetName   | String  | Yes<br><br><b>Note:</b> This parameter is not considered when publishing to external systems. | The name of the value set to publish to other Oracle Product Hub systems.<br><br>In Batch Mode, the value set name is derived for the selected value set, so there is no need to provide a name.<br><br>In List Mode, when publishing to other Oracle Product Hub systems, you must provide either the ValueSetId or the ValueSetName.                                                                                                                         |
| VersionSeqId   | Integer | Yes                                                                                           | Version for which the service is needed. If left blank, the default is the current effective version.                                                                                                                                                                                                                                                                                                                                                          |
| ChildValueSets | String  | No                                                                                            | Flag that indicates whether to publish child value sets associated with the selected value sets.                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name                                                                                                                          | Type    | Optional                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sync<br><br><b>Additional Information</b><br>: This appears as the "Publish to PIM Systems Only" checkbox in the batch publishing flow. | String  | No (batch mode),<br><br>No (non-batch mode)                                                  | Flag to indicate whether to publish to external systems or other Oracle Product Hub systems.<br><br>A value of Y indicates publishing to Oracle Product Hub systems.<br><br>A value of N indicates publishing to external systems.                                                                                                                                                                                                                                                                                                                                                                           |
| TriggerImport                                                                                                                           | String  | No<br><br><b>Note:</b> This parameter is not considered when publishing to external systems. | Flag to control whether to launch the 'EGO Import Metadata' concurrent program in the target Oracle Product Hub system or not. If it is set to 'N', the metadata interface tables in the target OPH systems are loaded with the data from the source OPH system. Users must manually launch this concurrent program to import the data in the interface tables into the production tables.<br><br>If it is set to 'Y', this concurrent program is automatically launched in the target OPH system, which then imports the data from the interface tables into the production tables.<br><br>Default Value: N |
| ReturnPayload                                                                                                                           | Boolean | Yes<br><br>Default: True                                                                     | Flag to control whether the output payload includes the details of the ICC(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### Multi Selection Parameters

| Parameter Name | Type   | Optional                                                                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LanguageCode   | String | Yes (batch mode)<br>Yes (non-batch mode)<br><br><b>Note:</b> This parameter is not considered when publishing to Oracle Product Hub systems. | <p>The language code in which to return the output.</p> <p>For example: US (for American English).</p> <p>Default: Return output in all installed languages.</p> <p><b>Note:</b> Repeat this parameter multiple times to return output in more than one language.</p> <p><b>When publishing to external systems:</b></p> <p>The language code in which to return the output, for example, US (for American English).</p> <p>Default: Return output in all installed languages.</p> <p><b>Note:</b> Repeat this parameter multiple times to return output in more than one language.</p> <p><b>When publishing to Oracle Product Hub systems:</b></p> <p>The value set translatable column</p> |



| Parameter Name | Type   | Optional                                                                                                                                      | Description                                                                                                                                                                                                                                                                                                                          |
|----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |        |                                                                                                                                               | values are published in all of the installed languages common between the source and target OPH systems.                                                                                                                                                                                                                             |
| LanguageName   | String | Yes (batch mode),<br>Yes (non-batch mode)<br><br><b>Note:</b> This parameter is not considered when publishing to Oracle Product Hub systems. | <b>When publishing to external systems:</b><br><br>The language name in which to return the output, for example, English.<br><br><b>When publishing to Oracle Product Hub systems:</b><br><br>Value set translatable column values are published in all of the installed languages common between the source and target OPH systems. |
| SystemCode     | String | Yes (batch mode)<br>No (non-batch mode)<br><br><b>Note:</b> This parameter is not considered when publishing to external systems.             | The system code for a target OPH system to which you plan to publish the ICC(s).<br><br>Default behavior: at least one target OPH system must be selected/provided.<br><br><b>Note:</b> Repeat this parameter multiple times when publishing to multiple systems.                                                                    |

| Parameter Name | Type   | Optional                                                                                                                          | Description                                                                                                                                                                                                                                                       |
|----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SystemName     | String | Yes (batch mode)<br>No (non-batch mode)<br><br><b>Note:</b> This parameter is not considered when publishing to external systems. | The system name for a target OPH system to which you plan to publish the ICC(s).<br><br>Default behavior: at least one target OPH system must be selected/provided.<br><br><b>Note:</b> Repeat this parameter multiple times when publishing to multiple systems. |

#### *FND Security Parameters*

| Parameter Name          | Type   | Optional                               | Description                                                                                                                                                                               |
|-------------------------|--------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESPONSIBILITY_N<br>AME | String | No (batch mode)<br>No (non-batch mode) | Provide the Responsibility name for the user.<br><br>For example:<br>EGO_DEVELOPMEN<br>T_MANAGER<br><br><b>Note:</b> In batch mode (using the UI flow), it is derived from the EBS Login. |

| Parameter Name               | Type   | Optional                                | Description                                                                                                                                                     |
|------------------------------|--------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESPONSIBILITY_A<br>PPL_NAME | String | No (batch mode)<br>No (non-batch mode)  | Application name for the above responsibility.<br><br>For example: EGO<br><br><b>Note:</b> In batch mode (using the UI flow), it is derived from the EBS Login. |
| SECURITY_GROUP_NAME          | String | Yes (batch mode)<br>No (non-batch mode) | Security group name.                                                                                                                                            |
| NLS_LANGUAGE                 | String | Yes (batch mode)<br>No (non-batch mode) | The current session language to use for the Web Service invocation.                                                                                             |

## Output

The output payload includes value set details based on the configurable parameters selected by the user.

### Payload when publishing to external systems:

- ListOfValueSets
  - Valueset
    - ValueSetId - Value set identifier.
    - ValueSetName - Name.
    - Description - Description.
    - DataType - Data type of the value set.
    - MaximumSize - Maximum size limit.
    - ValidationType - Validation type of the value set.
    - MinimumValue - Minimum value (applicable for only Number, Date, and

Date Time data types).

- **MaximumValue** - Maximum value (applicable for only Number, Date, and Date Time data types).
- **ParentValueSetName** - Parent value set name if this is a child value set.
- **ValuesetValue** - List of values for this value set (only for non-version value sets).
- **TranslatableAttributes**
- **ValuesetVersion** - List of versions for a value set with versions.
  - **VersionSeqId** - Sequence ID of the version.
  - **Description** - Description.
  - **StartActiveDate** - Effective start date.
  - **EndActiveDate** - Effective end date.
  - **VersionedVSValue** - List of values for this version.
- **TableInfo** - Validation information for a table type value set.
  - **TableName** - Database table name.
  - **WhereClause** - Where clause.
  - **ValueColumn** - Value column.
  - **IDColumn** - ID Column.
  - **MeaningColumn** - Meaning column.
- **Status** - Status information
  - **Error** - Error information
    - **Code** - Error code.
    - **Message** - Error message.
- **AdditionalInfo** - Additional info section when returnPayload input flag is set to false.

- SessionId - Session ID for invocation.
- EntityCount - Output entity count.

**Payload when publishing to Oracle Product Hub systems:**

- Status - Status information
  - Error - Error information
    - Code
    - Message
    - Language Code
    - System Code
  - Warning - Warning information
    - Code
    - Message
    - Language Code
    - System Code
- AdditionalInfo – Additional Info Section
  - SessionId
  - EntityCount

## Related Topics

Oracle Product Hub Web Services Overview, page G-1

Publishing Overview, *Oracle Product Hub User's Guide*

Defining Source Systems, page 12-5

Publishing Item Catalog Categories, page 3-45

Publishing Attribute Groups, page 4-54

Publishing Value Sets, page 4-16

Publishing Items and Structures, *Oracle Product Hub User's Guide*

Viewing the Publishing History, *Oracle Product Hub User's Guide*

## Attribute Group Web Service

The attribute group Web Service includes the getAGDetails method, which publishes the details of an attribute group and its associated entities to target Oracle Product Hub (OPH) systems based on input to these systems.

**Important:** Before using Publication Web Services, you must complete some manual post-installation steps. Complete the steps documented in My Oracle Support ID 1336146.1. This My Oracle Support ID also includes troubleshooting tips.

### Method: getAGDetails

The getAGDetails method provides an interface to extract information about attribute groups and their associated entities ( which include business entities, attributes, and value sets/child value sets associated to the attributes) based on input and then publish them to target OPH systems.

### Modes

You can invoke the getAGDetails method using two modes:

#### Batch Mode

- Retrieves the details of the attribute group(s) published as part of a batch from the PIM Publication user interface and publishes them to target OPH systems.
- Use this mode when you can directly provide the batch ID to use the attribute group services. The parameters provided while creating the batch take precedence over the values provided for the parameters below while invoking the service.

#### List Mode

- Passes a list of attribute group identifiers directly as input to the getAGDetails method.
- Optionally provides any number of attribute group IDs for invoking the services.

While invoking the attribute group services in either mode, provide the following WS-Security and Parameter inputs:

### **WS-Security**

| Parameter Name | Type   | Optional | Description                          |
|----------------|--------|----------|--------------------------------------|
| User Name      | String | NO       | EBS "Applications User name"         |
| Password       | String | NO       | Password for EBS "Applications User" |

### **Parameters**

| Parameter Name | Type    | Optional                                                        | Description                                                                                                                                                                                                                                                       |
|----------------|---------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BatchId        | Integer | No (batch mode)<br>Yes (non-batch mode)                         | The identifier of the Batch that was published from the PIM UI.                                                                                                                                                                                                   |
| AGId           | Integer | Yes (batch mode)<br>No (non-batch mode)                         | Attribute Group ID for which the details need to be published.<br><b>Note:</b> In Batch Mode, there is no need to provide the AGId. It is derived for the selected attribute group(s) in the batch. In List Mode, you must provide either the AGId or the AGName. |
| Agname         | String  | Yes (batch mode),<br>No (non-batch mode)                        | The name of the attribute group to be published.<br><b>Note:</b> In Batch Mode, there is no need to provide the AGname. It is derived for the selected attribute group(s) in the batch. In List Mode, you must provide either the AGId or the AGName.             |
| Valuesets      | String  | Yes (batch mode),<br>No (non-batch mode)<br>Default Value:<br>Y | Flag to control whether to publish the value sets associated with the user defined attributes to target OPH systems.                                                                                                                                              |

| Parameter Name                                              | Type   | Optional                                                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ChildValuesets                                              | String | Yes (batch mode),<br><br>No (non-batch mode)<br><br>Default Value:<br>Y                                         | Flag to control whether to publish the child value sets associated with the user defined attributes to target OPH systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Sync (In Batch flow - Publish to PIM Systems Only checkbox) | String | Yes (batch mode),<br><br>No (non-batch mode)<br><br>Default Value:<br>Y                                         | Flag to indicate whether to publish to external systems or OPH systems. Value Y indicates publishing to OPH systems. The attribute group Webservice supports publishing to OPH systems only.                                                                                                                                                                                                                                                                                                                                                                                  |
| Auto Import                                                 | String | Yes (batch mode),<br><br>No (non-batch mode)<br><br>Default Value:<br>N                                         | Flag to control whether to launch the 'EGO Import Metadata' concurrent program in the target OPH system or not.<br><br>If it is set to 'N', the metadata interface tables in the target OPH systems are loaded with the data from the source OPH system. Users must manually launch this concurrent program to import the data in the interface tables into the production tables.<br><br>If it is set to 'Y', this concurrent program is automatically launched in the target OPH system. The program imports the data from the interface tables into the production tables. |
| Return Payload                                              | String | No (non-batch mode)<br><br>Default Value:<br>N<br><br><b>Note:</b> This parameter is honored only in List mode. | Flag to control whether user wants to display the concurrent program errors and warnings that occur in the target OPH systems, as well as any additional information provided about the attribute groups.                                                                                                                                                                                                                                                                                                                                                                     |



### **Multi Selection Parameters**

| <b>Parameter Name</b> | <b>Type</b> | <b>Optional</b>                                                                                                                                                                                               | <b>Description</b>                                                                                                                                                           |
|-----------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LanguageCode          | String      | Yes (batch mode),<br>Yes (non-batch mode)<br><br>This parameter is not considered when publishing to OPH systems.                                                                                             | When publishing to OPH systems, attribute group translatable column values are published in all the installed languages common between the source and target OPH systems.    |
| LanguageName          | String      | Yes (batch mode),<br>Yes (non-batch mode)<br><br>This parameter is not considered when publishing to OPH systems.                                                                                             | When publishing to OPH systems, attribute group translatable column values are published in all of the installed languages common between the source and target OPH systems. |
| SystemCode            | String      | Yes (batch mode)<br><br>No (non-batch mode)<br><br>Default behavior: You must select at least one target OPH system.<br><br><b>Note:</b> Repeat this parameter multiple times to publish to multiple systems. | The system code for a target OPH system.                                                                                                                                     |

| Parameter Name | Type   | Optional                                                                                                                                                                                                            | Description                                 |
|----------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| SystemName     | String | Yes (batch mode),<br>No (non-batch mode)<br><br>Default behavior:<br>Select at least one<br>target OPH system.<br><br><b>Note:</b> Repeat this<br>parameter<br>multiple times to<br>publish to multiple<br>systems. | The system name for<br>a target OPH system. |

---

#### *FND Security Parameters*

| Parameter Name          | Type   | Optional                               | Description                                                                                                                                                                                                            |
|-------------------------|--------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESPONSIBILITY_N<br>AME | String | No (batch mode)<br>No (non-batch mode) | Provide the<br>Responsibility name<br>for the user. For<br>example:<br>EGO_DEVELOPMEN<br>T_MANAGER.<br><br><b>Note:</b> In batch<br>mode (UI flow),<br>the responsibility<br>name is derived<br>from the EBS<br>Login. |

---

| Parameter Name               | Type   | Optional                                | Description                                                                                                                                                      |
|------------------------------|--------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESPONSIBILITY_A<br>PPL_NAME | String | No (batch mode),<br>No (non-batch mode) | Application name for the above responsibility. For example: EGO<br><br><b>Note:</b> In batch mode (UI flow), the application name is derived from the EBS Login. |
| SECURITY_GROUP_<br>NAME      | String | Yes (batch mode)<br>No (non-batch mode) | Security group name.                                                                                                                                             |
| NLS_LANGUAGE                 | String | Yes (batch mode)<br>No (non-batch mode) | The current session language to use for the Web Service invocation.                                                                                              |

## Output

The output payload includes details based on the configurable parameters selected by the user.

### Payload when publishing to Oracle Product Hub systems:

- Status - Status information
  - Error - Error information
    - Code
    - Message
    - Language Code
    - System Code
  - Warning - Warning information
    - Code

- Message
- Language Code
- System Code
- AdditionalInfo – Additional Info Section
  - SessionId
  - EntityCount

## Related Topics

Oracle Product Hub Web Services Overview, page G-1

Publishing Overview, *Oracle Product Hub User's Guide*

Defining Source Systems, page 12-5

Publishing Value Sets, page 4-16

Publishing Attribute Groups, page 4-54

Publishing Item Catalog Categories, page 3-45

Publishing Items and Structures, *Oracle Product Hub User's Guide*

Viewing the Publishing History, *Oracle Product Hub User's Guide*

Controlling the Output Payload, page G-79

Error Reporting, page G-80

## Item Catalog Category Web Service

The item catalog category (ICC) Web Service includes the `getICCDetails` method, which returns the details of the ICC according to the parameters requested in the output.

Users can invoke the method on both ICC versions and non-version ICCs. It can also be used to publish an ICC and its associated entities to other Oracle Product Hub systems.

**Important:** Before using Publication Web Services, you must complete some manual post-installation steps. Complete the steps documented in My Oracle Support ID 1336146.1. This My Oracle Support ID also includes troubleshooting tips.

### Method: `getICCDetails`

This Web Service method provides an interface to extract the information about the ICC

and/or its hierarchy based on input. It can also be used to publish this data to other Oracle Product Hub systems.

## Modes

You can invoke the `getICCDetails` method using two modes:

### Batch Mode

- Retrieves the details of the ICC(s) published as part of a batch from the PIM Publication user interface.
- Use this mode when you can directly provide the batch ID to use the ICC services. The parameters provided while creating the batch take precedence over the values provided for the parameters below while invoking the service.

### List Mode

- Passes a list of ICC identifiers directly as input to the `getICCDetails` method.
- Optionally provides any number of ICC IDs for invoking the services.

While invoking the ICC services in either mode, provide the following WS-Security and Parameter inputs:

### WS-Security

| Parameter Name | Type   | Optional | Description                          |
|----------------|--------|----------|--------------------------------------|
| User Name      | String | NO       | EBS "Applications User name"         |
| Password       | String | NO       | Password for EBS "Applications User" |

### Parameters

| Parameter Name | Type    | Optional                                | Description                                                     |
|----------------|---------|-----------------------------------------|-----------------------------------------------------------------|
| BatchId        | Integer | No (batch mode)<br>Yes (non-batch mode) | The identifier of the Batch that was published from the PIM UI. |

| Parameter Name  | Type    | Optional                                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|---------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ICCID           | Integer | Yes (batch mode)<br><br>No (non-batch mode)                                                    | <p>The identifier of the ICC to be extracted/published.</p> <p><b>Note:</b> In Batch Mode, the ICCId is derived from the selected ICC.</p> <p>In List Mode:</p> <ul style="list-style-type: none"> <li>• When publishing to external systems, you must provide the ICCId.</li> <li>• When publishing to Oracle Product Hub systems, you must provide either the ICCId or the ICCName.</li> </ul> |
| ICCName         | String  | Yes<br><br><b>Note:</b> This parameter is not considered while publishing to external systems. | <p>The name of the ICC to be published to other Oracle Product Hub systems.</p> <p>In Batch Mode, the ICCName is derived from the selected ICC.</p> <p>In List Mode, provide either the ICCId or the ICCName.</p>                                                                                                                                                                                |
| VersionSequence | Integer | Yes (batch mode)<br><br>No (non-batch mode)                                                    | <p>Version for which the service is needed. If left blank, the default is the current effective version.</p>                                                                                                                                                                                                                                                                                     |
| UserDefAttrGrps | Boolean | Yes                                                                                            | <p>Flag to control whether to include the user defined attribute groups associated with the ICC(s) in the output payload or when publishing to Oracle Product Hub systems.</p> <p>Default Value : True</p>                                                                                                                                                                                       |

| Parameter Name | Type   | Optional                                                                                                      | Description                                                                                                                                                      |
|----------------|--------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Valuesets      | String | Yes<br><br><b>Note:</b> This parameter is not considered when publishing to external systems.                 | Flag to control whether to publish value sets associated with user defined attributes when publishing to Oracle Product Hub systems.<br><br>Default Value : True |
| ICCVersions    | String | Yes<br><br><b>Note:</b> This parameter is not considered when publishing to other Oracle Product Hub systems. | Flag that controls whether the version information for the ICC (s) is included in the output payload.<br><br>Default Value: True                                 |
| TransAttrs     | String | Yes<br><br><b>Note:</b> This parameter is not considered when publishing to other Oracle Product Hub systems. | Flag to control whether the transaction attributes for the ICC(s) are included in the output payload.<br><br>Default Value: True                                 |

| Parameter Name | Type   | Optional                                                                                                      | Description                                                                                                                    |
|----------------|--------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| ICCPages       | String | Yes<br><br><b>Note:</b> This parameter is not considered when publishing to external systems.                 | Flag that controls whether to publish pages associated with ICCs to Oracle Product Hub systems.<br><br>Default Value: True     |
| ICCFunctions   | String | Yes<br><br><b>Note:</b> This parameter is not considered when publishing to external systems.                 | Flag that controls whether to publish functions associated with ICCs to Oracle Product Hub systems.<br><br>Default Value: True |
| ICCStructure   | String | Yes<br><br><b>Note:</b> This parameter is not considered when publishing to other Oracle Product Hub systems. | Flag to control whether the structures for ICC(s) are included in the output payload.<br><br>Default Value: True               |



| Parameter Name                                                                                                    | Type   | Optional                                   | Description                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ParentlCCs                                                                                                        | String | Yes                                        | <p>Flag to control whether the parent hierarchy for the ICC(s) is included in the output payload or published to Oracle Product Hub systems.</p> <p>Default Value: True</p> <p><b>Note:</b> This parameter is honored only during "List" mode calls to the getlCCDetails method.</p> |
| ChildlCCs                                                                                                         | String | Yes                                        | <p>Flag to control whether the child hierarchy for ICC(s) is included in the output payload or published to Oracle Product Hub systems.</p> <p>Default Value: True</p> <p><b>Note:</b> This parameter is honored only during a "List" mode call to the getlCCDetails method.</p>     |
| Sync<br><br><b>Note:</b> This appears as the "Publish to PIM Systems Only" checkbox in the batch publishing flow. | String | No (batch mode)<br><br>No (non-batch mode) | <p>Flag to indicate whether to publish to external systems or other Oracle Product Hub systems.</p> <p>A value of Y indicates publishing to Oracle Product Hub systems.</p> <p>a value of N indicates publishing to external systems.</p>                                            |

| Parameter Name | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------|--------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Auto Import    | String | Yes      | <p>Flag to control whether to launch the 'EGO Import Metadata' concurrent program in the target Oracle Product Hub system or not.</p> <p>If it is set to 'N', the metadata interface tables in the target OPH systems are loaded with the data from the source OPH system. Users must manually launch this concurrent program to import the data in the interface tables into the production tables.</p> <p>If it is set to 'Y', this concurrent program is automatically launched in the target OPH system, which then imports the data from the interface tables into the production tables.</p> <p>Default Value: N</p> |
| ReturnPayload  | String | Yes      | <p>Publishing to external systems: Flag to control whether the output payload includes the ICC(s) details.</p> <p>Publishing to other Oracle Product Hub systems: Flag to control whether to display concurrent program errors and warnings occurring in the target OPH systems as well as any additional ICC information provided for publishing.</p> <p>Default Value: True</p>                                                                                                                                                                                                                                          |

### Multi Selection Parameters

| Parameter Name | Type   | Optional                                                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LanguageCode   | String | Yes (batch mode)<br><br>Yes (non-batch mode)<br><br><b>Note:</b> This parameter is not considered when publishing to Oracle Product Hub systems. | <b>When publishing to external systems:</b><br><br>The language code in which to return the output, for example, US (for American English).<br><br>Default: Return output in all installed languages.<br><br><b>Note:</b> Repeat this parameter multiple times to return output in more than one language.<br><br><b>When publishing to Oracle Product Hub systems:</b><br><br>When publishing to OPH systems, the attribute group translatable column values are published in all of the installed languages common between the source and target OPH systems. |

| Parameter Name | Type   | Optional                                                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LanguageName   | String | <p>Yes (batch mode),<br/>Yes (non-batch mode)</p> <p><b>Note:</b> This parameter is not considered when publishing to Oracle Product Hub systems.</p> | <p><b>When publishing to external systems:</b></p> <p>The language name in which to return the output, for example, English.</p> <p><b>When publishing to Oracle Product Hub systems:</b></p> <p>When publishing to OPH systems, attribute group translatable column values are published in all of the installed languages common between the source and target OPH systems.</p> |
| SystemCode     | String | <p>Yes (batch mode)<br/>No (non-batch mode)</p> <p><b>Note:</b> This parameter is not considered when publishing to external systems.</p>             | <p>The system code for a target OPH system to which you plan to publish the ICC(s)</p> <p>. Default behavior: at least one target OPH system must be selected/provided.</p> <p><b>Note:</b> Repeat this parameter multiple times when publishing to multiple systems.</p>                                                                                                         |

| Parameter Name | Type   | Optional                                                                                                                          | Description                                                                                                                                                                                                                                                          |
|----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SystemName     | String | Yes (batch mode)<br>No (non-batch mode)<br><br><b>Note:</b> This parameter is not considered when publishing to external systems. | The system name for the target OPH system to which you plan to publish the ICC(s) .<br><br>Default behavior: at least one target OPH system must be selected/provided.<br><br><b>Note:</b> Repeat this parameter multiple times when publishing to multiple systems. |

---

#### *FND Security Parameters*

| Parameter Name      | Type   | Optional                               | Description                                                                                                                                                                           |
|---------------------|--------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESPONSIBILITY_NAME | String | No (batch mode)<br>No (non-batch mode) | Provide the Responsibility name for the user.<br><br>For example:<br>EGO_DEVELOPMENT_MANAGER<br><br><b>Note:</b> In batch mode (using the UI flow), it is derived from the EBS Login. |

---

| Parameter Name               | Type   | Optional                                | Description                                                                                                                                                     |
|------------------------------|--------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESPONSIBILITY_A<br>PPL_NAME | String | No (batch mode)<br>No (non-batch mode)  | Application name for the above responsibility.<br><br>For example: EGO<br><br><b>Note:</b> In batch mode (using the UI flow), it is derived from the EBS Login. |
| SECURITY_GROUP_<br>NAME      | String | Yes (batch mode)<br>No (non-batch mode) | Security group name.                                                                                                                                            |
| NLS_LANGUAGE                 | String | Yes (batch mode)<br>No (non-batch mode) | The current session language to use for the Web Service invocation.                                                                                             |

## Output

The output payload includes details based on the configurable parameters selected by the user.

### Payload when publishing to external systems:

- ListOfICCs
  - ItemCatalogCategory
    - ItemCatalogGroupId - The ICC ID for which the details have been produced.
    - EnabledFlag - Flag indicating whether ICC is enabled.
    - InactiveDate - Inactive Date
    - ItemCreationAllowedFlag - Flag indicating ICC is available for item creation.
    - ParentCatalogGroupId - Parent ICC ID for the given ICC ID.
    - ParentCatalogGroupName - Parent ICC name for the given ICC ID.

- CategoryName - The concatenated name of the category formed from all segments.
- Segment1..20 - Segments containing the ICC Name.
- TranslatableAttributes
  - Language
  - Description
- ICCVersion
  - TransactionAttribute - Transaction attributes list and metadata for the ICC version.
  - ICCStructure - Structure associated with the ICC version and its metadata.
- UserDefAttrGrpAssociation - List of user defined attribute groups and their metadata associated to the ICC.
  - UserDefinedAttrGroup
    - TranslatableAttributes
  - UserDefAttrGrpBusEntity
  - UserDefinedAttribute - List of user defined attributes and their metadata.
    - TranslatableAttributes
    - Valueset
- Status - Status information
  - Error - Error information
    - Code
    - Message
- AdditionalInfo - Additional Info Section
  - SessionId

- EntityCount

**Payload when publishing to Oracle Product Hub systems:**

- Status - Status information
  - Error - Error information
    - Code
    - Message
    - Language Code
    - System Code
  - Warning - Warning information
    - Code
    - Message
    - Language Code
    - System Code
- AdditionalInfo – Additional Info Section
  - SessionId
  - EntityCount

## Related Topics

Publishing Overview, *Oracle Product Hub User's Guide*

Oracle Product Hub Web Services Overview, page G-1

Defining Source Systems, page 12-5

Publishing Item Catalog Categories, page 3-45

Controlling the Output Payload, page G-79

Error Reporting, page G-80

## Item Web Service

The item Web Service includes the getItemDetails and getStructureDetails methods, which return the details of the item according to the parameters requested in the



output.

**Important:** Before using Publication Web Services, you must complete some manual post-installation steps. Complete the steps documented in My Oracle Support ID 1336146.1. This My Oracle Support ID also includes troubleshooting tips.

## Method: getItemDetails

This Web Service method provides an interface to extract the information about the item and/or its hierarchy based on input.

### Modes

You can invoke the getItemDetails method using three modes:

#### Batch Mode

- Retrieves the details of the item(s) published as part of a batch from the PIM Publication user interface.
- Use this mode when you can directly provide the batch ID to use the item services. The parameters provided while creating the batch take precedence over the values provided for the parameters below while invoking the service.

#### Single Item Mode

- Use to pass item identifiers for a single item to the getItemDetails method.
- Use the basic input identifiers (InventoryItemId or InventoryItemName) and (OrganizationId or OrganizationCode) and (RevisionId or Revision or RevisionDate) in this mode.

#### List of Items Mode

- Apart from the single item mode, this passes a list of item identifiers directly as input to the getItemDetails method.
- Optionally provides any number of item identifiers for invoking the services.
- Use the list of basic input identifiers (InventoryItemId or InventoryItemName) and (OrganizationId or OrganizationCode) and (RevisionId or Revision or RevisionDate) in this mode.

While invoking the item services in either mode, provide the following WS-Security and Parameter inputs:

### ***WS-Security***

| <b>Parameter Name</b> | <b>Type</b> | <b>Optional</b> | <b>Description</b>                   |
|-----------------------|-------------|-----------------|--------------------------------------|
| User Name             | String      | NO              | EBS "Applications User name"         |
| Password              | String      | NO              | Password for EBS "Applications User" |

### ***Parameters***

| <b>itemQueryParameters</b> |             |                                                              |                                                                      |
|----------------------------|-------------|--------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Parameter Name</b>      | <b>Type</b> | <b>Optional</b>                                              | <b>Description</b>                                                   |
| BatchId                    | Integer     | No (batch mode)<br>Yes (list mode)<br>Yes (single item mode) | The batch identifier that was published from the PIM user interface. |
| InventoryItemId            | Integer     | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Item identifier for which to extract the details.                    |
| OrganizationId             | Integer     | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Organization identifier to which the publishing item belongs.        |
| OrganizationCode           | String      | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Organization code to which the publishing item belongs.              |

| Parameter Name             | Type    | Optional                                                                                                                                                                                                            | Description                                                                                                                                                                                                                       |
|----------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RevisionId                 | Integer | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode)                                                                                                                                                        | Revision ID of the publishing item.                                                                                                                                                                                               |
| Revision                   | String  | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode)                                                                                                                                                        | Revision code of the publishing item.                                                                                                                                                                                             |
| RevisionDate               | Date    | Yes (batch mode)<br>No (list mode)<br>No (single item mode)                                                                                                                                                         | Revision date of the publishing item. If the structure name is provided, this date is considered for explosion of the structure in list or single item mode.                                                                      |
| StructureName              | String  | Yes (batch mode)<br>No (list mode)<br>No (single item mode)                                                                                                                                                         | Structure name of the publishing item. All components of this structure are published along with the publishing item in list and single item mode.<br><br><b>Note:</b> For the primary structure, provide the value as "Primary". |
| <b>InventoryItem Name</b>  |         |                                                                                                                                                                                                                     |                                                                                                                                                                                                                                   |
| Segment1 through Segment19 | String  | Yes (batch mode)<br>Yes (List mode)<br>Single item mode:<br><ul style="list-style-type: none"> <li>• Yes – If InventoryItemI<br/>d is provided.</li> <li>• No – If InventoryItemI<br/>d is not provided.</li> </ul> | The name of each individual segment that comprises the Publishing item name.<br><br><b>Note:</b> The characteristics for all 19 segments are the same.                                                                            |

| Parameter Name               | Type    | Optional | Description                                                                                                                                                                                                                                    |
|------------------------------|---------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BomExploderParameters</b> |         |          |                                                                                                                                                                                                                                                |
| LevelsToExplode              | Integer | Yes      | Number of levels (0-60) to explode for a structure.<br>Default: 60                                                                                                                                                                             |
| ExplodeOption                | Integer | Yes      | The explode option for structure: <ol style="list-style-type: none"> <li>1. All</li> <li>2. Current</li> <li>3. Current and future. By default, current is considered.</li> </ol>                                                              |
| ExplodeStandard              | String  | Yes      | Whether to explode the standard structure or not. <ul style="list-style-type: none"> <li>• Y – Explodes the standard structure.</li> <li>• N – Do not explode the standard structure.</li> </ul> By default, standard structures are exploded. |
| <b>PublishEntities</b>       |         |          |                                                                                                                                                                                                                                                |
| OperationalAttributeGroups   | String  | Yes      | Used to turn operational attribute groups on and off in the payload. <ul style="list-style-type: none"> <li>• Y – turn on</li> <li>• N – turn off</li> </ul> Default Value: Y                                                                  |
| UserDefinedAttributeGroups   | String  | Yes      | Used to turn user defined attribute groups on and off in the payload. <ul style="list-style-type: none"> <li>• Y – turn on</li> <li>• N – turn off</li> </ul> Default Value: Y                                                                 |

| Parameter Name          | Type   | Optional | Description                                                                                                                                                                          |
|-------------------------|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ItemRevision            | String | Yes      | <p>Used to turn item revision details on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p>     |
| TransactionAttributes   | String | Yes      | <p>Used to turn transaction attributes on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p>    |
| RelatedItems            | String | Yes      | <p>Used to turn related items on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p>             |
| CustomerItems           | String | Yes      | <p>Used to turn customer items on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p>            |
| ManufacturerPartNumbers | String | Yes      | <p>Used to turn manufacturer part numbers on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> |

| Parameter Name                            | Type   | Optional | Description                                                                                                                                                                                                                                                                                 |
|-------------------------------------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GTINCrossReferences                       | String | Yes      | <p>Used to turn GTIN cross references on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p>                                                                                                            |
| AlternateCategoryAssignments              | String | Yes      | <p>Used to turn alternate category assignments on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p>                                                                                                   |
| SupplierAssignments                       | String | Yes      | <p>Used to turn supplier assignments on and off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p>                                                                                                             |
| ReturnPayload                             | String | Yes      | <p>Flag to control whether the output payload includes the details of the item(s).</p> <ul style="list-style-type: none"> <li>Y – includes the details of the item in the payload.</li> <li>N – does not include the details of the item in the payload.</li> </ul> <p>Default Value: Y</p> |
| <b>PublishOperationalAttribute Groups</b> |        |          |                                                                                                                                                                                                                                                                                             |

| Parameter Name | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|--------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ItemCatalog    | String | Yes      | <p>Used to turn the item catalog on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>ItemCatalog = N, then ItemCatalog is turned off in the payload.</li> <li>ItemCatalog = Y, then ItemCatalog is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>ItemCatalog = N, then ItemCatalog is turned off in the payload.</li> <li>ItemCatalog = Y, then ItemCatalog is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name           | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| InventoryCharacteristics | String | Yes      | <p>Used to turn InventoryCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>InventoryCharacteristics = N, then<br/>InventoryCharacteristics is turned off in the payload.</li> <li>InventoryCharacteristics = Y, then<br/>InventoryCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>InventoryCharacteristics = N, then<br/>InventoryCharacteristics is turned off in the payload.</li> <li>InventoryCharacteristics = Y, then<br/>InventoryCharacteristics is turned off in the payload</li> </ul> </li> </ul> |



| Parameter Name          | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PhysicalCharacteristics | String | Yes      | <p>Used to turn PhysicalCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>PhysicalCharacteristics = N, then PhysicalCharacteristics is turned off in the payload.</li> <li>PhysicalCharacteristics = Y, then PhysicalCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>PhysicalCharacteristics = N, then PhysicalCharacteristics is turned off in the payload.</li> <li>PhysicalCharacteristics = Y, then PhysicalCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name                 | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BillsOfMaterialCharacteristics | String | Yes      | <p>Used to turn BillsOfMaterialCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>BillsOfMaterialCharacteristics = N, then BillsOfMaterialCharacteristics is turned off in the payload.</li> <li>BillsOfMaterialCharacteristics = Y, then BillsOfMaterialCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>BillsOfMaterialCharacteristics = N, then BillsOfMaterialCharacteristics is turned off in the payload.</li> <li>BillsOfMaterialCharacteristics = Y, then BillsOfMaterialCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name                | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WorkInProcess Characteristics | String | Yes      | <p>Used to turn WorkInProcessCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>WorkInProcessCharacteristics = N, then WorkInProcessCharacteristics is turned off in the payload.</li> <li>WorkInProcessCharacteristics = Y, then WorkInProcessCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>WorkInProcessCharacteristics = N, then WorkInProcessCharacteristics is turned off in the payload.</li> <li>WorkInProcessCharacteristics = Y, then WorkInProcessCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name         | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CostingCharacteristics | String | Yes      | <p>Used to turn CostingCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>CostingCharacteristics = N, then CostingCharacteristics is turned off in the payload.</li> <li>CostingCharacteristics = Y, then CostingCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>CostingCharacteristics = N, then CostingCharacteristics is turned off in the payload.</li> <li>CostingCharacteristics = Y, then CostingCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name                    | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ProcessingLeadTimeCharacteristics | String | Yes      | <p>Used to turn ProcessingLeadTimeCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>ProcessingLeadTimeCharacteristics = N, then ProcessingLeadTimeCharacteristics is turned off in the payload.</li> <li>ProcessingLeadTimeCharacteristics = Y, then ProcessingLeadTimeCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>ProcessingLeadTimeCharacteristics = N, then ProcessingLeadTimeCharacteristics is turned off in the payload.</li> <li>ProcessingLeadTimeCharacteristics = Y, then ProcessingLeadTimeCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name          | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PlanningCharacteristics | String | Yes      | <p>Used to turn PlanningCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>PlanningCharacteristics = N, then PlanningCharacteristics is turned off in the payload.</li> <li>PlanningCharacteristics = Y, then PlanningCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>ProcessingLeadTimeCharacteristics = N, then ProcessingLeadTimeCharacteristics is turned off in the payload.</li> <li>ProcessingLeadTimeCharacteristics = Y, then ProcessingLeadTimeCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name            | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PurchasingCharacteristics | String | Yes      | <p>Used to turn PurchasingCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>PurchasingCharacteristics = N, then PurchasingCharacteristics is turned off in the payload.</li> <li>PurchasingCharacteristics = Y, then PurchasingCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>PurchasingCharacteristics = N, then PurchasingCharacteristics is turned off in the payload.</li> <li>PurchasingCharacteristics = Y, then PurchasingCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name           | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ReceivingCharacteristics | String | Yes      | <p>Used to turn ReceivingCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>ReceivingCharacteristics = N, then ReceivingCharacteristics is turned off in the payload.</li> <li>ReceivingCharacteristics = Y, then ReceivingCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>ReceivingCharacteristics = N, then ReceivingCharacteristics is turned off in the payload.</li> <li>ReceivingCharacteristics = Y, then ReceivingCharacteristics is turned off in the payload</li> </ul> </li> </ul> |



| Parameter Name                 | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OrderManagementCharacteristics | String | Yes      | <p>Used to turn OrderManagementCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>OrderManagementCharacteristics = N, then OrderManagementCharacteristics is turned off in the payload.</li> <li>OrderManagementCharacteristics = Y, then OrderManagementCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>OrderManagementCharacteristics = N, then OrderManagementCharacteristics is turned off in the payload.</li> <li>OrderManagementCharacteristics = Y, then OrderManagementCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name           | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| InvoicingCharacteristics | String | Yes      | <p>Used to turn InvoicingCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>InvoicingCharacteristics = N, then InvoicingCharacteristics is turned off in the payload.</li> <li>InvoicingCharacteristics = Y, then InvoicingCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>InvoicingCharacteristics = N, then InvoicingCharacteristics is turned off in the payload.</li> <li>InvoicingCharacteristics = Y, then InvoicingCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name            | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WebOptionsCharacteristics | String | Yes      | <p>Used to turn WebOptionsCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>WebOptionsCharacteristics = Y and either: <ul style="list-style-type: none"> <li>WebOptionsCharacteristics = N, then WebOptionsCharacteristics is turned off in the payload.</li> <li>WebOptionsCharacteristics = Y, then InvoicingCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>WebOptionsCharacteristics = N, then WebOptionsCharacteristics is turned off in the payload.</li> <li>WebOptionsCharacteristics = Y, then WebOptionsCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name         | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|--------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ServiceCharacteristics | String | Yes      | <p>Used to turn ServiceCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>ServiceCharacteristics = N, then<br/>ServiceCharacteristics is turned off in the payload.</li> <li>ServiceCharacteristics = Y, then<br/>ServiceCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>ServiceCharacteristics = N, then<br/>ServiceCharacteristics is turned off in the payload.</li> <li>ServiceCharacteristics = Y, then<br/>ServiceCharacteristics is turned off in the payload.</li> </ul> </li> </ul> |

| Parameter Name       | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AssetCharacteristics | String | Yes      | <p>Used to turn AssetCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>AssetCharacteristics = N, then AssetCharacteristics is turned off in the payload.</li> <li>AssetCharacteristics = Y, then AssetCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>AssetCharacteristics = N, then AssetCharacteristics is turned off in the payload.</li> <li>AssetCharacteristics = Y, then AssetCharacteristics is turned off in the payload</li> </ul> </li> </ul> |

| Parameter Name                 | Type    | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|---------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ProcessMfgCharacteristics      | String  | Yes      | <p>Used to turn ProcessMfgCharacteristics on or off in the payload.</p> <ul style="list-style-type: none"> <li>Y – turn on</li> <li>N – turn off</li> </ul> <p>Default Value: Y</p> <p>If OperationalAttributeGroups is Y, then only the value for this parameter is considered. For example, if:</p> <ul style="list-style-type: none"> <li>OperationalAttributeGroups = Y and either: <ul style="list-style-type: none"> <li>ProcessMfgCharacteristics = N, then ProcessMfgCharacteristics is turned off in the payload.</li> <li>ProcessMfgCharacteristics = Y, then ProcessMfgCharacteristics is turned on in the payload.</li> </ul> </li> <li>OperationalAttributeGroups = N and either: <ul style="list-style-type: none"> <li>ProcessMfgCharacteristics = N, then ProcessMfgCharacteristics is turned off in the payload.</li> <li>ProcessMfgCharacteristics = Y, then ProcessMfgCharacteristics is turned off in the payload</li> </ul> </li> </ul> |
| ListOfPublishedAttributeGroups |         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| UserDefinedAttributeGroups     |         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| AttributeId                    | Integer | -        | <p>The attribute ID identifies the transaction attribute to include in the payload. Use to configure the list of transaction attributes for a given revision to show in the payload. Use this to provide a list of attribute IDs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Parameter Name                                                                                      | Type   | Optional | Description                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AttributeName                                                                                       | String | -        | <p>The attribute internal name identifies the transaction attribute to include in the payload. Use this to provide a list of transaction attribute internal names. If neither AttributeId or AttributeName are provided, then all transaction attributes associated with the item for a given revision are published in the payload.</p> <p><b>Note:</b> AttributeName takes precedence over AttributeId.</p> |
| <b>ListOfLanguages</b>                                                                              |        |          |                                                                                                                                                                                                                                                                                                                                                                                                               |
| LanguageCode                                                                                        | String | Yes      | <p>The language code in which to return the translatable attributes in the payload.</p> <p>For example, US (for American English).</p> <p>By Default, the translatable attributes return in all installed languages.</p> <p><b>Note:</b> Repeat this parameter multiple times to return output in more than one language.</p>                                                                                 |
| LanguageName                                                                                        | String | Yes      | <p>The language name in which to return the translatable attributes in the payload.</p> <p>For example: English.</p>                                                                                                                                                                                                                                                                                          |
| <b>ItemsList</b>                                                                                    |        |          |                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>Important:</b> Provide the following parameters multiple times to publish multiple items.</p> |        |          |                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name   | Type    | Optional                                                     | Description                                             |
|------------------|---------|--------------------------------------------------------------|---------------------------------------------------------|
| InventoryItemId  | Integer | Yes (batch mode)<br>No (list mode)<br>Yes (single item mode) | Item ID for which to extract the details.               |
| OrganizationId   | Integer | Yes (batch mode)<br>No (list mode)<br>Yes (single item mode) | Organization ID to which the publishing item belongs.   |
| OrganizationCode | String  | Yes (batch mode)<br>No (list mode)<br>Yes (single item mode) | Organization code to which the publishing item belongs. |
| RevisionId       | Integer | Yes (batch mode)<br>No (list mode)<br>Yes (single item mode) | Revision ID of the publishing item.                     |
| Revision         | String  | Yes (batch mode)<br>No (list mode)<br>Yes (single item mode) | Revision code of the publishing item.                   |



| Parameter Name                                                                                                                   | Type   | Optional                                                                                                                                                                                                                                      | Description                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>InventoryItem Name</b><br><br><b>Important:</b><br>Provide the following parameters multiple times to publish multiple items. |        |                                                                                                                                                                                                                                               |                                                                                           |
| Segment1..<br>Segment20                                                                                                          | String | Batch mode: Yes<br><br>Single item mode: Yes<br><br>List of items mode: <ul style="list-style-type: none"> <li>• Yes – If InventoryItemId is provided for an item.</li> <li>• No – If InventoryItemId is not provided for an item.</li> </ul> | The individual segment's name that comprises the Publishing item name.                    |
| <b>FndSecurity</b>                                                                                                               |        |                                                                                                                                                                                                                                               |                                                                                           |
| RESPONSIBILITY_NAME                                                                                                              | String | No<br><br>(Mandatory for any mode other than batch mode.)                                                                                                                                                                                     | Provide the responsibility name for the user.<br><br>For example: EGO_DEVELOPMENT_MANAGER |
| RESPONSIBILITY_APPLICATION_NAME                                                                                                  | String | No<br><br>(Mandatory for any mode other than batch mode.)                                                                                                                                                                                     | Application name for the responsibility.<br><br>For example: EGO                          |

| Parameter Name      | Type   | Optional | Description                                                             |
|---------------------|--------|----------|-------------------------------------------------------------------------|
| SECURITY_GROUP_NAME | String | Yes      | Security group name.<br>For example: STANDARD                           |
| NLS_LANGUAGE        | String | Yes      | The current session language to be used for the Web Service invocation. |

## Output

The output payload includes item details based on the configurable parameters selected by the user.

### Payload

- ListOfItems
  - Item
    - MainCharacteristics - Main characteristics of the item.
      - MainCharTransalation - Translatable values item description and operating unit name.
    - ItemCatalog - Contains the item catalog to which the item belongs.
    - InventoryCharacteristics - Inventory characteristics of the item.
    - PhysicalCharacteristics - Physical characteristics of the item.
    - BillsOfMaterialCharacteristics - Bills of material characteristics of the item.
    - WorkInProgressCharacteristics - Work in process characteristics of the item.
    - CostingCharacteristics - Costing characteristics of the item.
    - ProcessingLeadTimeCharacteristics - Processing lead time characteristics of the item.
    - PlanningCharacteristics - Planning characteristics of the item.
    - PurchasingCharacteristics - Purchasing characteristics of the item.
    - ReceivingCharacteristics - Receiving characteristics of the item.

- OrderManagementCharacteristics - Order management characteristics of the item.
- InvoicingCharacteristics - Invoicing characteristics of the item.
- WebOptionsCharacteristics - Web options characteristics of the item.
- ServiceCharacteristics - Service characteristics of the item.
- AssetCharacteristics - Asset characteristics of the item.
- ProcessMfgCharacteristics - Process manufacturing characteristics.
- Organization - Organization details of the published item.
  - OrganizationTranslation - Translatable values organization name and operating unit name.
  - OrganizationAttributeGroup - Organization level attribute group.
    - ItemUDA - Organization level user defined attributes.
- AttributeGroup - Item level attribute groups.
  - ItemUDA - Item level user defined attributes.
- ItemRevision - Revision details of the item.
  - ItemRevisionTranslation - Translatable value of the revision description.
  - TransactionAttribute - Details of the transaction attributes for a revision.
  - RevisionAttributeGroup - Revision level attribute group.
    - ItemUDA - Revision level user defined attributes.
- CustomerItem - Customer items associated to the published item.
- ManufacturerPartNumber - Manufacturer part numbers associated to the published item.
- SupplierAssignment - Supplier assignments associated to the published item.
  - SupplierAttributeGroup - Supplier level attribute groups.

- ItemUDA -Supplier level user defined attributes.
- SupplierSiteAssignment - Supplier site assignments associated to the published item.
  - SupplierSiteAttributeGroup - Supplier site level attribute groups.
    - ItemUDA -Supplier site level user defined attributes.
- SupplierSiteOrganizationAssignment - Supplier site organization assignments associated to the published item.
  - SupplierSiteOrgAttributeGroup - Supplier site organization level attribute groups.
    - ItemUDA - Supplier site organization level user defined attributes.
- AlternateCategoryAssignment - Alternate category assignment associated to the published item.
  - AlternateCategoryTranslation - Translatable values of alternate catalog name and category description.
- RelatedItem - Related Items associated to the published item.
  - RelatedItemTranslations - Translatable values of the related item description.
- GTINCrossReference - GTIN cross references associated to the published item.
  - GTINCrossRefTranslation - Translatable values of the GTIN description.
- Status - Status information
  - Error - Details of the errors, if any, along with input identifiers.
    - Code - Error code.
    - Message - Error message.

## Method: getStructureDetails

This Web Service method provides an interface to extract the information about an item's structure based on input. The payload only contains the details about the

structure and its components. The payload does not contain item definition information for the components.

## Modes

You can invoke the `getStructureDetails` method using three modes. The mode used is derived from the XML elements provided.

### Batch Mode

- Retrieves the details of the structure payload for the items defined in a specific batch published from the PIM Publication user interface. Create the publication batch with the list of items and the structure name using PIM Publication user interface.
- Invoke this mode by providing the batch ID. The Web Service returns the structure payload for all items defined in that batch. The parameters provided while creating the batch take precedence over the values provided for the parameters below while invoking the service.

### Single Item Mode

- Use this web service to retrieve the structure payload for a given item without creating a publication batch. Provide the single item information directly in the input payload to retrieve the structure details.
- Use the basic input identifiers (`InventoryItemId` or `InventoryItemName`) and (`OrganizationId` or `OrganizationCode`) and (`RevisionId` or `Revision` or `RevisionDate`) in this mode.

### List of Items Mode

- You can also use this web service to retrieve the structure payload for a list of items without creating a publication batch. Apart from the single item mode, this passes a list of item identifiers directly as input to the `getStructureDetails` method.
- Optionally provides any number of item identifiers for invoking the services.
- Use the list of basic input identifiers (`InventoryItemId` or `InventoryItemName`) and (`OrganizationId` or `OrganizationCode`) and (`RevisionId` or `Revision` or `RevisionDate`) in this mode.

While invoking the item services in any mode, provide the following WS-Security and Parameter inputs:

### ***WS-Security***

| <b>Parameter Name</b> | <b>Type</b> | <b>Optional</b> | <b>Description</b>                           |
|-----------------------|-------------|-----------------|----------------------------------------------|
| User Name             | String      | NO              | EBS "Applications User name" (FND User Name) |
| Password              | String      | NO              | Password for EBS "Applications User"         |

### ***Parameters***

| <b>Parameter Name</b>    | <b>Type</b> | <b>Optional</b>                                              | <b>Description</b>                                                                                                                                                                                                                            |
|--------------------------|-------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| StructureQueryParameters | Complex     | Yes                                                          | Use this part of the XML document for single item mode.                                                                                                                                                                                       |
| InventoryItemId          | Integer     | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Use in single item mode to query the payload for a given inventory item ID.                                                                                                                                                                   |
| OrganizationId           | Integer     | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Use with the single item mode to query the payload for an item in a specific organization. Use an existing organization ID in the PIM instance. Use either the Organization Code or the Organization ID to retrieve a payload in this mode.   |
| OrganizationCode         | String      | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Use with the single item mode to query the payload for an item in a specific organization. Use an existing organization code in the PIM instance. Use either the Organization Code or the Organization ID to retrieve a payload in this mode. |

| Parameter Name             | Type    | Optional                                                     | Description                                                                                                                                                                                                                                                                                                       |
|----------------------------|---------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RevisionId                 | Integer | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Revision ID of the publishing item.                                                                                                                                                                                                                                                                               |
| Revision                   | String  | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Revision code of the publishing item.                                                                                                                                                                                                                                                                             |
| RevisionDate               | Date    | Yes (batch mode)<br>No (list mode)<br>No (single item mode)  | Use in single item mode to query the payload for a specific revision's structure of an item. Use a valid revision ID for the item provided in InventoryItemId or segments.<br>Provide one of the following fields as input for this mode: Revisionid /RevisionCode/RevisionDate.                                  |
| StructureName              | String  | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Use in single item mode to specify the name of the structure for which to query the payload. This is a mandatory input for single item mode.                                                                                                                                                                      |
| Segment1 through Segment15 | String  | Yes (batch mode)<br>Yes (list mode)<br>No (single item mode) | Use in single item mode to specify the name of the item for which to query the payload. Depending on the customer setup for item name segments, provide values for any of the 15 segment fields. Either a value in inventoryItemid or value(s) in one or more of these segments are required in single item mode. |

| Parameter Name             | Type    | Optional | Description                                                                                                                                                                                                                                                                                                       |
|----------------------------|---------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ListofLanguage Information | Complex | Yes      | Use this part of the XML input to provide the language information. Repeat this tag multiple times as needed to specify multiple language information.                                                                                                                                                            |
| LanguageCode               | String  | Yes      | Use in the single item mode to specify the language in which to return the payload. If no value is specified in both LanguageCode and LanguageName tags, the payload returns the information in all languages defined within the PIM instance.                                                                    |
| LanguageName               | String  | Yes      | Use in single item mode to specify the language in which to return the payload. If no value is specified in both LanguageCode and LanguageName tags, the payload returns the information in all languages defined within the PIM instance.                                                                        |
| LevelsToExplode            | Integer | Yes      | Use in single item mode to specify the number of levels to explode (0-60) for a structure.<br><br>Default: 60                                                                                                                                                                                                     |
| BomExploderParameters      | Complex | Yes      | Use to specify the BOM exploder behavior options as input. All the child elements control BOM explosion behavior.                                                                                                                                                                                                 |
| ExplodeOption              | Integer | Yes      | The explode options for the structure include: <ol style="list-style-type: none"> <li>1. All</li> <li>2. Current</li> <li>3. Current and future. By default, current is considered.</li> </ol> Default: Current                                                                                                   |
| ExplodeStandard            | String  | Yes      | Determines whether to explode the standard structure or not. <ul style="list-style-type: none"> <li>• Y – Explodes the standard structure.</li> <li>• N – Do not explode the standard structure.</li> </ul> By default, standard structures are exploded. The explosion only explodes the model and option class. |



| Parameter Name                 | Type    | Optional | Description                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|---------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PublishEntities                | Complex | Yes      | This specifies configurability options for the payload. By default, the payload includes all information about the structure. This enables users to pick and choose what information to include in the payload by providing values to the elements in this section. Valid values for any configurability options elements are Y or N. |
| StructureRevision              | String  | Yes      | Specify whether to include the structure revision information as part of the payload.                                                                                                                                                                                                                                                 |
| StructureHeaderAttributeGroups | String  | Yes      | Specify whether to include the structure attribute group information as part of the payload.                                                                                                                                                                                                                                          |
| StructureComponents            | String  | Yes      | Specify whether to include the structure components information as part of the payload.                                                                                                                                                                                                                                               |
| ReturnPayload                  | String  | Yes      | Flag that controls whether to include details in the output payload. For details, see Controlling the Output Payload, page G-79.                                                                                                                                                                                                      |
| PublishStructureAttributeGroup | Complex | Yes      | By default, the payload includes all attribute groups. However, you can specify the list of attribute groups to include. If attribute groups are specified, then only those attribute groups are included in the payload.                                                                                                             |
| AttributegroupId               | Integer | Yes      | UDA Attribute Group ID                                                                                                                                                                                                                                                                                                                |
| AttributeGroupName             | String  | Yes      | UDA attribute group name. Specify either the attribute group id or the attribute group name.                                                                                                                                                                                                                                          |
| PublishStructureComponents     | Complex | Yes      | Provides payload configurability options for structure component details. All configurability option elements accept Y or N values.<br><br>Default: Y, so, by default, the payload includes all information.                                                                                                                          |
| StructureReferenceDesignators  | String  | Yes      | Specify whether to include the reference designator in the payload.                                                                                                                                                                                                                                                                   |

| Parameter Name                        | Type    | Optional | Description                                                                                                                                                                                                             |
|---------------------------------------|---------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| StructureSubstituteComponents         | String  | Yes      | Specify whether to include the substitute component details in the payload.                                                                                                                                             |
| ComponentAttributeGroups              | String  | Yes      | Specifies whether to include the component UDAs in the payload.                                                                                                                                                         |
| ComponentExclusions                   | String  | Yes      | Specifies whether to include the component exclusions in the payload.                                                                                                                                                   |
| ComponentUDAOverrides                 | String  | Yes      | Specifies whether to include the component UDA override details in the payload.                                                                                                                                         |
| ValueSetExclusions                    | String  | Yes      | Specifies whether to include the value set exclusions in the payload.                                                                                                                                                   |
| ListofPublishComponentAttributeGroups | Complex | Yes      | By default, the payload includes all attribute groups. However, you can specify the list of attribute groups to include. If you specify attribute groups, then only those attribute groups are included in the payload. |
| AttributegroupId                      | Integer | Yes      | UDA attribute group ID                                                                                                                                                                                                  |
| AttributeGroupName                    | String  | Yes      | UDA attribute group name. You can specify either the attribute group ID or attribute group name.                                                                                                                        |
| BatchStructureQueryParameters         | Complex | Yes      | Use this part of the input payload when using batch mode.                                                                                                                                                               |
| BatchId                               | Integer | Yes      | Batch mode creates this batch ID during the item publication process. The return payload includes structure information for all items in this batch.                                                                    |
| Configurability                       | Complex | Yes      | All configurability options described for single item mode apply to batch mode.                                                                                                                                         |
| ListofItemStructureQueryParams        | Complex | Yes      | Use this part of the input payload when using list item mode.<br><br>All configurability options described for single item mode apply to the list item mode. The options apply to the whole list.                       |

| Parameter Name           | Type    | Optional                         | Description                                                                                                       |
|--------------------------|---------|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| InventoryItemId          | String  | Yes                              | Specifies in list item mode the inventory item ID in the list. You can either specify the ID or name.             |
| OrganizationId           | String  | Yes                              | Specify the organization for the item structure. You can either specify the organization ID or organization code. |
| OrganizationCode         | String  | Yes                              | Specify the organization for the item structure. You can either specify the organization ID or organization code. |
| RevisionId               | String  | Yes                              | Specify the revision of the item in the list. Provide either the revision ID or revision code.                    |
| Revision                 | String  | Yes                              | Specify the revision of the item in the list. Provide either the revision ID or revision code.                    |
| RevisionDate             | String  | Yes                              | Specify the revision date of the item in the list.                                                                |
| StructureName            | String  | Yes                              | Specify the structure name for which to return the payload.                                                       |
| Configurability          | Complex | Yes                              | The configurability options are same as single item mode.                                                         |
| FNDSecurity              | Complex | Yes                              | Specifies the FND securities information.                                                                         |
| RESPONSIBILITY_NAME      | String  | No<br>(Mandatory for list mode.) | Provide the responsibility name for the user.<br>For example: EGO_DEVELOPMENT_MANAGER                             |
| RESPONSIBILITY_APPL_NAME | String  | No<br>(Mandatory for list mode.) | Application name for the responsibility.<br>For example: EGO                                                      |
| SECURITY_GROUP_NAME      | String  | Yes                              | Security group name.<br>For example: STANDARD                                                                     |
| NLS_LANGUAGE             | String  | Yes                              | The current session language used for the Web Service invocation.                                                 |

## Output

The output payload includes structure details based on the configurable parameters selected by the user.

### Payload

- ListOfStructureHeaders
  - StructureHeader
    - AssemblyItemId - Inventory item identifier.
    - AssemblyItemName - Name of the assembly item.
    - Segment1..segment20 - Item name Segment1 through Segment20.
    - CommonAssemblyItemId - The inventory item ID of the common BOM assembly. This is applicable only for common BOMs.
    - CommonAssemblyItemName - Name of the common assembly item.
    - OrganizationId - Organization identifier.
    - OrganizationCode - Organization code.
    - SpecificAssemblyComment - Comment.
    - AssemblyType - 1 indicates a Manufacturing Bill. 2 indicates an Engineering Bill.
    - AlternateBomDesignator - Alternate designator code or structure name.
    - CommonBillSequenceId - Identifier of the common bill.
    - BillSequenceId - Bill Identifier.
    - CommonOrganizationId - Organization identifier of common bill.
    - CommonOrganizationCode - Organization Code for the common bill.
    - OperatingUnitId - Operating unit of the structure organization.
    - PendingFromEcn - Name of the ECN.
    - NextExplodeDate - Next date when pre-explosion will be refreshed.
    - ProjectId - Project identifier for Project Manufacturing.

- TaskId - Task identifier for Project Manufacturing.
- OriginalSystemReference - Original system that data for the current record has come from.
- StructureTypeId - Structure Type ID.
- ImplementationDate - Implementation date.
- ObjName - Object name identified if this is an assembly structure for an ICC structure.
- EffectivityControl - Specifies effectivity control for the structure.
- IsPreferred - Identifies if this is a preferred BOM.
- SourceBillSequenceId
- LastUpdateDate - Standard Who column.
- LastUpdateLogin - Standard Who column.
- AttributeCategory - Descriptive flexfield structure defining column.
- Attribute1..Attribute15 - Descriptive flexfield segment.
- Pk1Value..PK5Value - Additional primary key of the structure object.
- Language - Language code.
- PlanLevel - Plan level.
- StructureRevision - Structure revision details.
  - RevisionId - Assembly item revision ID.
  - Revision - Assembly item revision code.
  - RevisionLabel - Assembly item revision label.
  - RevisionReason - Revision reason.
  - ChangeNotice -Engineering change order number.
  - EcnInitiationDate - ECO Initiation date.
  - ImplementationDate - ECO implementation date.

- ImplementedSerialNumber - Implemented serial number.
- Description - Description
- LifecycleId - Lifecycle identifier
- CurrentPhaseId -Current Phase ID
- ObjectVersionNumber - Object version number tracks row updates.
- LastUpdateDate - Standard Who columns.
- ...
- LastUpdateLogin - Standard Who columns.
- EffectivityDate - Revision effectivity date.
- RevisedItemSequenceId - Revised item unique identifier.
- AttributeCategory - Descriptive flexfield structure defining column.
- Attribute1 - Descriptive flexfield segment.
- ...
- Attribute15 - Descriptive flexfield segment.
- RevisionDescription
  - LanguageCode - Language code
  - Description - Item revision translated description.
- StructureHeaderAttributeGroup
  - AttributegroupId - Identifier of user defined attribute group.
  - AttributeGroupName - User defined attribute group name.
  - ExtensionId - Row identifier in table BOM\_STRUCTURES\_EXT\_B.
  - StructureHeaderUDA
    - Name - Internal name of structure header attribute group.
    - CharValue - Character value of the UDA.

- ...
- TranslatedCharValue - Translated char value for an UDA
- HeaderTranslatableUDA - Translatable UDA details.
  - LanguageCode - Language code
  - TranslatedCharValue - Translated value of user defined attribute.
- HeaderUDADescription - Header UDA description.
  - Description - Description of user defined attribute.
  - LanguageCode - Language code.
- HeaderUDADisplayName
  - DisplayName - Display name of user defined attribute.
  - LanguageCode - Language code.
- HeaderAGDescription
  - Description - Description of attribute group.
  - LanguageCode - Language code.
- HeaderAGDisplayName
  - DisplayName - Display name of attribute group.
  - LanguageCode - Language code.
- StructureComponent - List of this assembly structure's direct child components with their operational attributes.
  - AssemblyItemId - Inventory Item Id of assembly item.
  - OrganizationId - Organization Id
  - CommonComponentSequenceId - Common component sequence ID.
  - ComponentSequenceId - Component sequence ID.

- OperationSeqNum - Operation sequence number to which this component belongs.
- ComponentItemId - Inventory Item ID for the component item.
- ComponentCode - Path of component in an assembly structure.
- ComponentItemName - Concatenated component item name.
- Segment1 - Individual Segment1 of component name.
- ...
- Segment20 - Individual Segment20 of component name.
- Description - Component item description.
- ItemSequenceNumber - Item number of this component on parent structure.
- PrimaryUomCode - Component UOM code.
- PrimaryUnitOfMeasure - Component UOM name.
- BasisType - Component basis type.
- ComponentQuantity - Component quantity.
- AutoRequestMaterial - Auto request flag.
- CurrentFuturePastFlag - Flag indicating if component is current or future effective.
- EffectivityDate - Effectivity date.
- DisableDate - Disable Date
- ImplementationDate - Implementation Date
- ChangeNotice - Change order name.
- PlanningFactor - Planning factor.
- ComponentYieldFactor - Yield factor.
- EnforceIntRequirements - Enforce Integer Requirements.



- IncludeInCostRollup - Include in cost rollup.
- BomItemType - BOM item type of component.
- WipSupplyType - WIP supply type.
- SupplySubinventory - Supply subinventory.
- SupplyLocatorId - Supply locator ID
- CheckAtp - Check ATP flag.
- Optional - Optional flag.
- MutuallyExclusiveOptions - Mutually exclusive options.
- LowQuantity - Low quantity.
- HighQuantity - High quantity.
- SoBasis - Sales order basis.
- ShippingAllowed - Shipping allowed.
- IncludeOnShipDocs - Include on shipping docs.
- RequiredToShip - Required to ship.
- TopItemId - Inventory item ID of top assembly.
- TopBillSequenceId - Bill sequence ID of top assembly.
- ParentBomItemType - Item type of parent assembly.
- ExplosionType - Explosion type.
- PlanLevel - Plan level.
- ExtendedQuantity - Extended quantity.
- SortOrder - Sort order for the hierarchy.
- CompBillSeqId - Component bill sequence ID.
- CompCommonBillSeqId - Common bill sequence ID.
- ParentSortOrder - Parent assembly sort order for the hierarchy.

- CompSourceBillSeqId - Component source bill sequence ID.
- SourceBillSequenceId - Source bill sequence ID.
- EcoForProduction - Indicates if this component on an Engineering Change Order is only for WIP jobs.
- OptionalOnModel - Optional on model.
- IncludeOnBillDocs - Include on billing documents.
- BaseItemId - Base item identifier.
- AtpComponentsFlag - Flag indicating whether an item has components requiring an ATP check.
- AtpFlag - Flag indicating whether to check ATP when item is ordered.
- PickComponentsFlag - Flag indicating to pick all shippable components.
- ReplenishToOrderFlag - Replenish to order flag.
- ShippableItemFlag - Shippable item flag.
- CustomerOrderFlag - Customer order flag.
- InternalOrderFlag - Internal order flag.
- CustomerOrderEnabledFlag - Customer Order Enabled Flag
- InternalOrderEnabledFlag - Internal Order Enabled Flag
- SoTransactionsFlag - Sales Order Transactions Flag
- LoopFlag - Flag indicating a loop in the hierarchy.
- ItemNum - Sequence number of item on parent structure.
- RexplodeFlag - ReExplode Flag
- SuggestedVendorName - Suggested vendor name
- VendorId - Vendor ID
- UnitPrice - Unit Price

- CreationDate - Standard Who column
- CreatedBy - Standard Who Column
- LastUpdateDate - Standard Who Column
- Context - Description flex field context
- Attribute1 - Descriptive flex field segment
- ...
- Attribute15 - Descriptive flex field segment
- QuantityRelated - Identifier to indicate if this component has quantity related reference designators.
- FromEndItemRevId - From End Item Revision ID
- ToEndItemRevId - To End Item Revision ID
- RevisionId - Revision ID
- Revision - Revision code
- StructureSubstituteComponents - List of substitute components for this structure component.
  - SubstituteComponentId - Inventory item ID of substitute component.
  - SubstituteCompItemName - Concatenated substitute component item name.
  - Segment1 - Individual segment1 of substitute component name
  - ...
  - Segment20 - Individual Segment20 of substitute component name.
  - SubstituteItemQuantity - Quantity of substitute items needed to replace the full component quantity.
  - ComponentSequenceId - Component Sequence ID of parent component.
  - AcdType - Type to indicate add or delete on an engineering change

order.

- ChangeNotice - Engineering change order number.
  - EnforceIntRequirements - Enforce integer requirements.
  - CommonComponentSequenceId - Common component sequence ID.
  - LastUpdateDate - Standard Who Column
  - ...
  - LastUpdateLogin - Standard Who Column
  - AttributeCategory - Descriptive flexfield structure defining column.
  - Attribute1 - Descriptive flexfield segment.
  - ...
  - Attribute15 - Descriptive flexfield segment
  - OriginalSystemReference - Original system from where the data for the current record came.
- StructureReferenceDesignators - List of reference designators for the parent component.
    - ComponentSequenceId - Component Sequence ID of the parent component.
    - ComponentReferenceDesignator - Component reference designator.
    - RefDesignatorComment - Reference designator comment.
    - CommonComponentSequenceId - Common component sequence ID.
    - ChangeNotice - Engineering change order number.
    - AcdType - Type to indicate add or delete on an engineering change order.
    - LastUpdateDate - Standard Who column

- ...
  - LastUpdateLogin - Standard Who Column
  - AttributeCategory - Descriptive flexfield structure defining column.
  - Attribute1 - Descriptive flexfield segment.
  - ...
  - Attribute15 - Descriptive flexfield segment.
  - OriginalSystemReference - Original system from where data for the current record came.
- ComponentAttributeGroup - List of attribute groups associated with this component.
    - AttributegroupId - Attribute Group ID
    - AttributeGroupName - Attribute group name
    - ExtensionId - Unique identifier for an attribute group and row.
  - ComponentUDA - List of attributes and their values associated with this attribute group.
    - Name - Attribute name
    - CharValue - Attribute character value
    - ...
    - DatetimeValue - Attribute datetime value
    - TranslatedCharValue - Attribute translated character value in the default language.
  - ComponentTranslatableUDA - Translatable attributes values in given languages.
    - LanguageCode - Language code.
    - TranslatedCharValue - Attribute translated character value in this language.

- ComponentUDADescription - List of component UDA descriptions in given languages.
  - Description - Translated description in given language.
  - LanguageCode - Language code.
- ComponentUDADisplayName - List of component UDA display names in given languages.
  - DisplayName - Translated display name in given language.
  - LanguageCode - Language code.
- ComponentAGDescription - List of component attribute group descriptions in given languages.
  - Description - Translated description in given language.
  - LanguageCode - Language code.
- ComponentAGDisplayName - List of component attribute group display names in given languages.
  - DisplayName - Translated display names in given languages.
  - LanguageCode - Language code.
- Status - Status information.
  - ListOfErrors - Error information.
    - Message - Error message.
    - Code - Error code.
    - LanguageCode - Language code of the error message in the payload.
    - InputIdentifier - The input identifier of the error record in the input payload.
  - ListOfWarnings - Warning information
    - Message - Warning message

- Code - Warning Code
  - LanguageCode - Language code in which the warning details are specified in the output payload.
- 
- AdditionalInfo - Additional info section when returnPayload input flag is set to false.
    - SessionId - Session ID for invocation.
    - EntityCount - Output entity count.
- 
- ContextInfo - The part of the XML document containing the context specific information for the top item in the Structure.
    - ComponentPath - Component path details.
      - ComponentItemList - List of items in the component path order.
        - ItemReference - Component item details in the component path order.
          - Name - Name of the item
          - Sequence - Sequence number specifying the component path for this component.
          - BomItemType - BOM item type of the component in the component path.
          - CatalogCategoryName - Item catalog category name of the component item.
    - ComponentCode - The internal component code of the component path.
    - OverrideAttrGrp - Override attribute group details.
      - AttributegroupId - Component attribute group name for which an override is defined in the above component path.
      - AttributeGroupName - Component attribute group name.
      - ExtensionId - Internal extension ID stored to refer the row in the extension table.

- ComponentCode - Component code that corresponds to the override component path.
  - ComponentOverrideUDA - Details about the override UDA columns.
    - Name - Name of the UDA.
    - CharValue - Char value.
    - ...
    - TranslatedCharValue - Translated values.
- ComponentTranslatableUDA - Translated UDA values.
  - ComponentExclusion Component - Exclusion details for the above component path. This exists only if the user defined an exclusion.
    - ComponentCode - Internal component code for the component path.
    - FromRevisionId - The effective revision from ID for the exclusion.
    - ToRevisionId - The effective revision to ID for the exclusion.
    - FromRevision - The effective revision from for the exclusion.
    - ToRevision - The effective revision to for the exclusion.
- ValuesetExclusion - Valueset exclusion details. This part of the payload includes information if the user defined any specific valueset exclusions in this component path.
  - ExcludedValue - Excluded value for the value set.
  - ComponentCode - Internal component code for the component path.
  - FromRevisionId - The effective revision from ID for the exclusion.
  - ToRevisionId - The effective revision to ID for the exclusion.
  - FromRevision - The effective revision from for the exclusion.
  - ToRevision - The effective revision to for the exclusion.



## Related Topics

Oracle Product Hub Web Services Overview, page G-1

Publishing Overview, *Oracle Product Hub User's Guide*

Publishing Items and Structures, *Oracle Product Hub User's Guide*

Controlling the Output Payload, page G-79

Error Reporting, page G-80

## Controlling the Output Payload

The output payload of Web Services when publishing to external systems can grow very large due to the number of entities and sub-entities returned. This can become a bottleneck in performance when the Web Service client consumes the payload.

To facilitate processing smaller portions of the entire payload at a time, use the optional parameter `ReturnPayload` as part of the input parameter list for every Web Service. When the Web Service client intends to process smaller payloads, the client invokes the Web Service method with the `ReturnPayload` parameter set to 'False'.

If the `ReturnPayload` parameter is set to 'False', the Web Service method does not return the payload and instead returns a set of parameters in the payload to use for retrieving the information from a database view.

### Example when `ReturnPayload = False`

The output returned contains the following information in `AdditionalInfo`:

```
<listOfStructureHeaders
  xmlns:ns0="http://xmlns.oracle.com/apps/bom/structure/service"
  xmlns="">
<AdditionalInfo>
<SessionId>7612</SessionId>
<EntityCount>0</EntityCount>
</AdditionalInfo>
.....
</listOfStructureHeaders>
```

Using the values of `SessionId` and `EntityCount`, the Web Service consumer can retrieve the payloads for individual entities from the database view `EGO_PUB_WS_OUTPUT_V` in the Oracle Product Hub instance. The columns of the database view `EGO_PUB_WS_OUTPUT_V` that you can use to retrieve the output payload in parts is provided below:

| Column Name | Description                                           |
|-------------|-------------------------------------------------------|
| SESSION_ID  | Session ID corresponding to this call to Web Service. |

| Column Name      | Description                                                |
|------------------|------------------------------------------------------------|
| WEB_SERVICE_NAME | Webservice method name.                                    |
| SEQUENCE_ID      | Sequence of the entity in the payload.                     |
| STATUS           | Marker column for consuming the Web Service entity.        |
| XMLCONTENT       | Output payload corresponding to one entity in the payload. |
| XMLCLOB          | Output in clob.                                            |
| CREATION_DATE    | Creation Date                                              |

Each row in the view corresponds to one entity returned by the Web Service method.

## Related Topics

Item Catalog Category Web Service, page G-18

Value Set Web Service, page G-2

Item Web Service, page G-30

Error Reporting, page G-80

## Error Reporting

Any errors in the execution of the Web Service are reported in the output payload Status section. The following two tables show the different errors that display when publishing to external systems or to Oracle Product Hub systems. The error information is contained in the format shown below.

### *Output Payload Status Section When Publishing to External Systems*

| Payload  | Description |
|----------|-------------|
| <Status> | -           |
| <Error>  | -           |

| Payload                                                                   | Description                    |
|---------------------------------------------------------------------------|--------------------------------|
| <code>&lt;Code&gt;EGO_INVALID_ITEM_ID&lt;/Code&gt;</code>                 | Error code                     |
| <code>    &lt;Message&gt;Invalid Inventory Item Id&lt;/Message&gt;</code> | Error message                  |
| <code>    &lt;InputIdentifier&gt;</code>                                  | Input that caused the error.   |
| <code>        &lt;Parameter&gt;</code>                                    | Parameters list for the input. |
| <code>            &lt;Name&gt;OrganizationId&lt;/Name&gt;</code>          | Parameter name                 |
| <code>            &lt;Value&gt;204&lt;/Value&gt;</code>                   | Parameter value                |
| <code>            ?..</code>                                              | -                              |
| <code>            ?..</code>                                              | -                              |
| <code>    &lt;/InputIdentifier&gt;</code>                                 | -                              |
| <code>&lt;/Error&gt;</code>                                               | -                              |
| <code>&lt;/Status&gt;</code>                                              | -                              |

***Output Payload Status Section When Publishing to Oracle Product Hub Systems***

| Payload                                                                                                                            | Description   |
|------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <code>&lt;Status&gt;</code>                                                                                                        | -             |
| <code>    &lt;Error&gt;</code>                                                                                                     | -             |
| <code>        &lt;Code&gt;EGO_SYNC_ENTITY_ERROR&lt;/Code&gt;</code>                                                                | Error code    |
| <code>            &lt;Message&gt;Entity VS115_0006 failed the import process in target system USER_ENTERED.&lt;/Message&gt;</code> | Error message |
| <code>            &lt;LanguageCode&gt;US&lt;/LanguageCode&gt;</code>                                                               | Language code |

| Payload                                                                                                                    | Description                                                                         |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <SystemCode>USER_ENTERED</SystemCode>                                                                                      | System code for the destination Oracle Product Hub system where the error occurred. |
| <InputIdentifier>                                                                                                          | Input that caused the error.                                                        |
| <Parameter>                                                                                                                | Parameters list for the input.                                                      |
| <Name>VS</Name>                                                                                                            | Parameter name                                                                      |
| <Value>VS115_0006</Value>                                                                                                  | Parameter value                                                                     |
| ?..                                                                                                                        | -                                                                                   |
| ?..                                                                                                                        | -                                                                                   |
| </InputIdentifier>                                                                                                         | -                                                                                   |
| </Error>                                                                                                                   | -                                                                                   |
| <Warning>                                                                                                                  | -                                                                                   |
| <Code>EGO_SYNC_ENTITY_WARNING</Code>                                                                                       | Warning code                                                                        |
| <Message>One of the entities associated with AG115_0006 failed the import process in target system USER_ENTERED.</Message> | Warning message                                                                     |
| <LanguageCode>US</LanguageCode>                                                                                            | Language code                                                                       |
| <SystemCode>USER_ENTERED</SystemCode>                                                                                      | System code for the destination Oracle Product Hub system where the error occurred. |
| <InputIdentifier>                                                                                                          | Input that caused the warning.                                                      |
| <Parameter>                                                                                                                | Parameters list for the input.                                                      |
| <Name>AG</Name>                                                                                                            | Parameter name                                                                      |
| <Value>AG115_0006</Value>                                                                                                  | Parameter value                                                                     |

| Payload            | Description |
|--------------------|-------------|
| ?..                | -           |
| ?..                | -           |
| </InputIdentifier> | -           |
| <Warning>          | -           |
| </Status>          | -           |

The Web Service client can extract and make use of this information to report errors returned from the Web Service call.

## Related Topics

[Oracle Product Hub Web Services Overview, page G-1](#)

[Publishing Overview, \*Oracle Product Hub User's Guide\*](#)

[Item Catalog Category Web Service, page G-18](#)

[Attribute Group Web Service, page G-12](#)

[Value Set Web Service, page G-2](#)

[Item Web Service, page G-30](#)



---

# Instructions for Restarting Concurrent Programs with a Manual Restart Mode

This appendix covers the following topics:

- Restarting Concurrent Programs with a Manual Restart Mode

## Restarting Concurrent Programs with a Manual Restart Mode

For Oracle E-Business concurrent programs, the **Restart Mode** attribute indicates how a program can be restarted in case of system failure.

The Restart Mode can be one of four values:

- **Resumable:** This program can be automatically restarted and can resume its processing at the point where it left off before the system failure.
- **Restartable:** This program can be automatically restarted but will go back to the beginning of its processing.
- **Manual:** An administrator must perform manual steps to restart this program. Product-specific documentation lists the steps required to restart any shipped programs with a Manual restart mode.
- **Unknown:** It is unknown whether this program can be restarted automatically or not. This value is the initial default setting.

**Important:** The Restart Mode for Oracle-provided programs should not be modified. If it were to be modified, the value may be overwritten without warning by future Oracle EBS software updates.

For more information about the Restart Mode, see the *Oracle E-Business Suite Setup Guide*.

For the following Oracle Product Hub concurrent programs with a Restart Mode of

Manual, an administrator must perform the steps specified prior to restarting the program following a system failure.

1. Import Catalog Items
2. EGO Spreadsheet Java Concurrent Program
3. Item/Item Revision User-Defined Attributes Concurrent Program
4. Item Org Import
5. EGO Item Import Batch Matching Concurrent Program
6. Ondemand Rule Execution Java Concurrent Program
7. Bulkload Program For Spread Sheet Upload
8. EGO Import Metadata

The manual restart steps are as follows:

1. Create a service request for data cleanup.
2. After data cleanup completes, rerun the program.



---

# Index

## A

---

- Attachment
  - change management, 10-5
  - review and approval, 10-3
  - set up repository, 9-3
- attribute group
  - adding actions, 4-52
- attributes
  - additional attributes, 4-38
  - seeded, 4-36
- Attributes
  - transaction, 4-33
- attribute usage, 4-44

## C

---

- catalog
  - assigning category, 5-9
  - assigning people, 11-12
- Catalogs
  - defining, 5-6
  - defining categories, 5-4
  - overview, 5-1
- category hierarchy
  - create, 5-10
- Category hierarchy
  - deleting, 5-11
  - updating, 5-11
- category mapping, 5-10
- Change categories, 7-1
  - defining, 7-2
  - display formats, 7-41

- function security, 7-5
- search criteria, 7-41
- Change category reports
  - defining, 7-42
- change headers
  - organization policies, 7-34
- change header types
  - associate attribute groups, 7-30
  - associate codes, 7-31
  - associate pages, 7-30
  - associate workflows, 7-31
  - configure, 7-31
  - create, 7-28
- change line types
  - associate pages, 7-38
  - create, 7-37
- Change management
  - text index, 7-39
- change management business event, 7-44
- Change Management Business Events
  - Describing Business Events, E-1
- Change management text index, 7-39
- Change management workflows
  - customize, 7-47
- Change policies, 3-16
- Classification codes
  - creating, 7-13
- content management
  - set up repository, 9-3
- creating parameter mappings, 4-52
- Creating statuses, 7-15
- Cross reference types, 3-57
- Customer users

- set up, 2-5
- custom privileges, 11-27

## D

---

- default options, 1-15
- Defining catalog categories, 5-4
- Defining catalogs, 5-6
- defining item numbers, 3-26
- Defining workflow templates, 7-17
- display formats
  - administrator level, 3-36
  - user level, 3-38
- Display formats, 3-35

## E

---

- E-business suite
  - overview of attachments, 10-1
- EGO\_DEFAULT\_OPTIONS table, 1-15
- expressions
  - rules, building, 6-16

## F

---

- folders
  - structure, 10-1

## H

---

- Header/Line type attributes, 7-25
- Header types
  - defining, 7-28

## I

---

- Import
  - attribute groups and attributes, 3-58
  - functions, 3-58
  - item catalog categories, 3-58
  - value sets, 3-58
- import formats, 3-40
- Import Workbench
  - match rules, 12-2
  - source system, 12-5
- independent values, 4-14
- Internal users, 2-4
- Item attachment
  - associate categories, 3-10
  - define categories, 3-9

- Item attribute
  - display attachment, 4-45
  - group security, 11-19
  - user-defined functions, 4-38

- Item attributes, 4-21

- item business events

  - technical information, B-1

- item business rules

  - copying item rules data to other instances, 6-43
  - creating, 6-4
  - including in other rule sets, 6-38
  - overview, 6-1
  - using on-demand rule set execution, 6-42
  - viewing rule use and attributes, 6-39

- Item catalog

  - associate attributes, 4-48
  - display formats, 3-35
  - import formats, 3-40
  - overview, 3-2
  - search criteria, 3-29
  - text index, 3-2

- item catalog category

  - adding attachment categories, 3-10
  - create, 3-2
  - creating pages, 3-28

- Item catalog category

  - associating attribute groups, 4-48
  - associating new item requests, 3-22

- Item change policy, 3-16

- Item number

  - generation rules, 3-26

- item request

  - define new item request, 3-19

- Item request

  - setting up, 3-18

- item status, 3-15

  - create, 3-15

  - edit, 3-15

- item templates

  - setting up, 3-12

## L

---

- libraries

  - Communications Product Details Library - Vertical, F-44

- Communications Services Billing Library - Vertical, F-25
- Product Management Library - Horizontal, F-23
- Sellable Product Information Library - Horizontal, F-1
- Line types
  - associate attribute groups, 7-35
  - Change line type attributes, 7-35
  - defining, 7-35

## **M**

---

- managing roles
  - add or remove members, 11-10
  - create group, 11-9
  - find group, 11-11
  - search company, 11-11
  - search group, 11-10
  - search people, 11-8
- managing roles , 11-6
- Match rules, 12-2

## **N**

---

- New item request
  - associate with an item catalog category, 3-22
  - define types, 3-18
  - define workflows, 3-21

## **O**

---

- Oracle applications technology
  - profile options, 1-3
  - set up, 1-1
  - set up prerequisites, 1-2
  - set up steps, 1-3
- Oracle isetup, 1-2

## **P**

---

- people and roles administration
  - creating roles, 11-3
  - deleting roles, 11-3
  - role mapping, 11-3
  - updating roles, 11-3
  - view roles, 11-1
- Priority codes
  - create, 7-13

- profile options
  - default options, 1-15
- programs
  - restarting concurrent programs, H-1
- publishing
  - attribute groups, 4-54
  - default source systems, 3-44
  - item catalog categories, 3-45
  - value sets, 4-16

## **R**

---

- Reason codes
  - creating, 7-13
- Report templates, 3-55
- repository
  - overview of setting up, 9-1
- Responsibilities and users
  - create responsibility, 2-1
  - define, 2-1
- Role based security, 11-11
- rule expressions
  - building, 6-16
- rules
  - copying item rules data to other instances, 6-43
  - creating, 6-4
  - including in other rule sets, 6-38
  - using on-demand rule set execution, 6-42
  - viewing rule use and attributes, 6-39
- rule sets
  - creating, 6-4

## **S**

---

- search criteria templates, 3-29
  - administrator level, 3-31
  - user level, 3-33
- security
  - catalog, 11-12
- Security, 11-11
- seeded attributes, 4-36
- seeded data
  - Communications Product Details Library - Vertical, F-44
  - Communications Services Billing Library - Vertical, F-25
  - Product Management Library - Horizontal, F-

23

Sellable Product Information Library -  
Horizontal, F-1

Source system, 12-5

Statuses

creating, 7-15

Structure names

defining, 8-5

Structures

associating a name with a type, 8-4

associating component attribute groups, 8-8

associating structure attribute groups, 8-6

creating structure names, 8-4

creating structure types, 8-2

display formats, 8-9

overview, 8-1

search criteria, 8-9

user-defined attributes, 8-6, 8-8

Structure types

associating structure names, 8-5

Supplier users

set up, 2-5

## T

---

table value sets, 4-12

identify, 4-13

Task templates

creating, 7-23

## U

---

user-defined attributes

create, 4-24

User-defined attributes, 4-21

reading data, C-1

value sets, 4-1

user-defined functions

create, 4-39

create parameter mapping, 4-41

User-defined functions, 4-38

## V

---

value sets

create, 4-1

versions, 4-7

## W

---

Web Services

attribute groups, G-12

error reporting, G-80

item catalog categories, G-18

items, G-30

Output Payload, G-79

overview, G-1

value sets, G-2

Workflows

change management, A-1

Workflow templates

defining, 7-17